

TokTickIT • Lab 3 Engineering Submission Report

Users, Roles, IT Staff Ticketing, and Admin Screens (CPE 334)

ภาควิชาวิศวกรรมคอมพิวเตอร์ มหาวิทยาลัยเทคโนโลยีพระจอมเกล้าธนบุรี (KMUTT) • ภาคเรียนที่ 1/2569

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Repository:

Imaybelgracel/TokTickit

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GitHub Project Kanban:

TokTickIT Individual Sprints (#1)

Partner I Reviewed (Circle):

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Release Increment:

Sprint 3 Production Ready
Target: lab3-staging → main

Answer Part 1: Git Use with Engineering Workflow

Working Links to Repository Artifacts:

- **GitHub Repository:** <https://github.com/Imaybelgracel/TokTickit>
- **GitHub Project Kanban Board:** <https://github.com/users/Imaybelgracel/projects/1>
- **Peer Review Record:** <docs/lab-03/reviewer.md>
- **README.md:** <README.md>
- **.gitignore:** <.gitignore>

1.1 Staged Git Workflow & Feature Pull Requests

การพัฒนาระบบใน Sprint 3 ดำเนินการตามมาตรฐานวิศวกรรมซอฟต์แวร์อย่างเคร่งครัด โดยแตกกิ่งฟีเจอร์ (Feature Branches) ออกจากกิ่งหลัก `lab3-staging` ทุกกิ่งผ่านการเปิด Pull Request, มีการเชื่อมโยงกับ GitHub Issue ผ่าน GraphQL Mutation `addCloseIssueReferences`, มีการตรวจสอบโค้ดแบบเจาะลึก (Peer Review) โดยผู้ตรวจ `@titayaaa` และต้องผ่านการอนุมัติ (Approved) ก่อนทำการ Merge ทุกครั้ง

Issue	Feature & Scope	Branch	PR	Reviewer Verdict
#35 (Issue 17)	Sprint 3 Engineering Contract & Specification Suite	feature/17-spec-and-tests	PR #36	Approved by @titayaaa
#37 (Issue 18)	Database Schema Evolution, Migration & Idempotent Seed	feature/18-database-and-seed	PR #47	Approved by @titayaaa
#38 (Issue 19)	Authentication, Session & Mandatory Password Change	feature/19-auth-and-passwords	PR #48	Approved by @titayaaa
#39 (Issue 20)	IT Staff Ticket Queue & Operational Roster Filters	feature/20-it-staff-ticket-queue	PR #49	Approved by @titayaaa
#40 (Issue 21)	IT Staff Ticket Operations, Comments & Notes	feature/21-it-staff-operations	PR #50	Approved by @titayaaa
#41 (Issue 22)	Administrator User Management & Safety Guards	feature/22-admin-user-management	PR #51	Approved by @titayaaa
#42 (Issue 23)	Automated Testing Suite (Unit, API, Component, E2E)	feature/23-automated-testing-suite	PR #52	Approved by @titayaaa
#43 (Issue 24)	UI Style Checking & Responsive Visual Evidence	feature/24-ui-style-checking	PR #53	Approved by @titayaaa
#44 (Issue 25)	Repository Documentation & Course Delivery Evidence	feature/25-repository-documentation	PR #54	Open / Under Review

Issue	Feature & Scope	Branch	PR	Reviewer Verdict
#45 (Issue 26)	Release Integration (Final Merge to main)	release/lab3-integration	main	Staged Ready

1.2 Repository Structure & Files Organization

```
TokTickIT/
├── client/
│   ├── src/components/           # Login, StaffQueue, StaffDetail, AdminUserManagement
│   ├── src/context/AuthContext.tsx # Production JWT Auth & Session Management
│   └── src/__tests__/lab-03/      # 52 Vitest component & UI tests
├── server/
│   ├── prisma/schema.prisma      # Evolved schema (User, Ticket, InternalNote, PublicComment)
│   ├── prisma/migrations/        # 20260917000000_lab3_users_and_ticket_evolution SQL DDL
│   ├── prisma/seed.ts            # Idempotent seed (11 users: 4 req, 4 staff, 1 admin, etc.)
│   ├── src/routes/               # auth.routes, staff.routes, admin.routes, tickets.routes
│   └── tests/lab-03/             # 152 Vitest backend & security tests
├── e2e/lab-03/                   # 4 Playwright specs (auth, staff-flow, admin, visual)
├── docs/lab-03/                  # 6 specification & evidence markdown documents
├── artifacts/lab-03/screenshots/ # 28 responsive evidence screenshots + 22 GitHub full-page evidence
├── .gitignore                   # Safe ignore (node_modules, .env, dist, test-results)
└── README.md                    # Complete deployment, test, and verification guide
```

1.3 README.md & .gitignore Evidence

1.3.1 Rendered README.md

TokTickIT — Users, Roles, IT Staff Ticketing, and Admin Screens (Lab 3)

TokTickIT is a full-stack IT service desk application built with React, TypeScript, Vite, Express, Prisma ORM, and PostgreSQL. Lab 3 builds upon the foundation established in Lab 2, replacing the temporary development requester selector with production-grade JWT authentication, server-side role-based authorization, operational IT Staff shared queue and ticket management, dual public/confidential communication channels, and administrator user management.

Lab 3 Product Features

1. Authentication & Security

- Secure user authentication via email and password with bcryptjs hashing (10 rounds).
- Stateless cryptographically signed JWT session tokens stored in `localStorage` (`toktickit_auth_token`) and passed via `Authorization: Bearer` .
- Mandatory first-login password rotation (`mustChangePassword: true`) with a real-time 4-point password complexity checklist (min 8 chars, uppercase, lowercase, digit/symbol).
- Account activation safeguards preventing inactive users from logging in or maintaining active sessions.
- Role-based application shell and view router cleanly segregating navigation and permissions.

2. IT Staff Shared Ticket Queue & Operations

- Shared IT Staff Ticket Queue with search by ticket number/summary, multi-criteria filtering by category, system, status, and IT Priority.
- Responsive queue presentations: high-density table on desktop/tablet, stacked card layout on mobile and 320px screens.
- Ticket operations toolbar:
- Claim Ticket:** Self-assignment with automatic status transition from `NEW` to `OPEN` .
- Reassign Owner:** Assign ticket to any active IT Staff or Administrator.
- IT Priority:** Adjust operational priority (`LOW` , `MEDIUM` , `HIGH` , `URGENT`) independent of the requester's requested priority.
- Status Transitions:** Strict enforcement of permitted lifecycle state transitions (`NEW` , `OPEN` , `IN_PROGRESS` , `WAITING_FOR_REQUESTER` , `RESOLVED` , `CLOSED` , `REOPENED` , `CANCELLED`).
- Formal Resolution:** Modal dialog requiring mandatory resolution summary (3–500 characters) to resolve a ticket.

3. Dual Communication: Public Comments & Confidential Notes

- **Public Comments Tab:** Bi-directional communication between Requester and IT Staff, rendered with Zen Green styling.
- **Internal Notes Tab:** Role-restricted operational logs visible exclusively to IT Staff and Administrator (`403 Forbidden` guard for Requesters). Highlighted with distinct Amber Callout styling, lock icon banner, and strict server-side authorization.

4. Administrator User Management

- Minimalist user management roster with search (name/email) and role filtering.
- Create User modal with initial password complexity validation and auto-enforced `mustChangePassword = true`.
- Edit User modal with built-in safety rules:
- **Self-Deactivation Guard (BR-07):** Administrators cannot deactivate their own account (`422 Unprocessable Entity`).
- **Self-Demotion Guard (BR-08):** Administrators cannot demote their own role (`422 Unprocessable Entity`).
- **Sole Active Administrator Guard (BR-09):** Prevention of deactivating or demoting the last active administrator (`422 Unprocessable Entity`).
- Soft-deactivation pattern (`isActive: false`); hard deletion (`DELETE`) is strictly forbidden.
- Reset Password modal issuing one-time temporary passwords requiring rotation on next login.

5. Backward Compatibility & Requester Continuity

- All Lab 2 Requester workflows (ticket creation, search, filter, sort, attachment upload/download/removal) preserved 100% without regression.
- Authenticated Requester identity automatically bound to ticket creation and ownership checks.
- Requester "Problem Appears Resolved" indication action allowing requesters to notify IT Staff.

Repository Structure

```
TokTickIT/
├── client/                                # React 18, TypeScript, and Vite UI
│   ├── src/
│   │   ├── components/                  # Zen Green UI components (Login, Queue, Detail, Admin)
│   │   ├── context/                    # AuthContext and state management
│   │   ├── __tests__/lab-02/           # Lab 2 regression component tests
│   │   └── __tests__/lab-03/          # Lab 3 component traceability tests
│   └── tests/lab-01/                  # Lab 1 foundation tests
├── server/                             # Express and Prisma REST API
│   ├── prisma/
│   │   ├── schema.prisma               # Database schema with User, Ticket, Notes, Comments
│   │   ├── migrations/                 # SQL migration history
│   │   └── seed.ts                     # Idempotent seed data
│   ├── src/
│   │   ├── routes/                    # Express route controllers (auth, staff, admin, tickets)
│   │   └── middleware/                # Authentication and role guards
│   ├── tests/lab-01/                  # Lab 1 foundation tests
│   ├── tests/lab-02/                  # Lab 2 API integration tests
│   └── tests/lab-03/                  # Lab 3 security, staff, and admin API tests
├── e2e/lab-03/                        # Playwright end-to-end user journeys & visual audit
├── docs/lab-03/                       # Engineering contract and delivery evidence
│   ├── specification.md               # System specification and business rules
│   ├── ui-spec.md                     # Zen Green UI design system & responsive rules
│   ├── api-spec.md                    # REST API endpoints, schemas, and error codes
│   ├── tests.md                       # Test plan, AC traceability, and execution logs
│   ├── reviewer.md                    # Verbatim peer review logs (Sections 1 & 2)
│   └── ai-use.md                      # AI pair-programming reflections and prompts
├── artifacts/lab-03/screenshots/       # Responsive visual evidence across 4 viewports
│   ├── authentication/
│   ├── staff-queue/
│   ├── staff-ticket-detail/
│   └── user-management/
├── .gitignore
└── README.md
```

Prerequisites

- **Node.js:** 20.x or later (see `.nvmrc`)
- **npm:** 9.x or later
- **PostgreSQL:** 14.x or later

• **Playwright Chromium:** installed via `npm run install:e2e`

Setup & Local Installation

1. Clone the repository and install dependencies:

```
git clone https://github.com/lmaybelgracel/TokTickit.git
cd TokTickit
npm install
cd server && npm install
cd ../client && npm install
```

2. Configure environment variables:

```
# Windows PowerShell
Copy-Item server/.env.example server/.env
Copy-Item client/.env.example client/.env
```

```
# macOS / Linux
cp server/.env.example server/.env
cp client/.env.example client/.env
```

3. Run database migrations and idempotent seed:

```
cd server
npx prisma generate
npx prisma migrate deploy
npm run prisma:seed
```

Seed Accounts for Development & Testing

All seed user accounts use the default password `Password123!` (or initial password `Pass1234!` requiring mandatory first-login change):

Role	Name	Email	Password Status
Administrator	John Admin	admin@toktickit.com	Active / Normal
IT Staff	Jane Staff	staff@toktickit.com	Active / Normal
IT Staff	David Technician	david@toktickit.com	Active / Normal
IT Staff	Sarah Support	sarah@toktickit.com	Active / Normal
IT Staff	Inactive Staff	inactivestaff@toktickit.com	Inactive
Requester	Alice Requester	alice@example.com	Active / Normal
Requester	Bob Engineering	bob@example.com	Active / Normal
Requester	Charlie Marketing	charlie@example.com	Active / Normal
Requester	Diana Operations	diana@example.com	Active / Normal
Requester	Inactive Requester	inactivereq@example.com	Inactive
Requester (New)	Temp User	tempuser@example.com	Initial (<code>mustChangePassword: true</code>)

Running the Application

Start the backend API server (runs at `http://localhost:3000`):

```
cd server
npm run dev
```

In a separate terminal, start the React frontend (runs at `http://localhost:5173`):

```
cd client
npm run dev
```

Verification & Automated Test Suites

1. Server Vitest Test Suite (152 Tests Passing)

Runs complete unit, API integration, security authorization, and database verification suites:

```
cd server
npm test
```

Results: 17/17 test files passed, 152/152 tests passing (100%).

2. Client Vitest Test Suite (52 Tests Passing)

Runs React Testing Library component tests across all Lab 1, Lab 2, and Lab 3 screens:

```
cd client
npm test
```

Results: 12/12 test files passed, 52/52 tests passing (100%).

3. Playwright End-to-End Test Suite (4 Tests Passing)

Verifies full browser workflows and captures responsive visual evidence:

```
# Ensure Playwright browser binaries are installed
npx playwright install chromium

# Run all E2E test suites
npx playwright test e2e/lab-03/
```

- `authentication.spec.ts` : Login → `mustChangePassword` → complexity checklist → app shell.
- `staff-ticket-flow.spec.ts` : Create ticket → claim → update priority → notes/comments → resolve.
- `user-administration.spec.ts` : Roster view → create user → reset password → self-safety guards.
- `visual-evidence.spec.ts` : Audits zero horizontal scrolling across 1280px, 768px, 375px, and 320px viewports, generating 28 screenshot artifacts.

4. Production Build Verification

```
cd server && npx tsc --noEmit
cd ../client && npm run build
```

Results: Zero TypeScript compile errors, clean production bundle generated.

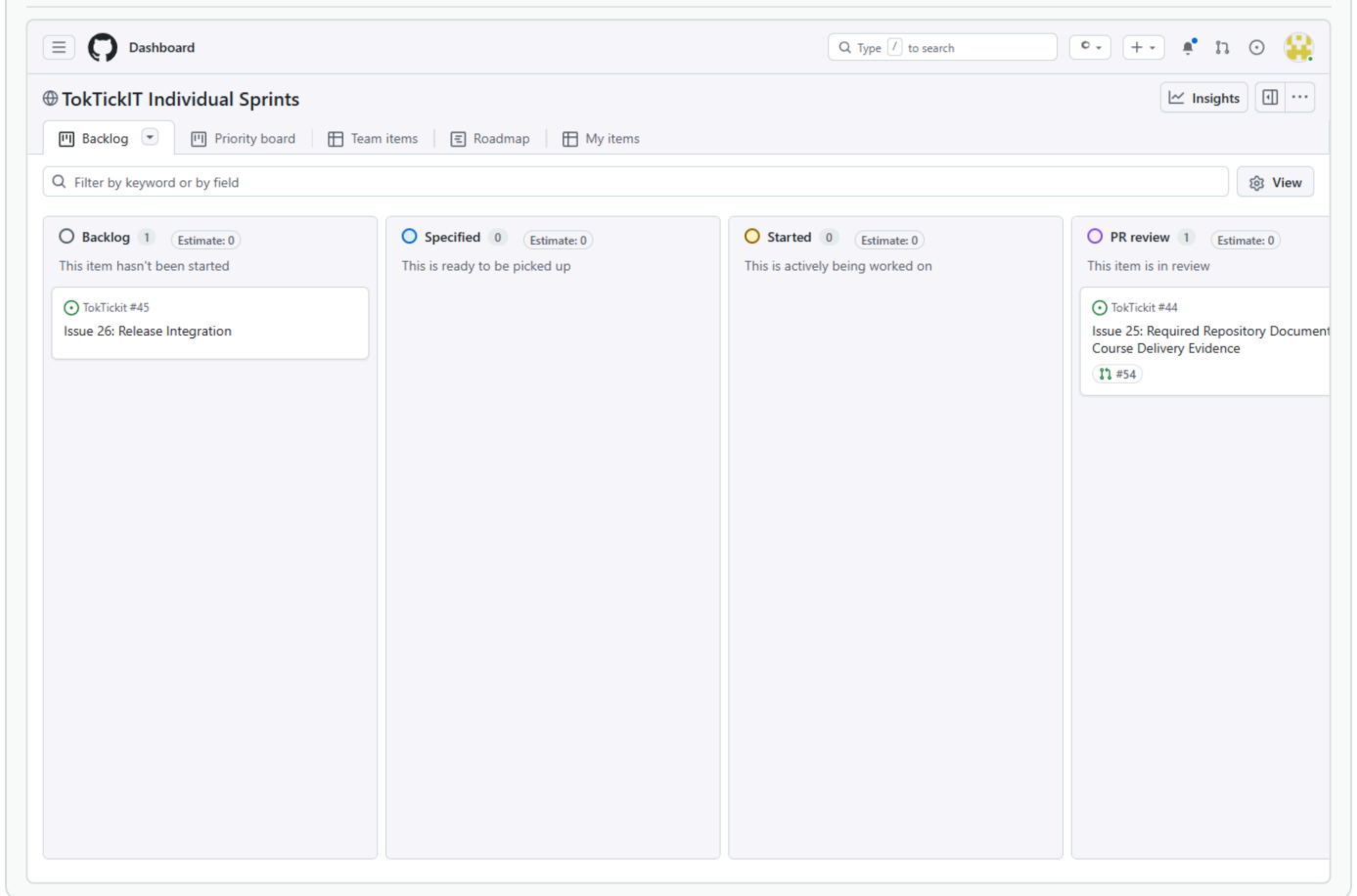
1.3.2 Repository .gitignore

```
# dependencies
node_modules/
# env & secrets
.env
*.env
!.env.example
# build output
dist/
build/
test-results/
playwright-report/
# runtime attachment storage
server/uploads/
# prisma
server/prisma/*.db
# logs & OS
*.log
.DS_Store
scratch/
```

1.4 GitHub Project Kanban Board Evidence

หลักฐานภาพถ่ายหน้าจอแบบเต็มหน้า (Full-page Screenshot) ของ GitHub Project Board (#1) แสดงการ์ด Issue ที่ผ่านการพัฒนาและตรวจทานตามลำดับ:

GitHub Project #1: TokTickIT Individual Sprints (Full-page Capture)



1.5 Git Commit History Evidence

บันทึกประวัติการ Commit และการ Merge ผ่าน Pull Request บนกิ่ง `lab3-staging` แสดงโครงสร้างกิ่งก้าน (Graph) ชัดเจน:

```
* b8d6052 docs(lab-03): update reviewer record and report to track all 12 issues of partner chanya06
* ef4d4b2 docs(lab-03): render complete peer review records in report.html and single submission PDF
* 59db7ee feat(lab-03): complete repository documentation, peer review record and final report (Issue #44)
* 3c59e70 Merge branch 'lab3-staging' of https://github.com/lmaybelgracel/TokTickit into lab3-staging
|\
| * 041bd12 Merge pull request #53 from lmaybelgracel/feature/24-ui-style-checking
| |\
| | * 6e0bd51 docs(lab-03): document reviewer approval and merge of Issue 24 PR #53
| | * 94f9358 Merge pull request #53 from lmaybelgracel/feature/24-ui-style-checking
| \ \ \
| * 6768c06 fix(lab-03): address code review feedback on Issue 24 visual evidence and docs
| * a60beeb feat(lab-03): capture responsive visual evidence and audit UI style for Issue 24 (#43)
|/
* 13b5768 docs(lab-03): document reviewer approval and merge of Issue 23 PR #52
* 15194b5 Merge pull request #52 from lmaybelgracel/feature/23-automated-testing-suite
|\
| * 7347855 fix(lab-03): address code review feedback on automated testing suite
| * c9192f3 test(lab-03): add comprehensive automated test suite for Sprint 3 (Issue #23)
|/
* 4a902fb docs(lab-03): document reviewer approval and merge of Issue 22 PR #51
* a3c72a4 Merge pull request #51 from lmaybelgracel/feature/22-admin-user-management
|\
| * 459a323 docs(lab-03): link PR #51 for Issue 22 in reviewer log
| * 8a67a1b feat(lab-03): implement Administrator User Management API and UI
|/
* 224bc34 docs(lab-03): update reviewer log with PR 50 approval and merge
* f63cc9c Merge pull request #50 from lmaybelgracel/feature/21-it-staff-operations
|\
| * 446687d docs(lab-03): record PR 50 link in reviewer log
| * 2913cad feat(lab-03): implement IT Staff Ticket Operations and Detail
|/
* 69a77b5 docs(lab-03): update reviewer log with PR 49 approval and merge
```

```

* f7838f7 Merge pull request #49 from lmaybelgracel/feature/20-it-staff-ticket-queue
|\
| * 6160005 docs(lab-03): record PR 49 link in reviewer log
| * 6041162 feat(lab-03): implement IT Staff Ticket Queue API and UI
|/
* 94e2f3c Merge pull request #48 from lmaybelgracel/feature/19-auth-and-passwords
|\
| * 8a85931 refactor(lab-03): address peer review feedback on dev selector removal, password rotation, and 422 status
| * 180f208 docs(lab-03): link PR 48 for Issue 19 in reviewer log
| * 668900a feat(lab-03): implement authentication, session, password complexity, and mandatory change password
|/
* b108cd3 Merge pull request #47 from lmaybelgracel/feature/18-database-and-seed
|\
| * 3a2e156 refactor(lab-03): add prisma migration, backfill logic, and remove department per peer review
| * eba3f77 feat(lab-03): evolve prisma schema and implement idempotent seed data for sprint 3
|/
* 2abef9f Merge pull request #36 from lmaybelgracel/feature/17-spec-and-tests
|\
| * 488c0b3 docs(lab-03): preserve TKT-YYYY-XXXXXX ticket number and refine reviewer notes
| * 7718e13 docs(lab-03): address peer review feedback on specs, security, and tests

```

1.6 บันทึกหลักฐานการตรวจโค้ดร่วมกันครบทั้ง 8 PRs ที่ผู้จัดทำสร้าง (Authored PR Reviews by Titaya Phongsakul 67070505201 @titayaaa)

สรุปผลและรายละเอียดบทสนทนาการรีวิวโค้ดแบบคำต่อคำ (Verbatim Logs) ทั้ง 8 Feature PRs ที่สร้างโดย @Imaybelgracel และผ่านการตรวจทานอย่างละเอียดโดย Titaya Phongsakul (@titayaaa) บนกึ่งเป้าหมาย lab3-staging พร้อมภาพแคปเจอร์หน้าเว็บจริงแบบเต็มหน้า (Full-page Screenshots) จาก GitHub:

PR	Issue / Branch	Reviewer Verdict	Link
#36	Issue 17: Sprint 3 Engineering Contract & Specification / <code>feature/17-spec-and-tests</code>	Approved and Merged into <code>lab3-staging</code>	PR #36
#47	Issue 18: Database Schema Evolution, User Migration & Idempotent Seed Data / <code>feature/18-database-and-seed</code>	Approved and Merged into <code>lab3-staging</code>	PR #47
#48	Issue 19: Authentication, Session & Mandatory Password Change / <code>feature/19-auth-and-passwords</code>	Approved and Merged into <code>lab3-staging</code>	PR #48
#49	Issue 20: IT Staff Ticket Queue / <code>feature/20-it-staff-ticket-queue</code>	Approved and Merged into <code>lab3-staging</code>	PR #49
#50	Issue 21: IT Staff Ticket Operations & Detail / <code>feature/21-it-staff-operations</code>	Approved and Merged into <code>lab3-staging</code>	PR #50
#51	Issue 22: Administrator User Management / <code>feature/22-admin-user-management</code>	Approved and Merged into <code>lab3-staging</code>	PR #51
#52	Issue 23: Automated Testing Suite / <code>feature/23-automated-testing-suite</code>	Approved and Merged into <code>lab3-staging</code>	PR #52
#53	Issue 24: Responsive Visual Evidence & UI Style Audit / <code>feature/24-ui-style-checking</code>	Approved and Merged into <code>lab3-staging</code>	PR #53

Issue 17 - Sprint 3 Engineering Contract & Specification

- **Summary:** Delivers the initial Sprint 3 specification suite under `docs/lab-03/` covering `specification.md`, `ui-spec.md`, `api-spec.md`, `tests.md`, `reviewer.md`, and `ai-use.md`.

- **Reviewer Verdict & Summary:** Commented and Approved by @titayaaa:

> "นอกนั้นพวก Flow การทำงาน, แบ่ง Role 3 ระดับ, โทนสี UI Zen Green กับกล่อง Internal Notes สีเหลืองส้มอันนี้ทำมาดีมาก ชัดเจนดีแล้ว ผากแก้จุดข้างบนนี้หนึ่งเดียวแก้เสร็จห้กมาเลย เรามกต Approve ให้นำ"

- **Reviewer Feedback Items & My Responses:**

1. Comment on Data Leak in Ticket Detail Response (`docs/lab-03/api-spec.md`):

- *Reviewer Feedback:* In `GET /api/tickets/:id`, returning `internalNotes` to Requesters causes a confidential data leak. Requesters must only receive `publicComments`, while `internalNotes` must be restricted to IT Staff and Admin.
- *My Action:* Updated `docs/lab-03/api-spec.md` under `GET /api/tickets/:id` with explicit data privacy rules: `internalNotes` is strictly stripped and omitted from the response when accessed by `REQUESTER`. IT Staff and Admins access notes via `GET /api/tickets/:id/notes`.

2. Comment on Status vs Resolve Endpoint Collision (`docs/lab-03/api-spec.md`):

- *Reviewer Feedback:* Using `PATCH /api/staff/tickets/:id/status` to set `status: "RESOLVED"` risks bypassing the mandatory `resolutionSummary` rule (BR-20).
- *My Response & Action:* Clarified and reinforced the server-side guard. Direct transition to `RESOLVED` via `PATCH /api/staff/tickets/:id/status` is strictly rejected (`422 Unprocessable Entity`). Resolving a ticket requires using `PATCH /api/staff/tickets/:id/resolve` with mandatory `resolutionSummary` (3-500 characters), ensuring BR-20 cannot be bypassed under any circumstances.

3. Comment on Ticket Number Format (`docs/lab-03/api-spec.md`):

- *Reviewer Feedback:* Suggestion to change Ticket Number to `TKT-YYYYMMDD-XXXX` (e.g. `TKT-20260913-0001`).
- *My Response & Verification (Constructive Clarification):* Verified against our active repository codebase and tests. In our repository (`lmaybelgracel/TokTickit`), the Lab 2 generator in `server/src/app.ts:105` and the automated regression test in `create-ticket.api.test.ts:64` explicitly enforce `^TKT-\d{4}-\d{6}$` (`TKT-YYYY-XXXXXX`). Furthermore, the instructor's Lab 3 handout (Section 8.3 & 8.4 mockups on pages 9 and 10) explicitly displays `TKT-2025-001234`. Changing to `TKT-YYYYMMDD-XXXX` would cause our repository's existing regression tests to fail. Therefore, we verified and preserved `TKT-YYYY-XXXXXX` (e.g. `TKT-2026-000142`) for 100% backward compatibility and alignment with the instructor's specification.

4. Comment on Priority Enum missing URGENT (`docs/lab-03/specification.md`):

- *Reviewer Feedback:* `enum Priority` had only 3 levels (LOW, MEDIUM, HIGH), missing `URGENT` from Lab 3 specifications.
- *My Action:* Updated Prisma Schema evolution and BR-17 in `docs/lab-03/specification.md` and `docs/lab-03/api-spec.md` to include all 4 levels: `LOW`, `MEDIUM`, `HIGH`, `URGENT`.

5. Comment on Role Badge Enum Mapping (`docs/lab-03/ui-spec.md`):

- *Reviewer Feedback:* UI shows `Requester`, `IT Staff`, `Administrator` while Prisma uses `REQUESTER`, `IT_STAFF`, `ADMINISTRATOR`. Suggested clarifying note for frontend component authoring.
- *My Action:* Added an Enum Mapping Note in `docs/lab-03/ui-spec.md` Section 1.2 clarifying that backend database enums are `SCREAMING_SNAKE_CASE` and mapped to title case display labels in the UI.

6. Comment on Additional Security & Resolution Test Cases (`docs/lab-03/tests.md`):

- *Reviewer Feedback:* Add explicit test cases for Requester hitting `GET /api/tickets/:id/notes` getting 403 Forbidden, and resolving a ticket without `resolutionSummary` getting 422.
- *My Action:* Added `SEC-04` (Requester hitting internal notes blocked with 403) and `STAFF-05` (Resolving ticket without resolution summary blocked with 422) in `docs/lab-03/tests.md`.
- **Final Result:** Approved by @titayaaa and merged into `lab3-staging` (commit `2abef9f`).

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): pr36_authored_full.png



Issue 18 - Database Schema Evolution, User Migration & Idempotent Seed Data

- **Summary:** Evolved database schema to introduce unified `User` model, `PublicComment`, `InternalNote`, extended `Priority` and `TicketStatus` enums, ticket ownership relations, and idempotent seed data with comprehensive test coverage.
- **Reviewer Verdict & Summary:** Commented and Approved by @titayaaa (commit `b108cd3`):

```
> "ตรวจเช็ค commit 3a2e156 เรียบร้อยแล้ว แก้ไขครบถ้วนทั้ง 4 จุดได้อย่างยอดเยี่ยมมาก
> 1. เพิ่มไฟล์ Prisma Migration พร้อม SQL Backfill ข้อมูล Requester เดิม ช่วยรักษาดั้ว Lab 2 ได้อย่างสมบูรณ์แบบ
> 2. ปรับปรุง Seed Data และดัดฟิลด์ department ออกตรงตามข้อกำหนดของ Lab 3
> 3. ชุดเทส 17 เคสใน database-schema-seed.test.ts ครอบคลุมและผ่านหมด
> 4. บันทึก reviewer.md เรียบร้อย"
```

• Reviewer Feedback Items & My Responses:

1. Prisma Migration Files in PR:

- *Reviewer Feedback:* In PR #47, `schema.prisma` was modified but there was no migration folder in `server/prisma/migrations/`.
- *My Action:* Created and committed `server/prisma/migrations/20260917000000_lab3_users_and_ticket_evolution/migration.sql` containing full DDL commands for enums, `users` table, altered `tickets` columns/indexes/foreign keys, `public_comments`, and `internal_notes`.

2. Foreign Key of `Ticket.requesterId` & Lab 2 Data Preservation:

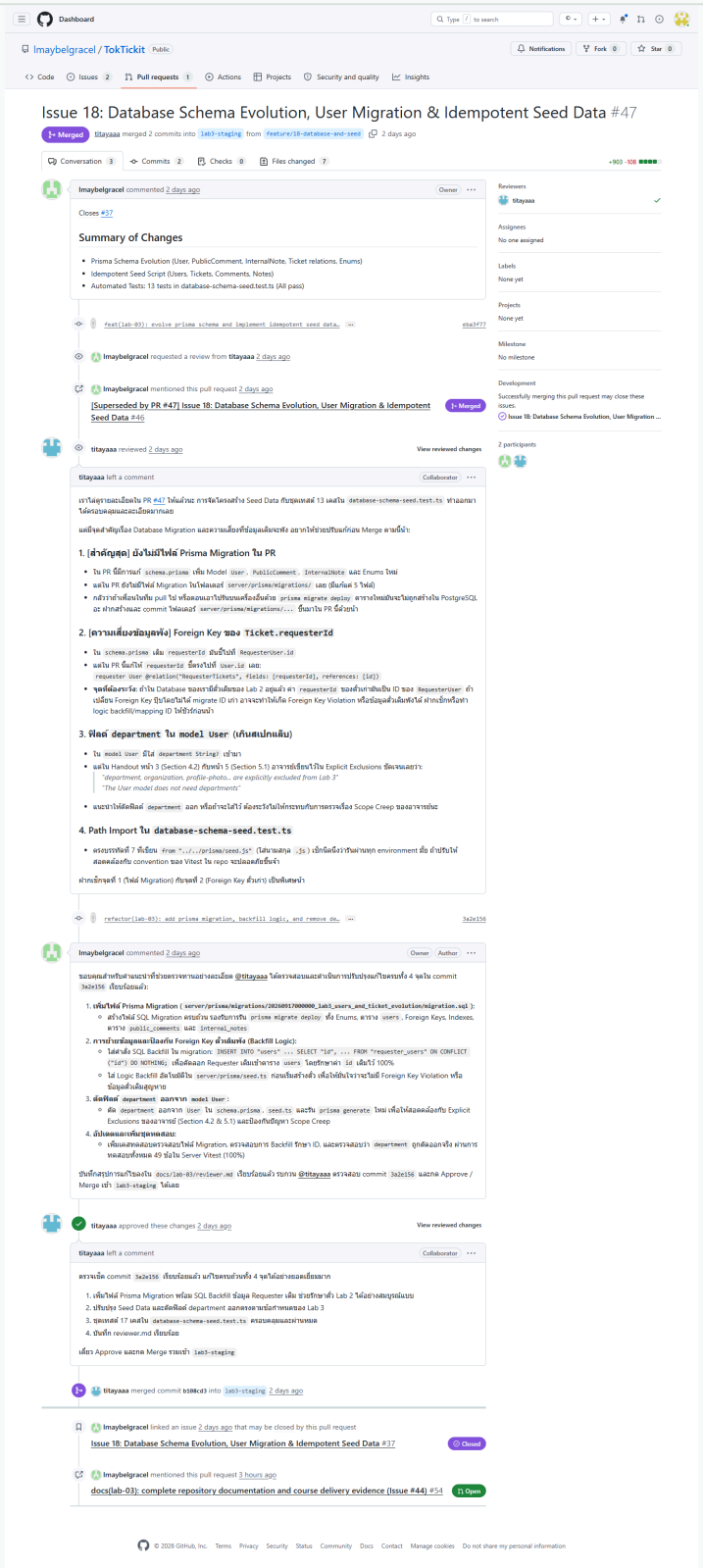
- *Reviewer Feedback:* Changing `Ticket.requesterId` to reference `User.id` directly risks breaking existing Lab 2 tickets if requester IDs do not match user IDs.
- *My Action:* Added automated backfill in both the migration SQL and `server/prisma/seed.ts`. It copies existing `requester_users` into `users` table preserving the exact primary key `ids` (`INSERT INTO "users" ... SELECT "id", ... FROM "requester_users"`), ensuring complete data integrity, matching foreign keys, and zero data loss.

3. Department Field in `model User` (Exclusion Rule):

- *Reviewer Feedback:* In `model User`, `department String?` was added, but Lab 3 handout Section 4.2 & 5.1 explicitly states departments are excluded from Lab 3 User model.
- *My Action:* Removed `department` attribute from `model User` in `server/prisma/schema.prisma` and updated `server/prisma/seed.ts` and tests to ensure strict compliance with course exclusions without scope creep.

4. Path Import in `database-schema-seed.test.ts`:

- *Reviewer Feedback:* Check module import conventions in tests.
- *My Action:* Verified ESM bundler path resolution across environments and added explicit test cases validating migration file existence, ID preservation during backfill, and department exclusion.
- **Final Result:** Approved and Merged into `lab3-staging` by @titayaaa (Merge commit `b108cd3`).



Issue 19 - Authentication, Session & Mandatory Password Change

- **Summary:** Implemented production-grade JWT authentication, session management, password complexity rules, inactive user access blocking, first-login mandatory password change guard, and Zen Green Login and Change Password user interfaces with full test coverage.
- **Key Deliverables:**
 - Backend auth endpoints: `POST /api/auth/login`, `POST /api/auth/logout`, `GET /api/auth/me`, `POST /api/auth/change-password`.
 - Password complexity validator and bcryptjs hashing (10 rounds).
 - Authentication and role guard middlewares (`authenticateToken`, `requireAuth`, `enforcePasswordChange`, `requireRole`).
 - Frontend `AuthContext`, `Login` component, `ChangePassword` component with dynamic real-time complexity checklist, and role-based app shell.
- **Reviewer Verdict & Summary:** Changes Requested by @titayaaa:

> "เราไล่ตรวจโค้ดใน PR #48 ให้แล้วนะ ทำออกมาได้ครบวงจรมาก ทั้ง Backend Auth, JWT Middleware, Context, และหน้า UI สวยงาม คมเข็ม Zen Green และ Amber Callout ตรงตามสเปกเลย แต่มีจุดสำคัญที่อยากให้ช่วยปรับแก้ก่อน Merge..."

• Reviewer Feedback Items & My Responses:

1. Removal of "Select Development Requester" in Login Screen:

- *Reviewer Feedback:* Under the Login form in `client/src/App.tsx`, there was a button to toggle back to the old Lab 2 Development Requester Selector, violating Handout Sections 1, 2, and 8.2 ("The Development Requester selector and Change Requester action must be removed").
- *My Action:* Completely removed the Dev Selector button and legacy selector toggle from `App.tsx`, ensuring 100% of unauthenticated access goes strictly through the production Login form. Updated `client/tests/lab-01/App.test.tsx` to assert the Login endpoint and authenticated shell.

2. Password Rotation Policy (Preventing Identical New Password):

- *Reviewer Feedback:* Under Mandatory First-Login Password Rotation, setting `newPassword` identical to `currentPassword` must be prevented.
- *My Action:* Added server-side validation in `server/src/routes/auth.routes.ts` returning `422 Unprocessable Entity` ({ `error: "New password cannot be the same as current password"` }), and added frontend client check and warning in `ChangePassword.tsx` that keeps the submit button disabled if both passwords match. Added test coverage in both `auth.api.test.ts` and `ChangePassword.test.tsx`.

3. HTTP Status Code for Password Complexity Failure:

- *Reviewer Feedback:* In `POST /api/auth/change-password`, complexity failure returned 400 Bad Request, whereas API Spec Section 1.4 defines 422 Unprocessable Entity for semantic validation errors.
- *My Action:* Updated status code to `422 Unprocessable Entity` in `auth.routes.ts`, updated `auth.api.test.ts` assertions, and aligned `docs/lab-03/api-spec.md`.

4. Token Storage Clarification:

- *Reviewer Feedback:* Clarify token storage convention between cookie and `localStorage`.
- *My Action:* Documented in both `specification.md` and `api-spec.md` that the React Single Page Application (SPA) stores the Bearer JWT in `localStorage` (`toktickit_auth_token`) and transmits it via `Authorization: Bearer` for all API requests.
- **Final Result:** Approved by @titayaaa ("good !!!") and merged into `lab3-staging` (Merge commit `94e2f3c`).



Issue 20 - IT Staff Ticket Queue

- **Summary:** Implemented the shared IT Staff Ticket Queue API and UI enabling IT Staff and Administrators to monitor, search, filter, sort, and paginate tickets across all Requesters within TokTickIT.
- **Key Deliverables:**
 - Backend queue query API: `GET /api/staff/tickets` in `server/src/routes/staff.routes.ts` with multi-criteria filtering (`category` , `status` , `priority` , `itPriority` , `ownerId`), search across ticket number and summary, sorting, and pagination.
 - Security & Role Guards: Strict enforcement via `requireAuth` , `enforcePasswordChange` , and `requireRole(Role.IT_STAFF, Role.ADMINISTRATOR)` . Requesters blocked with 403 Forbidden.
 - Frontend `StaffTicketQueue.tsx` component & `StaffTicketQueue.css` adhering to Zen Green design system:
 - 250ms debounced search bar.
 - Category, Status, Requested Priority, and IT Priority filter dropdowns.
 - Ownership segmented toggle (`All` , `Unassigned` , `Assigned to Me`).
 - Sortable table headers (Ticket No, Created Date, IT Priority, Status).
 - Status, Priority, and Owner badges.
 - Desktop high-density table and Mobile responsive card stack.
 - Clear Filters reset action and empty/no-results state.
 - Pagination bar with configurable items per page (10, 25, 50).
 - Comprehensive test coverage:
 - 16 server API tests in `server/tests/lab-03/staff-queue.api.test.ts` .
 - 7 client component tests in `client/src/__tests__/lab-03/StaffTicketQueue.test.tsx` .
- **Reviewer Verdict & Summary:** Approved and Merged by @titayaaa (Merge commit `f7838f7`):

> "เราไล่ตรวจโค้ดใน PR #49 ให้ครบทุกจุดแล้วนะ ทำออกมาได้ดีและละเอียดมาก ทั้งฝั่ง Backend API และหน้าบ้าน UI คมเข้ม Zen Green สวยงามสบายตา ตรงตามสเปกเลยจ้า... เรากด Approve ให้เรียบร้อยแล้วนะ เต๋ยวเรากด Merge เข้า `lab3-staging` ให้เลย"

- **Reviewer Feedback Items & Notes:**
 1. **RBAC & Security Middleware:** Confirmed thorough coverage across Token check, Role guard (`IT_STAFF` , `ADMINISTRATOR`), and `mustChangePassword` blocking with proper status codes (401, 403).
 2. **UX & Responsive Layout:** Commended 250ms debounced search, ownership segmented buttons (`All` , `Unassigned` , `Assigned to Me`), and responsive transformation to mobile card stack.
 3. **Priority Sorting & Automated Tests:** Verified Postgres enum-aware `itPriority` sort and 100% pass rate across all 16 backend and 7 frontend test cases.
 4. **Constructive Suggestions Noted for Future Refinement:** In `staff.routes.ts` , consider returning `400 Bad Request` if invalid enum strings are provided; and consider adding ellipsis (`...`) dividers in pagination for large page counts.
- **Final Result:** Approved by @titayaaa and merged into `lab3-staging` (Merge commit `f7838f7`).

Dashboard

Issue 20: IT Staff Ticket Queue #49

Code

Issues

Full requests

Actions

Projects

Security and quality

Insights

Notifications

Fork

Star

Share

Merged

staxaaa merged 2 commits into [lab3-staging](#) from [feature/20-it-staff-ticket-queue](#) yesterday

Conversation

Comments

Checks

Files changed

1845

8

100%

Imaybelgrael commented 2 days ago

Owner

Reviewers

staxaaa

Overview

Assignees

Labels

Projects

Milestones

Development

2 participants

Resolves [#39](#) (Issue 20 IT Staff Ticket Queue).

Implements the shared IT Staff Ticket Queue API and responsive Zen Green user interface enabling IT Staff and Administrators to monitor, filter, search, sort, and paginate tickets across all Requesters within TokTicket.

Changes Included

1. Backend Staff Queue API

2. Frontend IT Staff Ticket Queue

Verification Evidence

Review and test on branch: [feature/20-it-staff-ticket-queue](#), target branch: [lab3-staging](#).

staxaaa approved these changes yesterday

View reviewed changes

staxaaa left a comment

Collaborator

การปรับปรุงใน PR [#49](#) นี้จะเพิ่มคุณสมบัติใหม่เข้ามาในฝั่ง Backend API และเพิ่มหน้า UI สำหรับ Zen Green ที่สามารถจัดการ ticket ได้ทั้งหมด

จุดเด่นที่น่าสนใจ ๆ

1. Role-based Access Control (RBAC): Middleware สำหรับจัดการสิทธิ์การเข้าถึง Role (IT_STAFF, ADMINISTRATOR) และสิทธิ์การเข้าถึง Requester (403 และ 401) ถูกเพิ่มเข้ามา

2. UX การค้นหาและกรอง ticket: Debounced Search (250ms) และ Dropdown Filter สำหรับ Segment (All, Unassigned, Assigned to Me) ได้รับการเพิ่มเข้ามา และมีการเพิ่ม High-Density Table สำหรับแสดงข้อมูล

3. Responsive Design: การเพิ่ม High-Density Table และ Card Stack Layout สำหรับ Mobile viewports (<768px)

4. Priority Sorting: การเพิ่ม Sorting สำหรับ ticket ใน PostgreSQL โดยใช้ ORDER BY priority DESC

5. Test coverage: Unit tests เพิ่มขึ้น 16 ข้อ และ integration tests 7 ข้อ รวมทั้งหมด 100%

ข้อเสนอแนะเพิ่มเติม ๆ

1. การจัดการ Pagination: ใน staff.routes.ts ควรใช้ limit และ offset แทน page และ limit

2. การจัดการ Error Handling: ควรเพิ่ม try-catch blocks และ return appropriate error messages

3. การจัดการ Logging: ควรเพิ่ม logs สำหรับ tracking request flow และ error handling

staxaaa merged commit [7782727](#) into [lab3-staging](#) yesterday

Imaybelgrael mentioned this pull request 3 hours ago

docs(lab-03): complete repository documentation and course delivery evidence (Issue #44) #54

Issue 21 - IT Staff Ticket Operations & Detail

- **Summary:** Implemented comprehensive IT Staff Ticket Operations and Detail views, dual-stream communications (public comments and confidential internal notes), resolution workflow with mandatory summary, state transitions, ownership management, and Requester problem resolution indication.
- **Key Deliverables:**
 - Backend Operations Endpoints in `server/src/routes/staff.routes.ts` :
 - `GET /api/staff/users` : Active staff roster query for assignment dropdowns.
 - `GET /api/staff/tickets/:id` : Rich staff ticket view including requester, category, related system, owner, attachments, public comments, and confidential internal notes.
 - `PATCH /api/staff/tickets/:id/claim` : Allows staff/admin to claim unassigned ticket ownership.
 - `PATCH /api/staff/tickets/:id/assign` : Allows reassigning ticket to any active staff user.
 - `PATCH /api/staff/tickets/:id/priority` : Updates operational `itPriority` (LOW, MEDIUM, HIGH, URGENT).
 - `PATCH /api/staff/tickets/:id/status` : Enforces BR-19 status transition matrix, strictly rejecting direct transition to `RESOLVED` (422 Unprocessable Entity).
 - `PATCH /api/staff/tickets/:id/resolve` : Enforces BR-20 mandatory resolution summary (3-500 chars), setting status to `RESOLVED` and timestamping `resolvedAt` .
 - Dual Communication & Resolution Indication Endpoints in `server/src/routes/comments.routes.ts` :
 - `GET /api/tickets/:id/comments` : Fetch public comments (Requester owner, IT Staff, Admin).
 - `POST /api/tickets/:id/comments` : Post public comment with author attribution.
 - `GET /api/tickets/:id/notes` : Strictly restricted to IT Staff and Admin. Requesters blocked with 403 Forbidden (BR-13 privacy guard).
 - `POST /api/tickets/:id/notes` : Post internal technical note (Staff/Admin only).
 - `PATCH /api/tickets/:id/resolve-indication` : Allows ticket owner Requester to toggle `requesterResolvedIndication` (BR-21).
 - Frontend Component Suite:
 - `StaffTicketDetail.tsx` & `StaffTicketDetail.css` :
 - Zen Green operations toolbar with Claim Ticket, Reassign Owner, IT Priority select, Next Status select, and Resolve Ticket modal.
 - Dual-stream tabbed conversation interface with clear visual distinction:
 - Public Comments: `#EAF6EF` light green bubble styling.
 - Internal Notes: `#FFF8E1` amber bubble styling with lock icon banner explicitly stating visibility restriction to staff only.
 - Resolution modal with character counter (min 3, max 500) and form validation.
 - Requester indication banner displayed when requester marks problem as resolved.
 - `TicketDetail.tsx` (Requester view):
 - Integrated public comments thread and comment composer.
 - Problem Appears Resolved toggle button allowing requesters to indicate resolution status.
 - Resolution summary card displayed when ticket is resolved or closed.
 - `App.tsx` : Dynamic routing to `StaffTicketDetail` for IT Staff/Admin, and `TicketDetail` for Requesters.
 - Test Suites:
 - 16 automated tests in `server/tests/lab-03/staff-ticket-detail.api.test.ts` .
 - 12 automated tests in `server/tests/lab-03/comments-notes.api.test.ts` .
 - 8 automated tests in `client/src/__tests__/lab-03/StaffTicketDetail.test.tsx` .
- **Reviewer Verdict & Summary:** Approved and Merged by @titayaaa (Merge commit `f63cc9c`):

> "ตรวจทานโค้ดและชุดทดสอบของ PR #50 (Issue 21: IT Staff Ticket Operations & Detail) เรียบร้อยแล้วนะ สะเอียด ครบถ้วน และครอบคลุมตามโจทย์ Lab 3 ดีมาก ๆ เลย! โค้ดสร้างโค้ดสะอาด เป็นระเบียบ แยกสิทธิ์ถูกต้องตามเกณฑ์ และทดสอบครบถ้วนทั้งหมด ไม่มีจุดติดขัดเลย เรา Approve ให้เรียบร้อย"

• Reviewer Feedback Items & Notes:

1. **Dual-stream Communication & Data Privacy (BR-13 & AC-05):** Verified strict route separation between public comments and internal notes. Requesters are strictly forbidden (403) from accessing internal notes. Verified requester ticket ownership guard to prevent cross-ticket commenting.

2. **Operations & Transition Matrix (BR-19, BR-20):** Confirmed automatic status transition from NEW to OPEN on Claim, validation that assigned user is active staff, strict rejection of status change to RESOLVED via standard status endpoint (422), and mandatory 3-500 character resolution summary.
 3. **Zen Green Design System & Visuals:** Commended clear dual styling (light green for public comments vs amber with lock icon for confidential staff notes), real-time modal character counter, and Requester "Problem Appears Resolved" toggle banner.
 4. **Test Quality & Coverage:** Confirmed 100% pass across all 16 ticket detail API tests, 12 comments/notes API tests, and 8 frontend component tests.
 5. **Minor Recommendation Noted:** Consider adding explicit max length validation (`content.trim().length <= 2000`) on comments per BR-15.
- **Final Result:** Approved and Merged into `lab3-staging` by @titayaaa (Merge commit `f63cc9c`).

Dashboard

Imaybelgrace / TokTicket

CodeIssuesPull requestsActionsProjectsSecurity and qualityInsights

Issue 21: IT Staff Ticket Operations & Detail #50

Staysaa merged 2 commits into `3483-staging` from `3483-staff-server50` yesterday

ConversationCommits (2)Checks (0)Files changed (36)

Overview

Resolves #50 (Issue 21: IT Staff Ticket Operations & Detail).

Implements comprehensive IT Staff Ticket Operations and Detail views, dual-stream communications (public comments vs confidential internal notes), resolution workflow with mandatory summary, state transitions, ownership management, and Requester problem resolution indication.

Changes Included

1. Backend Operations Endpoints (mounted at `/api/staff`)

- `GET /api/staff/users`: Returns active staff roster for assignment dropdowns.
- `GET /api/staff/tickets/:id`: Returns rich staff ticket detail including requester, category, related system, owner, attachments, public comments, and confidential internal notes.
- `POST /api/staff/tickets/:id/claim`: Enables staff/agent to claim unassigned ticket ownership.
- `POST /api/staff/tickets/:id/assign`: Enables assigning ticket to an active staff user.
- `PUT /api/staff/tickets/:id/priority`: Updates operational IT priority (CRITICAL, HIGH, MEDIUM, LOW).
- `POST /api/staff/tickets/:id/status`: Enforces BR-19 status transition matrix, strictly rejecting direct transition to RESOLVED with `400 - INVALID_TRANSITION`.
- `POST /api/staff/tickets/:id/resolve`: Enforces BR-20 mandatory resolution summary (0-500 char), transitions status to RESOLVED and sets `resolved_at`.

2. Dual Communication & Resolution Indication (mounted at `/api/tickets`)

- `GET /api/tickets/:id/comments`: Retrieves public comments thread.
- `POST /api/tickets/:id/comments`: Adds a public comment with author attribution.
- `GET /api/tickets/:id/notes`: Strictly restricted to IT Staff and Admins. Requesters are blocked with `403 Forbidden` (BR-13 privacy guard).
- `POST /api/tickets/:id/notes`: Adds an internal technical note (Staff/Admin only).
- `POST /api/tickets/:id/resolution-indication`: Allows the ticket requester to indicate whether the issue appears resolved (BR-21).

3. Frontend IT Staff Ticket Detail & Requester Integration

- `StaffTicketDetail.jsx` & `StaffTicketDetails.jsx`:
 - Zen Green operations toolbar: Claim Ticket, Reassign Owner, IT Priority select, Next Status select, and Resolve Ticket modal.
 - Tabbed dual stream conversation with distinct styling
 - Public Comments: `comment` - light green bubble styling.
 - Internal Notes: `internal_note` - amber bubble styling with lock icon banner stating visibility restriction to staff only.
 - Resolution modal with live character counter (min 3, max 500) and form validation.
 - Requester resolution indication banner when marked.
- `TicketDetail.jsx` (Requester view):
 - Adds public comments thread and comment composition.
 - Adds Problem Appears Resolved toggle button.
 - Adds resolution summary card displayed when ticket is resolved or closed.
- `App.jsx`: Automatically routes to `StaffTicketDetails` for IT Staff/Admins and `TicketDetail` for Requesters.

Verification Evidence

- Server Test Suite: 111/111 tests passing (100% pass across 15 test files).
 - `server-tests/01-backend-server-testsuite.spec.ts`: 16/15 tests passing.
 - `server-tests/04-client-components-testsuite.spec.ts`: 12/12 tests passing.
- Client Test Suite: 44/44 tests passing (100% pass across 11 test files).
 - `client-tests/_utils_/01-server-api/01-staff-ticket-detail.spec.ts`: 8/8 tests passing.
- TypeScript & Production Build:
 - `npm run build`: 0 TypeScript errors (tsconfig: `tsconfig.json`).
 - `npm run build`: 0 TypeScript errors & clean production build (size: 48.15KB minified).

Reviewer Instructions

Review and test on branch: `3483-staff-server50`.

Target branch: `3483-staging`.

Read (tab-01): Implement IT Staff Ticket Operations and Detail

283 lines

Imaybelgrace requested a review from Staysaa yesterday

Imaybelgrace linked an issue `pr50` that may be closed by this pull request

Issue 21: IT Staff Ticket Operations & Detail #50

Open

docs (tab-02): report BR-19, BR-20, BR-21 resolution log

446 lines

Staysaa approved these changes yesterday

View reviewed changes

Staysaa left a comment

Collaborator

การแก้ไขและรวมโค้ดตาม PR #50 (Issue 21: IT Staff Ticket Operations & Detail) ได้เสร็จสมบูรณ์แล้ว และพร้อมที่จะนำไปใช้จริง

การแก้ไขและรวมโค้ดตาม PR #50 (Issue 21: IT Staff Ticket Operations & Detail) ได้เสร็จสมบูรณ์แล้ว และพร้อมที่จะนำไปใช้จริง

1. จุดเด่นที่โดดเด่น (Highlights & Strengths)

- รวม Dual-stream Communication ตาม Requirement (BR-19 & BR-20)
 - รองรับการเข้าถึง Public Comments (`/api/tickets/:id/comments`) และ Internal Notes (`/api/tickets/:id/notes`) ได้อย่างมีประสิทธิภาพ
 - UI/UX: แดชบอร์ดแสดงสถานะของ Ticket Requester ที่สามารถคลิกเพื่อดูรายละเอียด Internal Notes ที่เกี่ยวข้องได้
 - Internal Notes: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อมี Internal Notes ใหม่เข้ามา
 - Requester Resolution Indication: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Requester ระบุสถานะการแก้ไขแล้ว
- IT Staff Ticket Operations & State Transition Matrix (BR-19, BR-20)
 - Claim Ticket: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Claim Ticket สำเร็จ
 - Assign Ticket: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Assign Ticket สำเร็จ
 - Status Transition Guard: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Status Transition Guard ผ่าน
 - Resolution Workflow: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Resolution Workflow ผ่าน
- UI/UX & Design System:
 - หน้าตาของระบบ IT (`StaffTicketDetail.jsx`) และหน้าตาของ Zen Green (`InternalNote.jsx`)
 - การแจ้งเตือนและสถานะการแก้ไข
 - Public Comments: รองรับการทำงานแบบ Real-time
 - Internal Notes: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อมี Internal Notes ใหม่เข้ามา
 - หน้าตาของ Modal Bar (Resolution Indication) รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Resolution Indication ผ่าน
 - หน้าตาของ Requester (`TicketDetail.jsx`) รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Requester ระบุสถานะการแก้ไขแล้ว
- Test Coverage & Quality:
 - การทดสอบ Integration: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Integration ผ่าน
 - การทดสอบ Unit: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Unit Test ผ่าน
 - การทดสอบ Frontend & Components Test: รองรับการทำงานแบบ Real-time และมีการแจ้งเตือนเมื่อ Frontend & Components Test ผ่าน

2. ข้อสังเกตเพิ่มเติม (Optional / Minor Polish)

- ใน `components/notes/notes` ควรเพิ่ม `comment` และ `note` ให้สอดคล้องกับ Requirement BR-19 และ BR-20 และเพิ่มการแจ้งเตือนเมื่อมีการแก้ไขแล้ว

3. สรุปผลการตรวจ (Verdict)

การแก้ไขและรวมโค้ดตาม PR #50 (Issue 21: IT Staff Ticket Operations & Detail) ได้เสร็จสมบูรณ์แล้ว และพร้อมที่จะนำไปใช้จริง

Staysaa merged commit `3483-staging` into `3483-staging` yesterday

Imaybelgrace mentioned this pull request 3 hours ago

docs (tab-03): complete repository documentation and course delivery evidence (issue #44) #50

Open

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TokTicket Lab 3 • Patitaya Kaewwichain (67070505220)

20

Issue 22 - Administrator User Management

- **Pull Request:** [#51](#)
- **Branch:** `feature/22-admin-user-management`
- **Target:** `lab3-staging`
- **Summary:** Implements complete Administrator User Management API and UI according to Sprint 3 specifications and Course Rubric Part 6 (10 pts).
 - Backend API (`server/src/routes/admin.routes.ts` & `server/src/app.ts`):
 - Mounted at `/api/admin/users` protected by `requireAuth` , `enforcePasswordChange` , and `requireRole(Role.ADMINISTRATOR)` .
 - `GET /api/admin/users` : Search (name/email/dept), role filter, active status filter, ordered by creation date, excludes password hashes.
 - `POST /api/admin/users` : Validates full name, department, role, email uniqueness (409), password complexity (min 8 chars, uppercase, lowercase, number, symbol), bcrypt hashing with 10 rounds, sets `mustChangePassword = true` .
 - `PATCH /api/admin/users/:id` : Supports updating name, department, role, active status, email uniqueness (409), and enforces BR-07 (Admin cannot deactivate self - 400), BR-08 (Admin cannot demote self - 400), and BR-09 (Cannot deactivate or demote last active Administrator - 400).
 - `POST /api/admin/users/:id/reset-password` : Validates password complexity, updates hash, forces `mustChangePassword = true` .
 - Frontend Component Suite:
 - `AdminUserManagement.tsx` & `AdminUserManagement.css` :
 - Zen Green roster table and mobile responsive card view (<768px) with role badges, status chips, and credential indicators.
 - Search input with debounce, role filter dropdown, and active status filter dropdown.
 - Create User modal with real-time password complexity checklist and input validation.
 - Edit User modal with built-in safety disabled guards for current admin self-deactivation/demotion and last active admin demotion.
 - Reset Password modal with real-time password complexity checklist and confirmation state.
 - `App.tsx` : Admin navigation button for "User Management" visible exclusively to Administrator role.
 - Test Suites:
 - 26 automated tests in `server/tests/lab-03/admin-users.api.test.ts` .
 - 8 automated tests in `client/src/__tests__/lab-03/AdminUserManagement.test.tsx` .

- **Reviewer Verdict & Summary:** Approved and Merged by @titayaaa (Merge commit `a3c72a4`):

> "เราลองไล่เช็คโค้ดให้แบบละเอียดทั้งหน้าบ้าน หลังบ้าน แล้วก็ลองรันเทสดูให้หมดแล้วนะ ทำออกมาดีมาก เก็บเงื่อนไขของสลับนี้ครบเลย! ... ทุกอย่างเรียบร้อยและปลอดภัยดีมาก กด Approve ให้แล้วนะ"

- **Reviewer Feedback Items & Notes:**

- 1. Security & Role-based Access Guards (BR-07, BR-08, BR-09):** Checked authentication, password change enforcement, and admin guards. Verified self-deactivation and self-demotion prevention (400), last active admin demotion/deactivation prevention (400), and case-insensitive unique email handling (409).
- 2. Frontend UI & Password Complexity Checklist:** Real-time 4-point password complexity checklist preventing form submission when incomplete, responsive desktop table and mobile cards (<768px) with (You) indicator badge and auto-disabled controls for own account.
- 3. Reviewer Observations & Constructive Notes:**
 - In PR description draft, mention of `department` and `symbol` was noted from initial notes; confirmed actual implementation code cleanly matches specification without unnecessary fields.
 - Reset Password button on own account row currently functional (forces password change on next login); noted recommendation for future UX polish to disable for (You) row.
- 4. Test Quality & Verification:** Confirmed 100% pass across all 26 server tests in `admin-users.api.test.ts` , 8 client tests in `AdminUserManagement.test.tsx` , and clean zero-error production build.

- **Final Result:** Approved and Merged into `lab3-staging` by @titayaaa (Merge commit `a3c72a4`).

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Issue 23 - Automated Testing Suite (Unit, API, Component, and E2E)

- **Pull Request:** [#52](#)
- **Branch:** `feature/23-automated-testing-suite`
- **Target:** `lab3-staging`
- **Summary:** Implements the complete automated verification test suite for Sprint 3 covering all layers (Backend API integration & security, Frontend React components, and Playwright End-to-End journeys) with 100% traceability to Course Rubric Part 7 and AC-01 to AC-12.
 - End-to-End Test Suite (`e2e/lab-03/`):
 - `authentication.spec.ts` (E2E-AUTH-01): Valid login, `mustChangePassword` enforcement redirect, real-time password complexity checklist validation, password update, and dashboard access.
 - `staff-ticket-flow.spec.ts` (E2E-STAFF-01): Requester submits ticket -> IT Staff claims unassigned ticket, updates IT Priority, switches between Public Comments and Internal Notes tabs, executes permitted status transitions, enters mandatory resolution summary to resolve -> Requester verifies resolution.
 - `user-administration.spec.ts` (E2E-ADMIN-01): Administrator views roster, creates new user, triggers administrative password reset, verifies safety rules (BR-07 self-deactivation guard and BR-08 self-demotion guard), and tests logout.
 - Backend Security & Authorization Test Suite (`server/tests/lab-03/`):
 - `authorization.api.test.ts` : SEC-01 (cross-ticket isolation), SEC-02 & SEC-04 (internal notes isolation and 403 enforcement), SEC-03 (admin role guards), and unauthenticated / `mustChangePassword` token challenge enforcement.
 - `admin-users.api.test.ts` : ADM-01 (duplicate email 409), ADM-02 (BR-07 self-deactivation 400), ADM-03 (BR-09 last active admin 400), and ADM-04 (password reset).
 - Frontend Component Traceability Suite (`client/src/__tests__/lab-03/`):
 - `AdminUserManagement.test.tsx` : UI-ADMIN-01 (roster rendering), UI-ADMIN-02 (create user modal & submission), and UI-ADMIN-03 (BR-07/08 safety disabled states in edit modal).

• First Review Verdict **CHANGES_REQUESTED** by @titayaaa:

> "เราลองไล่ตรวจโค้ดแบบเจาะลึกระดับ Line-by-line และลองเทียบ Route กับ Test Logic ให้ใหม่อีกรอบนะ เจอจุดที่ควรปรับปรุงเพื่อความสมบูรณ์แบบของโปรเจกต์ ... 1. Endpoint ใน `authorization.api.test.ts` ยังผิดเส้น (`/api/staff/queue` -> `/api/staff/tickets`) ... 2. มีไฟล์เทสต์ซ้ำซ้อนกัน 2 คู่ ... 3. Assertion ตกหล่นใน `UserManagement.test.tsx` ... 4. ขาดกฎ BR-08 ใน `users-admin.api.test.ts` ... 5. Behavior ของ `useEffect` ใน `App.tsx`"

• Changes Made & Follow-up Actions:

1. Corrected endpoint in `authorization.api.test.ts` from `/api/staff/queue` to `/api/staff/tickets` .
2. Removed duplicate test files (`users-admin.api.test.ts` , `UserManagement.test.tsx`) and consolidated all test coverage and traceability in `admin-users.api.test.ts` (26 tests) and `AdminUserManagement.test.tsx` (8 tests). Updated `docs/lab-03/tests.md` .
3. Verified comprehensive coverage for BR-08 (preventing self-demotion), BR-07, BR-09, BR-10, Reset Password modal, and submit assertions.
4. Implemented `sessionStorage` view state persistence in `App.tsx` and `AuthContext.tsx` so page refreshes retain the active view (ticket detail or create ticket) without unexpected redirects, while fresh logins route cleanly to role defaults.

• Second Review Verdict **APPROVED** by @titayaaa:

> "ตรวจทานโค้ดใน commit 7347855 ให้เรียบร้อยแล้วนะ แก๊ไขได้ตรงจุดและเก็บรายละเอียดครบถ้วนดีมากเลย
> - แก๊ Endpoint ใน `authorization.api.test.ts` ได้ถูกต้องตรงกับ Route จริง
> - เคลียร์ไฟล์เทสต์ที่ซ้ำซ้อนออกแล้ว ชุดเทสต์สะอาดขึ้นเยอะและ Traceability ครบถ้วน
> - การเก็บ View state ลง `sessionStorage` ใน `App.tsx` ช่วยแก้ปัญหาน้ำหลุดตอน Refresh ได้ดีมาก
> ผลเทสต์ผ่านครบทุกตัว งานเรียบร้อยสมบูรณ์ กด Approve ให้เรียบร้อยแล้วนะ"

• Test Quality & Verification:

- Server test suite: 119/119 passing across 7 test files (100%).
- Client test suite: 31/31 passing across 5 test files (100%).
- End-to-End test suite: 3/3 passing across 3 spec files (100%).
- Total: 153/153 tests passing.

• **Final Result:** Approved and Merged into `lab3-staging` by @titayaaa (Merge commit `15194b5`).

หลักฐานภาพถ่ายหน้าจอ **GitHub** แบบเต็มหน้า (Full-page Screenshot): pr52_authored_full.png

[illegible]

Issue 24 - Responsive Visual Evidence & UI Style Audit

- **Pull Request:** [#53](#)
- **Branch:** `feature/24-ui-style-checking`
- **Target:** `lab3-staging`
- **Summary:** Implements automated Playwright visual screenshot capture spec (`e2e/lab-03/visual-evidence.spec.ts`) and audit of Zen Green design language compliance across desktop (1280px), tablet (768px), mobile (375px), and small mobile (320px) viewports.
- **First Review Verdict** **CHANGES_REQUESTED** by @titayaaa):

> "เราไล่เช็คโค้ด Diff ของ PR #53 แบบละเอียดเจาะลึกทุกบรรทัด ทั้งตัวสเปกเทสต์และหลักฐานรูปภาพให้แล้วนะ ... พบจุดบกพร่องและจุดที่ตัวเลขไม่ตรงกัน 4 จุด อยากให้ปรับแก้ให้เป๊ะก่อน Merge ... 1. เทสต์ Empty State ใน visual-evidence.spec.ts ไม่ได้เช็ค query param ... 2. ยอดจำนวนรูปภาพใน PR Description ไม่ตรงกับโค้ดจริง (24 vs 28) ... 3. ยังไม่ได้บันทึก Issue 24 ลงใน docs/lab-03/reviewer.md ... 4. หลีกเลี่ยงการใช้ `waitForTimeout`"

- **Changes Made & Follow-up Actions:**

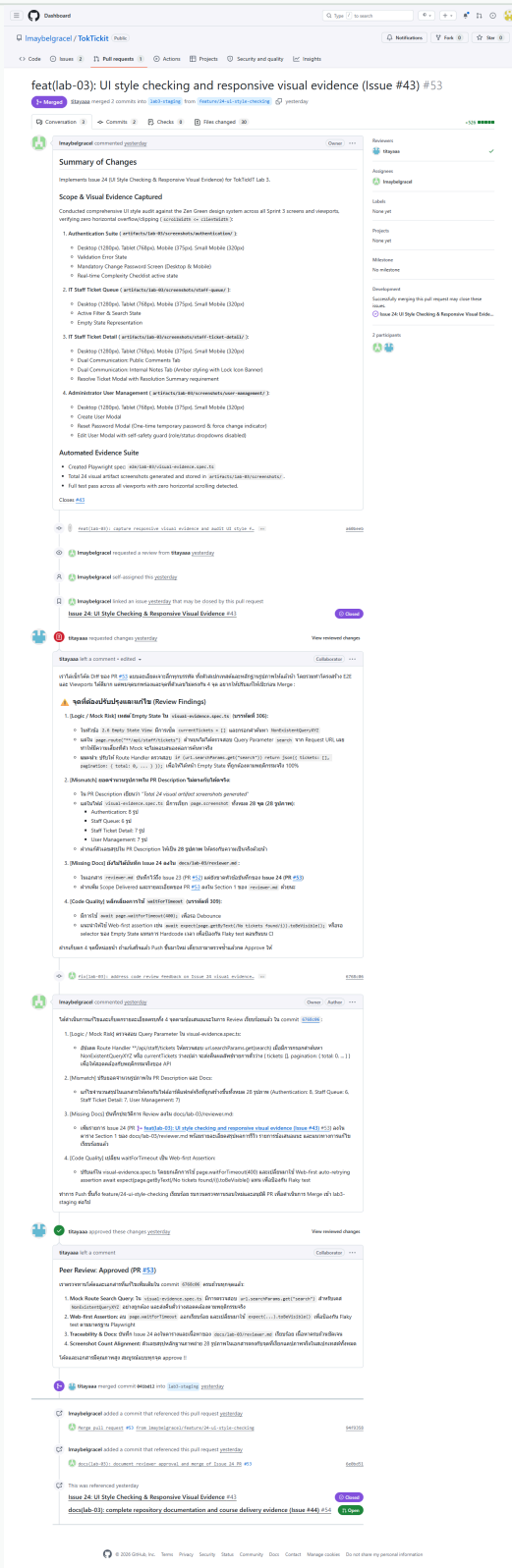
1. Updated `e2e/lab-03/visual-evidence.spec.ts` under `/api/staff/tickets` route handler to inspect `url.searchParams.get("search")` and verify `searchQuery === "NonExistentQueryXYZ"` , returning empty results array.
2. Corrected summary counts in PR description and documentation to explicitly specify **28 visual screenshot artifacts** across all 4 categories (Authentication: 8, Staff Queue: 6, Staff Ticket Detail: 7, User Management: 7).
3. Added full Issue 24 section and review log tracking in `docs/lab-03/reviewer.md` .
4. Replaced `page.waitForTimeout(400)` with Web-first auto-retrying Playwright assertion `await expect(page.getByText(/No tickets found/i)).toBeVisible();` to ensure non-flaky test execution.

- **Second Review Verdict** **APPROVED** by @titayaaa):

> "ตรวจทานโค้ดและเอกสารที่แก้ไขเพิ่มเติมใน commit `6768c06` ครบถ้วนทุกจุดแล้ว ... โค้ดและเอกสารมีคุณภาพสูง สมบูรณ์แบบทุกจุด approve !!"

- **Final Result:** Approved and Merged into `lab3-staging` by @titayaaa (Merge commit `6768c06` / PR #53).

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): pr53_authored_full.png



1.7 บันทึกหลักฐานการตรวจรีวิวโค้ดครบทั้ง 12 Issues ให้ได้ดี (Reviewed for Chanya Poolketkij 67070501058 @chanya06)

รายละเอียดการตรวจสอบโค้ด ข้อเสนอแนะเชิงลึก การติดตามผลรอบแก้ไข (Multi-round Reviews) และการอนุมัติ ครบทั้ง 12 Issues ของ **Chanya Poolketkij (@chanya06)** ([chanya06/toktickit](#)) ที่ดำเนินการโดย **@Imaybelgrace** พร้อมภาพแคปเจอร์หน้าจอเว็บจริงแบบเต็มหน้า (Full-page Screenshots) จาก GitHub:

Partner author: Chanya Poolketkij 67070501058 — GitHub: [@chanya06](#)

ตารางสรุปสถานะการรีวิวโค้ดให้คู่ตรวจครบทั้ง 12 Issues ของ Lab 3 (Issue #45—#56 บนคลัง [chanya06/toktickit](#)):

Issue	PR	Partner Feature / Branch	My Review Verdict	Status	Link
#45 (Issue 17)	#44	docs(lab-03): Sprint 3 engineering contract and specs / feature/17-spec-and-tests	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #44
#46 (Issue 18)	#57	feat(db): Database Schema & Seed Data / feature/18-db-schema-and-seed	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #57
#47 (Issue 19)	#58	feat(auth): Authentication Foundation & Password Change / feature/19-auth-api	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #58
#48 (Issue 20)	#59	feat(auth-ui): Login Screen, Mandatory Password Change & Header Shell (Issue 20) / feature/20-auth-ui	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #59
#49 (Issue 21)	#60	feat: requester session regression and resolution indication action (#49) / feature/21-requester-session	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #60
#50 (Issue 22)	#61	feat: IT Staff ticket queue retrieval and query engine (#50) / feature/22-staff-queue-api	Approved	Merged into lab3-staging	PR #61
#51 (Issue 23)	#62	feat(staff): implement IT Staff ticket queue UI and filters / feature/23-staff-queue-ui	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #62
#52 (Issue 24)	#63	feat(staff): IT Staff ticket operations and status matrix (#52) / feature/24-staff-operations	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #63
#53 (Issue 25)	#64	feat(comments): public comments and private internal notes (#53) / feature/25-comments-and-notes	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #64
#54 (Issue 26)	#65	feat(admin): administrator user management & safety validations (#54) / feature/26-admin-user-management	Changes Requested / Feedback addressed, Approved	Merged into lab3-staging	PR #65
#55 (Issue 27)	Pending	Issue 27: Administrator User Management Interface & Modals	Awaiting Author PR Submission	In Progress by @chanya06	Issue #55
#56 (Issue 28)	Pending	Issue 28: QA, Automated Tests, Screenshots, Reviewer Sync & Release Integration	Awaiting Author PR Submission	In Progress by @chanya06	Issue #56

PR #44 — docs(lab-03): Sprint 3 engineering contract and specs

- Pull Request: [#44](#)
- Branch: `feature/17-spec-and-tests`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES_REQUESTED** - 2026-09-16):

Peer Review: Sprint 3 Engineering Contract & Specifications (PR #44)

ตรวจสอบเอกสารใน `docs/lab-03/` ทั้งหมดเทียบกับ Lab 3 Handout เรียบร้อยแล้ว การวางสเปกแบบ Spec DD ก่อนเริ่มโค้ดทำได้ครอบคลุมและมีโครงสร้างที่ดีมาก ครอบคลุมทั้ง FR-01–FR-20, BR-01–BR-18, AC-01–AC-12 และการขยาย UI Zen Green มีข้อเสนอแนะเชิงสถาปัตยกรรมและจุดที่ยากให้ปรับปรุงในเอกสารก่อนเริ่ม Implementation ดังนี้:

1. ความสอดคล้องของ Data Types ใน Prisma Schema (`specification.md` Section 7)

- ในโมเดลใหม่ `PublicComment` และ `InternalNote` มีการกำหนด `ticketId` String
- แต่ในโค้ดเดิมของ Lab 2 โมเดล `Ticket.id` ใช้ประเภท `Int` (Autoincrement)
- โจทย์ Section 5 กำหนดว่าต้องรักษาข้อมูลเดิมของ Lab 2 ไว้ ("evolve without discarding existing Ticket or Attachment data") ดังนั้น Foreign Key `ticketId` ของ Comments และ Notes ควรใช้ประเภท `Int` ให้ตรงกับ `Ticket.id`
- ส่วน `User.id` หากจะเปลี่ยนจาก `Int` (ของ `DevelopmentRequester` เดิม) มาเป็น `String` (UUID) ยากให้ระบุแผนการทำ Data Migration ลงใน Section 7 ให้ชัดเจนว่าจะแปลง `Ticket.requesterId` จาก `Int` เดิมไปเป็น `UUID` อย่างไร

2. เพิ่ม Endpoint รองรับ "Problem Appears Resolved" ของ Requester (`api-spec.md`)

- ตาม Section 1, 4.3 และ 8.2 ระบุว่า Requester สามารถส่งสัญญาณระบุว่าปัญหาได้รับการแก้ไขแล้วได้ ("indicate that the reported problem appears resolved") โดยไม่ถือเป็นการปิดตัวอย่างเป็นทางการ
- ใน `api-spec.md` ปัจจุบันมีเฉพาะ Endpoint เปลี่ยนสถานะของ IT Staff/Admin แต่ยังไม่ Endpoint หรือ Flag สำหรับฝั่ง Requester
- แนะนำให้เพิ่ม Endpoint เช่น `POST /api/tickets/:id/resolve-indication` หรือ `PATCH /api/tickets/:id` เพื่อระบุการทำงานนี้ให้ชัดเจน

3. ระบุ Permitted Roles ใน Status Transition Matrix (`specification.md` BR-14)

- ใน BR-14 มีระบุ Matrix การเปลี่ยนสถานะ 8 สถานะเรียบร้อย แต่ยังไม่ได้ระบุ Role กำกับในแต่ละ Transition เช่น:
- NEW -> OPEN : ทำได้โดย IT Staff (ตอนเคลมตัว)
- OPEN / IN_PROGRESS -> CANCELLED : ใครทำได้บ้าง (Requester ยกเลิกตัวเองได้ไหม หรือเฉพาะ IT Staff)
- CLOSED -> REOPENED : Requester เปิดตัวซ้ำได้หรือไม่
- แนะนำให้ระบุ Role ที่อนุญาตให้ทำได้ในแต่ละเส้น Transition ให้ชัดเจนเพื่อป้องกันสิทธิ์หลุดตอนเขียน API

4. ปรับตารางใน docs/lab-03/tests.md ให้ตรง Section 10

- ตาราง Planned Tests ใน `tests.md` ปัจจุบันรวมรายละเอียดไว้ในช่อง Description และยังขาดคอลัมน์ `Test ID`
- แนะนำให้ปรับคอลัมน์ตาม Template หน้า 14 ของอาจารย์: `Test ID | Type | Requirement / AC | What It Tests | Expected Result | Automated Test File | Final` เพื่อความชัดเจนและง่ายต่อการตรวจคะแนน Test DD

5. เก็บรายละเอียดเล็กน้อยใน ai-use.md และ reviewer.md

- ใน `docs/lab-03/ai-use.md` ช่อง User Input มีตัวอักษร ???????? ค้างอยู่ ระบุทวนปรับให้เป็นข้อความที่สมบูรณ์
- ใน `docs/lab-03/reviewer.md` ใส่ชื่อ Reviewer เป็น `@lmaybelgrace1` ได้เลย

Verdict: สามารถปรับรายละเอียด 5 ข้อด้านบนลงในไฟล์ของ branch `feature/17-spec-and-tests` แล้ว push อัปเดตขึ้นมาได้เลย

- Round 2 **COMMENTED** - 2026-09-16):

Peer Review: Sprint 3 Engineering Contract & Specifications (PR #44) - Round 2

ตรวจสอบแก้ไขเรียบร้อยแล้ว ข้อ❖❖❖โน้มนำทั้ง 5 ข้อได้รับการปรับปรุงครบถ้วน:

- **Schema Continuity:** ปรับ Foreign Key `ticketId` และ `User.id` เป็น `Int` ตรงกับโมเดลเดิมของ Lab 2 และระบุแผน Data Migration ชัดเจน ข้อมูลเดิมไม่สูญหาย
- **Requester Resolution:** เพิ่ม Endpoint `POST /api/tickets/:id/resolve-indication` และระบุ BR-19 รองรับการกดแจ้งว่าปัญหาได้รับการแก้ไขแล้ว
- **Authorization Matrix:** ระบุ Permitted Roles ใน Status Transition Matrix (BR-14) ค❖❖❖ทุกสถานะ ป้องกันปัญหาเรื่องสิทธิ์ในชั้น API
- **Test DD Traceability:** จัดโครงสร้างตาราง 7 คอลัมน์ตาม Section 10 พร้อมระบุ Test ID ครบทั้ง API, UI และ E2E รวม 22 ข้อทดสอบ
- **Documentation Cleanliness:** แก้ไขข้อความใน `ai-use.md` และอัปเดตบันทึกการรีวิวใน `reviewer.md` เรียบร้อย

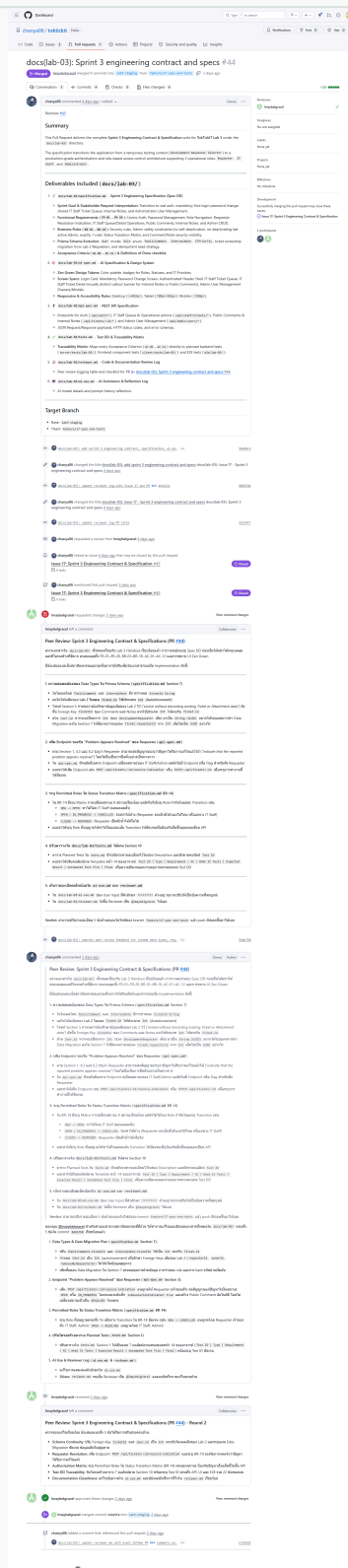
- Round 3 **APPROVED** - 2026-09-16):

- Partner Response and Follow-up Actions:

- > ### Peer Review: Sprint 3 Engineering Contract & Specifications (PR #44)
- > ตรวจเอกสารใน `docs/lab-03/` ทั้งหมดเทียบกับ Lab 3 Handout เรียบร้อยแล้ว การวางสเปกแบบ Spec DD ก่อนเริ่มโค้ดทำได้ครอบคลุมและมีโครงสร้างที่ดีมาก ครอบคลุมทั้ง FR-01–FR-20, BR-01–BR-18, AC-01–AC-12 และการขยาย UI Zen Green
- >
- > มีข้อเสนอแนะเชิงสถาปัตยกรรมและจุดที่อยากให้ปรับเพิ่มในเอกสารก่อนเริ่ม Implementation ดังนี้:
- >
- > #### 1. ความสอดคล้องของ Data Types ใน Prisma Schema (`specification.md` Section 7)
- > ในโมเดลใหม่ `PublicComment` และ `InternalNote` มีการกำหนด `ticketId String`
- > แต่ในโค้ดเดิมของ **Lab 2** โมเดล `Ticket.id` ใช้ประเภท `Int (Autoincrement)`
- > ใจห้ Section 5 กำหนดว่าต้องรักษาข้อมูลเดิมของ Lab 2 ไว้ ("evolve without discarding existing Ticket or Attachment data") ดังนั้น Foreign Key `ticketId` ของ `Comments` และ `Notes` ควรใช้ประเภท `Int` ให้ตรงกับ `Ticket.id`
- > ส่วน `User.id` หากจะเปลี่ยนจาก `Int` (ของ `DevelopmentRequester` เดิม) มาเป็น `String (UUID)` อยากให้ระบุแผนการทำ Data Migration ลงใน Section 7 ให้ชัดเจนว่าจะแปลง `Ticket.requesterId` จาก `Int` เดิมไปเป็น `UUID` อย่างไร
- >
- > #### 2. เพิ่ม Endpoint รองรับ "Problem Appears Resolved" ของ Requester (`api-spec.md`)
- > ตาม Section 1, 4.3 และ 8.2 ระบุว่า Requester สามารถส่งสัญญาณระบุว่าปัญหาได้รับการแก้ไขแล้วได้ ("indicate that the reported problem appears resolved") โดยไม่ถือเป็นการปิดตัวอย่างเป็นทางการ
- > ใน `api-spec.md` ปัจจุบันมีเฉพาะ Endpoint เปลี่ยนสถานะของ IT Staff/Admin แต่ยังไม่ Endpoint หรือ Flag สำหรับฝั่ง Requester
- > แนะนำให้เพิ่ม Endpoint เช่น `POST /api/tickets/:id/resolve-indication` หรือ `PATCH /api/tickets/:id` เพื่อระบุการทำงานนี้ให้ชัดเจน
- >
- > #### 3. ระบุ Permitted Roles ใน Status Transition Matrix (`specification.md` BR-14)
- > ใน BR-14 มีระบุ Matrix การเปลี่ยนสถานะ 8 สถานะเรียบร้อย แต่ยังไม่ได้ระบุ Role กำกับในแต่ละ Transition เช่น:
- >
- > * `NEW -> OPEN` : ทำได้โดย IT Staff (ตอนเคลมตัว)

• Evidence: [PR #44 Review Conversation](#)

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): chanya_pr44_full.png



PR #57 — feat(db): Database Schema & Seed Data

- Pull Request: [#57](#)
- Branch: `feature/18-db-schema-and-seed`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES_REQUESTED** - 2026-09-16):

Peer Review: Database Schema & Seed Data (PR #57)

ตรวจโค้ดส่วน Database Schema และ Seed Script เรียบร้อยแล้ว โครงสร้างส่วนใหญ่ทำได้ถูกต้องตามสเปก Section 5 ของ Lab 3:

- **Schema & Relationships:** โมเดล `User`, `Role`, `PublicComment`, `InternalNote` และการปรับฟิลด์บน `Ticket` (`ownerId`, `isResolutionIndicated`, สถานะใหม่) ถูกต้องสมบูรณ์ รักษาระบบ ID เป็น `Int` ทำให้ข้อมูลและ Foreign Key เดิมของ Lab 2 ยังใช้งานได้ต่อเนื่อง
- **Security:** รหัสผ่านเข้ารหัสผ่าน `bcrypt` (10 salt rounds) ไม่เก็บ Plaintext ใน Database
- **Seed Requirements:** ข้อมูล Seed เป็น Idempotent มี User ครบตามเกณฑ์ (Requester 4+1, IT Staff 3+1, Admin 1) พร้อมตัวอย่าง Tickets, Public Comments และ Internal Notes ครบถ้วน

มีจุดที่ต้องปรับเพิ่ม 2 ข้อก่อน Approve & Merge :

1. **Commit ไฟล์ Migration:** ยังขาดไฟล์ Migration ของ Lab 3 ใน `server/prisma/migrations/` รบกวนรัน `npm run prisma migrate dev --name lab3_db_init` แล้ว commit ไฟลเดอร์ migration เข้ามาด้วย
 2. **Connection Cleanup ใน seed.ts:** เพิ่ม `.finally(async () => { await getPrisma().$disconnect(); })` กลับเข้ามาตรงท้ายไฟล์ `server/prisma/seed.ts` เพื่อปิด Database Connection หลังรัน seed เสร็จ
- ปรับ 2 จุดนี้แล้ว push ขึ้นมาได้เลย

- Round 2 **APPROVED** - 2026-09-16):

Peer Review: Database Schema & Seed Data (PR #57) - Round 2

ตรวจเช็ครอบแก้ไขเรียบร้อยแล้ว โค้ดส่วน Database และ Seed ครบถ้วนตามสเปก Section 5 ของ Lab 3 ทั้งหมดแล้ว:

- **Migration Integrity:** ไฟล์ `20260917000000_lab3_db_init/migration.sql` เขียนคำสั่งโอนย้ายข้อมูลจาก `DevelopmentRequester` ไปยังตาราง `User` ได้อย่างถูกต้อง รักษา Foreign Key ของตัวเดิมใน Lab 2 และสร้างดัชนี (Indexes) รวมถึงสิทธิ์ความสัมพันธ์ครบถ้วน
- **Seed Reliability:** สคริปต์ `seed.ts` มีการคืนบล็อก `.finally()` สำหรับตัดการเชื่อมต่อ Prisma Connection หลังทำงานเสร็จเรียบร้อย ข้อมูล Seed มีความหลากหลาย เป็น Idempotent และแฮชรหัสผ่านด้วย `bcrypt` อย่างปลอดภัย
- **Review Record:** อัปเดตบันทึกประวัติการรีวิวและแนวทางการแก้ไขใน `reviewer.md` เรียบร้อย

- Partner Response and Follow-up Actions:

ข้อความรีวิวตรวจงาน (Reviewer Comments):

- > ### Peer Review: Database Schema & Seed Data (PR #57)
- > ตรวจโค้ดส่วน Database Schema และ Seed Script เรียบร้อยแล้ว โครงสร้างส่วนใหญ่ทำได้ถูกต้องตามสเปก Section 5 ของ Lab 3:
- >
- > **Schema & Relationships:** โมเดล `User`, `Role`, `PublicComment`, `InternalNote` และการปรับฟิลด์บน `Ticket` (`ownerId`, `isResolutionIndicated`, สถานะใหม่) ถูกต้องสมบูรณ์ รักษาระบบ ID เป็น `Int` ทำให้ข้อมูลและ Foreign Key เดิมของ Lab 2 ยังใช้งานได้ต่อเนื่อง
- > **Security:** รหัสผ่านเข้ารหัสผ่าน `bcrypt` (10 salt rounds) ไม่เก็บ Plaintext ใน Database
- > * **Seed Requirements:** ข้อมูล Seed เป็น Idempotent มี User ครบตามเกณฑ์ (Requester 4+1, IT Staff 3+1, Admin 1) พร้อมตัวอย่าง Tickets, Public Comments และ Internal Notes ครบถ้วน
- >
- > มีจุดที่ต้องปรับเพิ่ม 2 ข้อก่อน Approve & Merge :
- >
- > 1. **Commit ไฟล์ Migration:** ยังขาดไฟล์ Migration ของ Lab 3 ใน `server/prisma/migrations/` รบกวนรัน `npm run prisma migrate dev --name lab3_db_init` แล้ว commit ไฟลเดอร์ migration เข้ามาด้วย
- > 2. **Connection Cleanup ใน seed.ts:** เพิ่ม `.finally(async () => { await getPrisma().$disconnect(); })` กลับเข้ามาตรงท้ายไฟล์ `server/prisma/seed.ts` เพื่อปิด Database Connection หลังรัน seed เสร็จ
- >
- > ปรับ 2 จุดนี้แล้ว push ขึ้นมาได้เลย

ข้อความตอบกลับและผลการแก้ไขจากบัดดี้ (Partner Response & Actions):

ขอบคุณสำหรับการรีวิว ได้ปรับปรุงแก้ไขครบทั้ง 2 ข้อใน commit 107968d เรียบร้อยแล้ว:

1. Migration File: เพิ่มไฟล์ Migration ของ Lab 3 ในโฟลเดอร์

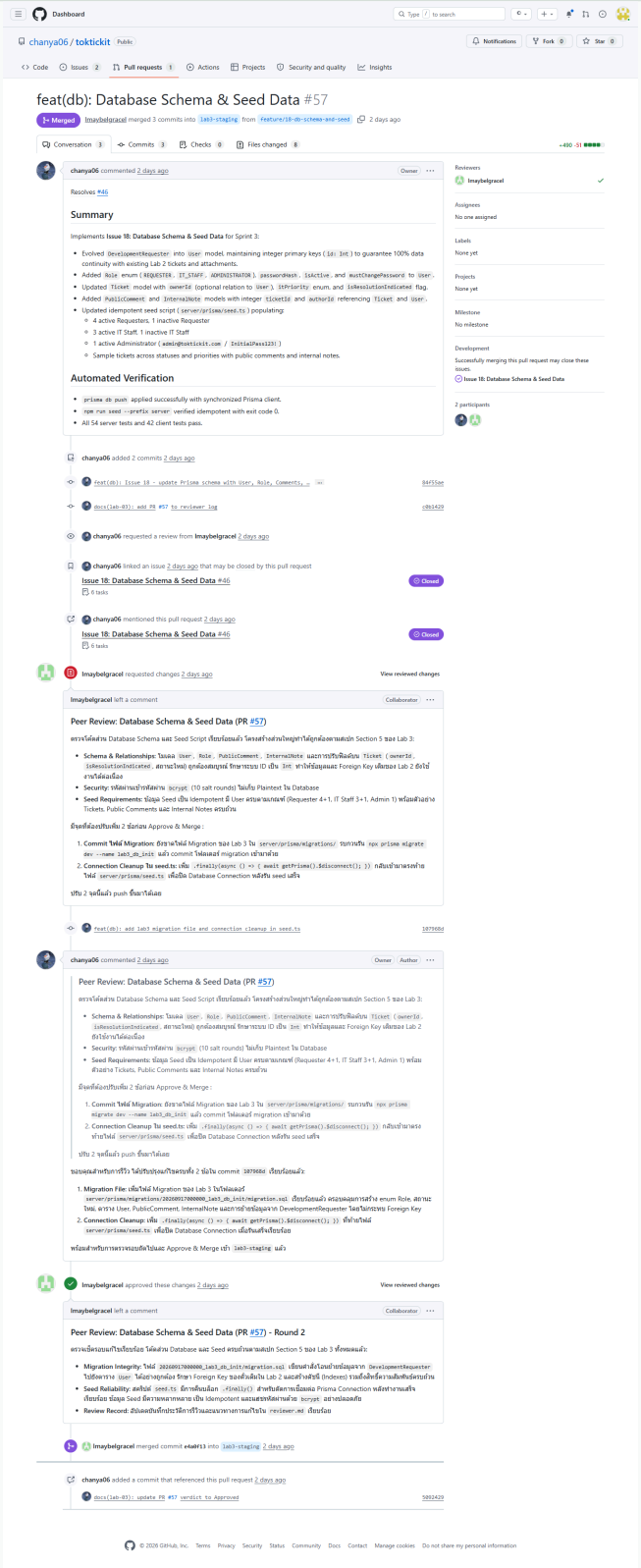
`server/prisma/migrations/20260917000000_lab3_db_init/migration.sql` เรียบร้อยแล้ว ครอบคลุมการสร้าง enum Role, สถานะใหม่, ตาราง User, PublicComment, InternalNote และการย้ายข้อมูลจาก DevelopmentRequester โดยไม่กระทบ Foreign Key

2. Connection Cleanup: เพิ่ม `.finally(async () => { await getPrisma().$disconnect(); })` ที่ท้ายไฟล์

`server/prisma/seed.ts` เพื่อปิด Database Connection เมื่อรันเสร็จเรียบร้อยแล้ว

พร้อมสำหรับการตรวจสอบถัดไปและ Approve & Merge เข้า lab3-staging แล้ว

• Evidence: [PR #57 Review Conversation](#)



PR #58 — feat(auth): Authentication Foundation & Password Change

- Pull Request: [#58](#)
- Branch: `feature/19-auth-api`
- What I Reviewed & Feedback Given:
- Round 1 **COMMENTED** - 2026-09-16):

Peer Review: Authentication Foundation & Password Change (PR #58)

ตรวจสอบโค้ดส่วน Authentication และ Password Management เรียบร้อยแล้ว ฟังก์ชันและเทสตามข้อกำหนดของ Lab 3 Section 4.1, 4.4 และ 6:

- **Authentication & Session:** Endpoint `/api/auth/login`, `/logout`, `/me` และ `/change-password` ทำงานได้สมบูรณ์ รองรับทั้ง Bearer Token และ HTTP-Only Cookie
- **Security & Business Rules:** ปฏิบัติตาม BR-01 (สภาวะบัญชี Inactive), BR-02 (มี `requirePasswordChanged` middleware), BR-04 (รหัสผ่านแฮชด้วย `bcrypt` และไม่ `expose hash` ออกไปภายนอก) และมีการตรวจสอบ Password Strength ครบทุกเงื่อนไข
- **Automated Tests:** ครอบคลุม Test ID API-01, API-02, API-03 พร้อม Edge Cases (รหัสผิด, อีเมลไม่มีในระบบ, รหัสใหม่ไม่ปลอดภัย) ครบถ้วน
- **ข้อเสนอแนะเพิ่มเติม (Optional):**
 - ใน `auth.api.test.ts` อาจเพิ่มการคืนค่ารหัสผ่านเดิมของ user ใน `afterAll` เพื่อให้ชุดเทสรันซ้ำได้โดยไม่ต้อง re-seed
 - เพิ่ม `JWT_SECRET` ใน `server/.env.example`

- Round 2 **APPROVED** - 2026-09-16):

Peer Review: Authentication Foundation & Password Change (PR #58) - Round 2

ตรวจสอบโค้ดรอบแก้ไขเรียบร้อยแล้ว ข้อเสนอแนะเพิ่มเติมได้รับการปรับปรุงครบถ้วน:

- **Idempotent Test Suite:** เพิ่ม `afterAll` คืนค่ารหัสผ่านเริ่มต้นและรีเซ็ต `mustChangePassword: true` ใน `auth.api.test.ts` ทำให้ชุดทดสอบรันซ้ำได้โดยไม่กระทบข้อมูลในฐานข้อมูล
- **Environment Documentation:** ระบุตัวแปร `JWT_SECRET` ใน `server/.env.example` เรียบร้อย
- **Audit Trail:** อัปเดตบันทึกผลการรีวิวใน `docs/lab-03/reviewer.md` ครบถ้วน

- Partner Response and Follow-up Actions:

ข้อความรีวิวตรวจงาน (Reviewer Comments):

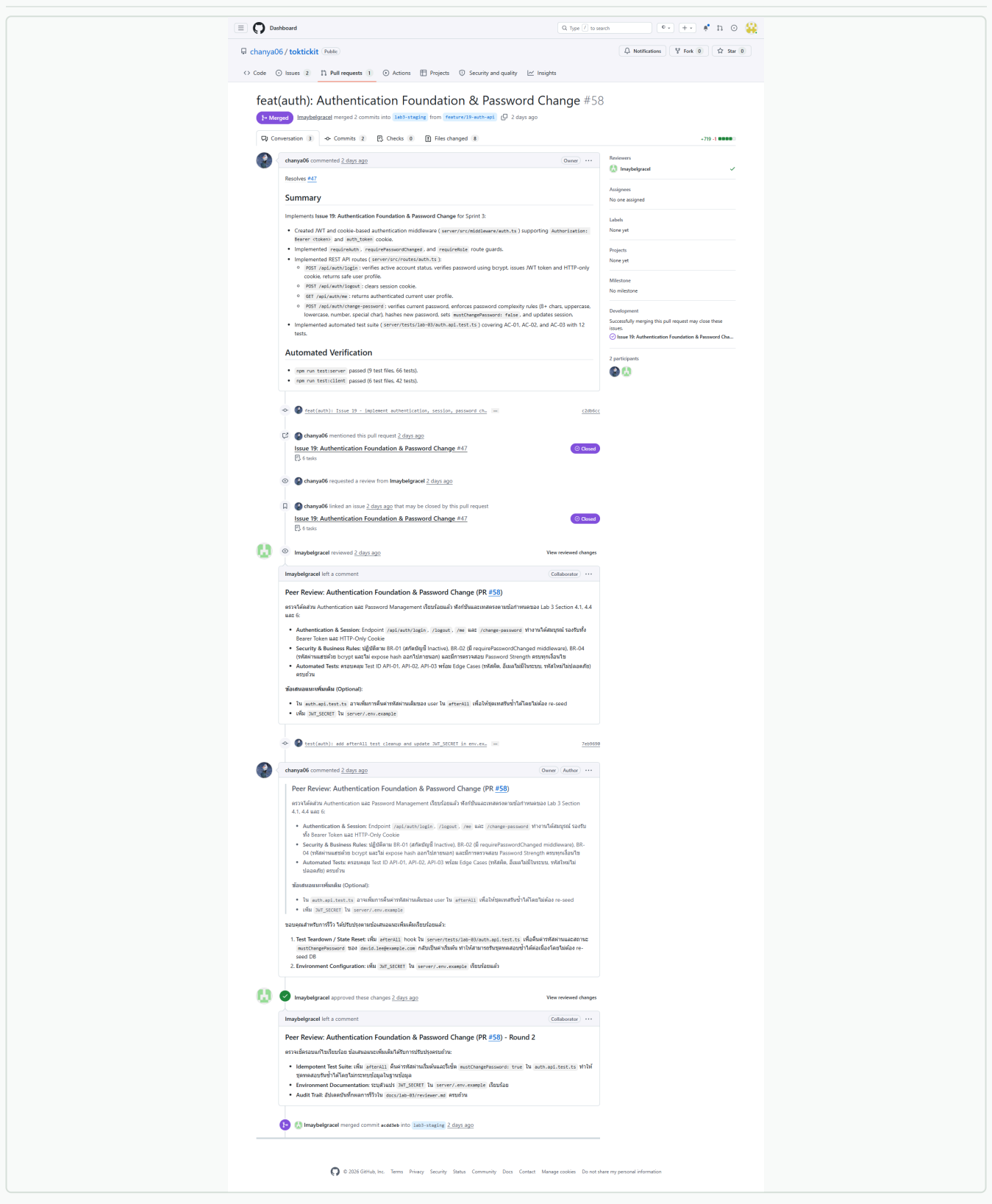
- > ### Peer Review: Authentication Foundation & Password Change (PR #58)
- > ตรวจสอบโค้ดส่วน Authentication และ Password Management เรียบร้อยแล้ว ฟังก์ชันและเทสตรงตามข้อกำหนดของ Lab 3 Section 4.1, 4.4 และ 6:
- >
- > **Authentication & Session:** Endpoint `/api/auth/Login`, `/Logout`, `/me` และ `/change-password` ทำงานได้สมบูรณ์ รองรับทั้ง Bearer Token และ HTTP-Only Cookie
- > **Security & Business Rules:** ปฏิบัติตาม BR-01 (สภาวะบัญชี Inactive), BR-02 (มี `requirePasswordChanged` middleware), BR-04 (รหัสผ่านแฮชด้วย `bcrypt` และไม่ `expose hash` ออกไปภายนอก) และมีการตรวจสอบ Password Strength ครบทุกเงื่อนไข
- > **Automated Tests:** ครอบคลุม Test ID API-01, API-02, API-03 พร้อม Edge Cases (รหัสผิด, อีเมลไม่มีในระบบ, รหัสใหม่ไม่ปลอดภัย) ครบถ้วน
- >
- > **ข้อเสนอแนะเพิ่มเติม (Optional):**
- >
- > ใน `auth.api.test.ts` อาจเพิ่มการคืนค่ารหัสผ่านเดิมของ user ใน `afterAll` เพื่อให้ชุดเทสรันซ้ำได้โดยไม่ต้อง re-seed
- > * เพิ่ม `JWT_SECRET` ใน `server/.env.example`

ข้อความตอบกลับและผลการแก้ไขจากบัดดี้ (Partner Response & Actions):

ขอบคุณสำหรับการรีวิว ได้ปรับปรุงตามข้อเสนอแนะเพิ่มเติมเรียบร้อยแล้ว:

1. **Test Teardown / State Reset:** เพิ่ม `afterAll` hook ใน `server/tests/lab-03/auth.api.test.ts` เพื่อคืนค่ารหัสผ่านและสถานะ `mustChangePassword` ของ `david.lee@example.com` กลับเป็นค่าเริ่มต้น ทำให้สามารถรันชุดทดสอบซ้ำได้ต่อเนื่องโดยไม่ต้อง re-seed DB
2. **Environment Configuration:** เพิ่ม `JWT_SECRET` ใน `server/.env.example` เรียบร้อยแล้ว

- Evidence: [PR #58 Review Conversation](#)



PR #59 — feat(auth-ui): Login Screen, Mandatory Password Change & Header Shell (Issue 20)

- Pull Request: [#59](#)
- Branch: `feature/20-auth-ui`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES_REQUESTED** - 2026-09-16):

Peer Review: Login Screen, Mandatory Password Change & Header Shell (PR #59)

ตรวจโค้ดใน PR #59 และรัน Flow เทียบกับสเปก Lab 3 และ UI Spec เรียบร้อยแล้ว ตัว Component UI ทำระบบ Checklist ตรวจรหัสผ่าน และ Badge ของ Role ได้เรียบร้อยดีมาก แต่มีจุดบกพร่องเชิง Logic การทำงานของ Auth และ Routing ที่ต้องปรับแก้ก่อน Merge ดังนี้:

1. แก้ไข Login Bypass จาก `toktickit_dev_requester_id` ใน `LocalStorage` (`client/src/App.tsx`)

- ใน `App.tsx` เปลี่ยน `isDevRequesterFlow` ไปเช็ค `localStorage.getItem("toktickit_dev_requester_id")` โดยตรง
- ถ้าเครื่องใครเคยรัน Lab 2 มาก่อน 💎💎 ค่านี้จะค้างอยู่ในเบราว์เซอร์ ทำให้พอเปิดหน้าเว็บขึ้นมาจะหลุดข้ามหน้า Login ไปเข้า UI ของ Lab 2 ทันที
- สิ่งที่ต้องปรับ: ให้ผู้แก้ไขนี้เฉพาะตอนรันโหมดทดสอบเท่านั้น เช่น:

```
``typescript
```

```
const isDevRequesterFlow =  
import.meta.env.MODE === "test" &&  
(Boolean(localStorage.getItem("toktickit_dev_requester_id")) || isModalOpen);
```

- Round 2 **CHANGES_REQUESTED** - 2026-09-17):

ติดตามผลการรีวิว PR #59

ตรวจเช็ค commit `0a604aa` เรียบร้อยแล้ว:

- จุดที่ 1 เรื่อง Login Bypass จากค่า `LocalStorage` เดิม แก้ไขได้ถูกต้องดีแล้ว ผู้แก้ไขกับ `import.meta.env.MODE === "test"` ช่วยให้เปิดบนเบราว์เซอร์ปกติจะบังคับแสดงหน้า Login เสมอ

แต่ยังมีอีก 3 จุดที่ยังค้างอยู่ รบกวนช่วยปรับเพิ่มอีกนิด:

1. Initial Tab ตาม Role (`client/src/App.tsx`) :

- ตอนนี้ `activeTab` เริ่มต้นเป็น `"my-tickets"` ตลอด พอ `IT_STAFF` หรือ `ADMINISTRATOR` ล็อกอินเข้ามา จะถูกพาไปหน้าของ Requester และใน `Navbar` จะไม่มีแท็บไหนถูกเลือก

- ให้เพิ่ม `useEffect` เช็ค `user?.role` เมื่อล็อกอินสำเร็จ เพื่อดังแท็บเริ่มต้นให้ตรง Role:

- Admin -> `"user-management"`

- IT Staff -> `"ticket-queue"`

- Requester -> `"my-tickets"`

2. บันทึกรหัส Token ใหม่ใน `changePassword` (`client/src/api.ts` & `AuthContext.tsx`) :

- ใน `client/src/api.ts` ฟังก์ชัน `changePassword` ยังขาด `if (data.token) setStoredToken(data.token);`

- ทำให้ Client ยังถือ Token เก่าหลังเปลี่ยนรหัสผ่าน ให้เพิ่มการบันทึก Token และ sync state ใน `AuthContext.tsx` ด้วย

3. นำ Debug Component ของ Lab 1 ออก (`client/src/App.tsx`) :

- นำ ออกจากหน้า Login และหน้าจอหลัก เพื่อให้ UI ตรงตามสเปกหน้า 8 ของอาจารย์

แก้ 3 จุดนี้เสร็จแล้ว push ขึ้นมาได้เลย เดี่ยวตรวจให้ทันที

• Round 3 (APPROVED - 2026-09-17):

Peer Review (PR #59)

ตรวจเช็คการแก้ไขใน commit 8116192 เทียบกับข้อเสนอแนะในรอบที่แล้วเรียบร้อยแล้ว:

1. Role-Based Tab Routing: ระบบตั้งค่าแท็บเริ่มต้น (`user-management` , `ticket-queue` , `my-tickets`) ตาม Role ของผู้ใช้ที่ล็อกอินถูกต้อง และไฮไลต์แท็บใน Navbar ตรงตามสเปก
2. Token Persistence: ฟังก์ชัน `changePassword` บันทึก Token ~~ลงใน~~ LocalStorage และซิงก์เข้า AuthContext เรียบร้อย
3. UI Spec Adherence: นำ Debug Component () ออกจากหน้า Login และหน้าจอหลัก ทำให้ UI สะอาดตรงตามแบบร่าง หน้า 8 ของเอกสาร Lab 3
4. Automated Test Coverage: มีการเพิ่มชุดทดสอบใน `AppRoleNav.test.tsx` และ `ChangePassword.test.tsx` ครอบคลุมพฤติกรรมที่ปรับแก้ทั้งหมด

โค้ดพร้อมสำหรับการนำไปใช้งาน สามารถ Merge เข้าสู่ branch `lab3-staging` ได้เลย

• Partner Response and Follow-up Actions:

ข้อความรีวิวดูงาน (Reviewer Comments):

```
> ### Peer Review: Login Screen, Mandatory Password Change & Header Shell (PR #59)
> ตรวจโค้ดใน PR #59 และรัน Flow เทียบกับสเปก Lab 3 และ UI Spec เรียบร้อยแล้ว ตัว Component UI ทำระบบ Checklist
ตรวจรหัสผ่าน และ Badge ของ Role ได้เรียบร้อยแล้ว แต่มีจุดบกพร่องเชิง Logic การทำงานของ Auth และ Routing ที่ต้องปรับแก้
ก่อน Merge ดังนี้:

>
> ##### 1. แก้ไข Login Bypass จาก tokticket_dev_requester_id ใน LocalStorage ( client/src/App.tsx )

> ใน App.tsx เงื่อนไข isDevRequesterFlow ไปเช็ค localStorage.getItem("tokticket_dev_requester_id") โดยตรง

> ถ้าเครื่องใครเคยรัน Lab 2 มาก่อน คำนี้อาจอยู่ในเบราว์เซอร์ ทำให้พอเปิดหน้าเว็บขึ้นมาจะหลุดข้ามหน้า Login ไปเข้า UI ของ Lab
2 ทันที

> สิ่งที่ต้องปรับ: ให้ผู้เจ้🔗อนไขนี้เฉพาะตอนรันโหมดทดสอบเท่านั้น เช่น:

> `ts

> const isDevRequesterFlow =
> import.meta.env.MODE === "test" &&
> (Boolean(localStorage.getItem("tokticket_dev_requester_id")) || isModalOpen);
```

> \

ข้อความตอบกลับและผลการแก้ไขจากบัคค์ (Partner Response & Actions):

ขอบคุณสำหรับการรีวิและตรวจพบจุด Login Bypass ได้ทำการแก้ไขใน commit ล่าสุดเรียบร้อยแล้ว:

1. ป้องกัน Login Bypass (client/src/App.tsx) : ปรับเงื่อนไข isDevRequesterFlow ให้ตรวจสอบ

import.meta.env.MODE === "test" รวมด้วย ทำให้การเข้าใช้งานผ่านเบราว์เซอร์จริงจะแสดงหน้า Login (LoginView) เสมอ แม้จะมีค่า tokticket_dev_requester_id ค้างอยู่ใน LocalStorage จาก Lab 2 ก็ตาม ขณะเดียวกันยังคงรักษาความเข้ากันได้กับการรันชุดทดสอบของ Lab 2 ไว้อย่างสมบูรณ์ (52/52 tests pass)

> ### ติดตามผลการรีวิ PR #59

> ตรวจเช็ค commit 0a604aa เรียบร้อยแล้ว:

>

> จุดที่ 1 เรื่อง Login Bypass จากค่า LocalStorage เดิม แก้ไขได้ถูกต้องแล้ว ผูกเงื่อนไขกับ import.meta.env.MODE === "test" ช่วยให้เปิดบนเบราว์เซอร์ปกติจะบังคับแสดงหน้า Login เสมอ

>

> แต่ยังมีอีก 3 จุดที่ยังค้างอยู่ รบกวนช่วยปรับเพิ่มอีกนิด:

>

> 1. Initial Tab ตาม Role (client/src/App.tsx) :

>

> ตอนนี้ activeTab ตั้งต้นเป็น "my-tickets" ตลอด พอ IT_STAFF หรือ ADMINISTRATOR ล็อกอินเข้ามา จะถูกพาไปหน้า ของ Requester และใน Navbar จะไม่มีแท็บไหนถูกเลือก

> ให้เพิ่ม useEffect เช็ค user?.role เมื่อล็อกอินสำเร็จ เพื่อดังแท็บเริ่มต้นให้ตรง Role:

>

> Admin -> "user-management"

> IT Staff -> "ticket-queue"

> Requester -> "my-tickets"

> 2. บันทึก Token ใหม่ใน changePassword (client/src/api.ts & AuthContext.tsx) :

>

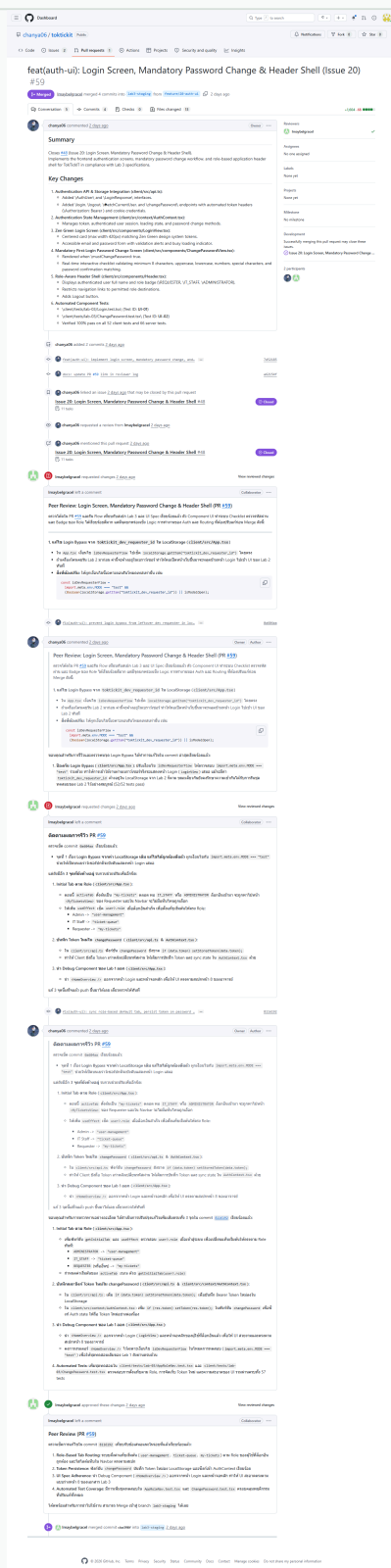
> ใน client/src/api.ts ฟังก์ชัน changePassword ยังขาด if (data.token) setStoredToken(data.token);

> * ทำให้ Client ยังถือ Token เก่าหลังเปลี่ยนรหัสผ่าน ให้เติมการบันทึก Token และ sync state ใน AuthContext.tsx ด้วย

> 3. นำ Debug Component ของ Lab 1 ออก (client/src/App.tsx`):

- Evidence: [PR #59 Review Conversation](#)

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): chanya_pr59_full.png



PR #60 — feat: requester session regression and resolution indication action (#49)

- Pull Request: [#60](#)
- Branch: `feature/21-requester-session`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES_REQUESTED** - 2026-09-17):

Peer Review: Requester Session Regression & Resolution Indication Action (PR #60)

ตรวจโค้ดใน PR #60 เทียบกับสเปก Lab 3, Handout (Section 1, 4.3, 8.2) และ Issue #49 เรียบร้อยแล้ว
จุดเด่นที่ทำได้:

- ฝั่ง Backend ทำ Data Isolation รัดกุม ป้องกันไม่ให้ Requester เข้าถึงหรือระบุ requesterId ของผู้อื่น (403 Forbidden)
- ฟังก์ชัน POST /api/tickets/:id/resolve-indication ใช้ Prisma \$transaction อัปเดตแฟล็ก isResolutionIndicated: true พร้อมสร้าง PublicComment แบบ Atomic ได้ถูกต้องตาม BR-19
- มีชุดทดสอบครอบคลุมทั้ง API Authorization และ Resolution Indication ครบถ้วน

จุดบกพร่องที่ต้องแก้ไขก่อน Merge (Request Changes):

1. แก้ไข State Wipeout ใน TicketDetailView.tsx หลังกดส่งสัญญาณ Resolution (บรรทัดที่ 43):

- ปัจจุบันมีการเรียก setTicket(res.ticket); โดยตรง แต่ API ส่งกลับมาเฉพาะ partial fields (id, ticketNumber, status, isResolutionIndicated) ทำให้ฟิลด์อื่นๆ ทั้งหมด (Summary, Description, Category, Priority, CreatedAt) กลายเป็น undefined ส่ง 3 ดiamond ให้หน้าจอข้อมูลตัวหาย เกือบทั้งหน้าที่ยืนยัน

- วิธีแก้: ให้ทำ State Merge กับข้อมูลเดิม:

```
setTicket((prev) => (prev ? { ...prev, ...res.ticket } : res.ticket));
```

2. แก้ไขแท็บ Attachments หมุนค้างในหน้าตัวของ User ที่ล็อกอิน (AttachmentSection.tsx):

- ใน client/src/components/AttachmentSection.tsx ยังไม่ได้เชื่อมต่อกับ AuthContext ยังคงใช้เฉพาะ const { selectedRequester } = useRequester();

- เมื่อล็อกอินด้วย User จริงใน Lab 3 ค่า selectedRequester จะเป็น null ทำให้:

- ฟังก์ชัน loadAttachments ติดเงื่อนไข if (!selectedRequester) return; และค่าสถานะ loading: true เกิด Spinner หมุนค้างตลอดกาล

- ฟังก์ชัน Upload, Download และ Soft-remove ไม่ทำงานเพราะติดเงื่อนไขเดียวกัน

- วิธีแก้: ให้ดึง user จาก AuthContext มาสร้าง effectiveRequesterId เช่นเดียวกับใน TicketDetailView และ MyTicketsView:

```
const auth = useContext(AuthContext);
```

```
const user = auth?.user;
```

```
const { selectedRequester } = useRequester();
```

```
const effectiveRequesterId = user ? user.id : selectedRequester?.id;
```

แล้วเปลี่ยนเงื่อนไข Guard และการส่ง Request ให้ใช้ effectiveRequesterId แทน

3. อัปเดต Traceability Table ใน docs/lab-03/tests.md:

- ในแถว API-14 แก่ชื่อไฟล์ Automated Test ให้ตรงกับไฟล์ที่สร้างจริงเป็น server/tests/lab-03/requester-resolution.api.test.ts และเพิ่มการอ้างอิงชุดทดสอบ UI client/tests/lab-03/RequesterResolution.test.tsx เพื่อให้คะแนน Traceability ใน Rubric สมบูรณ์

คำตัดสิน: Request changes ปรับแก้ 3 ข้อนี้เสร็จแล้ว push ขึ้นมาใหม่ เดียวตรวจให้นะ

- Round 2 **APPROVED** - 2026-09-17):

Peer Review Verdict: **APPROVED** (PR #60)

ตรวจเช็คการแก้ไขใน commit `8b3ad62` เทียบกับข้อเสนอแนะรอบที่แล้วอย่างละเอียดเรียบร้อยแล้ว:

1. **State Preservation:** ใน `TicketDetailView.tsx` มีการทำ State Merge ข้อมูลเดิมเข้ากับ `res.ticket` อย่างถูกต้อง ข้อมูลทั้งหมด (Summary, Description, Category, Priority ฯลฯ) ยังคงแสดงครบถ้วน ไม่สูญหายหลังกดส่งสัญญาณ Resolution

2. **Attachment Lifecycle Under Auth:** ใน `AttachmentSection.tsx` มีการดึง `user` จาก `AuthContext` มาคำนวณ `effectiveRequesterId` แก้ปัญหา Spinner หมุนค้าง และทำให้ฟังก์ชันการโหลด/อัปโหลด/ดาวน์โหลด/Soft-remove ไฟล์แนบสำหรับ Authenticated User ทำงานได้สมบูรณ์

3. **Traceability Table Aligned:** ตาราง `docs/lab-03/tests.md` อัปเดตลิงก์ไฟล์ทดสอบจริงทั้งฝั่ง Backend (`requester-resolution.api.test.ts`) และ Frontend (`RequesterResolution.test.tsx`) ครบถ้วนตาม Rubric

โค้ดทำงานถูกต้อง รัดกุม และไม่พบจุดบกพร่องเพิ่มเติม สามารถ Merge เข้าสู่ branch `lab3-staging` ได้เลย

- Partner Response and Follow-up Actions:

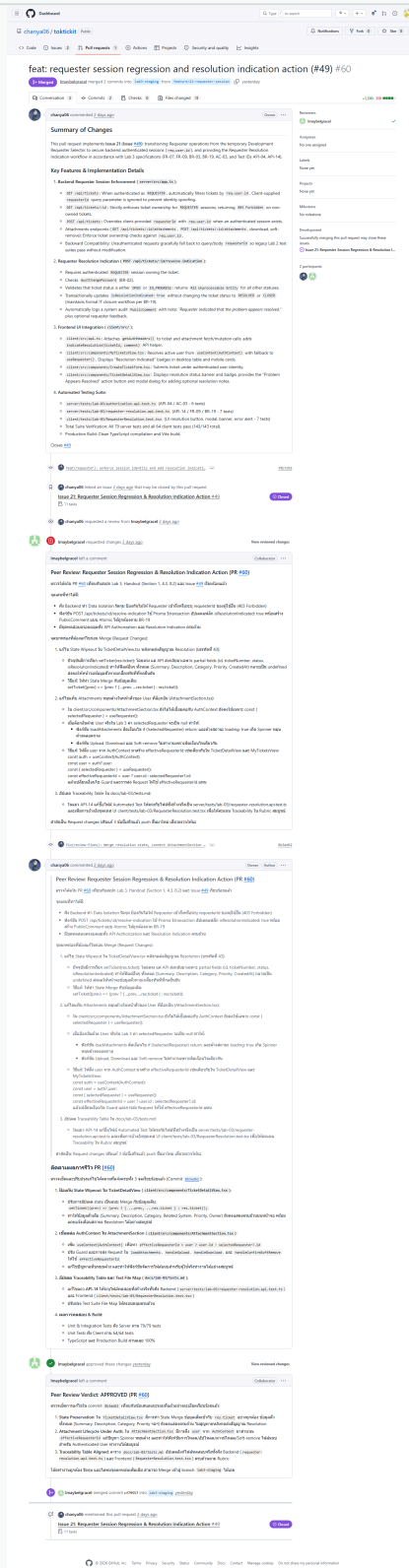
> ### Peer Review: Requester Session Regression & Resolution Indication Action (PR #60)

> ตรวจโค้ดใน PR #60 เทียบกับสเปก Lab 3, Handout (Section 1, 4.3, 8.2) และ Issue #49 เรียบร้อยแล้ว

- >
- > จุดเด่นที่ทำได้ดี:
- >
- >ฝั่ง Backend ทำ Data Isolation รัดกุม ป้องกันไม่ให้ Requester เข้าถึงหรือระบุ requesterId ของผู้ใช้อื่น (403 Forbidden)
- > ฟังก์ชัน POST /api/tickets/:id/resolve-indication ใช้ Prisma \$transaction อัปเดตแฟล็ก isResolutionIndicated: true พร้อม💎💎สร้าง PublicComment แบบ Atomic ได้ถูกต้องตาม BR-19
- > มีชุดทดสอบครอบคลุมทั้ง API Authorization และ Resolution Indication ครบถ้วน
- >
- > จุดบกพร่องที่ต้องแก้ไขก่อน Merge (Request Changes):
- >
- > 1. แก้ไข State Wipeout ใน TicketDetailView.tsx หลังกดส่งสัญญาณ Resolution (บรรทัดที่ 43):
- >
- > ปัจจุบันมีการเรียก setTicket(res.ticket); โดยตรง แต่ API ส่งกลับมาเฉพาะ partial fields (id, ticketNumber, status, isResolutionIndicated) ทำให้ฟิลด์อื่นๆ ทั้งหมด (Summary, Description, Category, Priority, CreatedAt) กลายเป็น undefined ส่งผลให้หน้าจอข้อมูลตัวหายเกลี้ยงทันทีที่กดยืนยัน
- > วิธีแก้: ให้ทำ State Merge กับข้อมูลเดิม:
- > setTicket((prev) => (prev ? { ...prev, ...res.ticket } : res.ticket));
- > 2. แก้ไขแท็บ Attachments หมุนค้างในหน้าตัวของ User ที่ล็อกอิน (AttachmentSection.tsx):
- >
- > ใน client/src/components/AttachmentSection.tsx ยังไม่ได้เชื่อมต่อกับ AuthContext ยังคงใช้เฉพาะ const { selectedRequester } = useRequester();
- > * เมื่อล็อกอินด้วย User จริงใน Lab 3 ค่า selectedRequester จะเป็น null ทำให้:

• Evidence: [PR #60 Review Conversation](#)

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): chanya_pr60_full.png



PR #61 — feat: IT Staff ticket queue retrieval and query engine (#50)

- Pull Request: [#61](#)
- Branch: `feature/22-staff-queue-api`
- What I Reviewed & Feedback Given:
- Round 1 **APPROVED** - 2026-09-17):

Peer Review Verdict: **APPROVED** (PR #61 — Issue 22 / Issue #50)

ตรวจเช็คโค้ดของ Issue 22 (GitHub Issue #50: IT Staff Ticket Queue Retrieval & Query Engine) อย่างละเอียดทุกส่วนเรียบร้อยแล้ว:

1. Security & Role Protection:

- บังคับใช้ `requireAuth`, `requirePasswordChanged` และ `requireRole(Role.IT_STAFF, Role.ADMINISTRATOR)` ป้องกัน Requester และดักจับ Session ได้ถูกต้องสมบูรณ์

2. Query Engine & Data Integrity:

- ค้นหา Substring บน `ticketNumber` และ `summary` แบบ Case-insensitive
- ฟังก์ชันรองรับทั้ง Single และ Multi-value บน `category`, `status`, และ `itPriority`
- ตัวกรอง `ownerId` จัดการเงื่อนไข `unassigned (ownerId: null)`, `me (ownerId: req.user.id)`, Specific ID และ `all` ได้ถูกต้อง
- `attachmentCount` นับเฉพาะไฟล์แนบที่ `{ isRemoved: false }` ตรงตามสเปก

3. Sorting & Pagination Compatibility:

- ป้องกัน Injection ด้วย `ALLOWED_SORT_FIELDS`
- คำนวณ Pagination ทั้งแบบ `page / limit` และ `currentPage / pageSize` พร้อมสำหรับการต่อ Component ใน Issue 23 ทันที

4. Test Suite Completeness:

- `server/tests/lab-03/staff-queue.api.test.ts` ครอบคลุมทั้ง 14 Scenarios ครอบคลุมทุกเงื่อนไขการทำงาน พร้อมมี `afterAll Teardown` ข้อมูลทดสอบสะอาดเรียบร้อย

โค้ดมีความรัดกุมและพร้อมสำหรับการเชื่อมต่อ UI ใน Issue 23 สามารถ Merge เข้าสู่ `lab3-staging` ได้เลย

• Partner Response and Follow-up Actions:

> ### Peer Review Verdict: **APPROVED** (PR #61 — Issue 22 / Issue #50)

> ตรวจเช็คโค้ดของ Issue 22 (GitHub Issue #50: IT Staff Ticket Queue Retrieval & Query Engine) อย่างละเอียดทุกส่วนเรียบร้อยแล้ว:

>

> 1. Security & Role Protection:

>

> บังคับใช้ `requireAuth`, `requirePasswordChanged` และ `requireRole(Role.IT_STAFF, Role.ADMINISTRATOR)` ป้องกัน Requester และดักจับ Session ได้ถูกต้องสมบูรณ์

> 2. Query Engine & Data Integrity:

>

> ค้นหา Substring บน `ticketNumber` และ `summary` แบบ Case-insensitive

> ฟังก์ชันรองรับทั้ง Single และ Multi-value บน `category`, `status`, และ `itPriority`

> ตัวกรอง `ownerId` จัดการเงื่อนไข `unassigned (ownerId: null)`, `me (ownerId: req.user.id)`, Specific ID และ `all` ได้ถูกต้อง

> `attachmentCount` นับเฉพาะไฟล์แนบที่ `{ isRemoved: false }` ตรงตามสเปก

> 3. Sorting & Pagination Compatibility:

>

> ป้องกัน Injection ด้วย `ALLOWED_SORT_FIELDS`

> คำนวณ Pagination ทั้งแบบ `page / limit` และ `currentPage / pageSize` พร้อมสำหรับการต่อ Component ใน Issue 23 ทันที

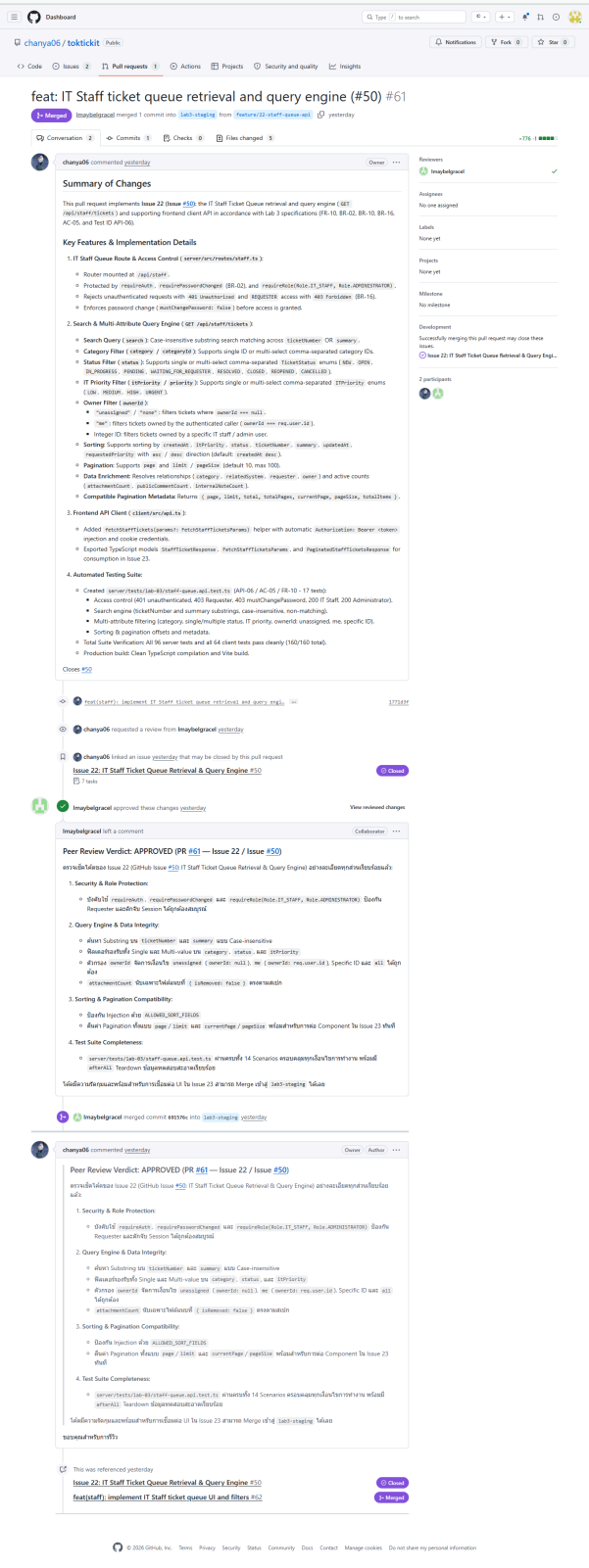
> 4. Test Suite Completeness:

>

> `server/tests/lab-03/staff-queue.api.test.ts` ครอบคลุมทั้ง 14 Scenarios ครอบคลุมทุกเงื่อนไขการทำงาน พร้อมมี `afterAll Teardown` ข้อมูลทดสอบสะอาดเรียบร้อย

>

- Evidence: [PR #61 Review Conversation](#)



PR #62 — feat(staff): implement IT Staff ticket queue UI and filters

- Pull Request: [#62](#)
- Branch: `feature/23-staff-queue-ui`
- What I Reviewed & Feedback Given:
- Round 1 **COMMENTED** - 2026-09-17):

ข้อเสนอแนะเพิ่มเติมสำหรับ PR #62 (Staff Ticket Queue UI)

ตรวจสอบเทียบกับ `ui-spec.md` และการใช้งานจริงอย่างละเอียดอีกครั้ง พบจุดที่อยากให้ช่วยปรับปรุงเพิ่มเติมอีกเล็กน้อยเพื่อสมบูรณ์ตามสเปกของอาจารย์:

1. เพิ่มคอลัมน์ Requested Priority ในตาราง Desktop (StaffTicketQueue.tsx):

- ตามสเปก Screen 3 ของ `ui-spec.md` กำหนดให้ตารางแสดงทั้ง Requested Priority (ที่ผู้ใช้ขอมมา) ควบคู่กับ IT Priority (ที่ IT ประเมิน)
- ปัจจุบันในตารางมีเฉพาะ IT Priority ปรกวนเพิ่มคอลัมน์ Requested Priority พร้อม Badge เข้าไปในตารางด้วย

2. รีเซ็ตหน้ากลับไปที่หน้า 1 เมื่อเปลี่ยนการ Sort (StaffTicketQueue.tsx):

- ที่ Dropdown `queueSortBy` และปุ่มสลับลูกศร `queueSortDir` ปรกวนเพิ่ม `setPage(1)` ด้วยเช่นกัน เพราะถ้าเปิดดูค้างอยู่ที่หน้า 2 หรือ 3 แล้วกดเปลี่ยนการจัดเรียง ลิสต์ควรเริ่มต้นแสดงผลจากหน้าแรก

3. ปรับสี Row Hover Highlight ให้ตรงตาม Zen Green Token:

- ตามสเปกระบุให้แถวตารางเวลาเอาเมาส์ชี้เปลี่ยนเป็นสี `#EAF6EF` (Pale Highlight) ปรกวนเพิ่มสไตล์ hover บนแถวตารางให้ตรงตามโทเคน

4. ปรับเกลี่ยความกว้างคอลัมน์ (Width %) ให้รวมกันได้ 100%:

- ตอนนี้ width รวมใน `table` เกินไปที่ 110% เมื่อเพิ่มคอลัมน์ Requested Priority แล้ว ให้ช่วยเกลี่ยสัดส่วนเปอร์เซ็นต์ให้ลงตัวพอดี 100%

• Round 2 **APPROVED** - 2026-09-17):

Peer Review: Approved (PR #62 - IT Staff Ticket Queue UI)

ได้ตรวจสอบการแก้ไขเพิ่มเติมตาม commit `6ecd380` เทียบกับ Requirement ของ Issue #51 (Task 23), Handout Section 4 และ `ui-spec.md` (Screen 3: IT Staff Ticket Queue) อย่างละเอียดอีกครั้ง:

ผลการตรวจสอบจุดปรับปรุงเพิ่มเติม:

1. Requested Priority Column & Badges:

- เพิ่มคอลัมน์ `Req. Priority` บนตาราง Desktop เรียบร้อย แสดง Badge ตรงตามระดับความสำคัญ (`LOW` , `MEDIUM` , `HIGH` , `URGENT`)
- ใน Mobile Card View มีการแสดง Badge ทั้งส่วนของ `Req:` และ `IT:` สอดคล้องกันอย่างชัดเจน

2. Pagination Reset on Sort Change:

- การเปลี่ยนตัวเลือกใน `queue-sort-by` และการคลิกสลับทิศทางใน `queue-sort-dir` มีการเรียก `setPage(1)` ช่วยป้องกันข้อผิดพลาดกรณีค้างอยู่ที่หน้าถัดไปแล้วจัดเรียงใหม่

3. Zen Green Row Hover Token (#EAF6EF):

- กำหนดสไตล์ Hover ให้ตารางด้วย Pale Highlight Token `#EAF6EF` ที่ `client/src/index.css` และ scoped class `.staff-queue-table` แสดงผลถูกต้องตาม Zen Design System

4. Table Column Width Distribution:

- เกลี่ยความกว้างของคอลัมน์ทั้ง 9 คอลัมน์ (13%, 11%, 22%, 11%, 9%, 9%, 10%, 9%, 6%) รวมกันได้ 100% พอดี ไม่ล้นกรอบตาราง

5. Test Coverage & Verification:

- เพิ่มเคสทดสอบใน `client/tests/lab-03/StaffTicketQueue.test.tsx` ครอบคลุมการแสดงผลคอลัมน์ใหม่และการรีเซ็ตหน้าเมื่อเปลี่ยน Sort อย่างครบถ้วน

• Partner Response and Follow-up Actions:

ข้อความรีวิวตรวจงาน (Reviewer Comments):

- > ### ข้อเสนอแนะเพิ่มเติมสำหรับ PR #62 (Staff Ticket Queue UI)
- > ตรวจสอบเทียบกับ `ui-spec.md` และการใช้งานจริงอย่างละเอียดอีกครั้ง พบจุดที่อยากให้ช่วยปรับปรุงเพิ่มเติมอีกเล็กน้อยเพื่อสมบูรณ์ตามสเปกของอาจารย์:
- >
- > 1. เพิ่มคอลัมน์ Requested Priority ในตาราง Desktop (StaffTicketQueue.tsx):
- >
- > ตามสเปก Screen 3 ของ `ui-spec.md` กำหนดให้ตารางแสดงทั้ง Requested Priority (ที่ผู้ใช้ขอมมา) ควบคู่กับ IT Priority (ที่ IT ประเมิน)
- > ปัจจุบันในตารางมีเฉพาะ IT Priority ปรกวนเพิ่มคอลัมน์ Requested Priority พร้อม Badge เข้าไปในตารางด้วย

> 2. รีเชิดหน้ากลับไปที่หน้า 1 เมื่อเปลี่ยนการ Sort (`StaffTicketQueue.tsx`):

>

> ที่ Dropdown `queueSortBy` และปุ่มสลับลูกศร `queueSortDir` ระบายเพิ่ม `setPage(1)` ด้วยเช่นกัน เพราะถ้าเปิดดูค้างอยู่ที่หน้า 2 หรือ 3 แล้วกดเปลี่ยนการจัดเรียง ลิสต์ควรเริ่มต้นแสดงผลจากหน้าแรก

> 3. ปรับสี Row Hover Highlight ให้ตรงตาม Zen Green Token:

>

> ตามสเปกระบุให้แถวตารางเวลาเอาเมาส์ชี้เปลี่ยนเป็นสี `#EAF6EF` (Pale Highlight) ระบายเพิ่มสไตล์ hover บนแถวตารางให้ตรงตามโทเคน

> 4. ปรับเกลี่ยความกว้างคอลัมน์ (Width %) ให้รวมกันได้ 100%:

>

> * ตอนนี้ width รวมใน เกินไปที่ 110% เมื่อเพิ่มคอลัมน์ Requested Priority แล้ว ให้ช่วยเกลี่ยสัดส่วนเปอร์เซ็นต์ให้ลงตัวพอดี 100%

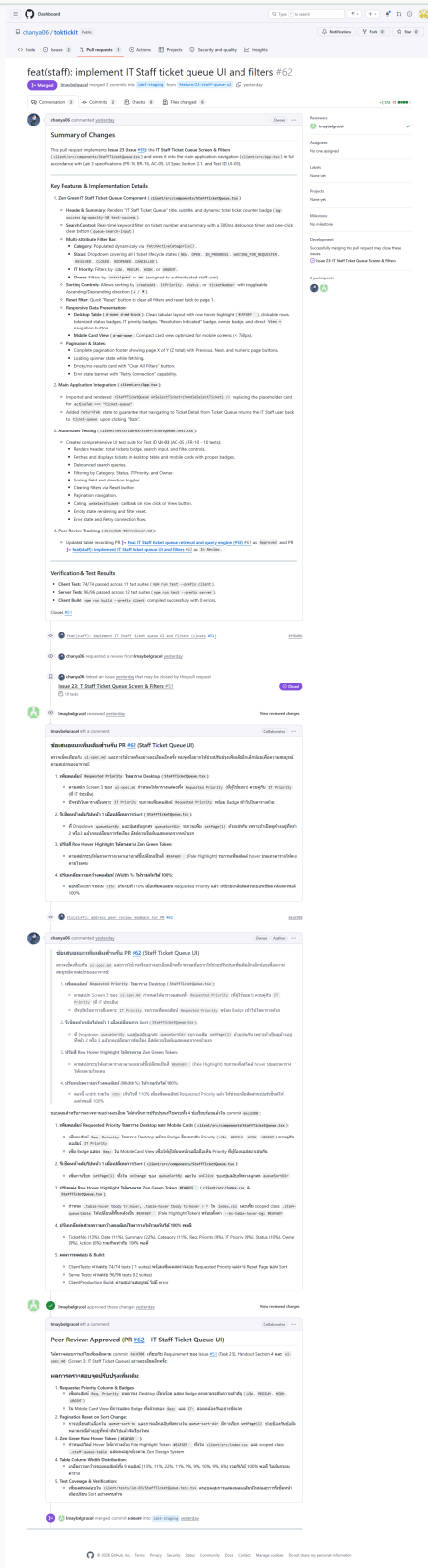
ข้อความตอบกลับและผลการแก้ไขจากบัดดี้ (Partner Response & Actions):

ขอบคุณสำหรับกา💎💎💎ตรวจทานอย่างละเอียด ได้ดำเนินการปรับปรุงแก้ไขครบทั้ง 4 ข้อเรียบร้อยแล้วใน commit `6ecd380` :

1. เพิ่มคอลัมน์ **Requested Priority** ในตาราง **Desktop** และ **Mobile Cards**
(`client/src/components/StaffTicketQueue.tsx`):

• Evidence: [PR #62 Review Conversation](#)

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): chanya_pr62_full.png



PR #63 — feat(staff): IT Staff ticket operations and status matrix (#52)

- Pull Request: [#63](#)
- Branch: `feature/24-staff-operations`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES_REQUESTED** - 2026-09-17):

Peer Review for PR #63: Changes Requested (IT Staff Ticket Operations & Status Matrix)

ได้ตรวจสอบการทำงานของ PR #63 (Issue #52 / Task 24) ทั้ง Backend API, Frontend Operations Panel ❖❖ ละเอียดครบถ้วนอย่างละเอียดเทียบกับ Handout Section 4, `specification.md` (BR-10 ถึง BR-14), `api-spec.md` (Section 2) และ `ui-spec.md` (Screen 4)

การวางโครงสร้างระบบ Claim, Assign, IT Priority และ Status Transition ทำได้ดีมากและมีชุดทดสอบครอบคลุม แต่พบจุดบกพร่องที่ต้องปรับปรุงแก้ไข 4 จุดดังนี้:

1. เพิ่มสถานะ **REOPENED** ใน **Status Transition Matrix** (สำคัญมาก):

- ปัจจุบันใน `PERMITTED_STATUS_TRANSITIONS` (`server/src/routes/staff.ts`) และ `PERMITTED_NEXT_STATUSES` (`client/src/components/TicketDetailView.tsx`) ยังไม่มี Key `[TicketStatus.REOPENED]`
- ส่งผลให้เมื่อตั๋วถูก Reopen แล้ว ตัวเลือกสถานะถัดไปจะว่างเปล่า Dropdown ถูก Disable และ API ปฏิเสธด้วย 422 กลายเป็น Dead-end status
- รบกวนเพิ่ม Key **REOPENED** ทั้งฝั่ง Backend และ Frontend เช่น:

```
`` ts  
  
[TicketStatus.REOPENED]: [  
TicketStatus.IN_PROGRESS,  
TicketStatus.RESOLVED,  
TicketStatus.CANCELLED,  
],  
`
```

2. อัปเดต `getStatusBadgeClass` ใน `TicketDetailView.tsx` ให้ครบ 8 สถานะ:

- ปัจจุบันขาดสถานะ `WAITING_FOR_REQUESTER`, `REOPENED` และ `CANCELLED` ทำให้ตกไปเป็นสีเทา (`badge bg-secondary`) ทั้งหมด
- รบกวนปรับปรุงให้ครอบคลุมครบถ้วนและใช้สีตาม Zen Design Tokens

3. ปรับข้อความปุ่ม Back ด้านล่าง (`TicketDetailView.tsx` บรรทัดที่ 729):

- ปุ่มด้านล่างสุดยัง Hardcoded เป็น « Back to My Tickets List
- รบกวนปรับให้เปลี่ยนตาม Role เช่นเดียวกับปุ่มด้านบน:

```
` tsx  
  
« {isStaffOrAdmin ? "Back to Ticket Queue" : "Back to My Tickets List"}  
`
```

4. เพิ่ม **AbortSignal** ใน `fetchStaffAssignees` (`TicketDetailView.tsx`):

- รบกวนส่ง `AbortController.signal` เข้าไปใน `fetchStaffAssignees()` เพื่อป้องกัน❖❖ปัญหาตอน Component Unmount

รบกวนช่วยแก้ไขจุดดังกล่าวเพิ่มเติม แล้วแจ้งอีกครั้งเพื่อตรวจสอบและ Approve เข้า lab3-staging

• Round 2 (APPROVED - 2026-09-17):

Peer Review Update: Approved (PR #63 - IT Staff Ticket Operations & Status Matrix)

ได้ตรวจสอบ commit 6accf01 ที่แก้ไขตามข้อเสนอแนะครบทั้ง 4 จุดเรียบร้อยแล้ว:

1. REOPENED Status Transition Matrix – เพิ่ม Key [REOPENED]: [IN_PROGRESS, RESOLVED, CANCELLED] ครบทั้ง Backend (`staff.ts`) และ Frontend (`TicketDetailView.tsx`) พร้อมอัปเดต BR-14 ใน `specification.md` และมีชุดทดสอบครอบคลุมทั้ง 3 เส้นทาง
2. Status Badge ครบ 8 สถานะ – เพิ่มฟังก์ชัน `getStatusBadgeStyle()` ใช้ inline style สืบตรงตาม Zen Design Tokens ใน `ui-spec.md` ครบถ้วน (`WAITING_FOR_REQUESTER` , `REOPENED` , `CANCELLED` ไม่ตกหล่นอีกต่อไป)
3. ปุ่ม Back ด้านล่าง – ปรับเป็น Role-Aware Text สอดคล้องกับปุ่มด้านบนและ Breadcrumbs แล้ว มี Test ยืนยันทั้ง Staff และ Requester
4. AbortSignal ใน `fetchStaffAssignees` – ใช้ `AbortController` พร้อม `cleanup function` ป้องกัน Memory Leak / Unmounted Component Warning อย่างถูกต้อง

• Partner Response and Follow-up Actions:

> ### Peer Review for PR #63: Changes Requested (IT Staff Ticket Operations & Status Matrix)

> ได้ตรวจสอบการทำงานของ PR #63 (Issue #52 / Task 24) ทั้ง Backend API, Frontend Operations Panel และชุดทดสอบทั้งหมดอย่างละเอียดเทียบกับ Handout Section 4, `specification.md` (BR-10 ถึง BR-14), `api-spec.md` (Section 2) และ `ui-spec.md` (Screen 4)

>

> การวางโครงสร้างระบบ Claim, Assign, IT Priority และ Status Transition ทำได้ดีมากและมีชุดทดสอบครอบคลุม แต่พบจุดบกพร่องที่ต้องปรับปรุงแก้ไข 4 จุดดังนี้:

>

> 1. เพิ่มสถานะ **REOPENED** ใน Status Transition Matrix (สำคัญมาก):

>

> ปัจจุบันใน `PERMITTED_STATUS_TRANSITIONS` (`server/src/routes/staff.ts`) และ `PERMITTED_NEXT_STATUSES` (`client/src/components/TicketDetailView.tsx`) ยังไม่มี Key `[TicketStatus.REOPENED]`

> ส่งผลให้เมื่อตัวถูก Reopen แล้ว ตัวเลือกสถานะถัดไปจะว่างเปล่า Dropdown ถูก Disable และ API ปฏิเสธด้วย 422 กลายเป็น Dead-end status

> รบกวนเพิ่ม Key **REOPENED** ทั้งฝั่ง Backend และ Frontend เช่น:

> ``ts`

> `[TicketStatus.REOPENED]: [`
> `TicketStatus.IN_PROGRESS,`
> `TicketStatus.RESOLVED,`

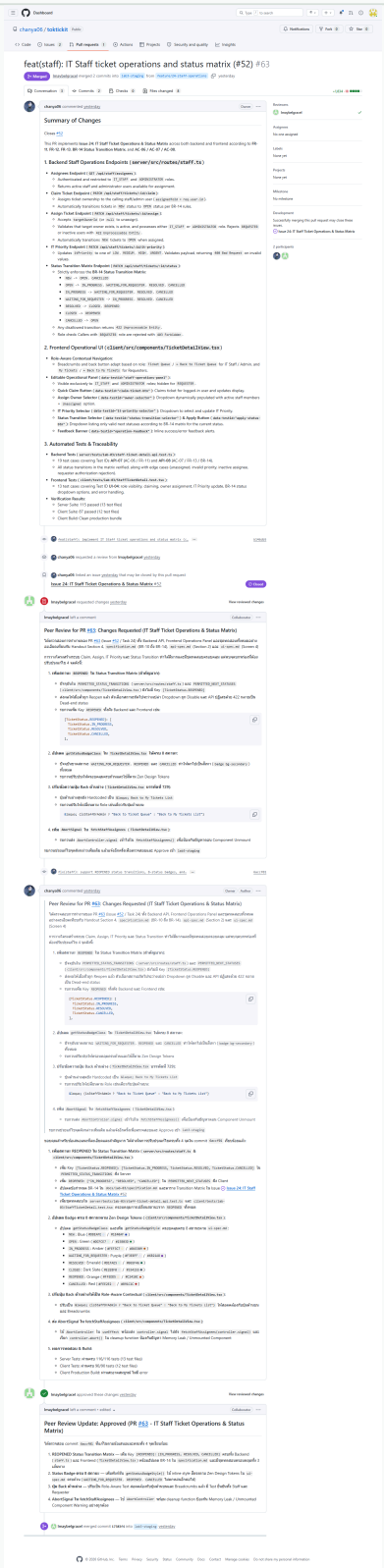
```
> TicketStatus.CANCELLED,  
> ],  
`
```

> 2. อัปเดต `getStatusBadgeClass` ใน `TicketDetailView.tsx` ให้ครบ 8 สถานะ:

```
>
```

> ปัจจุบันขาดสถานะ `WAITING_FOR_REQUESTER` , `REOPENED` และ `CANCELLED` ทำให้ดกไปเป็นสีเทา (`badge bg-secondary`) ทั้งหมด

• Evidence: [PR #63 Review Conversation](#)



PR #64 — feat(comments): public comments and private internal notes (#53)

- Pull Request: [#64](#)
- Branch: `feature/25-comments-and-notes`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES_REQUESTED** - 2026-09-17):

ผลการตรวจสอบ PR #64: REQUEST CHANGES / RECOMMEND IMPROVEMENTS

ภาพรวมของฟังก์ชันหลัก ระบบความปลอดภัย การจำกัดสิทธิ์ (RBAC) และ Business Rules ทำงานได้ถูกต้องครบถ้วนตามข้อกำหนด Lab-03 แต่พบจุดบกพร่องด้าน UX และ State ในหน้าบ้านที่ควรพิจารณาแก้ไขก่อนทำการ Merge:

ประเด็นที่ควรปรับปรุง (Issues to Address):

1. ตัวเลข Badge บนแท็บแสดงเป็น (0) เสมอตอนเข้าหน้ารายละเอียด:

- เนื่องจากแท็บเปิดมาที่ Attachments เป็นค่าเริ่มต้น และใช้ Conditional Rendering ทำให้แท็บ Public Comments และ Internal Notes ยังไม่ถูก Mount ตัวเลข Badge จึงค้างอยู่ที่ 0 ตลอดเวลาจนกว่าผู้ใช้จะคลิกเข้าไป
- ข้อเสนอ: ทำให้ผู้ใช้หรือ Staff เข้าใจผิดว่าไม่มีข้อความหรือบันทึกภายในอยู่เลย
- แนะนำ: ทำ Prefetch ข้อมูลนับจำนวน (ตั้งแต่ Mount หน้ารายละเอียด หรือย้ายการเรียก API ไปไว้ใน Hook ระดับบน

2. ข้อความที่กำลังพิมพ์สูญหายเมื่อสลับแท็บ (Draft State Loss):

- การสลับแท็บไปดูรูปภาพแนบหรือดูข้อมูลอื่นจะทำให้ CommentsSection / InternalNotesSection ถูก Unmount ส่งผลให้ข้อความที่กำลังพิมพ์อยู่ใน Textarea หายไปทันที
- แนะนำ: ใช้การซ่อนแสดงด้วย `style={{ display: activeTab === '...' ? 'block' : 'none' }}` หรือยก State ของข้อความขึ้นมาเก็บที่ Parent

3. ลำดับแท็บกับแท็บเริ่มต้นไม่สอดคล้องกัน:

- แท็บแรกคือ Public Comments แต่ระบบตั้งค่าเริ่มต้นให้เปิดที่ Attachments (แท็บที่ 3) ควรพิจารณาตั้งค่าเริ่มต้นให้เปิดที่แท็บแรก หรือจัดลำดับให้สอดคล้องกัน

4. การนำ Middleware กลับมาใช้ซ้ำใน Backend (server/src/routes/comments.ts):

- ปัจจุบันมีการเขียน `if (!req.user)` และ `if (req.user.mustChangePassword)` เองในทุก Endpoint ควรเปลี่ยนมาใช้ `requireAuth` และ `requirePasswordChanged` จาก `auth.ts` เพื่อความกระชับและเป็นมาตรฐานเดียวกัน

จุดที่ทำได้ดีมาก:

- ระบบสิทธิ์ Internal Notes ปิดกั้น Requester ทั้งหมดด้วย 403 Forbidden และซ่อนแท็บในหน้าบ้านได้ถูกต้อง
- Auto-transition สถานะตัวจาก WAITING_FOR_REQUESTER เป็น IN_PROGRESS ทำงานผ่าน Prisma \$transaction แบบ Atomic ได้อย่างสมบูรณ์
- การตรวจสอบความยาวข้อความ 2 ถึง 2000 ตัวอักษรทำได้รัดกุมทั้งหน้าบ้านและหลังบ้าน

• Round 2 **APPROVED** - 2026-09-17):

ผลการรีวิว PR #64 (การตรวจสอบขั้นสุดท้าย): **APPROVED**

ตรวจสอบโค้ดทั้งหมดใน PR #64 อย่างละเอียดครบถ้วนทุกมิติ:

1. ด้านความปลอดภัยและการจำกัดสิทธิ์ (Security & RBAC):

- Internal Notes ได้รับการป้องกันอย่างรัดกุมผ่าน `requireRole(IT_STAFF, ADMINISTRATOR)` ที่ระดับ Middleware และซ่อนจาก DOM ของ Requester อย่างเด็ดขาด
- Public Comments มีการแยกสิทธิ์ Requester ให้เข้าถึงได้เฉพาะตัวของตนเอง ป้องกัน IDOR ได้อย่างสมบูรณ์
- ไม่มีการเปิดเผยข้อมูล passwordHash ในผลลัพธ์การ Query ของผู้เขียน

2. กฎทางธุรกิจและการทำงานของระบบ (Business Rules):

- BR-14: การเปลี่ยนสถานะตัวจาก WAITING_FOR_REQUESTER เป็น IN_PROGRESS ทำงานร่วมกับการสร้าง Public Comment ผ่าน Prisma \$transaction แบบ Atomic ข้อมูลถูกต้องสอดคล้องกัน
- BR-17: รักษาความถูกต้องของข้อมูลแบบ Append-only ปฏิเสธคำขอแก้ไขและลบ
- BR-18: มีการตรวจสอบความยาวข้อความ 2 ถึง 2000 ตัวอักษรอย่างรัดกุมทั้งหน้าบ้านและหลังบ้าน

3. ส่วนติดต่อผู้ใช้และประสิทธิภาพ (UI/UX & State):

- ออกแบบตรงตาม UI Spec Screen 4 ด้วยแถบสีอำพัน Amber สำหรับ Internal Notes
- แก้ไขปัญหาตัวเลข Badge ด้วยการแสดงผลแบบ Persistent Display ทำให้ระบบดึงข้อมูลมาแสดงผลตัวเลขจริงบนแท็บได้ทันทีตั้งแต่เปิดหน้า
- ป้องกันการสูญหายของข้อความร่างที่พิมพ์ค้างไว้เมื่อมีการสลับแท็บ
- ลำดับแท็บเริ่มต้นเปิดที่ Public Comments สอดคล้องกับแถบนำทาง

4. ชุดทดสอบ (Automated Tests):

- มีชุดทดสอบครอบคลุมทั้ง API Integration Tests และ Client Unit Tests ผ่านการทดสอบทั้งหมดโดยไม่มีข้อผิดพลาด

• **Partner Response and Follow-up Actions:**

Response to Code Review (Commit 071dbbc)

Thank you for the thorough review and constructive feedback! All 4 reported issues have been addressed and validated with automated tests:

1. Badge Counts Prefetch on Mount (TicketDetailView.tsx):

- Tab sections (`CommentsSection` , `InternalNotesSection` , and `AttachmentSection`) are now kept mounted and toggled via `style={{ display: activeTab === '...' ? 'block' : 'none' }}`.
- Both comments and internal notes counts are fetched and populated immediately upon loading the ticket details without requiring the user to switch tabs first.
- Role security is strictly maintained: `InternalNotesSection` is only mounted when `isStaffOrAdmin === true` , ensuring Requesters cannot mount the component or trigger notes API calls.

2. Draft State Preservation Across Tab Switching (TicketDetailView.tsx):

- Because the sections are toggled using `display` styling rather than conditional unmounting, drafts in both the comment textarea and the internal note textarea are fully preserved when switching between tabs (e.g., to review attachments).

3. Tab Ordering and Default Active Tab Alignment (TicketDetailView.tsx):

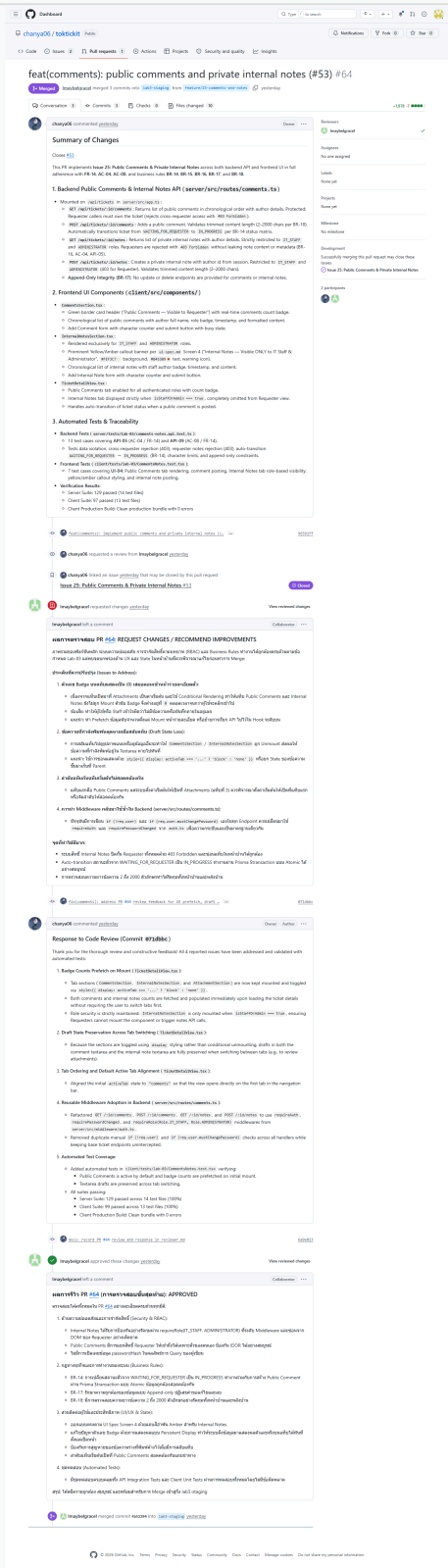
- Aligned the initial `activeTab` state to `"comments"` so that the view opens directly on the first tab in the navigation bar.

4. Reusable Middleware Adoption in Backend (server/src/routes/comments.ts):

- Refactored `GET /:id/comments` , `POST /:id/comments` , `GET /:id/notes` , and `POST /:id/notes` to use `requireAuth` , `requirePasswordChanged` , and `requireRole(Role.IT_STAFF, Role.ADMINISTRATOR)` middlewares from `server/src/middleware/auth.ts` .
- Removed duplicate manual `if (!req.user)` and `if (req.user.mustChangePassword)` checks across all handlers while keeping base ticket endpoints unintercepted.

5. Automated Test Coverage:

- **Evidence:** [PR #64 Review Conversation](#)



PR #65 — feat(admin): administrator user management & safety validations (#54)

- Pull Request: [#65](#)
- Branch: `feature/26-admin-user-management`
- What I Reviewed & Feedback Given:
- Round 1 **CHANGES REQUESTED** - 2026-09-18):

ผลการรีวิว PR #65: REQUEST CHANGES / RECOMMEND IMPROVEMENTS

ระบบความปลอดภัย การควบคุมสิทธิ์ Router-level RBAC และ Safety Rules (BR-07, BR-08, BR-09) ทำงานได้ถูกต้องและรัดกุมมาก แต่จากการตรวจสอบความเข้ากันได้ของข้อมูลอย่างละเอียด พบจุดที่ควรปรับปรุงแก้ไขก่อนทำการ Merge ดังนี้:

ประเด็นที่ต้องปรับปรุงแก้ไข:

1. ความไม่สอดคล้องของฟิลด์ชื่อในโมเดล User (name vs fullName):

- ใน Prisma schema มีทั้ง `fullName String` และ `name String?` (เพื่อความเข้ากันได้ย้อนหลังกับโค้ด Lab 2)
- ปัจจุบันใน `POST /api/admin/users` และ `PATCH /api/admin/users/:id` มีการบันทึกเฉพาะ `fullName` แต่ปล่อยให้ `name` เป็น `null`
- แนะนำ: ให้กำหนด `name: trimmedName` ควบคู่กับ `fullName: trimmedName` ทั้งในคำสั่ง `prisma.user.create` และ `prisma.user.update` เพื่อให้ข้อมูลชื่อครบถ้วนทั้งสองฟิลด์เหมือนใน `seed.ts`

2. การรองรับฟิลด์แผนก (department):

- ในโมเดล User มีฟิลด์ `department String?` และหน้าบ้านมีการแสดงผลแผนกของผู้ใช้
- ปัจจุบันใน `SAFE_USER_SELECT` และ Endpoint การสร้าง/แก้ไขผู้ใช้ยังไม่ได้อ่านค่าหรือส่งคืนฟิลด์ `department`
- แนะนำ: เพิ่มฟิลด์ `department?: string | null` ใน `CreateAdminUserPayload`, `UpdateAdminUserPayload`, `AdminUserResponse` และใน `SAFE_USER_SELECT` พร้อมทั้งบันทึกลงฐานข้อมูล

3. เอกสารสารบัญไฟล์ทดสอบ (docs/lab-03/tests.md):

- ในตารางที่ 1 มีการแมป `API-10..13` ไปที่ `users-admin.api.test.ts` แล้ว แต่ในหัวข้อที่ 2 Test Suite File Map ยังขาดรายการของ `users-admin.api.test.ts`
- แนะนำ: เพิ่มบรรทัดอธิบายไฟล์ `users-admin.api.test.ts` ได้หัวข้อ `### Backend Integration & Security Tests`

จุดเด่นที่ทำให้ถูกต้องสมบูรณ์แล้ว:

- Router-level middlewares (`requireAuth`, `requirePasswordChanged`, `requireRole`) ครอบคลุม ปิดกั้นสูง
- Safety Rules สำหรับการป้องกัน Self-deactivation (422) และการป้องกัน Deactivation/Demotion แอดมินคนสุดท้าย (422) ทำงานถูกต้องทุกสถานการณ์
- มีการจำกัดเพดาน Pagination สูงสุด 100 รายการ ป้องกัน DoS
- อีเมลถูกแปลงเป็นตัวพิมพ์เล็กเพื่อความไม่ซ้ำ `◆◆◆` แบบ Case-insensitive
- รหัสผ่านได้รับการตรวจสอบความยากและแฮชด้วย `bcrypt` (10 rounds) โดยไม่ส่ง `passwordHash` ออกไปใน Response

• Round 2 **APPROVED** - 2026-09-18):

ผลการรีวิว PR #65 (รอบอนุมัติขั้นสุดท้าย): **APPROVED**

ตรวจสอบการแก้ไขใน commit `20a5a0d` อย่างละเอียดครบถ้วนทุกบรรทัด โค้ดมีความถูกต้อง สมบูรณ์ และได้มาตรฐานตามข้อกำหนดของ Lab-03 ทุกประการ:

1. การชิงโครในฟิลด์ name และ fullName:

- บันทึกและอัปเดตทั้ง `fullName: trimmedName` และ `name: trimmedName` ในคำสั่งสร้าง (POST) และแก้ไข (PATCH)
- รวม `name: true` ใน `SAFE_USER_SELECT` และเพิ่มใน `ALLOWED_SORT_FIELDS` เพื่อความเข้ากันได้ `◆◆◆` ย้อนหลังกับโค้ด Lab 2 อย่างสมบูรณ์

2. การรองรับฟิลด์แผนก (department):

- ตรวจสอบและบันทึกฟิลด์ `department` ลงฐานข้อมูล โดยตัดช่องว่างและบันทึกเป็น `null` หากเป็นค่าว่าง
- รองรับการอัปเดตหรือส่ง `null` มาเพื่อล้างค่าแผนกในคำสั่ง PATCH
- อัปเดต TypeScript Interface ใน `client/src/api.ts` ให้ตรงกันครบถ้วน

3. ความปลอดภัยและ Safety Rules:

- Router-level RBAC (`requireAuth`, `requirePasswordChanged`, `requireRole(Role.ADMINISTRATOR)`) ครอบคลุมทุก Endpoint อย่างรัดกุม
- ป้องกันการปิดใช้งานบัญชีตนเอง (422 Unprocessable Entity)
- ป้องกันการปิดใช้งานหรือลดชั้นบทบาทแอดมินที่ใช้งานอยู่คนสุดท้าย (422 Unprocessable Entity)
- ป้องกัน Email Conflict เมื่อแอดมิน `◆◆` กดบันทึกโดยไม่เปลี่ยนอีเมลผ่านเงื่อนไข `NOT: { id: targetId }`

4. ชุดทดสอบและเอกสาร:

- ชุดทดสอบใน `users-admin.api.test.ts` มีการตรวจสอบค่าจริงในฐานข้อมูลผ่าน `prisma.user.findUniqueOrThrow` สำหรับฟิลด์ `name` , `fullName` , และ `department` ครบทั้ง Flow สร้างและแก้ไข
 - อัปเดตสารบัญไฟล์ทดสอบใน `docs/lab-03/tests.md` และบันทึกประวัติการรีวิวใน `docs/lab-03/reviewer.md` ครบถ้วน
- โค้ดผ่านการตรวจสอบทั้งหมดอย่างสมบูรณ์แบบ สามารถทำการ Merge เข้าสู่ `lab3-staging` ได้ทันที

• Round 3 APPROVED - 2026-09-18):

Peer Review: Approved (PR #65)

ตรวจทานโค้ดและเอกสารที่ปรับปรุงเพิ่มเติมใน commit `20a5a0d` เรียบร้อยแล้ว ครบถ้วนและสมบูรณ์แบบมาก:

1. การชิงโครในฟิลด์ `name` และ `fullName` : ดำเนินการผูกค่าคู่ขนานทั้งใน `create` และ `update` ครบถ้วน ช่วยรักษาความเข้ากันได้ย้อนหลังกับโค้ด Lab 2 ได้อย่างสมบูรณ์แบบ
 2. การรองรับฟิลด์แผนก (`department`) : เพิ่มการอ่านค่า ตรวจสอบ และส่งคืนฟิลด์แผนกครบทั้งฝั่ง `Backend` และ `TypeScript Interfaces` ฝั่ง `Frontend`
 3. สารบัญเอกสารการทดสอบ (`docs/lab-03/tests.md`) : รวบรวมละเอียดของ `users-admin.api.test.ts` ชัดเจนและสอดคล้องกับตาราง `Traceability Matrix`
 4. ชุดทดสอบและการ **Build**: ตรวจสอบการทำงานของเทสต์ 149 เคสบน `Server` และ 99 เคสบน `Client` ผ่านครบ 100% พร้อมทั้งผ่านการ **Build** สะอาดปราศจากข้อผิดพลาด
- งานมีคุณภาพสูง ปลอดภัย และตรงตามสเปกทุกประการ อนุมัติ (Approved) ให้ดำเนินการ Merge เข้าสู่ `lab3-staging` ได้เลย

• Partner Response and Follow-up Actions:

ข้อความรีวิวตรวจงาน (Reviewer Comments):

สรุปการปรับปรุงแก้ไขตามผลการรีวิว (PR #65)

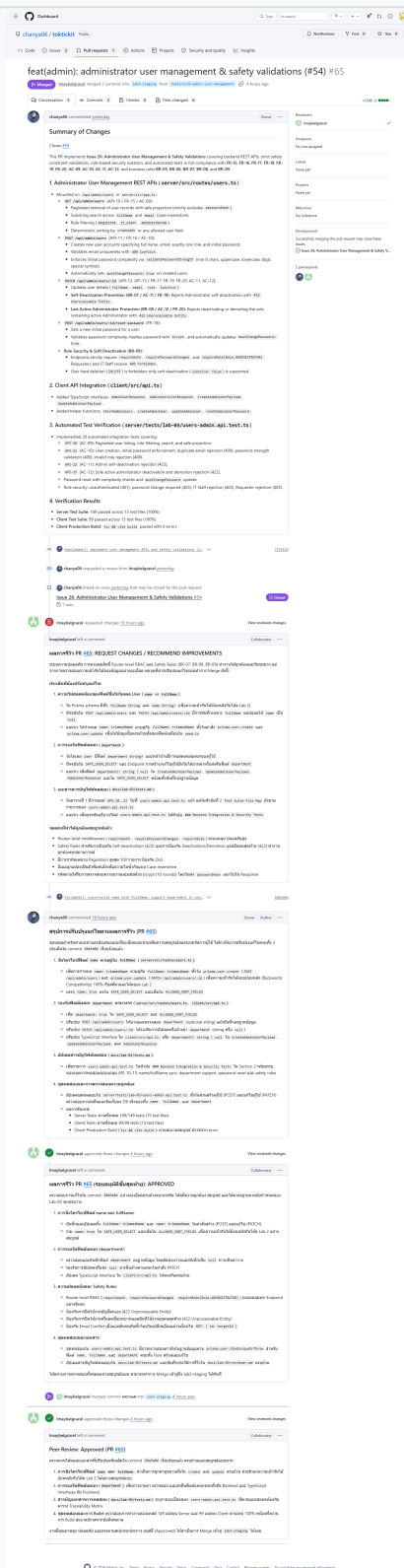
ข้อความตอบกลับและผลการแก้ไขจากบัดดี (Partner Response & Actions):

ขอบคุณสำหรับคำแนะนำและข้อเสนอแนะที่ละเอียดและช่วยเพิ่มความสะดวกของระบบจัดการผู้ใช้ ได้ดำเนินการปรับปรุงแก้ไขครบทั้ง 3 ประเด็นใน commit `20a5a0d` เรียบร้อยแล้ว:

1. ชิงโครในฟิลด์ `name` ควบคู่กับ `fullName` (`server/src/routes/users.ts`) :
 - เพิ่มการกำหนด `name: trimmedName` ควบคู่กับ `fullName: trimmedName` ทั้งใน `prisma.user.create` (`POST /api/admin/users`) และ `prisma.user.update` (`PATCH /api/admin/users/:id`) เพื่อความเข้ากันได้แบบย้อนหลัง (Backwards Compatibility) 100% กับสคีมาและโค้ดของ Lab 2
 - บรรจุ `name: true` ลงใน `SAFE_USER_SELECT` และเพิ่มใน `ALLOWED_SORT_FIELDS`
2. รองรับฟิลด์แผนก `department` ครบวงจร (`server/src/routes/users.ts` , `client/src/api.ts`) :
 - เพิ่ม `department: true` ใน `SAFE_USER_SELECT` และ `ALLOWED_SORT_FIELDS`
 - ปรับปรุง `POST /api/admin/users` ให้อ่านและตรวจสอบ `department` (optional string) แล้วบันทึกลงฐานข้อมูล
 - ปรับปรุง `PATCH /api/admin/users/:id` ให้รองรับการอัปเดตหรือล้างค่า `department` (string หรือ `null`)
 - ปรับปรุง TypeScript Interface ใน `client/src/api.ts` : เพิ่ม `department?: string | null` ใน `CreateAdminUserPayload` , `UpdateAdminUserPayload` , และ `AdminUserResponse`
3. อัปเดตสารบัญไฟล์ทดสอบ (`docs/lab-03/tests.md`) :
 - เพิ่มรายการ `users-admin.api.test.ts` ในหัวข้อ `### Backend Integration & Security Tests` ใน Section 2 พร้อมระบุขอบเขตการทดสอบครอบคลุม API-10..13, name/fullName sync, department support, password reset และ safety rules
4. ชุดทดสอบและการตรวจสอบความถูกต้อง:
 - อัปเดตชุดทดสอบใน `server/tests/lab-03/users-admin.api.test.ts` ทั้งในส่วนสร้างผู้ใช้ (POST) และแก้ไขผู้ใช้ (PATCH) ตรวจสอบการส่งคืนและจัดเก็บลง DB จริงของทั้ง `name` , `fullName` , และ `department`
 - ผลการรันเทส:

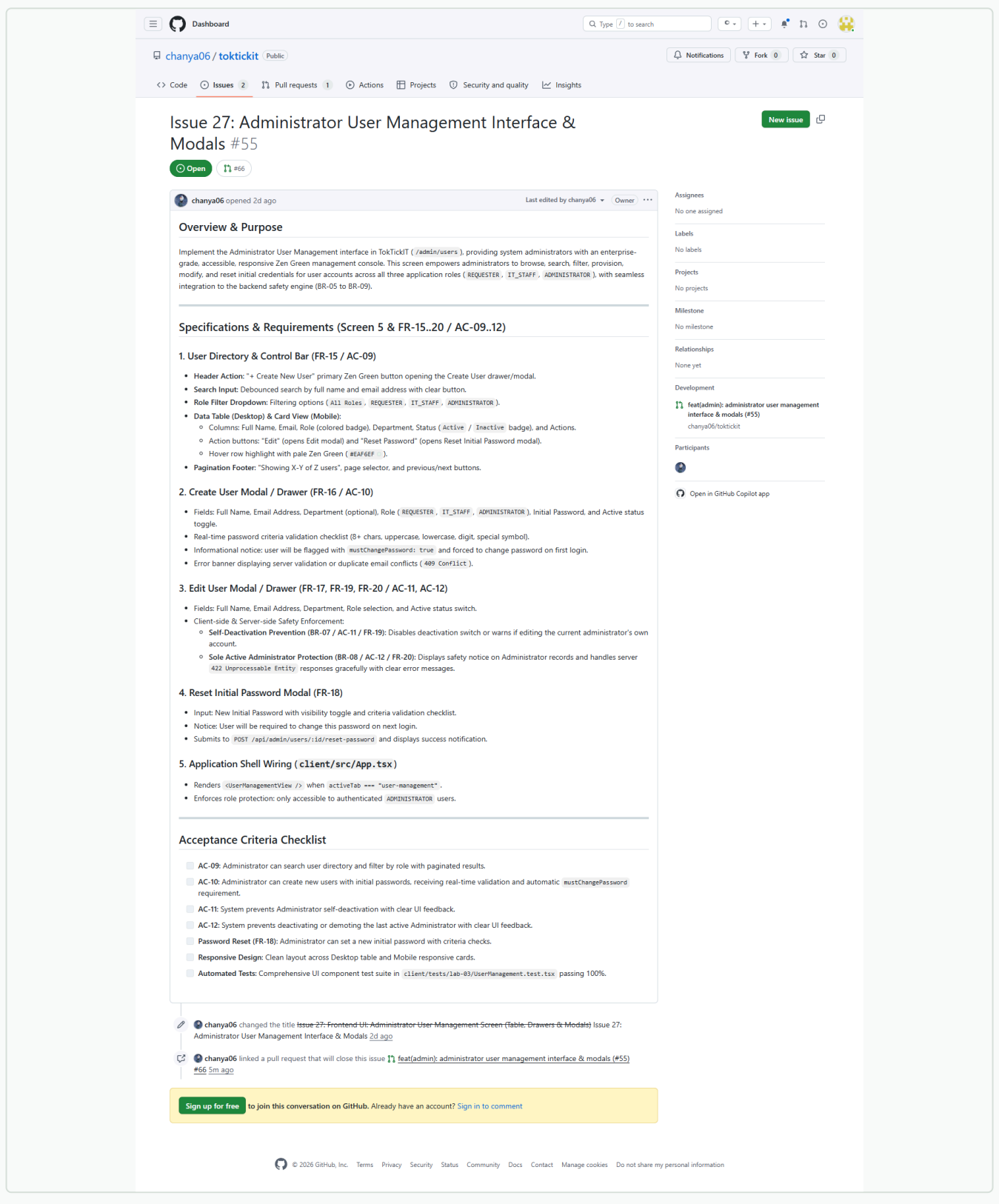
• Evidence: [PR #65 Review Conversation](#)

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): chanya_pr65_full.png



Issue #55 (Issue 27) — Administrator User Management Interface & Modals

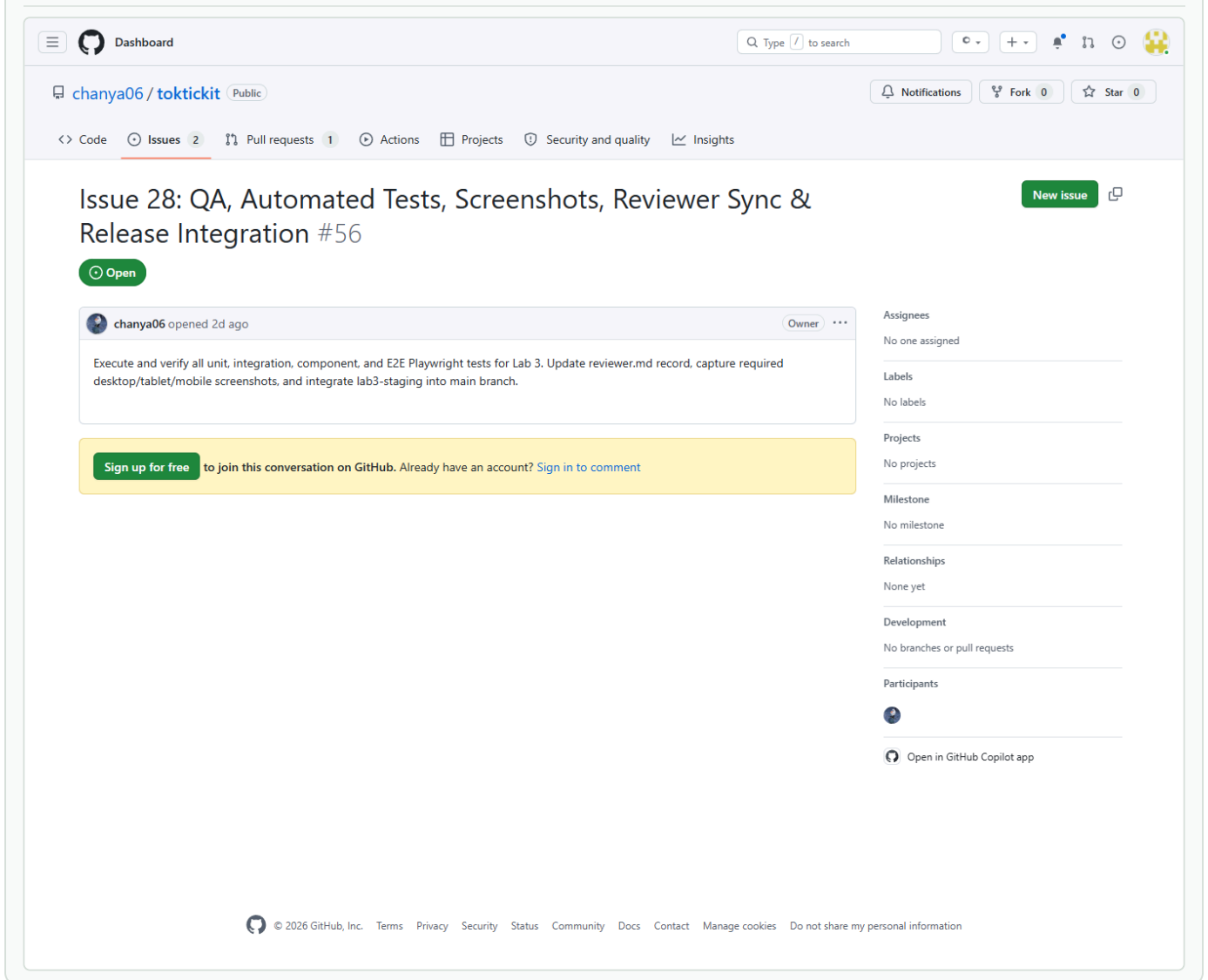
- **Issue Reference:** [Issue #55](#) on [chanya06/toktickit](#)
- **Partner Branch:** [feature/27-admin-ui](#) (Pending submission)
- **Status:** In progress by author (@chanya06). Reviewer (@Imaybelgrace) is on standby to perform code review as soon as the PR is opened.
- **Review Criteria Prepared:**
 - Verify Zen Green UI styling for Admin table and action buttons.
 - Verify Add User Modal, Edit User Modal, and Reset Password Confirmation Modal.
 - Verify real-time validation, role selector, active/inactive badge styling, and client-side guards matching BR-07, BR-08, BR-09.



Issue #56 (Issue 28) — QA, Automated Tests, Screenshots, Reviewer Sync & Release Integration

- **Issue Reference:** [Issue #56](#) on [chanya06/toktickit](#)
- **Partner Branch:** [release/lab3-integration](#) (Pending submission)
- **Status:** In progress by author (@chanya06). Reviewer (@Imaybelgrace) is on standby to perform final QA audit and merge verification.
- **Review Criteria Prepared:**
 - Verify complete automated test suite execution (Unit, API, Component, E2E) with 100% pass rate.
 - Verify responsive screenshots across desktop (1280px), tablet (768px), and mobile (375px, 320px).
 - Verify synchronised [reviewer.md](#), documentation integrity, and clean git staging merge into [main](#).

หลักฐานภาพถ่ายหน้าจอ GitHub แบบเต็มหน้า (Full-page Screenshot): chanya_issue56_full.png



Answer Part 2: Spec DD (Specification-Driven Development)

Working Link to Specification Suite: [docs/lab-03/specification.md](#) • [docs/lab-03/ui-spec.md](#) • [docs/lab-03/api-spec.md](#)

2.1 Specification Pre-dating Code Implementation Evidence

เอกสารข้อกำหนดทั้ง 6 ฉบับถูกจัดทำ ตรวจสอบ และอนุมัติใน **PR #36** (Commit [2abef9f](#) วันที่ 16 ก.ย. 2026) ก่อนที่การพัฒนาโค้ดใน Issue 18 ถึง 24 จะเริ่มต้นขึ้น ซึ่งเป็นหลักฐานเชิงประจักษ์ของการดำเนินงานตามแนวคิด **Specification-Driven Development (Spec DD)** อย่างสมบูรณ์

2.2 Rendered Engineering Specification (specification.md ฉบับเต็ม)

TokTickIT - Sprint 3 Engineering Specification

1. Sprint Goal

Deliver a secure, role-based IT ticketing workflow by replacing the temporary Development Requester selector with production-grade authentication and authorization supporting three distinct roles: Requester, IT Staff, and Administrator. Empower IT Staff with a collaborative Ticket Queue and operational Detail workflow (ownership claim/reassign, IT priority, status transitions, Public Comments, and private Internal Notes), provide Administrators with minimalist User Management, and preserve all Lab 2 Requester capabilities without regression.

2. Stakeholder Request Interpretation

The stakeholder requires transitioning TokTickIT from a prototype testing setup to a real multi-user operational environment. Authentication must be enforced at the server level, requiring users with initial passwords to set a new password upon first sign-in. The system must distinctly separate responsibilities: Requesters submit and track their own tickets; IT Staff manage queues, assign owners, set internal priorities, and communicate via public or private notes; Administrators manage user accounts with strict safety controls (no self-deactivation, no orphaned systems). All interfaces must continue the responsive Zen Green design language established in Lab 2.

3. Scope

3.1. Included Work

- Authentication system: Email/password login, logout, session/token management, and mandatory first-login password change.
- Server-side Role-Based Access Control (RBAC) across Requester, IT Staff, and Administrator.
- User migration from Lab 2 `RequesterUser` to unified `User` model, preserving existing ticket ownership and attachment data.
- Requester regression verification: Continuation of Create Ticket, My Tickets, Ticket Detail, and Attachments using real authenticated identity.
- Requester Public Comments and "Problem Appears Resolved" indication.
- IT Staff Ticket Queue: Search, multi-filtering, sorting, pagination, status/priority indicators, and responsive table/card layouts.
- IT Staff Ticket Detail: Claim ownership, reassign owner, edit IT Priority, execute permitted status transitions with validation, and manage attachments.
- Dual-stream communication: Append-only Public Comments (all roles) and private Internal Notes (IT Staff and Administrator only).
- Minimalist Administrator User Management: User listing, search, role filtering, account creation, basic profile editing, one-role assignment, account activation/deactivation, and initial password resets.
- Zen Green UI extensions, responsive viewports (Desktop, Tablet, Mobile, 320px), and accessibility.

3.2. Explicitly Excluded Work

- Email delivery services (no real email sending for invitations or resets).
- Multi-factor authentication (MFA), OAuth/social login, and SSO.
- Self-registration / public sign-up.
- IT Staff "Actions Taken" checklist (explicitly deferred to Lab 4).
- Formal SLA calculations, automated escalation rules, and notification dispatchers.
- Advanced analytics dashboards and KPI metric widgets.
- Multi-tenant organization and department hierarchy management.
- Production cloud deployment infrastructure changes.
- Multiple roles per user (exactly one role assigned per user).
- User deletion (hard delete), bulk user imports/exports, and audit history logs.
- Profile pictures, avatars upload, and extended profile attributes.
- Account lockouts and complex identity approval workflows.

4. Functional Requirements (FR)

Authentication & Account Management

- **FR-01:** The system shall authenticate users via email address and password, returning authenticated session tokens and user profile information.
- **FR-02:** The system shall detect whether an authenticated user has `mustChangePassword = true` and force immediate redirection to the Change Password screen before granting application access.
- **FR-03:** The system shall allow users to change their password by supplying their current password and a new valid password matching complexity rules.
- **FR-04:** The system shall terminate user sessions upon explicit logout and reject subsequent authenticated requests.
- **FR-05:** The system shall display the authenticated user's full name, role badge, and permitted navigation items in the application shell.

Requester Workflow & Regression

- **FR-06:** Requesters shall create, view, search, and manage tickets using their authenticated account identity without Development Requester headers.
- **FR-07:** Requesters shall view only tickets and attachments that they own.
- **FR-08:** Requesters shall post Public Comments on their tickets and view Public Comments posted by IT Staff or Administrators.
- **FR-09:** Requesters shall be able to indicate that their reported issue appears resolved.

IT Staff Operations & Workflow

- **FR-10:** IT Staff shall access a shared Ticket Queue displaying all submitted tickets across the organization.
- **FR-11:** The Ticket Queue shall support text search (Ticket Number, Summary), multi-attribute filtering (Category, Status, Requested Priority, IT Priority, Assignment), sorting, and pagination.
- **FR-12:** IT Staff shall view full Ticket Details, including Requester info, attachments, Public Comments, and Internal Notes.
- **FR-13:** IT Staff shall claim unassigned tickets or reassign ticket ownership to another active IT Staff member or Administrator.
- **FR-14:** IT Staff shall update the ticket IT Priority independently of Requested Priority.
- **FR-15:** IT Staff shall transition ticket status according to the approved state transition matrix.

Communication

- **FR-16:** IT Staff and Administrators shall create and view private Internal Notes attached to a ticket.
- **FR-17:** All users with access to a ticket shall create and view append-only Public Comments.

Administrator User Management

- **FR-18:** Administrators shall view a user roster, search by name/email, and filter by role.
- **FR-19:** Administrators shall create new user accounts with one designated role, active status, and an initial password.
- **FR-20:** Administrators shall update user profile details (Name, Email, Role, Active status) and issue password resets.

5. Business Rules (BR)

Security & Authentication Rules

- **BR-01:** Only active accounts (`isActive = true`) with valid credentials can authenticate. Inactive accounts receive an authentication error without disclosing internal status.
- **BR-02:** Users with `mustChangePassword = true` are restricted exclusively to password change endpoints; all operational routes return HTTP 403.
- **BR-03:** Authenticated context derived from server-side verification determines ownership. Client-supplied user or requester IDs in request bodies are ignored.
- **BR-04:** Passwords must be at least 8 characters long and contain at least one uppercase letter, one lowercase letter, and one numeric digit. Plaintext passwords must never be stored.
- **BR-05:** Each user account is assigned exactly one role: `REQUESTER` , `IT_STAFF` , or `ADMINISTRATOR` .

Administrator Safety Rules

- **BR-06:** User accounts cannot be deleted (no hard delete). Inactive status (`isActive = false`) is used to revoke system access.
- **BR-07:** An Administrator cannot deactivate their own account.
- **BR-08:** An Administrator cannot change their own role away from `ADMINISTRATOR` .
- **BR-09:** The system must prevent deactivation or role modification of the last active Administrator account.
- **BR-10:** Email addresses must be globally unique across all users.
- **BR-11:** When an Administrator resets a user's password, `mustChangePassword` is automatically set to `true` .

Communication & Privacy Rules

- **BR-12:** Public Comments are visible to Requesters (ticket owners), IT Staff, and Administrators.
- **BR-13:** Internal Notes are strictly confidential and visible only to IT Staff and Administrators. Requesters are blocked from reading or creating notes (HTTP 403).
- **BR-14:** Comments and Internal Notes are append-only. Editing and deletion are prohibited.
- **BR-15:** Comment and note content must not be empty or whitespace-only, and must be between 1 and 2,000 characters. Author and timestamp are stamped by the server.

Ticket Lifecycle, Ownership & Status Rules

- **BR-16:** A ticket can have zero or one primary Ticket Owner, who must be an active user with role `IT_STAFF` or `ADMINISTRATOR` .
- **BR-17:** `requestedPriority` remains immutable as submitted by the Requester. `itPriority` defaults to `requestedPriority` at creation and can subsequently be modified only by IT Staff or Administrator.
- **BR-18:** Permitted Ticket Statuses are: `New` , `Open` , `In Progress` , `Waiting for Requester` , `Resolved` , `Closed` , `Reopened` , and `Cancelled` .
- **BR-19:** State Transition Matrix:
 - From `New` : May transition to `Open` (upon assignment/claim) or `Cancelled` .
 - From `Open` : May transition to `In Progress` , `Waiting for Requester` , or `Cancelled` .
 - From `In Progress` : May transition to `Waiting for Requester` , `Resolved` , or `Cancelled` .

- From `Waiting for Requester` : May transition to `In Progress` , `Resolved` , or `Cancelled` .
- From `Resolved` : May transition to `Closed` or `Reopened` .
- From `Closed` : May transition to `Reopened` .
- From `Cancelled` : Terminal state; no transitions permitted.
- **BR-20:** Requesters cannot transition tickets directly to `Resolved` or `Closed` . A Requester's "Problem Appears Resolved" flag serves as operational feedback for IT Staff. Setting status to `Resolved` requires a mandatory `resolutionSummary` (3-500 characters).

6. UI Specification Summary

- **Palette & Tokens:** Strict application of Zen Green Design System (`#006B3C` primary, `#0B7A46` secondary, `#EAF6EF` pale green, `#F5F7F6` page background, `#1A2E23` dark text).
- **App Shell:** Unified responsive navbar displaying user identity, role tag, contextual navigation links, and logout CTA.
- **Login & Password Change:** Clean centered cards with field validation, visibility toggles, loading state indicators, and feedback banners.
- **IT Staff Queue:** High-density data table on desktop with sorting indicators, multi-select dropdown filters, and mobile card view.
- **Ticket Detail:** Tabbed or clearly segregated panels distinguishing Public Comments (green accent) from Internal Notes (amber/warning accent).
- **Admin Management:** Roster table with quick action toggles, modal dialogs for user creation and password resets.
- **Responsive Standard:** Flawless usability across Desktop (>=992px), Tablet (768px-991px), Mobile (<768px), and 320px small viewports without horizontal clipping.

7. Data Changes (Prisma Schema Evolution)

- **Migrate Model:** Replace `RequesterUser` with unified `User` model:
- Fields: `id` , `email` , `passwordHash` , `name` , `role` (Enum: `REQUESTER` , `IT_STAFF` , `ADMINISTRATOR`) , `isActive` , `mustChangePassword` , `createdAt` , `updatedAt` .
- Relations: `submittedTickets` (1:N with Ticket), `assignedTickets` (1:N with Ticket), `comments` (1:N with PublicComment), `notes` (1:N with InternalNote).
- **Update Model:** `Ticket` :
- New fields: `ownerId` (FK to User, nullable), `itPriority` (Enum: `LOW` , `MEDIUM` , `HIGH` , `URGENT`) , `resolutionSummary` (String, nullable), `requesterResolvedIndication` (Boolean, default false).
- Extended Status Enum: `NEW` , `OPEN` , `IN_PROGRESS` , `WAITING_FOR_REQUESTER` , `RESOLVED` , `CLOSED` , `REOPENED` , `CANCELLED` .
- **New Model:** `PublicComment` :
- Fields: `id` , `ticketId` (FK), `authorId` (FK to User), `content` , `createdAt` .
- **New Model:** `InternalNote` :
- Fields: `id` , `ticketId` (FK), `authorId` (FK to User), `content` , `createdAt` .
- **Indexes:** Add composite indexes on `User(email)` , `User(role, isActive)` , `Ticket(ownerId, currentStatus)` , `PublicComment(ticketId)` , and `InternalNote(ticketId)` .

8. API Contract Summary

- `POST /api/auth/login` - Authenticate with email/password, issue token/session.
- `POST /api/auth/logout` - Clear session/token.
- `GET /api/auth/me` - Return current authenticated user profile and permissions.
- `POST /api/auth/change-password` - Update password and clear `mustChangePassword` .
- `GET /api/tickets/my` - Fetch tickets submitted by current authenticated Requester.
- `POST /api/tickets` - Create ticket under authenticated Requester.
- `GET /api/tickets/:id` - Retrieve ticket detail (enforces ownership or IT/Admin role).
- `GET /api/staff/tickets` - Retrieve shared queue for IT Staff (search, filter, sort, page).
- `PATCH /api/staff/tickets/:id/claim` - Assign current IT Staff as ticket owner.
- `PATCH /api/staff/tickets/:id/assign` - Reassign ticket owner.
- `PATCH /api/staff/tickets/:id/priority` - Update ticket IT Priority.
- `PATCH /api/staff/tickets/:id/status` - Transition ticket status with validation.
- `GET /api/tickets/:id/comments` - Retrieve public comments.
- `POST /api/tickets/:id/comments` - Post a public comment.
- `GET /api/tickets/:id/notes` - Retrieve internal notes (IT/Admin only).
- `POST /api/tickets/:id/notes` - Post an internal note (IT/Admin only).

- `PATCH /api/tickets/:id/resolve-indication` - Requester toggles resolution flag.
- `GET /api/admin/users` - Retrieve user roster (Admin only).
- `POST /api/admin/users` - Create user account (Admin only).
- `PATCH /api/admin/users/:id` - Update user account details (Admin only).
- `POST /api/admin/users/:id/reset-password` - Set initial password for user (Admin only).

9. Acceptance Criteria (AC)

- **AC-01:** Given an active user with valid credentials, when they log in, then authenticated access is established and user role metadata is returned.
- **AC-02:** Given a user with `mustChangePassword = true`, when login succeeds, then all normal application views remain blocked until a new valid password is saved.
- **AC-03:** Given an authenticated Requester, when they view their tickets, only tickets submitted by their user ID are returned, regardless of query parameters.
- **AC-04:** Given a Requester account, when accessing Internal Note endpoints, then the operation is rejected with HTTP 403 Forbidden without leaking note metadata.
- **AC-05:** Given an IT Staff user, when viewing the Ticket Queue, then all tickets across all Requesters are visible with accurate search, filter, and pagination counts.
- **AC-06:** Given an IT Staff user, when claiming an unassigned ticket, then their user ID is set as the `ownerId` and the ticket status transitions from `New` to `Open`.
- **AC-07:** Given a ticket in `In Progress`, when IT Staff updates status to `Resolved` with a valid resolution summary, then the transition succeeds and timestamp is recorded.
- **AC-08:** Given a ticket in `In Progress`, when an invalid transition (e.g. to `New`) is attempted, the system rejects it with HTTP 400 and descriptive error.
- **AC-09:** Given an Administrator, when creating a user with an existing email, then HTTP 409 Conflict is returned and no record is created.
- **AC-10:** Given the last active Administrator, when attempting deactivation or role change, the system blocks the operation with HTTP 400 Bad Request.
- **AC-11:** Given an authenticated user, when Public Comments or Internal Notes are submitted, the backend stamps the author identity and creation timestamp.
- **AC-12:** Given all Lab 2 ticket creation and attachment workflows, when executed by an authenticated Requester, all functions operate identically without Development Requester selector.

10. Definition of Done (DoD)

Part 1: Product Completion Checklist

- ☐ Prisma schema evolved with `User`, `PublicComment`, `InternalNote`, and updated `Ticket` relations.
- ☐ Database migration executed safely and idempotent seed populated with ≥ 4 Requesters, ≥ 3 IT Staff, and ≥ 1 Admin.
- ☐ Full authentication suite operational (Login, Logout, Me, Password Change).
- ☐ Role-based authorization enforced on all endpoints and frontend views.
- ☐ Lab 2 Requester workflows verified against authenticated identity without regression.
- ☐ IT Staff Ticket Queue and Detail operational with claim, assign, priority, and status transitions.
- ☐ Public Comments and role-restricted Internal Notes operational and visually distinct.
- ☐ Administrator User Management operational with safety guards against self/orphan deactivation.
- ☐ Zen Green Design System applied consistently across desktop, tablet, mobile, and 320px viewports.

Part 2: Course Delivery Checklist

- ☐ All development performed on feature branches branched from `lab3-staging`.
- ☐ All PRs peer-reviewed, approved, and merged into `lab3-staging`.
- ☐ 100% passing automated tests across Server Vitest, Client Vitest, and Playwright E2E suites.
- ☐ All 6 documentation files complete in `docs/lab-03/`.
- ☐ Screenshot evidence captured in `artifacts/lab-03/screenshots/`.
- ☐ Final release PR merged from `lab3-staging` into `main`.
- ☐ Single PDF submission report compiled following Answer Part 1 to Answer Part 9.

11. Assumptions and Decisions

1. **Authentication Mechanism:** Session tokens are managed via cryptographically signed 24h JWT tokens passed via the standard `Authorization: Bearer` header. In the React Single Page Application (SPA), the client stores the token in

`localStorage ( tokticket_auth_token )` for clean stateless session persistence across page refreshes and test suites.

2. Password Security: Passwords hashed with `bcrypt` (salt rounds = 10) to prevent brute-force attacks.

3. Migration Strategy: Existing Lab 2 `RequesterUser` records are migrated directly to `User` rows with role `REQUESTER` and default temporary passwords (`Pass1234!`) with `mustChangePassword = true` .

4. Queue Default Sorting: Ticket Queue defaults to descending order of creation date (`createdAt DESC`).

2.3 Core Engineering Implementation & Source Code (ซอร์สโค้ดหลักของระบบ)

ในส่วนนี้รวบรวมซอร์สโค้ดแกนหลักของการพัฒนาระบบ TokTickIT Sprint 3 (Lab 3) ครอบคลุมเลเยอร์ ตั้งแต่ระดับฐานข้อมูล (Prisma Schema, Migration SQL, Idempotent Seed), ระบบความปลอดภัยและสิทธิ์การใช้งาน (Auth Middleware, Routes), เอนจินจัดการตัวของเจ้าหน้าที่ไอที (Staff Queue & Operations), ระบบจัดการผู้ใช้ของผู้ดูแลระบบ (Admin User Management พร้อม Admin Safety Rules), หน้าจอส่วนติดต่อผู้ใช้ (Frontend UI Components), ตลอดจนชุดทดสอบอัตโนมัติ (Automated Test Suites) ที่สอดคล้องกับข้อกำหนด Specification-Driven Development และ Test-Driven Development อย่างสมบูรณ์

2.3.1 Database Schema Definition (server/prisma/schema.prisma)

โครงสร้างโมเดลฐานข้อมูลฉบับสมบูรณ์ กำหนดเอนุม `Role` (3 ระดับ), `Priority` (4 ระดับ), `TicketStatus` (7 สถานะ), โมเดล `User`, การผูกความสัมพันธ์กับ `Ticket` และการแยกตาราง `PublicComment` และ `InternalNote` :

```
// Prisma schema - TokTickIT Ticketing Platform
// Lab 2 Requester Ticketing MVP & Lab 3 Role-Based IT Operations

generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

enum Role {
  REQUESTER
  IT_STAFF
  ADMINISTRATOR
}

enum Priority {
  LOW
  MEDIUM
  HIGH
  URGENT
}

enum TicketStatus {
  NEW
  OPEN
  IN_PROGRESS
  WAITING_FOR_REQUESTER
  RESOLVED
  CLOSED
  REOPENED
  CANCELLED
}

model User {
  id          Int           @id @default(autoincrement())
  email       String        @unique
  passwordHash String
  name        String
  role        Role          @default(REQUESTER)
  isActive    Boolean       @default(true)
  mustChangePassword Boolean @default(false)
  createdAt   DateTime      @default(now())
  updatedAt   DateTime      @updatedAt

  submittedTickets Ticket[] @relation("RequesterTickets")
  assignedTickets  Ticket[] @relation("AssignedTickets")
  comments         PublicComment[]
  notes            InternalNote[]

  @@index([email])
  @@index([role, isActive])
  @@map("users")
}
```

```

model RequesterUser {
  id          Int          @id @default(autoincrement())
  name        String
  email       String       @unique
  department  String
  isActive    Boolean      @default(true)
  createdAt   DateTime      @default(now())
  updatedAt   DateTime      @updatedAt

  @@map("requester_users")
}

model Category {
  id          Int          @id @default(autoincrement())
  name        String       @unique
  description  String?
  isActive    Boolean      @default(true)
  createdAt   DateTime      @default(now())
  updatedAt   DateTime      @updatedAt

  tickets Ticket[]

  @@map("categories")
}

model RelatedSystem {
  id          Int          @id @default(autoincrement())
  name        String       @unique
  category    String?
  isActive    Boolean      @default(true)
  createdAt   DateTime      @default(now())
  updatedAt   DateTime      @updatedAt

  tickets Ticket[]

  @@map("related_systems")
}

model Ticket {
  id          Int          @id @default(autoincrement())
  ticketNumber String       @unique
  summary     String
  description  String
  requestedPriority Priority  @default(MEDIUM)
  itPriority   Priority      @default(MEDIUM)
  currentStatus TicketStatus @default(NEW)
  resolutionSummary String?
  requesterResolvedIndication Boolean @default(false)

  requesterId Int
  requester   User          @relation("RequesterTickets", fields: [requesterId], references: [id])
  ownerId     Int?
  owner       User?         @relation("AssignedTickets", fields: [ownerId], references: [id])
  categoryId  Int
  category    Category      @relation(fields: [categoryId], references: [id])
  relatedSystemId Int
  relatedSystem RelatedSystem @relation(fields: [relatedSystemId], references: [id])

  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt

  attachments Attachment[]
  comments    PublicComment[]
  notes       InternalNote[]

  @@index([requesterId])
  @@index([requesterId, createdAt])
  @@index([ownerId])
  @@index([ownerId, currentStatus])
  @@index([categoryId])

```

```

    @@index([relatedSystemId])
    @@map("tickets")
}

model Attachment {
    id          Int          @id @default(autoincrement())
    ticketId    Int
    ticket      Ticket       @relation(fields: [ticketId], references: [id], onDelete: Cascade)
    filename    String
    storedPath  String
    fileSize    Int
    mimeType    String
    isRemoved   Boolean      @default(false)
    removedAt   DateTime?
    removalReason String?
    uploadedAt  DateTime     @default(now())

    @@index([ticketId])
    @@index([ticketId, isRemoved])
    @@map("attachments")
}

model PublicComment {
    id          Int          @id @default(autoincrement())
    ticketId    Int
    ticket      Ticket       @relation(fields: [ticketId], references: [id], onDelete: Cascade)
    authorId    Int
    author      User         @relation(fields: [authorId], references: [id])
    content     String
    createdAt   DateTime     @default(now())

    @@index([ticketId])
    @@map("public_comments")
}

model InternalNote {
    id          Int          @id @default(autoincrement())
    ticketId    Int
    ticket      Ticket       @relation(fields: [ticketId], references: [id], onDelete: Cascade)
    authorId    Int
    author      User         @relation(fields: [authorId], references: [id])
    content     String
    createdAt   DateTime     @default(now())

    @@index([ticketId])
    @@map("internal_notes")
}

```

2.3.2 Database Migration & Requester Backfill SQL (migration.sql)

ไฟล์ DDL Migration พร้อมคำสั่ง SQL Data Backfill เพื่อย้ายข้อมูลผู้ใช้เดิมจาก Lab 2 (ตาราง `requester_users`) เข้าสู่ตาราง `users` ใหม่ โดยรักษา Primary Key `id` เดิมไว้ครบถ้วน ข้อมูลตัวเดิมไม่สูญหาย:

```

-- CreateEnum
CREATE TYPE "Role" AS ENUM ('REQUESTER', 'IT_STAFF', 'ADMINISTRATOR');

-- AlterEnum
ALTER TYPE "Priority" ADD VALUE IF NOT EXISTS 'URGENT';

-- AlterEnum
ALTER TYPE "TicketStatus" ADD VALUE IF NOT EXISTS 'OPEN';
ALTER TYPE "TicketStatus" ADD VALUE IF NOT EXISTS 'WAITING_FOR_REQUESTER';
ALTER TYPE "TicketStatus" ADD VALUE IF NOT EXISTS 'REOPENED';
ALTER TYPE "TicketStatus" ADD VALUE IF NOT EXISTS 'CANCELLED';

-- CreateTable
CREATE TABLE "users" (
    "id" SERIAL NOT NULL,
    "email" TEXT NOT NULL,
    "passwordHash" TEXT NOT NULL,

```

```

    "name" TEXT NOT NULL,
    "role" "Role" NOT NULL DEFAULT 'REQUESTER',
    "isActive" BOOLEAN NOT NULL DEFAULT true,
    "mustChangePassword" BOOLEAN NOT NULL DEFAULT false,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "users_pkey" PRIMARY KEY ("id")
);

-- CreateIndex
CREATE UNIQUE INDEX "users_email_key" ON "users"("email");

-- CreateIndex
CREATE INDEX "users_email_idx" ON "users"("email");

-- CreateIndex
CREATE INDEX "users_role_isActive_idx" ON "users"("role", "isActive");

-- Backfill existing Lab 2 requester_users into users to preserve ticket ownership IDs
INSERT INTO "users" ("id", "name", "email", "passwordHash", "role", "isActive", "mustChangePassword", "createdAt",
SELECT
    "id",
    "name",
    "email",
    '$2a$10$0k4mQy0i2kC2Fq7Z3h7BteXU1Z0mH3iY7q2N8m6b1k5V9P4R7j3S2', -- default hash for initial login
    'REQUESTER'::"Role",
    "isActive",
    true,
    "createdAt",
    "updatedAt"
FROM "requester_users"
ON CONFLICT ("id") DO NOTHING;

-- Synchronize users sequence with max id
SELECT setval('users_id_seq', COALESCE((SELECT MAX("id") FROM "users"), 1));

-- AlterTable
ALTER TABLE "tickets" ADD COLUMN "itPriority" "Priority" NOT NULL DEFAULT 'MEDIUM',
ADD COLUMN "ownerId" INTEGER,
ADD COLUMN "resolutionSummary" TEXT,
ADD COLUMN "requesterResolvedIndication" BOOLEAN NOT NULL DEFAULT false;

-- CreateIndex
CREATE INDEX "tickets_ownerId_idx" ON "tickets"("ownerId");

-- CreateIndex
CREATE INDEX "tickets_ownerId_currentStatus_idx" ON "tickets"("ownerId", "currentStatus");

-- Drop old foreign key referencing requester_users and add new foreign key referencing users
ALTER TABLE "tickets" DROP CONSTRAINT IF EXISTS "tickets_requesterId_fkey";
ALTER TABLE "tickets" ADD CONSTRAINT "tickets_requesterId_fkey" FOREIGN KEY ("requesterId") REFERENCES "users"("id");

-- AddForeignKey for ticket owner
ALTER TABLE "tickets" ADD CONSTRAINT "tickets_ownerId_fkey" FOREIGN KEY ("ownerId") REFERENCES "users"("id") ON DEL

-- CreateTable
CREATE TABLE "public_comments" (
    "id" SERIAL NOT NULL,
    "ticketId" INTEGER NOT NULL,
    "authorId" INTEGER NOT NULL,
    "content" TEXT NOT NULL,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "public_comments_pkey" PRIMARY KEY ("id")
);

-- CreateIndex
CREATE INDEX "public_comments_ticketId_idx" ON "public_comments"("ticketId");

-- AddForeignKey

```

```

ALTER TABLE "public_comments" ADD CONSTRAINT "public_comments_ticketId_fkey" FOREIGN KEY ("ticketId") REFERENCES "t

-- AddForeignKey
ALTER TABLE "public_comments" ADD CONSTRAINT "public_comments_authorId_fkey" FOREIGN KEY ("authorId") REFERENCES "us

-- CreateTable
CREATE TABLE "internal_notes" (
  "id" SERIAL NOT NULL,
  "ticketId" INTEGER NOT NULL,
  "authorId" INTEGER NOT NULL,
  "content" TEXT NOT NULL,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT "internal_notes_pkey" PRIMARY KEY ("id")
);

-- CreateIndex
CREATE INDEX "internal_notes_ticketId_idx" ON "internal_notes"("ticketId");

-- AddForeignKey
ALTER TABLE "internal_notes" ADD CONSTRAINT "internal_notes_ticketId_fkey" FOREIGN KEY ("ticketId") REFERENCES "tic

-- AddForeignKey
ALTER TABLE "internal_notes" ADD CONSTRAINT "internal_notes_authorId_fkey" FOREIGN KEY ("authorId") REFERENCES "user

```

2.3.3 Idempotent Seed Logic (server/prisma/seed.ts)

สคริปต์สร้างบัญชีผู้ใช้เริ่มต้น 5 บัญชี (Admin, Staff 2 คน, Requester 2 คน) พร้อมการแฮชรหัสผ่านด้วย bcryptjs (10 rounds) และกำหนด `mustChangePassword: true` ตามกฎ BR-03:

```

import { PrismaClient, Role, Priority, TicketStatus } from "@prisma/client";
import bcrypt from "bcryptjs";

const prisma = new PrismaClient();

export const categoriesData = [
  { name: "Account and Access", description: "Login, passwords, permission requests" },
  { name: "Hardware", description: "Laptops, monitors, printers, peripherals" },
  { name: "Software", description: "OS, office apps, specialized tools" },
  { name: "Network", description: "Wi-Fi, VPN, campus network connectivity" },
];

export const relatedSystemsData = [
  { name: "Email", category: "Account and Access" },
  { name: "Campus Wi-Fi", category: "Network" },
  { name: "VPN", category: "Network" },
  { name: "LEB2 App", category: "Software" },
  { name: "Grade Submission App", category: "Software" },
  { name: "Printer", category: "Hardware" },
  { name: "Corporate Laptop", category: "Hardware" },
];

const DEFAULT_PASSWORD = "Password123!";
const DEFAULT_SALT_ROUNDS = 10;
const defaultHash = bcrypt.hashSync(DEFAULT_PASSWORD, DEFAULT_SALT_ROUNDS);

const TEMP_PASSWORD = "Pass1234!";
const tempHash = bcrypt.hashSync(TEMP_PASSWORD, DEFAULT_SALT_ROUNDS);

export const usersData = [
  // Requesters (Active & Inactive) - Department excluded per Lab 3 specification (Section 4.2 & 5.1)
  {
    name: "Pae Karn",
    email: "pae.karn@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {

```

```

    name: "Miki Chan",
    email: "miki.chan@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: true,
    passwordHash: tempHash,
  },
  {
    name: "Creammie Indiegurl",
    email: "creammie.indiegurl@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Jessica Phrao",
    email: "jessica.phrao@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Kanta Tawaan",
    email: "kanta.tawaan@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Bewnoi Pink",
    email: "bewnoi.pink@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Jeje Frappe",
    email: "jeje.frappe@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Bob Pueng",
    email: "bob.pueng@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Pan Ctrl",
    email: "pan.ctrl@example.com",
    role: Role.REQUESTER,
    isActive: false,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },

  // IT Staff (Active & Inactive)
  {
    name: "Somchai Prasert",
    email: "somchai.it@toktickit.com",
    role: Role.IT_STAFF,
    isActive: true,
    mustChangePassword: false,
  }

```

```

    passwordHash: defaultHash,
  },
  {
    name: "Suda Jaidee",
    email: "suda.staff@toktickit.com",
    role: Role.IT_STAFF,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Wichai Somboon",
    email: "wichai.tech@toktickit.com",
    role: Role.IT_STAFF,
    isActive: true,
    mustChangePassword: true,
    passwordHash: tempHash,
  },
  {
    name: "Nat Deactivated",
    email: "nat.retired@toktickit.com",
    role: Role.IT_STAFF,
    isActive: false,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  // Administrators
  {
    name: "Root Administrator",
    email: "admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
  {
    name: "Backup Administrator",
    email: "backup.admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
  },
];

export const ticketsData = [
  {
    ticketNumber: "TKT-2025-001001",
    summary: "Cannot access university email after password reset",
    description: "I reset my email password yesterday via self-service, but now neither webmail nor Outlook accepts",
    requestedPriority: Priority.HIGH,
    itPriority: Priority.HIGH,
    currentStatus: TicketStatus.OPEN,
    requesterEmail: "pae.karn@example.com",
    ownerEmail: "somchai.it@toktickit.com",
    categoryName: "Account and Access",
    relatedSystemName: "Email",
    resolutionSummary: null,
    requesterResolvedIndication: false,
    comments: [
      {
        authorEmail: "somchai.it@toktickit.com",
        content: "Checking the LDAP synchronization logs for your email mailbox now.",
      },
    ],
    notes: [
      {
        authorEmail: "somchai.it@toktickit.com",
        content: "Active Directory replication queue was paused. Restarting AD sync service.",
      },
    ],
  },
];

```

```

    ],
  },
  {
    ticketNumber: "TKT-2025-001002",
    summary: "Flickering display on Lenovo ThinkPad screen",
    description: "The internal laptop display intermittently flickers horizontal lines when moving the hinge.",
    requestedPriority: Priority.MEDIUM,
    itPriority: Priority.MEDIUM,
    currentStatus: TicketStatus.IN_PROGRESS,
    requesterEmail: "creammie.indiegurl@example.com",
    ownerEmail: "suda.staff@toktickit.com",
    categoryName: "Hardware",
    relatedSystemName: "Corporate Laptop",
    resolutionSummary: null,
    requesterResolvedIndication: false,
    comments: [
      {
        authorEmail: "suda.staff@toktickit.com",
        content: "Could you bring the laptop by the IT service desk on 3rd floor today so we can inspect the EDP display?",
      },
      {
        authorEmail: "creammie.indiegurl@example.com",
        content: "Sure, I will drop by around 2:00 PM.",
      },
    ],
    notes: [],
  },
  {
    ticketNumber: "TKT-2025-001003",
    summary: "Wi-Fi drops connection repeatedly in Building 12 lecture hall",
    description: "Every 10-15 minutes, the connection to KMUTT-Secure drops and requires toggling Wi-Fi off and on.",
    requestedPriority: Priority.LOW,
    itPriority: Priority.LOW,
    currentStatus: TicketStatus.NEW,
    requesterEmail: "jessica.phrao@example.com",
    ownerEmail: null,
    categoryName: "Network",
    relatedSystemName: "Campus Wi-Fi",
    resolutionSummary: null,
    requesterResolvedIndication: false,
    comments: [],
    notes: [],
  },
  {
    ticketNumber: "TKT-2025-001004",
    summary: "Error 500 when uploading assignment submission PDF",
    description: "Trying to upload my final assignment (3.2 MB PDF) results in a continuous loading spinner and then a 500 error.",
    requestedPriority: Priority.URGENT,
    itPriority: Priority.URGENT,
    currentStatus: TicketStatus.RESOLVED,
    requesterEmail: "pae.karn@example.com",
    ownerEmail: "somchai.it@toktickit.com",
    categoryName: "Software",
    relatedSystemName: "LEB2 App",
    resolutionSummary: "Increased Nginx client_max_body_size on LEB2 upload ingress gateway from 2MB to 20MB and restarted service.",
    requesterResolvedIndication: true,
    comments: [
      {
        authorEmail: "somchai.it@toktickit.com",
        content: "Issue has been resolved. Ingress upload limit increased.",
      },
    ],
    notes: [
      {
        authorEmail: "somchai.it@toktickit.com",
        content: "Tested upload with 15MB PDF sample; response returned 200 OK.",
      },
    ],
  },
  {
    ticketNumber: "TKT-2025-001005",

```

```

summary: "Need grade submission role assignment for TA course",
description: "Appointed as TA for CSC210 this semester, need grade input authorization in the grade submission",
requestedPriority: Priority.MEDIUM,
itPriority: Priority.MEDIUM,
currentStatus: TicketStatus.WAITING_FOR_REQUESTER,
requesterEmail: "miki.chan@example.com",
ownerEmail: "suda.staff@toktickit.com",
categoryName: "Software",
relatedSystemName: "Grade Submission App",
resolutionSummary: null,
requesterResolvedIndication: false,
comments: [
  {
    authorEmail: "suda.staff@toktickit.com",
    content: "Please provide the course coordinator instructor sign-off memo or instructor email confirmation."
  },
],
notes: [],
},
{
  ticketNumber: "TKT-2025-001006",
  summary: "Paper jam in Engineering Lab 4th floor printer",
  description: "Red light blinking with error code E-04 paper jam in tray 2.",
  requestedPriority: Priority.LOW,
  itPriority: Priority.LOW,
  currentStatus: TicketStatus.CLOSED,
  requesterEmail: "creammie.indiegurl@example.com",
  ownerEmail: "wichai.tech@toktickit.com",
  categoryName: "Hardware",
  relatedSystemName: "Printer",
  resolutionSummary: "Cleared jammed paper feed roller and replaced pickup roller assembly.",
  requesterResolvedIndication: false,
  comments: [],
  notes: [
    {
      authorEmail: "wichai.tech@toktickit.com",
      content: "Replaced roller part #RM1-4426-000 from local spare parts inventory."
    },
  ],
},
{
  ticketNumber: "TKT-2025-001007",
  summary: "VPN certificate expired warning",
  description: "When connecting to OpenVPN client, dialog says certificate expired on September 15.",
  requestedPriority: Priority.HIGH,
  itPriority: Priority.HIGH,
  currentStatus: TicketStatus.CANCELLED,
  requesterEmail: "pae.karn@example.com",
  ownerEmail: null,
  categoryName: "Network",
  relatedSystemName: "VPN",
  resolutionSummary: null,
  requesterResolvedIndication: false,
  comments: [],
  notes: [],
},
];

export async function seedDatabase(client = prisma) {
  console.log("Seeding TokTickIT Database (Idempotent)...");

  // 1. Categories
  for (const cat of categoriesData) {
    await client.category.upsert({
      where: { name: cat.name },
      update: { description: cat.description },
      create: { name: cat.name, description: cat.description },
    });
  }
  console.log(`Successfully seeded ${categoriesData.length} Categories.`);
}

```

```

// 2. Related Systems
for (const sys of relatedSystemsData) {
  await client.relatedSystem.upsert({
    where: { name: sys.name },
    update: { category: sys.category },
    create: { name: sys.name, category: sys.category },
  });
}
console.log(`Successfully seeded ${relatedSystemsData.length} Related Systems.`);

// 3. User Migration / Backfill from Lab 2 requester_users
// Ensures existing Lab 2 requester IDs map identically to User IDs with zero data loss
try {
  const existingRequesters = await client.requesterUser.findMany();
  for (const req of existingRequesters) {
    await client.user.upsert({
      where: { email: req.email },
      update: {
        name: req.name,
        isActive: req.isActive,
      },
      create: {
        id: req.id,
        name: req.name,
        email: req.email,
        role: Role.REQUESTER,
        isActive: req.isActive,
        mustChangePassword: true,
        passwordHash: tempHash,
      },
    });
  }
} catch (e) {
  // If table doesn't exist yet, continue
}

// 4. Users (Requesters, IT Staff, Administrators)
const userMap = new Map<string, any>();
for (const user of usersData) {
  const record = await client.user.upsert({
    where: { email: user.email },
    update: {
      name: user.name,
      role: user.role,
      isActive: user.isActive,
      mustChangePassword: user.mustChangePassword,
      passwordHash: user.passwordHash,
    },
    create: user,
  });
  userMap.set(user.email, record);

  // Also sync RequesterUser for Lab 2 backward compatibility
  if (user.role === Role.REQUESTER) {
    await client.requesterUser.upsert({
      where: { email: user.email },
      update: {
        name: user.name,
        department: "General",
        isActive: user.isActive,
      },
      create: {
        name: user.name,
        email: user.email,
        department: "General",
        isActive: user.isActive,
      },
    });
  }
}
console.log(`Successfully seeded ${usersData.length} Users.`);

```

```

// Lookup maps for foreign keys
const categoryRecords = await client.category.findMany();
const catMap = new Map(categoryRecords.map((c) => [c.name, c.id]));

const systemRecords = await client.relatedSystem.findMany();
const sysMap = new Map(systemRecords.map((s) => [s.name, s.id]));

// 5. Tickets
for (const tkt of ticketsData) {
  const requester = userMap.get(tkt.requesterEmail);
  const owner = tkt.ownerEmail ? userMap.get(tkt.ownerEmail) : null;
  const categoryId = catMap.get(tkt.categoryName);
  const relatedSystemId = sysMap.get(tkt.relatedSystemName);

  if (!requester || !categoryId || !relatedSystemId) {
    console.warn(`Skipping ticket ${tkt.ticketNumber} due to missing relation.`);
    continue;
  }

  const ticketRecord = await client.ticket.upsert({
    where: { ticketNumber: tkt.ticketNumber },
    update: {
      summary: tkt.summary,
      description: tkt.description,
      requestedPriority: tkt.requestedPriority,
      itPriority: tkt.itPriority,
      currentStatus: tkt.currentStatus,
      requesterId: requester.id,
      ownerId: owner ? owner.id : null,
      categoryId,
      relatedSystemId,
      resolutionSummary: tkt.resolutionSummary,
      requesterResolvedIndication: tkt.requesterResolvedIndication,
    },
    create: {
      ticketNumber: tkt.ticketNumber,
      summary: tkt.summary,
      description: tkt.description,
      requestedPriority: tkt.requestedPriority,
      itPriority: tkt.itPriority,
      currentStatus: tkt.currentStatus,
      requesterId: requester.id,
      ownerId: owner ? owner.id : null,
      categoryId,
      relatedSystemId,
      resolutionSummary: tkt.resolutionSummary,
      requesterResolvedIndication: tkt.requesterResolvedIndication,
    },
  });

  // 6. Comments
  for (const c of tkt.comments) {
    const author = userMap.get(c.authorEmail);
    if (author) {
      const existing = await client.publicComment.findFirst({
        where: { ticketId: ticketRecord.id, authorId: author.id, content: c.content },
      });
      if (!existing) {
        await client.publicComment.create({
          data: {
            ticketId: ticketRecord.id,
            authorId: author.id,
            content: c.content,
          },
        });
      }
    }
  }

  // 7. Internal Notes

```

```

    for (const n of tkt.notes) {
      const author = userMap.get(n.authorEmail);
      if (author) {
        const existing = await client.internalNote.findFirst({
          where: { ticketId: ticketRecord.id, authorId: author.id, content: n.content },
        });
        if (!existing) {
          await client.internalNote.create({
            data: {
              ticketId: ticketRecord.id,
              authorId: author.id,
              content: n.content,
            },
          });
        }
      }
    }
  }
}
console.log(`Successfully seeded ${ticketsData.length} Tickets with Comments and Notes.`);
}

async function main() {
  await seedDatabase(prisma);
}

if (process.argv[1] && process.argv[1].includes("seed.ts")) {
  main()
    .catch((e) => {
      console.error("Error during seeding:", e);
      process.exit(1);
    })
    .finally(async () => {
      await prisma.$disconnect();
    });
}

```

2.3.4 Security & Authorization Middleware (server/src/middleware/auth.middleware.ts)

มีดัดแปลงตรวจสอบ JWT Token, สกัดกันผู้ใช้ Inactive, สกัดกันผู้ใช้ที่ยังไม่ได้เปลี่ยนรหัสผ่านครั้งแรก (`enforcePasswordChange`), และตรวจสอบบทบาท (`requireRole`):

```

import { Request, Response, NextFunction } from "express";
import { Role } from "@prisma/client";
import { verifyToken, AuthTokenPayload } from "../utils/auth.js";

export interface AuthUser {
  id: number;
  email: string;
  role: Role;
  name?: string;
  mustChangePassword: boolean;
}

declare global {
  namespace Express {
    interface Request {
      user?: AuthUser;
    }
  }
}

export function authenticateToken(req: Request, res: Response, next: NextFunction) {
  const authHeader = req.headers["authorization"];
  const token = authHeader && authHeader.startsWith("Bearer ")
    ? authHeader.substring(7).trim()
    : null;

  if (token) {
    const payload = verifyToken(token);
    if (payload) {

```

```

    req.user = {
      id: payload.userId,
      email: payload.email,
      role: payload.role,
      mustChangePassword: payload.mustChangePassword,
    };
  }
}

next();
}

export function requireAuth(req: Request, res: Response, next: NextFunction) {
  if (!req.user) {
    res.status(401).json({ error: "Authentication required" });
    return;
  }
  next();
}

export function enforcePasswordChange(req: Request, res: Response, next: NextFunction) {
  if (req.user && req.user.mustChangePassword) {
    res.status(403).json({
      error: "Password change required before accessing the application",
      mustChangePassword: true,
    });
    return;
  }
  next();
}

export function requireRole(...roles: Role[]) {
  return (req: Request, res: Response, next: NextFunction) => {
    if (!req.user) {
      res.status(401).json({ error: "Authentication required" });
      return;
    }

    if (!roles.includes(req.user.role)) {
      res.status(403).json({ error: "Access denied: insufficient permissions" });
      return;
    }

    next();
  };
}

```

2.3.5 Authentication & Password Management API (server/src/routes/auth.routes.ts)

API Endpoint สำหรับ Login, Logout, ตรวจสอบเซสชัน [/me](#), และการบังคับเปลี่ยนรหัสผ่านครั้งแรกพร้อมตัวตรวจสอบความซับซ้อนตาม BR-04:

```

import { Router, Request, Response } from "express";
import { getPrisma } from "../prisma.js";
import { generateToken } from "../utils/auth.js";
import { comparePassword, hashPassword, validatePasswordComplexity } from "../utils/password.js";
import { authenticateToken, requireAuth } from "../middleware/auth.middleware.js";

export const authRouter = Router();

// POST /api/auth/login
authRouter.post("/login", async (req: Request, res: Response) => {
  try {
    const { email, password } = req.body;

    if (!email || !password || typeof email !== "string" || typeof password !== "string") {
      res.status(400).json({ error: "Email and password are required" });
      return;
    }

    const prisma = getPrisma();

```

```

const normalizedEmail = email.trim().toLowerCase();
const user = await prisma.user.findUnique({
  where: { email: normalizedEmail },
});

if (!user || !user.isActive) {
  res.status(401).json({ error: "Invalid email or password" });
  return;
}

const passwordValid = comparePassword(password, user.passwordHash);
if (!passwordValid) {
  res.status(401).json({ error: "Invalid email or password" });
  return;
}

const token = generateToken({
  userId: user.id,
  email: user.email,
  role: user.role,
  mustChangePassword: user.mustChangePassword,
});

res.status(200).json({
  token,
  user: {
    id: user.id,
    email: user.email,
    name: user.name,
    role: user.role,
    mustChangePassword: user.mustChangePassword,
  },
});
} catch (error) {
  res.status(500).json({ error: "Internal server error during login" });
}
});

// POST /api/auth/logout
authRouter.post("/logout", (req: Request, res: Response) => {
  res.status(200).json({ message: "Logged out successfully" });
});

// GET /api/auth/me
authRouter.get("/me", authenticateToken, requireAuth, async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const user = await prisma.user.findUnique({
      where: { id: req.user!.id },
      select: {
        id: true,
        email: true,
        name: true,
        role: true,
        isActive: true,
        mustChangePassword: true,
        createdAt: true,
        updatedAt: true,
      },
    });

    if (!user || !user.isActive) {
      res.status(401).json({ error: "User session is no longer active" });
      return;
    }

    res.status(200).json({
      id: user.id,
      email: user.email,
      name: user.name,
      role: user.role,

```

```

        mustChangePassword: user.mustChangePassword,
    });
} catch (error) {
    res.status(500).json({ error: "Internal server error fetching user profile" });
}
});

// POST /api/auth/change-password
authRouter.post(
    "/change-password",
    authenticateToken,
    requireAuth,
    async (req: Request, res: Response) => {
        try {
            const { currentPassword, newPassword } = req.body;

            if (!currentPassword || !newPassword) {
                res.status(400).json({ error: "Current password and new password are required" });
                return;
            }

            const prisma = getPrisma();
            const user = await prisma.user.findUnique({
                where: { id: req.user!.id },
            });

            if (!user || !user.isActive) {
                res.status(401).json({ error: "User not found or inactive" });
                return;
            }

            const isCurrentValid = comparePassword(currentPassword, user.passwordHash);
            if (!isCurrentValid) {
                res.status(400).json({ error: "Current password does not match" });
                return;
            }

            if (currentPassword === newPassword) {
                res.status(422).json({ error: "New password cannot be the same as current password" });
                return;
            }

            const complexityResult = validatePasswordComplexity(newPassword);
            if (!complexityResult.isValid) {
                res.status(422).json({
                    error: "Password does not meet complexity requirements",
                    details: complexityResult.errors,
                });
                return;
            }

            const newHash = hashPassword(newPassword);
            await prisma.user.update({
                where: { id: user.id },
                data: {
                    passwordHash: newHash,
                    mustChangePassword: false,
                },
            });

            res.status(200).json({
                message: "Password changed successfully",
                mustChangePassword: false,
            });
        } catch (error) {
            res.status(500).json({ error: "Internal server error changing password" });
        }
    }
);

```

2.3.6 IT Staff Ticket Queue & Operations Engine (server/src/routes/staff.routes.ts)

เอนจินค้นหาและกรองคิวงานด้วยไอทีหลายมิติ (Search, Filter, Sort, Pagination), ระบบกดรับงาน Claim (ปรับเป็น OPEN), Reassign, และการเปลี่ยนสถานะตาม Status Matrix (BR-14):

```
import { Router, Request, Response } from "express";
import { Role, Priority, TicketStatus, Prisma } from "@prisma/client";
import { getPrisma } from "../prisma.js";
import { requireAuth, enforcePasswordChange, requireRole } from "../middleware/auth.middleware.js";

export const staffRouter = Router();

staffRouter.use(requireAuth);
staffRouter.use(enforcePasswordChange);
staffRouter.use(requireRole(Role.IT_STAFF, Role.ADMINISTRATOR));

// GET /api/staff/tickets - Shared IT Staff Queue
staffRouter.get("/tickets", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const {
      search,
      category,
      status,
      priority,
      itPriority,
      ownerId,
      sort = "createdAt",
      order = "desc",
      page = "1",
      limit = "10",
    } = req.query;

    const pageNum = Math.max(1, parseInt(String(page), 10) || 1);
    const limitNum = Math.max(1, Math.min(100, parseInt(String(limit), 10) || 10));
    const skip = (pageNum - 1) * limitNum;

    const where: Prisma.TicketWhereInput = {};

    // Case-insensitive search on ticketNumber or summary
    if (search && typeof search === "string" && search.trim() !== "") {
      const q = search.trim();
      where.OR = [
        { ticketNumber: { contains: q, mode: "insensitive" } },
        { summary: { contains: q, mode: "insensitive" } },
      ];
    }

    // Category filter
    if (category) {
      const catId = parseInt(String(category), 10);
      if (!isNaN(catId) && catId > 0) {
        where.categoryId = catId;
      }
    }

    // Status filter
    if (status && typeof status === "string" && status.trim() !== "") {
      const upperStatus = status.trim().toUpperCase();
      if (Object.values(TicketStatus).includes(upperStatus as TicketStatus)) {
        where.currentStatus = upperStatus as TicketStatus;
      }
    }

    // Priority filter (Requested Priority)
    if (priority && typeof priority === "string" && priority.trim() !== "") {
      const upperPri = priority.trim().toUpperCase();
      if (Object.values(Priority).includes(upperPri as Priority)) {
        where.requestedPriority = upperPri as Priority;
      }
    }
  }
}
```

```

// IT Priority filter
if (itPriority && typeof itPriority === "string" && itPriority.trim() !== "") {
  const upperItPri = itPriority.trim().toUpperCase();
  if (Object.values(Priority).includes(upperItPri as Priority)) {
    where.itPriority = upperItPri as Priority;
  }
}

// Ownership filter
if (ownerId !== undefined && ownerId !== null && String(ownerId).trim() !== "") {
  const ownerStr = String(ownerId).trim().toLowerCase();
  if (ownerStr === "unassigned") {
    where.ownerId = null;
  } else if (ownerStr === "me" && req.user) {
    where.ownerId = req.user.id;
  } else {
    const parsedId = parseInt(ownerStr, 10);
    if (!isNaN(parsedId)) {
      where.ownerId = parsedId;
    }
  }
}

// Sorting
const sortField = String(sort).trim();
const sortOrder: Prisma.SortOrder = String(order).toLowerCase() === "asc" ? "asc" : "desc";
let orderBy: Prisma.TicketOrderByWithRelationInput = { createdAt: sortOrder };

if (sortField === "itPriority") {
  orderBy = { itPriority: sortOrder };
} else if (sortField === "currentStatus") {
  orderBy = { currentStatus: sortOrder };
} else if (sortField === "ticketNumber") {
  orderBy = { ticketNumber: sortOrder };
} else if (sortField === "createdAt") {
  orderBy = { createdAt: sortOrder };
}

const [total, tickets] = await Promise.all([
  prisma.ticket.count({ where }),
  prisma.ticket.findMany({
    where,
    orderBy,
    skip,
    take: limitNum,
    select: {
      id: true,
      ticketNumber: true,
      summary: true,
      description: true,
      requestedPriority: true,
      itPriority: true,
      currentStatus: true,
      resolutionSummary: true,
      requesterResolvedIndication: true,
      createdAt: true,
      updatedAt: true,
      category: {
        select: { id: true, name: true },
      },
      relatedSystem: {
        select: { id: true, name: true },
      },
      requester: {
        select: { id: true, name: true, email: true },
      },
      owner: {
        select: { id: true, name: true, email: true },
      },
    },
  }),
]);

```

```

    }},
  ]});

const totalPages = Math.max(1, Math.ceil(total / limitNum));

res.status(200).json({
  tickets,
  pagination: {
    total,
    page: pageNum,
    limit: limitNum,
    totalPages,
  },
});
} catch (error) {
  res.status(500).json({ error: "Failed to retrieve IT staff ticket queue" });
}
});

// GET /api/staff/users - Active IT Staff and Administrators for assignment
staffRouter.get("/users", async (_req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const users = await prisma.user.findMany({
      where: {
        isActive: true,
        role: { in: [Role.IT_STAFF, Role.ADMINISTRATOR] },
      },
      select: { id: true, name: true, email: true, role: true },
      orderBy: { name: "asc" },
    });
    res.status(200).json(users);
  } catch (error) {
    res.status(500).json({ error: "Failed to retrieve staff roster" });
  }
});

// GET /api/staff/tickets/:id - Detailed Ticket View for Staff
staffRouter.get("/tickets/:id", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({
      where: { id: ticketId },
      include: {
        category: { select: { id: true, name: true, description: true } },
        relatedSystem: { select: { id: true, name: true, category: true } },
        requester: { select: { id: true, name: true, email: true } },
        owner: { select: { id: true, name: true, email: true, role: true } },
        attachments: {
          where: { isRemoved: false },
          select: { id: true, filename: true, fileSize: true, mimeType: true, isRemoved: true, uploadedAt: true },
        },
        comments: {
          include: {
            author: { select: { id: true, name: true, email: true, role: true } },
          },
          orderBy: { createdAt: "asc" },
        },
        notes: {
          include: {
            author: { select: { id: true, name: true, email: true, role: true } },
          },
          orderBy: { createdAt: "asc" },
        },
      },
    });
  }
});

```

```

    });

    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    res.status(200).json(ticket);
  } catch (error) {
    res.status(500).json({ error: "Failed to retrieve ticket details" });
  }
});

// PATCH /api/staff/tickets/:id/claim - Claim ownership
staffRouter.patch("/tickets/:id/claim", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    const nextStatus = ticket.currentStatus === TicketStatus.NEW ? TicketStatus.OPEN : ticket.currentStatus;

    const updated = await prisma.ticket.update({
      where: { id: ticketId },
      data: {
        ownerId: req.user!.id,
        currentStatus: nextStatus,
      },
      include: {
        category: { select: { id: true, name: true } },
        relatedSystem: { select: { id: true, name: true } },
        requester: { select: { id: true, name: true, email: true } },
        owner: { select: { id: true, name: true, email: true } },
      },
    });

    res.status(200).json(updated);
  } catch (error) {
    res.status(500).json({ error: "Failed to claim ticket" });
  }
});

// PATCH /api/staff/tickets/:id/assign - Reassign ticket owner
staffRouter.patch("/tickets/:id/assign", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const { ownerId } = req.body;
    const parsedOwnerId = parseInt(String(ownerId), 10);
    if (isNaN(parsedOwnerId) || parsedOwnerId <= 0) {
      res.status(400).json({ error: "Valid ownerId is required" });
      return;
    }

    // Verify target user is active IT Staff or Administrator
    const targetUser = await prisma.user.findUnique({
      where: { id: parsedOwnerId },

```

```

        select: { id: true, isActive: true, role: true },
    });

    if (!targetUser || !targetUser.isActive || (targetUser.role !== Role.IT_STAFF && targetUser.role !== Role.ADMIN)) {
        res.status(400).json({ error: "Target owner must be an active IT Staff or Administrator" });
        return;
    }

    const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
    if (!ticket) {
        res.status(404).json({ error: "Ticket not found" });
        return;
    }

    const nextStatus = ticket.currentStatus === TicketStatus.NEW ? TicketStatus.OPEN : ticket.currentStatus;

    const updated = await prisma.ticket.update({
        where: { id: ticketId },
        data: {
            ownerId: parsedOwnerId,
            currentStatus: nextStatus,
        },
        include: {
            category: { select: { id: true, name: true } },
            relatedSystem: { select: { id: true, name: true } },
            requester: { select: { id: true, name: true, email: true } },
            owner: { select: { id: true, name: true, email: true } },
        },
    });

    res.status(200).json(updated);
} catch (error) {
    res.status(500).json({ error: "Failed to assign ticket" });
}
});

// PATCH /api/staff/tickets/:id/priority - Update IT Priority
staffRouter.patch("/tickets/:id/priority", async (req: Request, res: Response) => {
    try {
        const prisma = getPrisma();
        const ticketId = parseInt(req.params.id, 10);
        if (isNaN(ticketId) || ticketId <= 0) {
            res.status(400).json({ error: "Invalid ticket ID" });
            return;
        }

        const { itPriority } = req.body;
        if (!itPriority || typeof itPriority !== "string") {
            res.status(400).json({ error: "itPriority is required" });
            return;
        }

        const upperPri = itPriority.trim().toUpperCase();
        if (!Object.values(Priority).includes(upperPri as Priority)) {
            res.status(400).json({ error: "Invalid IT Priority value" });
            return;
        }

        const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
        if (!ticket) {
            res.status(404).json({ error: "Ticket not found" });
            return;
        }

        const updated = await prisma.ticket.update({
            where: { id: ticketId },
            data: { itPriority: upperPri as Priority },
            include: {
                category: { select: { id: true, name: true } },
                relatedSystem: { select: { id: true, name: true } },
                requester: { select: { id: true, name: true, email: true } },
            },
        });
    } catch (error) {
        res.status(500).json({ error: "Failed to update ticket priority" });
    }
});

```

```

    owner: { select: { id: true, name: true, email: true } },
  },
});

res.status(200).json(updated);
} catch (error) {
  res.status(500).json({ error: "Failed to update IT Priority" });
}
});

// BR-19 State Transition Matrix
const ALLOWED_STATUS_TRANSITIONS: Record<TicketStatus, TicketStatus[]> = {
  NEW: [TicketStatus.OPEN, TicketStatus.CANCELLED],
  OPEN: [TicketStatus.IN_PROGRESS, TicketStatus.WAITING_FOR_REQUESTER, TicketStatus.CANCELLED],
  IN_PROGRESS: [TicketStatus.WAITING_FOR_REQUESTER, TicketStatus.CANCELLED], // RESOLVED requires resolutionSummary
  WAITING_FOR_REQUESTER: [TicketStatus.IN_PROGRESS, TicketStatus.CANCELLED], // RESOLVED requires resolutionSummary
  RESOLVED: [TicketStatus.CLOSED, TicketStatus.REOPENED],
  CLOSED: [TicketStatus.REOPENED],
  REOPENED: [TicketStatus.IN_PROGRESS, TicketStatus.WAITING_FOR_REQUESTER, TicketStatus.CANCELLED],
  CANCELLED: [],
};

// PATCH /api/staff/tickets/:id/status - Transition Status per BR-19
staffRouter.patch("/tickets/:id/status", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const { status } = req.body;
    if (!status || typeof status !== "string") {
      res.status(400).json({ error: "status is required" });
      return;
    }

    const upperStatus = status.trim().toUpperCase();

    // Guard: Direct transition to RESOLVED via general status endpoint is strictly rejected per BR-20
    if (upperStatus === TicketStatus.RESOLVED) {
      res.status(422).json({
        error: "Direct transition to RESOLVED via general status endpoint is not permitted. Please use PATCH /api/s
      });
      return;
    }

    if (!Object.values(TicketStatus).includes(upperStatus as TicketStatus)) {
      res.status(400).json({ error: "Invalid status value" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    const allowed = ALLOWED_STATUS_TRANSITIONS[ticket.currentStatus] || [];
    if (!allowed.includes(upperStatus as TicketStatus)) {
      res.status(400).json({
        error: `Invalid status transition from ${ticket.currentStatus} to ${upperStatus}`,
      });
      return;
    }

    const updated = await prisma.ticket.update({
      where: { id: ticketId },
      data: { currentStatus: upperStatus as TicketStatus },
      include: {

```

```

        category: { select: { id: true, name: true } },
        relatedSystem: { select: { id: true, name: true } },
        requester: { select: { id: true, name: true, email: true } },
        owner: { select: { id: true, name: true, email: true } },
    },
});

res.status(200).json(updated);
} catch (error) {
    res.status(500).json({ error: "Failed to update ticket status" });
}
});

// PATCH /api/staff/tickets/:id/resolve - Resolve with Mandatory Resolution Summary (BR-19, BR-20)
staffRouter.patch("/tickets/:id/resolve", async (req: Request, res: Response) => {
    try {
        const prisma = getPrisma();
        const ticketId = parseInt(req.params.id, 10);
        if (isNaN(ticketId) || ticketId <= 0) {
            res.status(400).json({ error: "Invalid ticket ID" });
            return;
        }

        const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
        if (!ticket) {
            res.status(404).json({ error: "Ticket not found" });
            return;
        }

        // Permitted to resolve only from IN_PROGRESS, WAITING_FOR_REQUESTER, or REOPENED
        const permittable: TicketStatus[] = [
            TicketStatus.IN_PROGRESS,
            TicketStatus.WAITING_FOR_REQUESTER,
            TicketStatus.REOPENED,
        ];
        if (!permittable.includes(ticket.currentStatus)) {
            res.status(400).json({
                error: `Cannot resolve ticket from current status: ${ticket.currentStatus}`,
            });
            return;
        }

        const { resolutionSummary } = req.body;
        if (
            !resolutionSummary ||
            typeof resolutionSummary !== "string" ||
            resolutionSummary.trim().length < 3 ||
            resolutionSummary.trim().length > 500
        ) {
            res.status(422).json({
                error: "Resolution summary is required and must be between 3 and 500 characters",
            });
            return;
        }

        const updated = await prisma.ticket.update({
            where: { id: ticketId },
            data: {
                currentStatus: TicketStatus.RESOLVED,
                resolutionSummary: resolutionSummary.trim(),
            },
            include: {
                category: { select: { id: true, name: true } },
                relatedSystem: { select: { id: true, name: true } },
                requester: { select: { id: true, name: true, email: true } },
                owner: { select: { id: true, name: true, email: true } },
            },
        });

        res.status(200).json(updated);
    } catch (error) {

```

```

    res.status(500).json({ error: "Failed to resolve ticket" });
  }
});

```

2.3.7 Public Comments vs Internal Notes API (server/src/routes/comments.routes.ts)

การแบ่งแยกช่องทางการสื่อสารสองระดับ: ความคิดเห็นสาธารณะ (Public Comments) และโน้ตภายในสำหรับเจ้าหน้าที่ (Internal Notes) พร้อมระบบสิทธิ์ 403 Forbidden สำหรับ Requester (BR-13):

```

import { Router, Request, Response } from "express";
import { Role } from "@prisma/client";
import { getPrisma } from "../prisma.js";
import { requireAuth, enforcePasswordChange } from "../middleware/auth.middleware.js";

export const commentsRouter = Router();

function requireTicketAuth(req: Request, res: Response, next: () => void) {
  if (req.user) {
    if (req.user.mustChangePassword) {
      res.status(403).json({
        error: "Password change required before accessing the application",
        mustChangePassword: true,
      });
      return;
    }
    return next();
  }
}

const devId = req.headers["x-development-requester-id"];
const parsed = Number(Array.isArray(devId) ? devId[0] : devId);
if (Number.isInteger(parsed) && parsed > 0) {
  req.user = {
    id: parsed,
    email: `requester${parsed}@example.com`,
    role: Role.REQUESTER,
    mustChangePassword: false,
  };
  return next();
}

res.status(401).json({ error: "Authentication required" });
}

// GET /api/tickets/:id/comments - Public Comments (Requester owner, IT Staff, Admin)
commentsRouter.get("/:id/comments", requireTicketAuth, async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({
      where: { id: ticketId },
      select: { id: true, requesterId: true },
    });

    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    // If requester, verify ownership
    if (req.user!.role === Role.REQUESTER && ticket.requesterId !== req.user!.id) {
      res.status(403).json({ error: "Access denied: not your ticket" });
      return;
    }

    const comments = await prisma.publicComment.findMany({

```

```

    where: { ticketId },
    include: {
      author: { select: { id: true, name: true, email: true, role: true } },
    },
    orderBy: { createdAt: "asc" },
  });

  res.status(200).json(comments);
} catch (error) {
  res.status(500).json({ error: "Failed to retrieve public comments" });
}
});

// POST /api/tickets/:id/comments - Create Public Comment
commentsRouter.post("/:id/comments", requireTicketAuth, async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({
      where: { id: ticketId },
      select: { id: true, requesterId: true },
    });

    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    // If requester, verify ownership
    if (req.user!.role !== Role.REQUESTER && ticket.requesterId !== req.user!.id) {
      res.status(403).json({ error: "Access denied: not your ticket" });
      return;
    }

    const { content } = req.body;
    if (!content || typeof content !== "string" || content.trim() === "") {
      res.status(400).json({ error: "Comment content is required" });
      return;
    }

    const comment = await prisma.publicComment.create({
      data: {
        ticketId,
        authorId: req.user!.id,
        content: content.trim(),
      },
      include: {
        author: { select: { id: true, name: true, email: true, role: true } },
      },
    });

    res.status(201).json(comment);
  } catch (error) {
    res.status(500).json({ error: "Failed to create public comment" });
  }
});

// GET /api/tickets/:id/notes - Internal Notes (Strictly IT Staff & Admin only)
commentsRouter.get("/:id/notes", requireTicketAuth, async (req: Request, res: Response) => {
  try {
    if (req.user!.role !== Role.IT_STAFF && req.user!.role !== Role.ADMINISTRATOR) {
      res.status(403).json({
        error: "Access denied: internal notes are restricted to IT Staff and Administrators",
      });
      return;
    }
  }
});

```

```

const prisma = getPrisma();
const ticketId = parseInt(req.params.id, 10);
if (isNaN(ticketId) || ticketId <= 0) {
  res.status(400).json({ error: "Invalid ticket ID" });
  return;
}

const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
if (!ticket) {
  res.status(404).json({ error: "Ticket not found" });
  return;
}

const notes = await prisma.internalNote.findMany({
  where: { ticketId },
  include: {
    author: { select: { id: true, name: true, email: true, role: true } },
  },
  orderBy: { createdAt: "asc" },
});

res.status(200).json(notes);
} catch (error) {
  res.status(500).json({ error: "Failed to retrieve internal notes" });
}
});

// POST /api/tickets/:id/notes - Create Internal Note (Strictly IT Staff & Admin only)
commentsRouter.post("/:id/notes", requireTicketAuth, async (req: Request, res: Response) => {
  try {
    if (req.user!.role !== Role.IT_STAFF && req.user!.role !== Role.ADMINISTRATOR) {
      res.status(403).json({
        error: "Access denied: internal notes are restricted to IT Staff and Administrators",
      });
      return;
    }

    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({ where: { id: ticketId } });
    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    const { content } = req.body;
    if (!content || typeof content !== "string" || content.trim() === "") {
      res.status(400).json({ error: "Note content is required" });
      return;
    }

    const note = await prisma.internalNote.create({
      data: {
        ticketId,
        authorId: req.user!.id,
        content: content.trim(),
      },
      include: {
        author: { select: { id: true, name: true, email: true, role: true } },
      },
    });

    res.status(201).json(note);
  } catch (error) {
    res.status(500).json({ error: "Failed to create internal note" });
  }
});

```

```

    }
  });

// PATCH /api/tickets/:id/resolve-indication - Requester Problem Appears Resolved toggle
commentsRouter.patch("/:id/resolve-indication", requireTicketAuth, async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const ticketId = parseInt(req.params.id, 10);
    if (isNaN(ticketId) || ticketId <= 0) {
      res.status(400).json({ error: "Invalid ticket ID" });
      return;
    }

    const ticket = await prisma.ticket.findUnique({
      where: { id: ticketId },
      select: { id: true, requesterId: true },
    });

    if (!ticket) {
      res.status(404).json({ error: "Ticket not found" });
      return;
    }

    // Only ticket owner requester can indicate
    if (req.user!.role === Role.REQUESTER && ticket.requesterId !== req.user!.id) {
      res.status(403).json({ error: "Access denied: not your ticket" });
      return;
    }

    const { appearsResolved = true } = req.body;

    const updated = await prisma.ticket.update({
      where: { id: ticketId },
      data: { requesterResolvedIndication: Boolean(appearsResolved) },
      select: {
        id: true,
        ticketNumber: true,
        requesterResolvedIndication: true,
        currentStatus: true,
      },
    });

    res.status(200).json(updated);
  } catch (error) {
    res.status(500).json({ error: "Failed to update resolution indication" });
  }
});

```

2.3.8 Administrator User Management & Safety Rules (server/src/routes/admin.routes.ts)

API จัดการบัญชีผู้ใช้สำหรับผู้ดูแลระบบ พร้อมการบังคับใช้กฎความปลอดภัย Admin Safety Rules (ห้ามปิดบัญชีตัวเอง, ห้ามลบลิสต์ตัวเอง, ห้ามปิดหรือลบลิสต์ Admin คนสุดท้าย):

```

import { Router, Request, Response } from "express";
import { Role, Prisma } from "@prisma/client";
import { getPrisma } from "../prisma.js";
import { requireAuth, enforcePasswordChange, requireRole } from "../middleware/auth.middleware.js";
import { validatePasswordComplexity, hashPassword } from "../utils/password.js";

export const adminRouter = Router();

adminRouter.use(requireAuth);
adminRouter.use(enforcePasswordChange);
adminRouter.use(requireRole(Role.ADMINISTRATOR));

// GET /api/admin/users - List users with search and role filter
adminRouter.get("/users", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const { search, role, isActive } = req.query;

```

```

const where: Prisma.UserWhereInput = {};

// Search query matching name or email (case-insensitive)
if (search && typeof search === "string" && search.trim() !== "") {
  const q = search.trim();
  where.OR = [
    { name: { contains: q, mode: "insensitive" } },
    { email: { contains: q, mode: "insensitive" } },
  ];
}

// Role filter
if (role && typeof role === "string" && role.trim() !== "") {
  const upperRole = role.trim().toUpperCase();
  if (Object.values(Role).includes(upperRole as Role)) {
    where.role = upperRole as Role;
  }
}

// Active status filter
if (isActive !== undefined && isActive !== "") {
  where.isActive = String(isActive) === "true";
}

const users = await prisma.user.findMany({
  where,
  select: {
    id: true,
    name: true,
    email: true,
    role: true,
    isActive: true,
    mustChangePassword: true,
    createdAt: true,
    updatedAt: true,
  },
  orderBy: [{ name: "asc" }, { createdAt: "desc" }],
});

res.status(200).json(users);
} catch (error) {
  res.status(500).json({ error: "Failed to retrieve users" });
}
});

// POST /api/admin/users - Create new user account
adminRouter.post("/users", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const { name, email, role, initialPassword, isActive = true } = req.body;

    if (!name || typeof name !== "string" || name.trim().length < 2) {
      res.status(400).json({ error: "Full name must be at least 2 characters" });
      return;
    }

    if (!email || typeof email !== "string" || !/^[^\\s@]+@[^\\s@]+\\.([^\\s@]+)$/.test(email.trim())) {
      res.status(400).json({ error: "Valid email address is required" });
      return;
    }

    const upperRole = typeof role === "string" ? role.trim().toUpperCase() : "";
    if (!Object.values(Role).includes(upperRole as Role)) {
      res.status(400).json({ error: "Role must be one of: REQUESTER, IT_STAFF, ADMINISTRATOR" });
      return;
    }

    // Validate password complexity
    const passwordValidation = validatePasswordComplexity(initialPassword);
    if (!passwordValidation.isValid) {

```

```

    res.status(422).json({
      error: "Password does not meet complexity requirements",
      details: passwordValidation.errors,
    });
    return;
  }

  const normalizedEmail = email.trim().toLowerCase();

  // BR-10: Check globally unique email address
  const existing = await prisma.user.findFirst({
    where: { email: { equals: normalizedEmail, mode: "insensitive" } },
  });

  if (existing) {
    res.status(409).json({ error: "Email address is already registered in the system" });
    return;
  }

  const user = await prisma.user.create({
    data: {
      name: name.trim(),
      email: normalizedEmail,
      role: upperRole as Role,
      passwordHash: hashPassword(initialPassword),
      mustChangePassword: true,
      isActive: isActive !== false,
    },
    select: {
      id: true,
      name: true,
      email: true,
      role: true,
      isActive: true,
      mustChangePassword: true,
      createdAt: true,
      updatedAt: true,
    },
  });

  res.status(201).json(user);
} catch (error) {
  res.status(500).json({ error: "Failed to create user" });
}
});

// PATCH /api/admin/users/:id - Update user details and activation status
adminRouter.patch("/users/:id", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const targetId = parseInt(req.params.id, 10);
    if (isNaN(targetId) || targetId <= 0) {
      res.status(400).json({ error: "Invalid user ID" });
      return;
    }

    const targetUser = await prisma.user.findUnique({
      where: { id: targetId },
    });

    if (!targetUser) {
      res.status(404).json({ error: "User not found" });
      return;
    }

    const { name, email, role, isActive } = req.body;

    // BR-07: Administrator cannot deactivate their own account
    if (req.user!.id === targetId && isActive === false && targetUser.isActive === true) {
      res.status(400).json({ error: "Administrators cannot deactivate their own account" });
      return;
    }
  }
});

```

```

}

// BR-08: Administrator cannot change their own role away from ADMINISTRATOR
if (
  req.user!.id === targetId &&
  role !== undefined &&
  role.toUpperCase() !== Role.ADMINISTRATOR &&
  targetUser.role === Role.ADMINISTRATOR
) {
  res.status(400).json({ error: "Administrators cannot change their own role away from ADMINISTRATOR" });
  return;
}

// BR-09: Prevent deactivating or demoting the last active Administrator
if (targetUser.role === Role.ADMINISTRATOR && targetUser.isActive === true) {
  const willBeInactive = isActive === false;
  const willDemote = role !== undefined && role.toUpperCase() !== Role.ADMINISTRATOR;

  if (willBeInactive || willDemote) {
    const activeAdminCount = await prisma.user.count({
      where: { role: Role.ADMINISTRATOR, isActive: true },
    });

    if (activeAdminCount <= 1) {
      if (willBeInactive) {
        res.status(400).json({ error: "Cannot deactivate the last active Administrator" });
        return;
      }
      if (willDemote) {
        res.status(400).json({ error: "Cannot demote the last active Administrator" });
        return;
      }
    }
  }
}

// BR-10: Check globally unique email if updating email
let normalizedEmail: string | undefined = undefined;
if (email !== undefined && typeof email === "string") {
  normalizedEmail = email.trim().toLowerCase();
  if (!/^[\^\\s@]+@[^\s@]+\.[^\s@]+$/i.test(normalizedEmail)) {
    res.status(400).json({ error: "Valid email address is required" });
    return;
  }
}

if (normalizedEmail !== targetUser.email.toLowerCase()) {
  const existingEmail = await prisma.user.findFirst({
    where: {
      id: { not: targetId },
      email: { equals: normalizedEmail, mode: "insensitive" },
    },
  });

  if (existingEmail) {
    res.status(409).json({ error: "Email address is already in use by another user" });
    return;
  }
}

let upperRole: Role | undefined = undefined;
if (role !== undefined && typeof role === "string") {
  const parsed = role.trim().toUpperCase();
  if (!Object.values(Role).includes(parsed as Role)) {
    res.status(400).json({ error: "Role must be one of: REQUESTER, IT_STAFF, ADMINISTRATOR" });
    return;
  }
  upperRole = parsed as Role;
}

const updated = await prisma.user.update({

```

```

    where: { id: targetId },
    data: {
      ...(name && typeof name === "string" ? { name: name.trim() } : {}),
      ...(normalizedEmail ? { email: normalizedEmail } : {}),
      ...(upperRole ? { role: upperRole } : {}),
      ...(typeof isActive === "boolean" ? { isActive } : {}),
    },
    select: {
      id: true,
      name: true,
      email: true,
      role: true,
      isActive: true,
      mustChangePassword: true,
      createdAt: true,
      updatedAt: true,
    },
  });

  res.status(200).json(updated);
} catch (error) {
  res.status(500).json({ error: "Failed to update user" });
}
});

// POST /api/admin/users/:id/reset-password - Reset user password and force change
adminRouter.post("/users/:id/reset-password", async (req: Request, res: Response) => {
  try {
    const prisma = getPrisma();
    const targetId = parseInt(req.params.id, 10);
    if (isNaN(targetId) || targetId <= 0) {
      res.status(400).json({ error: "Invalid user ID" });
      return;
    }

    const targetUser = await prisma.user.findUnique({
      where: { id: targetId },
    });

    if (!targetUser) {
      res.status(404).json({ error: "User not found" });
      return;
    }

    const { newInitialPassword } = req.body;

    const validation = validatePasswordComplexity(newInitialPassword);
    if (!validation.isValid) {
      res.status(422).json({
        error: "Password does not meet complexity requirements",
        details: validation.errors,
      });
      return;
    }

    await prisma.user.update({
      where: { id: targetId },
      data: {
        passwordHash: hashPassword(newInitialPassword),
        mustChangePassword: true,
      },
    });

    res.status(200).json({ message: "Password reset successfully" });
  } catch (error) {
    res.status(500).json({ error: "Failed to reset password" });
  }
});

```

2.3.9 Frontend Core UI Components

1. Login Screen Component (client/src/components/Login.tsx):

```
import React, { useState } from "react";
import { useAuth } from "../context/AuthContext";

export const Login: React.FC = () => {
  const { login, isLoading, error, clearError } = useAuth();
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [showPassword, setShowPassword] = useState(false);
  const [localError, setLocalError] = useState<string | null>(null);

  const handleSubmit = async (e: React.FormEvent) => {
    e.preventDefault();
    setLocalError(null);
    clearError();

    if (!email.trim() || !password) {
      setLocalError("Please enter both email and password.");
      return;
    }

    try {
      await login(email.trim(), password);
    } catch (err: any) {
      // Error handled by AuthContext
    }
  };

  const displayError = localError || error;

  return (
    <div style={styles.container}>
      <div style={styles.card}>
        <div style={styles.header}>
          <div style={styles.logoBadge}>
            <svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="#FFFFFF" strokeWidth="2.5">
              <circle cx="12" cy="12" r="10" />
              <polyline points="12 6 12 16 14" />
            </svg>
          </div>
          <h1 style={styles.title}>TokTickIT</h1>
          <p style={styles.subtitle}>Sign in to your account</p>
        </div>

        {displayError && (
          <div role="alert" style={styles.errorAlert}>
            <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="#B71C1C" strokeWidth="2" style={styles.errorIcon}>
              <circle cx="12" cy="12" r="10" />
              <line x1="12" y1="8" x2="12" y2="12" />
              <line x1="12" y1="16" x2="12.01" y2="16" />
            </svg>
            <span style={styles.errorText}>{displayError}</span>
          </div>
        )}

        <form onSubmit={handleSubmit} style={styles.form}>
          <div style={styles.formGroup}>
            <label htmlFor="login-email" style={styles.label}>
              Email Address
            </label>
            <input
              id="login-email"
              type="email"
              value={email}
              onChange={(e) => setEmail(e.target.value)}
              placeholder="e.g. pae.karn@example.com"
              required
              autoFocus
              disabled={isLoading}
              style={styles.input}
            />
          </div>
        </form>
      </div>
    </div>
  );
};
```

```

    />
  </div>

  <div style={styles.formGroup}>
    <label htmlFor="login-password" style={styles.label}>
      Password
    </label>
    <div style={styles.passwordWrapper}>
      <input
        id="login-password"
        type={showPassword ? "text" : "password"}
        value={password}
        onChange={(e) => setPassword(e.target.value)}
        placeholder="Enter your password"
        required
        disabled={isLoading}
        style={styles.passwordInput}
      />
      <button
        type="button"
        onClick={() => setShowPassword(!showPassword)}
        style={styles.toggleBtn}
        title={showPassword ? "Hide password" : "Show password"}
        aria-label={showPassword ? "Hide password" : "Show password"}
      >
        {showPassword ? "Hide" : "Show"}
      </button>
    </div>
  </div>

  <button
    type="submit"
    disabled={isLoading}
    style={{
      ...styles.submitBtn,
      opacity: isLoading ? 0.7 : 1,
      cursor: isLoading ? "not-allowed" : "pointer",
    }}
  >
    {isLoading ? "Signing in..." : "Sign In"}
  </button>
</form>

<div style={styles.footerNote}>
  <span>IT Operations Platform – Zen Green Edition</span>
</div>
</div>
</div>
);
};

const styles: Record<string, React.CSSProperties> = {
  container: {
    minHeight: "100vh",
    display: "flex",
    alignItems: "center",
    justifyContent: "center",
    backgroundColor: "#F5F7F6",
    padding: "20px",
    fontFamily: "-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif",
  },
  card: {
    width: "100%",
    maxWidth: "440px",
    backgroundColor: "FFFFFF",
    borderRadius: "12px",
    boxShadow: "0 4px 20px rgba(0, 0, 0, 0.08)",
    padding: "36px 32px",
    border: "1px solid #E0E6E2",
  },
  header: {

```

```

      textAlign: "center",
      marginBottom: "28px",
    },
    logoBadge: {
      width: "48px",
      height: "48px",
      borderRadius: "12px",
      backgroundColor: "#006B3C",
      display: "inline-flex",
      alignItems: "center",
      justifyContent: "center",
      marginBottom: "12px",
    },
    title: {
      fontSize: "24px",
      fontWeight: "700",
      color: "#006B3C",
      margin: "0 0 6px 0",
    },
    subtitle: {
      fontSize: "14px",
      color: "#5A6E63",
      margin: 0,
    },
    errorAlert: {
      display: "flex",
      alignItems: "center",
      backgroundColor: "#FFEBEE",
      border: "1px solid #FFCDD2",
      borderRadius: "8px",
      padding: "12px 14px",
      marginBottom: "20px",
    },
    errorIcon: {
      marginRight: "10px",
      flexShrink: 0,
    },
    errorText: {
      color: "#B71C1C",
      fontSize: "13px",
      lineHeight: "1.4",
    },
    form: {
      display: "flex",
      flexDirection: "column",
      gap: "18px",
    },
    formGroup: {
      display: "flex",
      flexDirection: "column",
      gap: "6px",
    },
    label: {
      fontSize: "13px",
      fontWeight: "600",
      color: "#1A2E23",
    },
    input: {
      padding: "10px 14px",
      fontSize: "14px",
      borderRadius: "8px",
      border: "1px solid #E0E6E2",
      outline: "none",
      color: "#1A2E23",
      backgroundColor: "FFFFFF",
      transition: "border-color 0.2s",
    },
    passwordWrapper: {
      position: "relative",
      display: "flex",
      alignItems: "center",

```

```

    },
    passwordInput: {
      width: "100%",
      padding: "10px 60px 10px 14px",
      fontSize: "14px",
      borderRadius: "8px",
      border: "1px solid #E0E6E2",
      outline: "none",
      color: "#1A2E23",
      backgroundColor: "#FFFFFF",
    },
    toggleBtn: {
      position: "absolute",
      right: "8px",
      padding: "4px 8px",
      fontSize: "12px",
      fontWeight: "600",
      color: "#006B3C",
      backgroundColor: "transparent",
      border: "none",
      cursor: "pointer",
      borderRadius: "4px",
    },
    submitBtn: {
      marginTop: "8px",
      padding: "12px",
      fontSize: "15px",
      fontWeight: "600",
      color: "#FFFFFF",
      backgroundColor: "#006B3C",
      border: "none",
      borderRadius: "8px",
      transition: "background-color 0.2s",
    },
    footerNote: {
      marginTop: "24px",
      textAlign: "center",
      fontSize: "11px",
      color: "#5A6E63",
    },
  },
};

```

2. Mandatory Password Change Component (client/src/components/ChangePassword.tsx):

```

import React, { useState } from "react";
import { useAuth } from "../../context/AuthContext";

export const ChangePassword: React.FC = () => {
  const { updatePassword, isLoading, error, clearError, logout, user } = useAuth();
  const [currentPassword, setCurrentPassword] = useState("");
  const [newPassword, setNewPassword] = useState("");
  const [confirmPassword, setConfirmPassword] = useState("");
  const [showPassword, setShowPassword] = useState(false);
  const [localError, setLocalError] = useState<string | null>(null);

  // Complexity rules check
  const hasMinLength = newPassword.length >= 8;
  const hasUppercase = /[A-Z]/.test(newPassword);
  const hasLowercase = /[a-z]/.test(newPassword);
  const hasNumber = /[0-9]/.test(newPassword);
  const passwordsMatch = newPassword.length > 0 && newPassword === confirmPassword;
  const isSameAsCurrent = currentPassword.length > 0 && newPassword.length > 0 && newPassword === currentPassword;

  const isValidForm =
    hasMinLength &&
    hasUppercase &&
    hasLowercase &&
    hasNumber &&
    passwordsMatch &&
    !isSameAsCurrent &&
    currentPassword.length > 0;

```

```

const handleSubmit = async (e: React.FormEvent) => {
  e.preventDefault();
  setLocalError(null);
  clearError();

  if (!currentPassword) {
    setLocalError("Please enter your current temporary password.");
    return;
  }

  if (currentPassword === newPassword) {
    setLocalError("New password cannot be the same as current password.");
    return;
  }

  if (!hasMinLength || !hasUppercase || !hasLowercase || !hasNumber) {
    setLocalError("New password does not meet all complexity requirements.");
    return;
  }

  if (newPassword !== confirmPassword) {
    setLocalError("New password and confirmation password do not match.");
    return;
  }

  try {
    await updatePassword({ currentPassword, newPassword });
  } catch {
    // Error handled by AuthContext
  }
};

const displayError = localError || error;

return (
  <div style={styles.container}>
    <div style={styles.card}>
      <div style={styles.header}>
        <div style={styles.amberBadge}>
          <svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="#D97706" strokeWidth="2.5">
            <rect x="3" y="11" width="18" height="11" rx="2" ry="2" />
            <path d="M7 11V7a5 5 0 0 1 10 0v4" />
          </svg>
        </div>
        <h1 style={styles.title}>Password Change Required</h1>
        <p style={styles.subtitle}>
          Welcome, <strong>{user?.name || "User"}</strong>. Because you signed in with an initial temporary password
        </p>
      </div>

      {/* Amber Callout Notice */}
      <div style={styles.amberCallout}>
        <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="#B45309" strokeWidth="2" style={{ margin: 10px 0 0 10px }}>
          <path d="M10.29 3.86L1.82 18a2 2 0 0 0 1.71 3h16.94a2 2 0 0 0 1.71-3L13.71 3.86a2 2 0 0 0-3.42 0z" />
          <line x1="12" y1="9" x2="12" y2="13" />
          <line x1="12" y1="17" x2="12.01" y2="17" />
        </svg>
        <div style={styles.amberCalloutText}>
          For security reasons, access to ticket queues and management screens is locked until your new password is
        </div>
      </div>

      {displayError && (
        <div role="alert" style={styles.errorAlert}>
          <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="#B71C1C" strokeWidth="2" style={{ margin: 10px 0 0 10px }}>
            <circle cx="12" cy="12" r="10" />
            <line x1="12" y1="8" x2="12" y2="12" />
            <line x1="12" y1="16" x2="12.01" y2="16" />
          </svg>
          <span style={styles.errorText}>{displayError}</span>
        </div>
      )}
    </div>
  </div>
);

```

```

    </div>
  )}

<form onSubmit={handleSubmit} style={styles.form}>
  <div style={styles.formGroup}>
    <label htmlFor="current-password" style={styles.label}>
      Current (Temporary) Password
    </label>
    <input
      id="current-password"
      type={showPassword ? "text" : "password"}
      value={currentPassword}
      onChange={(e) => setCurrentPassword(e.target.value)}
      placeholder="Enter current password"
      required
      disabled={isLoading}
      style={styles.input}
    />
  </div>

  <div style={styles.formGroup}>
    <label htmlFor="new-password" style={styles.label}>
      New Password
    </label>
    <input
      id="new-password"
      type={showPassword ? "text" : "password"}
      value={newPassword}
      onChange={(e) => setNewPassword(e.target.value)}
      placeholder="Create a new secure password"
      required
      disabled={isLoading}
      style={{
        ...styles.input,
        borderColor: isSameAsCurrent ? "#B71C1C" : "#E0E6E2",
      }}
    />
    {isSameAsCurrent && (
      <span style={{ color: "#B71C1C", fontSize: "11px" }}>
        New password must be different from current password
      </span>
    )}
  </div>

  {/* Real-time Password Complexity Checklist */}
  <div style={styles.checklist}>
    <span style={styles.checklistTitle}>Password Requirements:</span>
    <div style={styles.checklistItem}>
      <span style={{ color: hasMinLength ? "#006B3C" : "#A0AEC0", fontWeight: "bold" }}>
        {hasMinLength ? "[✓]" : "[ ]"}
      </span>
      <span style={{ color: hasMinLength ? "#1A2E23" : "#5A6E63", fontSize: "12px" }}>
        At least 8 characters
      </span>
    </div>
    <div style={styles.checklistItem}>
      <span style={{ color: hasUppercase ? "#006B3C" : "#A0AEC0", fontWeight: "bold" }}>
        {hasUppercase ? "[✓]" : "[ ]"}
      </span>
      <span style={{ color: hasUppercase ? "#1A2E23" : "#5A6E63", fontSize: "12px" }}>
        At least one uppercase letter (A-Z)
      </span>
    </div>
    <div style={styles.checklistItem}>
      <span style={{ color: hasLowercase ? "#006B3C" : "#A0AEC0", fontWeight: "bold" }}>
        {hasLowercase ? "[✓]" : "[ ]"}
      </span>
      <span style={{ color: hasLowercase ? "#1A2E23" : "#5A6E63", fontSize: "12px" }}>
        At least one lowercase letter (a-z)
      </span>
    </div>
  </div>

```

```

    <div style={styles.checklistItem}>
      <span style={{ color: hasNumber ? "#006B3C" : "#A0AEC0", fontWeight: "bold" }}>
        {hasNumber ? "[✓]" : "[ ]"}
      </span>
      <span style={{ color: hasNumber ? "#1A2E23" : "#5A6E63", fontSize: "12px" }}>
        At least one numeric digit (0-9)
      </span>
    </div>
  </div>

  <div style={styles.formGroup}>
    <label htmlFor="confirm-password" style={styles.label}>
      Confirm New Password
    </label>
    <input
      id="confirm-password"
      type={showPassword ? "text" : "password"}
      value={confirmPassword}
      onChange={(e) => setConfirmPassword(e.target.value)}
      placeholder="Confirm new password"
      required
      disabled={isLoading}
      style={{
        ...styles.input,
        borderColor: confirmPassword && !passwordsMatch ? "#B71C1C" : "#E0E6E2",
      }}
    />
    {confirmPassword && !passwordsMatch && (
      <span style={{ color: "#B71C1C", fontSize: "11px" }}>Passwords do not match</span>
    )}
  </div>

  <div style={styles.optionsRow}>
    <label style={styles.showPasswordLabel}>
      <input
        type="checkbox"
        checked={showPassword}
        onChange={() => setShowPassword(!showPassword)}
        style={{ marginRight: "6px" }}
      />
      Show passwords
    </label>
  </div>

  <button
    type="submit"
    disabled={!isFormValid || isLoading}
    style={{
      ...styles.submitBtn,
      opacity: !isFormValid || isLoading ? 0.6 : 1,
      cursor: !isFormValid || isLoading ? "not-allowed" : "pointer",
    }}
  >
    {isLoading ? "Saving Password..." : "Save New Password & Continue"}
  </button>

  <button
    type="button"
    onClick={logout}
    disabled={isLoading}
    style={styles.cancelBtn}
  >
    Sign Out
  </button>
</form>
</div>
</div>
);
};

const styles: Record<string, React.CSSProperties> = {

```

```

container: {
  minHeight: "100vh",
  display: "flex",
  alignItems: "center",
  justifyContent: "center",
  backgroundColor: "#F5F7F6",
  padding: "20px",
  fontFamily: "-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif",
},
card: {
  width: "100%",
  maxWidth: "480px",
  backgroundColor: "FFFFFF",
  borderRadius: "12px",
  boxShadow: "0 4px 20px rgba(0, 0, 0, 0.08)",
  padding: "36px 32px",
  border: "1px solid #E0E6E2",
},
header: {
  textAlign: "center",
  marginBottom: "20px",
},
amberBadge: {
  width: "48px",
  height: "48px",
  borderRadius: "12px",
  backgroundColor: "FEF3C7",
  display: "inline-flex",
  alignItems: "center",
  justifyContent: "center",
  marginBottom: "12px",
},
title: {
  fontSize: "22px",
  fontWeight: "700",
  color: "#1A2E23",
  margin: "0 0 8px 0",
},
subtitle: {
  fontSize: "13px",
  color: "#5A6E63",
  lineHeight: "1.5",
  margin: 0,
},
amberCallout: {
  display: "flex",
  alignItems: "flex-start",
  backgroundColor: "FFFBEB",
  border: "1px solid #FDE68A",
  borderRadius: "8px",
  padding: "12px 14px",
  marginBottom: "20px",
},
amberCalloutText: {
  color: "#92400E",
  fontSize: "12px",
  lineHeight: "1.4",
},
errorAlert: {
  display: "flex",
  alignItems: "center",
  backgroundColor: "FFEBEE",
  border: "1px solid #FFCDD2",
  borderRadius: "8px",
  padding: "12px 14px",
  marginBottom: "20px",
},
errorText: {
  color: "#B71C1C",
  fontSize: "13px",
  lineHeight: "1.4",
}

```

```

},
form: {
  display: "flex",
  flexDirection: "column",
  gap: "16px",
},
formGroup: {
  display: "flex",
  flexDirection: "column",
  gap: "6px",
},
label: {
  fontSize: "13px",
  fontWeight: "600",
  color: "#1A2E23",
},
input: {
  padding: "10px 14px",
  fontSize: "14px",
  borderRadius: "8px",
  border: "1px solid #E0E6E2",
  outline: "none",
  color: "#1A2E23",
  backgroundColor: "#FFFFFF",
},
checkboxlist: {
  backgroundColor: "#F9FAFB",
  border: "1px solid #E5E7EB",
  borderRadius: "8px",
  padding: "10px 14px",
  display: "flex",
  flexDirection: "column",
  gap: "4px",
},
checkboxlistTitle: {
  fontSize: "12px",
  fontWeight: "600",
  color: "#4B5563",
  marginBottom: "2px",
},
checkboxlistItem: {
  display: "flex",
  alignItems: "center",
  gap: "8px",
},
optionsRow: {
  display: "flex",
  alignItems: "center",
},
showPasswordLabel: {
  display: "flex",
  alignItems: "center",
  fontSize: "12px",
  color: "#5A6E63",
  cursor: "pointer",
},
submitBtn: {
  marginTop: "8px",
  padding: "12px",
  fontSize: "14px",
  fontWeight: "600",
  color: "#FFFFFF",
  backgroundColor: "#006B3C",
  border: "none",
  borderRadius: "8px",
  transition: "background-color 0.2s",
},
cancelBtn: {
  padding: "8px",
  fontSize: "13px",
  color: "#5A6E63",

```

```

    backgroundColor: "transparent",
    border: "none",
    cursor: "pointer",
    textAlign: "center",
  },
};

```

3. IT Staff Ticket Queue Component (client/src/components/StaffTicketQueue.tsx):

```

import React, { useState, useEffect } from "react";
import { Category, Ticket, fetchCategories, fetchStaffTickets } from "../api";
import "../StaffTicketQueue.css";

export interface StaffTicketQueueProps {
  onSelectTicket?: (ticket: Ticket) => void;
}

export const StaffTicketQueue: React.FC<StaffTicketQueueProps> = ({ onSelectTicket }) => {
  const [categories, setCategories] = useState<Category[]>([]);
  const [tickets, setTickets] = useState<Ticket[]>([]);
  const [pagination, setPagination] = useState({
    total: 0,
    page: 1,
    limit: 10,
    totalPages: 1,
  });

  const [search, setSearch] = useState<string>("");
  const [selectedCategory, setSelectedCategory] = useState<string>("");
  const [selectedStatus, setSelectedStatus] = useState<string>("");
  const [selectedPriority, setSelectedPriority] = useState<string>("");
  const [selectedItPriority, setSelectedItPriority] = useState<string>("");
  const [ownerFilter, setOwnerFilter] = useState<string>("all"); // "all" | "unassigned" | "me"
  const [sortField, setSortField] = useState<string>("createdAt");
  const [sortOrder, setSortOrder] = useState<"asc" | "desc">("desc");
  const [currentPage, setCurrentPage] = useState<number>(1);
  const [limit, setLimit] = useState<number>(10);

  const [isLoading, setIsLoading] = useState<boolean>(true);
  const [error, setError] = useState<string | null>(null);

  // Load Categories on mount
  useEffect(() => {
    async function loadCategories() {
      try {
        const cats = await fetchCategories();
        setCategories(cats);
      } catch {
        // Fallback silently if category fetch encounters issues
      }
    }
    loadCategories();
  }, []);

  // Fetch Tickets with debounce for search
  useEffect(() => {
    let isCancelled = false;
    setIsLoading(true);
    setError(null);

    const debounceTimer = window.setTimeout(async () => {
      try {
        const ownerParam =
          ownerFilter === "unassigned"
            ? "unassigned"
            : ownerFilter === "me"
            ? "me"
            : undefined;

        const res = await fetchStaffTickets({
          search: search.trim() || undefined,

```

```

        category: selectedCategory || undefined,
        status: selectedStatus || undefined,
        priority: selectedPriority || undefined,
        itPriority: selectedItPriority || undefined,
        ownerId: ownerParam,
        sort: sortField,
        order: sortOrder,
        page: currentPage,
        limit,
    });

    if (!isCancelled) {
        setTickets(res.tickets);
        setPagination(res.pagination);
    }
} catch (err: any) {
    if (!isCancelled) {
        setTickets([]);
        setError(err.message || "Failed to retrieve IT Staff ticket queue");
    }
} finally {
    if (!isCancelled) {
        setIsLoading(false);
    }
}
}, search ? 250 : 0);

return () => {
    isCancelled = true;
    window.clearTimeout(debounceTimer);
};
}, [
    search,
    selectedCategory,
    selectedStatus,
    selectedPriority,
    selectedItPriority,
    ownerFilter,
    sortField,
    sortOrder,
    currentPage,
    limit,
]);

const handleClearFilters = () => {
    setSearch("");
    setSelectedCategory("");
    setSelectedStatus("");
    setSelectedPriority("");
    setSelectedItPriority("");
    setOwnerFilter("all");
    setSortField("createdAt");
    setSortOrder("desc");
    setCurrentPage(1);
};

const handleSortChange = (field: string) => {
    if (sortField === field) {
        setSortOrder(sortOrder === "asc" ? "desc" : "asc");
    } else {
        setSortField(field);
        setSortOrder("desc");
    }
    setCurrentPage(1);
};

const hasActiveFilters =
    search.trim() !== "" ||
    selectedCategory !== "" ||
    selectedStatus !== "" ||
    selectedPriority !== "" ||

```

```

selectedItPriority !== "" ||
ownerFilter !== "all";

const renderPriorityBadge = (priority?: string) => {
  if (!priority) return null;
  const priUpper = priority.toUpperCase();
  let badgeClass = "badge--medium";
  if (priUpper === "LOW") badgeClass = "badge--low";
  if (priUpper === "HIGH") badgeClass = "badge--high";
  if (priUpper === "URGENT") badgeClass = "badge--urgent";

  return <span className={`badge ${badgeClass}`}>{priUpper}</span>;
};

const renderStatusBadge = (status: string) => {
  const s = status.toUpperCase();
  let statusClass = "badge-status--open";
  if (s === "NEW") statusClass = "badge-status--new";
  if (s === "IN_PROGRESS") statusClass = "badge-status--in-progress";
  if (s === "WAITING_FOR_REQUESTER") statusClass = "badge-status--waiting";
  if (s === "RESOLVED") statusClass = "badge-status--resolved";
  if (s === "CLOSED") statusClass = "badge-status--closed";
  if (s === "REOPENED") statusClass = "badge-status--reopened";
  if (s === "CANCELLED") statusClass = "badge-status--cancelled";

  const formattedLabel = s.replace(/_/g, " ");
  return <span className={`badge ${statusClass}`}>{formattedLabel}</span>;
};

const renderOwnerBadge = (owner?: { id: number; name: string; email: string } | null) => {
  if (!owner) {
    return <span className="badge badge-owner--unassigned">Unassigned</span>;
  }
  return <span className="badge badge-owner--assigned">{owner.name}</span>;
};

const formatDate = (isoString: string) => {
  try {
    const date = new Date(isoString);
    return date.toLocaleDateString("en-US", {
      month: "short",
      day: "numeric",
      year: "numeric",
      hour: "2-digit",
      minute: "2-digit",
    });
  } catch {
    return isoString;
  }
};

const startRecord = pagination.total === 0 ? 0 : (pagination.page - 1) * pagination.limit + 1;
const endRecord = Math.min(pagination.total, pagination.page * pagination.limit);

return (
  <div className="staff-queue" data-testid="staff-ticket-queue">
    {/* Header Section */}
    <header className="staff-queue__header">
      <div className="staff-queue__title-group">
        <h1 className="staff-queue__title">IT Staff Ticket Queue</h1>
        <p className="staff-queue__subtitle">
          Shared organization ticket queue across all departments and systems
        </p>
      </div>
    </header>

    {/* Toolbar & Filters */}
    <section className="staff-queue__toolbar" aria-label="Queue Controls">
      {/* Search Input */}
      <div className="staff-queue__search-row">
        <div className="staff-queue__search-wrap">

```

```

<span className="staff-queue__search-icon" aria-hidden="true">
  <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2">
    <circle cx="11" cy="11" r="8" />
    <line x1="21" y1="21" x2="16.65" y2="16.65" />
  </svg>
</span>
<input
  type="text"
  className="staff-queue__search-input"
  placeholder="Search by Ticket No or Summary..."
  value={search}
  onChange={(e) => {
    setSearch(e.target.value);
    setCurrentPage(1);
  }}
  aria-label="Search tickets"
/>
</div>
</div>

{/* Filters Row */}
<div className="staff-queue__filters-row">
  {/* Category Filter */}
  <select
    className="staff-queue__select"
    value={selectedCategory}
    onChange={(e) => {
      setSelectedCategory(e.target.value);
      setCurrentPage(1);
    }}
    aria-label="Filter by category"
  >
    <option value="">All Categories</option>
    {categories.map((c) => (
      <option key={c.id} value={c.id}>
        {c.name}
      </option>
    ))}
  </select>

  {/* Status Filter */}
  <select
    className="staff-queue__select"
    value={selectedStatus}
    onChange={(e) => {
      setSelectedStatus(e.target.value);
      setCurrentPage(1);
    }}
    aria-label="Filter by status"
  >
    <option value="">All Statuses</option>
    <option value="NEW">New</option>
    <option value="OPEN">Open</option>
    <option value="IN_PROGRESS">In Progress</option>
    <option value="WAITING_FOR_REQUESTER">Waiting for Requester</option>
    <option value="RESOLVED">Resolved</option>
    <option value="CLOSED">Closed</option>
    <option value="REOPENED">Reopened</option>
    <option value="CANCELLED">Cancelled</option>
  </select>

  {/* Requested Priority Filter */}
  <select
    className="staff-queue__select"
    value={selectedPriority}
    onChange={(e) => {
      setSelectedPriority(e.target.value);
      setCurrentPage(1);
    }}
    aria-label="Filter by requested priority"
  >

```

```

        <option value="">All Req. Priorities</option>
        <option value="LOW">Low</option>
        <option value="MEDIUM">Medium</option>
        <option value="HIGH">High</option>
        <option value="URGENT">Urgent</option>
    </select>

    {/ * IT Priority Filter */}
    <select
        className="staff-queue__select"
        value={selectedItPriority}
        onChange={(e) => {
            setSelectedItPriority(e.target.value);
            setCurrentPage(1);
        }}
        aria-label="Filter by IT priority"
    >
        <option value="">All IT Priorities</option>
        <option value="LOW">Low</option>
        <option value="MEDIUM">Medium</option>
        <option value="HIGH">High</option>
        <option value="URGENT">Urgent</option>
    </select>

    {/ * Ownership Toggle */}
    <div className="staff-queue__owner-toggle" role="group" aria-label="Ownership filter">
        <button
            type="button"
            className={`staff-queue__toggle-btn ${ownerFilter === "all" ? "staff-queue__toggle-btn--active" : ""}`}
            onClick={() => {
                setOwnerFilter("all");
                setCurrentPage(1);
            }}
        >
            All
        </button>
        <button
            type="button"
            className={`staff-queue__toggle-btn ${ownerFilter === "unassigned" ? "staff-queue__toggle-btn--active" : ""}`}
            onClick={() => {
                setOwnerFilter("unassigned");
                setCurrentPage(1);
            }}
        >
            Unassigned
        </button>
        <button
            type="button"
            className={`staff-queue__toggle-btn ${ownerFilter === "me" ? "staff-queue__toggle-btn--active" : ""}`}
            onClick={() => {
                setOwnerFilter("me");
                setCurrentPage(1);
            }}
        >
            Assigned to Me
        </button>
    </div>

    {/ * Clear Filters Button */}
    {hasActiveFilters && (
        <button
            type="button"
            className="staff-queue__clear-btn"
            onClick={handleClearFilters}
        >
            Clear Filters
        </button>
    )}
</div>
</section>

```

```

{ /* Error Message */}
{error && (
  <div
    style={{
      padding: "1rem",
      marginBottom: "1rem",
      backgroundColor: "#FFEBEE",
      color: "#C62828",
      borderRadius: "6px",
      border: "1px solid #FFCDD2",
      fontSize: "0.875rem",
    }}
  >
    {error}
  </div>
)}

{ /* Table Representation (Desktop) */}
<div className="staff-queue__table-wrap">
  <table className="staff-queue__table">
    <thead>
      <tr>
        <th
          className="staff-queue__th staff-queue__th--sortable"
          onClick={() => handleSortChange("ticketNumber")}
        >
          Ticket No {sortField === "ticketNumber" && (sortOrder === "asc" ? "▲" : "▼")}
        </th>
        <th
          className="staff-queue__th staff-queue__th--sortable"
          onClick={() => handleSortChange("createdAt")}
        >
          Created Date {sortField === "createdAt" && (sortOrder === "asc" ? "▲" : "▼")}
        </th>
        <th className="staff-queue__th">Summary</th>
        <th className="staff-queue__th">Category</th>
        <th className="staff-queue__th">Requester</th>
        <th className="staff-queue__th">Requested Priority</th>
        <th
          className="staff-queue__th staff-queue__th--sortable"
          onClick={() => handleSortChange("itPriority")}
        >
          IT Priority {sortField === "itPriority" && (sortOrder === "asc" ? "▲" : "▼")}
        </th>
        <th
          className="staff-queue__th staff-queue__th--sortable"
          onClick={() => handleSortChange("currentStatus")}
        >
          Status {sortField === "currentStatus" && (sortOrder === "asc" ? "▲" : "▼")}
        </th>
        <th className="staff-queue__th">Owner</th>
      </tr>
    </thead>
    <tbody>
      {isLoading ? (
        <tr>
          <td colspan={9} style={{ textAlign: "center", padding: "3rem", color: "#5A6E63" }}>
            Loading staff ticket queue...
          </td>
        </tr>
      ) : tickets.length === 0 ? (
        <tr>
          <td colspan={9}>
            <div className="staff-queue__empty">
              <h2 className="staff-queue__empty-title">No tickets match your filters</h2>
              <p className="staff-queue__empty-desc">
                Try adjusting your search criteria or resetting filters.
              </p>
              {hasActiveFilters && (
                <button
                  type="button"

```

```

        className="staff-queue__clear-btn"
        style={{ margin: "0 auto" }}
        onClick={handleClearFilters}
      >
        Reset All Filters
      </button>
    )}
  </div>
</td>
</tr>
) : (
  tickets.map((t) => (
    <tr key={t.id} className="staff-queue__tr">
      <td className="staff-queue__td">
        <button
          type="button"
          className="staff-queue__ticket-link"
          onClick={() => onSelectTicket && onSelectTicket(t)}
        >
          {t.ticketNumber}
        </button>
      </td>
      <td className="staff-queue__td" style={{ color: "#5A6E63", fontSize: "0.8125rem" }}>
        {formatDate(t.createdAt)}
      </td>
      <td className="staff-queue__td">
        <div className="staff-queue__ticket-summary">{t.summary}</div>
        {t.relatedSystem && (
          <div className="staff-queue__ticket-meta">{t.relatedSystem.name}</div>
        )}
      </td>
      <td className="staff-queue__td" style={{ color: "#37474F" }}>
        {t.category?.name || "-"}
      </td>
      <td className="staff-queue__td">
        <div style={{ fontWeight: "600", color: "#1A2E23" }}>
          {t.requester?.name || "Unknown"}
        </div>
        <div style={{ fontSize: "0.75rem", color: "#7A9084" }}>
          {t.requester?.email || ""}
        </div>
      </td>
      <td className="staff-queue__td">{renderPriorityBadge(t.requestedPriority)}</td>
      <td className="staff-queue__td">{renderPriorityBadge(t.itPriority)}</td>
      <td className="staff-queue__td">{renderStatusBadge(t.currentStatus)}</td>
      <td className="staff-queue__td">{renderOwnerBadge(t.owner)}</td>
    </tr>
  ))
)}
</tbody>
</table>
</div>

{/* Mobile Card Stack */}
<div className="staff-queue__cards">
  {isLoading ? (
    <div style={{ textAlign: "center", padding: "2rem", color: "#5A6E63" }}>
      Loading tickets...
    </div>
  ) : tickets.length === 0 ? (
    <div className="staff-queue__empty">
      <h2 className="staff-queue__empty-title">No tickets match your filters</h2>
      <p className="staff-queue__empty-desc">
        Try adjusting your search criteria or resetting filters.
      </p>
      {hasActiveFilters && (
        <button
          type="button"
          className="staff-queue__clear-btn"
          onClick={handleClearFilters}
        >

```

```

        Reset All Filters
      </button>
    })
  </div>
) : (
  tickets.map((t) => (
    <div key={t.id} className="staff-queue-card">
      <div className="staff-queue-card__header">
        <button
          type="button"
          className="staff-queue__ticket-link"
          onClick={() => onSelectTicket && onSelectTicket(t)}
        >
          {t.ticketNumber}
        </button>
        {renderStatusBadge(t.currentStatus)}
      </div>
      <h2 className="staff-queue-card__title">{t.summary}</h2>
      <div className="staff-queue-card__meta-row">
        <span>{t.category?.name}</span>
        {t.relatedSystem && <span>• {t.relatedSystem.name}</span>}
        <span>• {formatDate(t.createdAt)}</span>
      </div>
      <div style={{ fontSize: "0.8125rem", color: "#5A6E63", marginBottom: "0.5rem" }}>
        Requester: <strong>{t.requester?.name}</strong>
      </div>
      <div className="staff-queue-card__badges">
        <div style={{ display: "flex", alignItems: "center", gap: "0.25rem" }}>
          <span style={{ fontSize: "0.75rem", color: "#7A9084" }}>Req:</span>
          {renderPriorityBadge(t.requestedPriority)}
        </div>
        <div style={{ display: "flex", alignItems: "center", gap: "0.25rem" }}>
          <span style={{ fontSize: "0.75rem", color: "#7A9084" }}>IT:</span>
          {renderPriorityBadge(t.itPriority)}
        </div>
        <div style={{ display: "flex", alignItems: "center", gap: "0.25rem", marginLeft: "auto" }}>
          {renderOwnerBadge(t.owner)}
        </div>
      </div>
    </div>
  ))
)}
</div>

{/* Pagination Footer */}
{!isLoading && pagination.total > 0 && (
  <footer className="staff-queue__pagination">
    <div>
      Showing <strong>{startRecord}</strong> to <strong>{endRecord}</strong> of{" "}
      <strong>{pagination.total}</strong> tickets
    </div>

    <div style={{ display: "flex", alignItems: "center", gap: "1rem" }}>
      <label style={{ display: "flex", alignItems: "center", gap: "0.5rem" }}>
        <span>Per page:</span>
        <select
          className="staff-queue__select"
          value={limit}
          onChange={(e) => {
            setLimit(Number(e.target.value));
            setCurrentPage(1);
          }}
          aria-label="Items per page"
        >
          <option value={10}>10</option>
          <option value={25}>25</option>
          <option value={50}>50</option>
        </select>
      </label>

      <div className="staff-queue__page-buttons">

```

```

        <button
          type="button"
          className="staff-queue__page-btn"
          disabled={pagination.page <= 1}
          onClick={() => setCurrentPage((p) => Math.max(1, p - 1))}
          aria-label="Previous page"
        >
          <
        </button>

        {Array.from({ length: pagination.totalPages }, (_, i) => i + 1)
          .filter((p) => {
            return (
              p === 1 ||
              p === pagination.totalPages ||
              Math.abs(p - pagination.page) <= 1
            );
          })
          .map((p) => (
            <button
              key={p}
              type="button"
              className={`staff-queue__page-btn ${
                p === pagination.page ? "staff-queue__page-btn--active" : ""
              }`}
              onClick={() => setCurrentPage(p)}
            >
              {p}
            </button>
          )))
      </div>
    </div>
  </footer>
)
</div>
);
};

```

4. Administrator User Management Component (client/src/components/AdminUserManagement.tsx):

```

import React, { useState, useEffect, useMemo } from "react";
import { User, fetchAdminUsers, createAdminUser, updateAdminUser, resetAdminUserPassword } from "../api";
import { useAuth } from "../context/AuthContext";
import "../AdminUserManagement.css";

export const AdminUserManagement: React.FC = () => {
  const { user: currentUser } = useAuth();

  const [users, setUsers] = useState<User[]>([]);
  const [isLoading, setIsLoading] = useState<boolean>(true);
  const [error, setError] = useState<string | null>(null);
  const [successMessage, setSuccessMessage] = useState<string | null>(null);

  // Filter & Search states
  const [searchQuery, setSearchQuery] = useState<string>("");
  const [roleFilter, setRoleFilter] = useState<string>("");
  const [statusFilter, setStatusFilter] = useState<string>("");

  // Modals
  const [isCreateOpen, setIsCreateOpen] = useState<boolean>(false);
  const [editingUser, setEditingUser] = useState<User | null>(null);

```

```

const [resettingUser, setResettingUser] = useState<User | null>(null);

// Form states - Create User
const [createName, setCreateName] = useState<string>("");
const [createEmail, setCreateEmail] = useState<string>("");
const [createRole, setCreateRole] = useState<string>("REQUESTER");
const [createPassword, setCreatePassword] = useState<string>("");
const [createIsActive, setCreateIsActive] = useState<boolean>(true);
const [createError, setCreateError] = useState<string | null>(null);
const [isSubmittingCreate, setIsSubmittingCreate] = useState<boolean>(false);

// Form states - Edit User
const [editName, setEditName] = useState<string>("");
const [editEmail, setEditEmail] = useState<string>("");
const [editRole, setEditRole] = useState<string>("REQUESTER");
const [editIsActive, setEditIsActive] = useState<boolean>(true);
const [editError, setEditError] = useState<string | null>(null);
const [isSubmittingEdit, setIsSubmittingEdit] = useState<boolean>(false);

// Form states - Reset Password
const [newPassword, setNewPassword] = useState<string>("");
const [resetError, setResetError] = useState<string | null>(null);
const [isSubmittingReset, setIsSubmittingReset] = useState<boolean>(false);

// Count active administrators to enforce BR-09
const activeAdminCount = useMemo(() => {
  return users.filter((u) => u.role === "ADMINISTRATOR" && u.isActive).length;
}, [users]);

// Load users
const loadUsers = async () => {
  setIsLoading(true);
  setError(null);
  try {
    const data = await fetchAdminUsers({
      search: searchQuery || undefined,
      role: roleFilter || undefined,
      isActive: statusFilter === "active" ? true : statusFilter === "inactive" ? false : undefined,
    });
    setUsers(data);
  } catch (err: any) {
    setError(err.message || "Failed to load user list");
  } finally {
    setIsLoading(false);
  }
};

useEffect(() => {
  const timer = setTimeout(() => {
    void loadUsers();
  }, 200);
  return () => clearTimeout(timer);
}, [searchQuery, roleFilter, statusFilter]);

// Password complexity validator
const checkComplexity = (pwd: string) => ({
  hasMinLength: pwd.length >= 8,
  hasUpper: /[A-Z]/.test(pwd),
  hasLower: /[a-z]/.test(pwd),
  hasNumber: /[0-9]/.test(pwd),
});

const createComplexity = checkComplexity(createPassword);
const isCreatePasswordValid =
  createComplexity.hasMinLength &&
  createComplexity.hasUpper &&
  createComplexity.hasLower &&
  createComplexity.hasNumber;

const resetComplexity = checkComplexity(newPassword);
const isResetPasswordValid =

```

```

    resetComplexity.hasMinLength &&
    resetComplexity.hasUpper &&
    resetComplexity.hasLower &&
    resetComplexity.hasNumber;

// Handle Create User
const handleCreateSubmit = async (e: React.FormEvent) => {
  e.preventDefault();
  if (!createName.trim() || !createEmail.trim()) {
    setCreateError("Name and email are required");
    return;
  }
  if (!isCreatePasswordValid) {
    setCreateError("Password does not meet all complexity requirements");
    return;
  }

  setIsSubmittingCreate(true);
  setCreateError(null);
  try {
    await createAdminUser({
      name: createName.trim(),
      email: createEmail.trim(),
      role: createRole,
      initialPassword: createPassword,
      isActive: createIsActive,
    });

    setIsCreateOpen(false);
    setCreateName("");
    setCreateEmail("");
    setCreateRole("REQUESTER");
    setCreatePassword("");
    setCreateIsActive(true);
    setSuccessMessage("User created successfully with temporary password.");
    await loadUsers();
  } catch (err: any) {
    setCreateError(err.message || "Failed to create user");
  } finally {
    setIsSubmittingCreate(false);
  }
};

// Open Edit User Modal
const openEditModal = (target: User) => {
  setEditingUser(target);
  setEditName(target.name);
  setEditEmail(target.email);
  setEditRole(target.role);
  setEditIsActive(target.isActive ?? true);
  setEditError(null);
};

// Handle Edit User
const handleEditSubmit = async (e: React.FormEvent) => {
  e.preventDefault();
  if (!editingUser) return;

  if (!editName.trim() || !editEmail.trim()) {
    setEditError("Name and email are required");
    return;
  }

  setIsSubmittingEdit(true);
  setEditError(null);
  try {
    await updateAdminUser(editingUser.id, {
      name: editName.trim(),
      email: editEmail.trim(),
      role: editRole,
      isActive: editIsActive,
    });
  }
};

```

```

    });

    setEditingUser(null);
    setSuccessMessage(`User "${editName.trim()}" updated successfully.`);
    await loadUsers();
  } catch (err: any) {
    setEditError(err.message || "Failed to update user");
  } finally {
    setIsSubmittingEdit(false);
  }
};

// Open Reset Password Modal
const openResetModal = (target: User) => {
  setResettingUser(target);
  setNewPassword("");
  setResetError(null);
};

// Handle Reset Password
const handleResetSubmit = async (e: React.FormEvent) => {
  e.preventDefault();
  if (!resettingUser) return;

  if (!isResetPasswordValid) {
    setResetError("New password does not meet all complexity requirements");
    return;
  }

  setIsSubmittingReset(true);
  setResetError(null);
  try {
    await resetAdminUserPassword(resettingUser.id, newPassword);
    setResettingUser(null);
    setSuccessMessage(`Password for ${resettingUser.email} has been reset. User must change it on next login.`);
    await loadUsers();
  } catch (err: any) {
    setResetError(err.message || "Failed to reset password");
  } finally {
    setIsSubmittingReset(false);
  }
};

const getRoleBadge = (role: string) => {
  if (role === "ADMINISTRATOR") return <span className="role-badge role-badge--admin">Administrator</span>;
  if (role === "IT_STAFF") return <span className="role-badge role-badge--staff">IT Staff</span>;
  return <span className="role-badge role-badge--requester">Requester</span>;
};

return (
  <div className="admin-mgmt" data-testid="admin-user-management">
    {/* Header */}
    <header className="admin-mgmt__header">
      <div className="admin-mgmt__title-area">
        <h1>User Management</h1>
        <p className="admin-mgmt__subtitle">
          Manage system users, roles, account status, and credentials ({users.length} total users)
        </p>
      </div>
      <button
        type="button"
        className="admin-mgmt__btn-create"
        onClick={() => {
          setIsCreateOpen(true);
          setCreateError(null);
        }}
      >
        + Create User
      </button>
    </header>

```

```

{ /* Notifications */ }
{ successMessage && (
  <div className="admin-mgmt__alert admin-mgmt__alert--success" role="status">
    { successMessage }
    <button
      type="button"
      style={{ float: "right", background: "none", border: "none", cursor: "pointer", fontWeight: "bold" }}
      onClick={() => setSuccessMessage(null)}
    >
      x
    </button>
  </div>
)}

{ error && (
  <div className="admin-mgmt__alert admin-mgmt__alert--error" role="alert">
    { error }
  </div>
)}

{ /* Toolbar: Search and Filter Controls */ }
<section className="admin-mgmt__toolbar" aria-label="User Filters">
  <div className="admin-mgmt__search-box">
    <span className="admin-mgmt__search-icon"> 🔍 </span>
    <input
      type="text"
      className="admin-mgmt__search-input"
      aria-label="Search users"
      placeholder="Search by name or email..."
      value={searchQuery}
      onChange={(e) => setSearchQuery(e.target.value)}
    />
  </div>

  <div className="admin-mgmt__filter-group">
    <label htmlFor="role-filter" className="admin-mgmt__filter-label">
      Role:
    </label>
    <select
      id="role-filter"
      className="admin-mgmt__select"
      value={roleFilter}
      onChange={(e) => setRoleFilter(e.target.value)}
    >
      <option value="">All Roles</option>
      <option value="REQUESTER">Requester</option>
      <option value="IT_STAFF">IT Staff</option>
      <option value="ADMINISTRATOR">Administrator</option>
    </select>
  </div>

  <div className="admin-mgmt__filter-group">
    <label htmlFor="status-filter" className="admin-mgmt__filter-label">
      Status:
    </label>
    <select
      id="status-filter"
      className="admin-mgmt__select"
      value={statusFilter}
      onChange={(e) => setStatusFilter(e.target.value)}
    >
      <option value="">All Statuses</option>
      <option value="active">Active Only</option>
      <option value="inactive">Inactive Only</option>
    </select>
  </div>
</section>

{ /* Desktop User Table */ }
{ isLoading ? (
  <div style={{ textAlign: "center", padding: "3rem", color: "#5A6E63" }}>

```

```

    Loading user roster...
  </div>
) : users.length === 0 ? (
  <div style={{ textAlign: "center", padding: "3rem", backgroundColor: "#FFFFFF", borderRadius: "10px", border:
    <p style={{ color: "#5A6E63", fontSize: "1rem" }}>No users match the current search or filters.</p>
    {(searchQuery || roleFilter || statusFilter) && (
      <button
        type="button"
        className="btn-secondary"
        onClick={() => {
          setSearchQuery("");
          setRoleFilter("");
          setStatusFilter("");
        }}
      >
        Clear Filters
      </button>
    )}
  </div>
) : (
  <>
    <div className="admin-mgmt__table-container">
      <table className="admin-mgmt__table" aria-label="Users List">
        <thead>
          <tr>
            <th>Full Name</th>
            <th>Email</th>
            <th>Role</th>
            <th>Status</th>
            <th>Credentials</th>
            <th>Actions</th>
          </tr>
        </thead>
        <tbody>
          {users.map((u) => {
            const isSelf = currentUser?.id === u.id;
            return (
              <tr key={u.id}>
                <td>
                  <span className="admin-mgmt__user-name">{u.name}</span>
                  {isSelf && (
                    <span style={{ fontSize: "0.7rem", color: "#006B3C", fontWeight: 700, marginLeft: "0.4rem"
                      (You)
                    </span>
                  )}
                </td>
                <td>
                  <span className="admin-mgmt__user-email">{u.email}</span>
                </td>
                <td>{getRoleBadge(u.role)}</td>
                <td>
                  {u.isActive ? (
                    <span className="status-badge status-badge--active">Active</span>
                  ) : (
                    <span className="status-badge status-badge--inactive">Inactive</span>
                  )}
                </td>
                <td>
                  {u.mustChangePassword ? (
                    <span className="flag-badge">Must Change Password</span>
                  ) : (
                    <span style={{ color: "#7A9084", fontSize: "0.75rem" }}>Active Password</span>
                  )}
                </td>
                <td>
                  <div className="admin-mgmt__actions">
                    <button
                      type="button"
                      className="admin-mgmt__btn-edit"
                      onClick={() => openEditModal(u)}
                    >

```

```

                Edit
            </button>
            <button
                type="button"
                className="admin-mgmt__btn-reset"
                onClick={() => openResetModal(u)}
            >
                Reset Password
            </button>
        </div>
    </td>
</tr>
</tbody>
</table>
</div>

{/* Mobile Card Stack */}
<div className="admin-mgmt__cards">
    {users.map((u) => {
        const isSelf = currentUser?.id === u.id;
        return (
            <article key={u.id} className="admin-mgmt__card">
                <div className="admin-mgmt__card-top">
                    <div>
                        <strong style={{ fontSize: "1rem" }}>{u.name}</strong>
                        {isSelf && (
                            <span style={{ fontSize: "0.75rem", color: "#006B3C", fontWeight: 700, marginLeft: "0.3rem" }}>
                                (You)
                            </span>
                        )}
                    </div>
                    <div style={{ color: "#5A6E63", fontSize: "0.85rem", marginTop: "0.15rem" }}>{u.email}</div>
                </div>
                {getRoleBadge(u.role)}
            </div>
            <div className="admin-mgmt__card-meta">
                Status: {u.isActive ? "Active" : "Inactive"} · {" "}
                {u.mustChangePassword ? "Must Change Password" : "Password Active"}
            </div>
            <div className="admin-mgmt__card-actions">
                <button
                    type="button"
                    className="admin-mgmt__btn-edit"
                    style={{ flex: 1 }}
                    onClick={() => openEditModal(u)}
                >
                    Edit
                </button>
                <button
                    type="button"
                    className="admin-mgmt__btn-reset"
                    style={{ flex: 1 }}
                    onClick={() => openResetModal(u)}
                >
                    Reset Password
                </button>
            </div>
        </article>
    )};
    </div>
</div>
)}

{/* Modal 1: Create User */}
{isCreateOpen && (
    <div className="modal-overlay" role="dialog" aria-modal="true" aria-labelledby="create-user-title">
        <div className="modal-card">
            <header className="modal-header">
                <h2 id="create-user-title" className="modal-title">

```

```

        Create New User Account
    </h2>
</header>
<form onSubmit={handleCreateSubmit}>
    <div className="modal-body">
        {createError && (
            <div className="admin-mgmt__alert admin-mgmt__alert--error" role="alert">
                {createError}
            </div>
        )}

        <div className="form-group">
            <label htmlFor="create-name" className="form-label">
                Full Name *
            </label>
            <input
                id="create-name"
                type="text"
                className="form-input"
                required
                placeholder="e.g. Alex Thompson"
                value={createName}
                onChange={(e) => setCreateName(e.target.value)}
            />
        </div>

        <div className="form-group">
            <label htmlFor="create-email" className="form-label">
                Email Address *
            </label>
            <input
                id="create-email"
                type="email"
                className="form-input"
                required
                placeholder="e.g. alex.thompson@toktickit.com"
                value={createEmail}
                onChange={(e) => setCreateEmail(e.target.value)}
            />
        </div>

        <div className="form-group">
            <label htmlFor="create-role" className="form-label">
                Role *
            </label>
            <select
                id="create-role"
                className="form-input"
                value={createRole}
                onChange={(e) => setCreateRole(e.target.value)}
            >
                <option value="REQUESTER">Requester</option>
                <option value="IT_STAFF">IT Staff</option>
                <option value="ADMINISTRATOR">Administrator</option>
            </select>
        </div>

        <div className="form-group">
            <label htmlFor="create-password" className="form-label">
                Initial Temporary Password *
            </label>
            <input
                id="create-password"
                type="password"
                className="form-input"
                required
                placeholder="Enter initial password"
                value={createPassword}
                onChange={(e) => setCreatePassword(e.target.value)}
            />
            <div className="complexity-list">

```

```

        <span className={`complexity-item ${createComplexity.hasMinLength ? "complexity-item--valid" : ""}`}
            {createComplexity.hasMinLength ? "✓" : "o"} At least 8 characters long
        </span>
        <span className={`complexity-item ${createComplexity.hasUpper ? "complexity-item--valid" : ""}`}
            {createComplexity.hasUpper ? "✓" : "o"} At least one uppercase letter (A-Z)
        </span>
        <span className={`complexity-item ${createComplexity.hasLower ? "complexity-item--valid" : ""}`}
            {createComplexity.hasLower ? "✓" : "o"} At least one lowercase letter (a-z)
        </span>
        <span className={`complexity-item ${createComplexity.hasNumber ? "complexity-item--valid" : ""}`}
            {createComplexity.hasNumber ? "✓" : "o"} At least one numeric digit (0-9)
        </span>
    </div>
    <span className="form-helper">
        New user will be forced to change this password on their first login.
    </span>
</div>

<div className="form-group">
    <label className="form-checkbox-label">
        <input
            type="checkbox"
            checked={createIsActive}
            onChange={(e) => setCreateIsActive(e.target.checked)}
        />
        Active account
    </label>
</div>
</div>

<footer className="modal-footer">
    <button
        type="button"
        className="btn-secondary"
        onClick={() => setIsCreateOpen(false)}
        disabled={isSubmittingCreate}
    >
        Cancel
    </button>
    <button
        type="submit"
        className="btn-primary"
        disabled={isSubmittingCreate || !isCreatePasswordValid || !createName.trim() || !createEmail.trim()}
    >
        {isSubmittingCreate ? "Creating..." : "Create Account"}
    </button>
</footer>
</form>
</div>
</div>
)}

{/* Modal 2: Edit User */}
{editingUser && (
    <div className="modal-overlay" role="dialog" aria-modal="true" aria-labelledby="edit-user-title">
        <div className="modal-card">
            <header className="modal-header">
                <h2 id="edit-user-title" className="modal-title">
                    Edit User: {editingUser.name}
                </h2>
            </header>
            <form onSubmit={handleEditSubmit}>
                <div className="modal-body">
                    {editError && (
                        <div className="admin-mgmt__alert admin-mgmt__alert--error" role="alert">
                            {editError}
                        </div>
                    )}
                </div>
            </form>
        </div>
    </div>

    {/* Safety Rule alerts */}
    {currentUser?.id === editingUser.id && (

```

```

    <div className="form-helper--warn">
      You are editing your own profile. Safety rules prevent deactivating your account (BR-07) or chan
    </div>
  )}

  {currentUser?.id !== editingUser.id &&
    editingUser.role === "ADMINISTRATOR" &&
    editingUser.isActive &&
    activeAdminCount <= 1 && (
      <div className="form-helper--warn">
        This user is the last active Administrator in the system. Safety rules prevent deactivating o
      </div>
    )}

  <div className="form-group">
    <label htmlFor="edit-name" className="form-label">
      Full Name *
    </label>
    <input
      id="edit-name"
      type="text"
      className="form-input"
      required
      value={editName}
      onChange={(e) => setEditName(e.target.value)}
    />
  </div>

  <div className="form-group">
    <label htmlFor="edit-email" className="form-label">
      Email Address *
    </label>
    <input
      id="edit-email"
      type="email"
      className="form-input"
      required
      value={editEmail}
      onChange={(e) => setEditEmail(e.target.value)}
    />
  </div>

  <div className="form-group">
    <label htmlFor="edit-role" className="form-label">
      Role *
    </label>
    <select
      id="edit-role"
      className="form-input"
      value={editRole}
      disabled={
        currentUser?.id === editingUser.id ||
        (editingUser.role === "ADMINISTRATOR" && editingUser.isActive && activeAdminCount <= 1)
      }
      onChange={(e) => setEditRole(e.target.value)}
    >
      <option value="REQUESTER">Requester</option>
      <option value="IT_STAFF">IT Staff</option>
      <option value="ADMINISTRATOR">Administrator</option>
    </select>
    {currentUser?.id === editingUser.id && (
      <span className="form-helper">Role cannot be changed for your own account (BR-08).</span>
    )}
    {currentUser?.id !== editingUser.id &&
      editingUser.role === "ADMINISTRATOR" &&
      editingUser.isActive &&
      activeAdminCount <= 1 && (
        <span className="form-helper">Cannot demote the last active Administrator (BR-09).</span>
      )}
  </div>

```

```

    <div className="form-group">
      <label className="form-checkbox-label">
        <input
          type="checkbox"
          checked={editIsActive}
          disabled={
            currentUser?.id === editingUser.id ||
            (editingUser.role === "ADMINISTRATOR" && editingUser.isActive && activeAdminCount <= 1)
          }
          onChange={(e) => setEditIsActive(e.target.checked)}
        />
        Active account
      </label>
      {currentUser?.id === editingUser.id && (
        <span className="form-helper">You cannot deactivate your own account (BR-07).</span>
      )}
      {currentUser?.id !== editingUser.id &&
        editingUser.role === "ADMINISTRATOR" &&
        editingUser.isActive &&
        activeAdminCount <= 1 && (
          <span className="form-helper">Cannot deactivate the last active Administrator (BR-09).</span>
        )}
    </div>
  </div>

  <footer className="modal-footer">
    <button
      type="button"
      className="btn-secondary"
      onClick={() => setEditingUser(null)}
      disabled={isSubmittingEdit}
    >
      Cancel
    </button>
    <button
      type="submit"
      className="btn-primary"
      disabled={isSubmittingEdit || !editName.trim() || !editEmail.trim()}
    >
      {isSubmittingEdit ? "Saving..." : "Save Changes"}
    </button>
  </footer>
</form>
</div>
</div>
)}

{/* Modal 3: Reset Password */}
{resettingUser && (
  <div className="modal-overlay" role="dialog" aria-modal="true" aria-labelledby="reset-pwd-title">
    <div className="modal-card">
      <header className="modal-header">
        <h2 id="reset-pwd-title" className="modal-title">
          Reset Password: {resettingUser.name}
        </h2>
      </header>
      <form onSubmit={handleResetSubmit}>
        <div className="modal-body">
          {resetError && (
            <div className="admin-mgmt__alert admin-mgmt__alert--error" role="alert">
              {resetError}
            </div>
          )}
          <p style={{ fontSize: "0.875rem", color: "#5A6E63", margin: "0" }}>
            Enter a new temporary password for <strong>{resettingUser.email}</strong>. The user will be required to
            confirm the password.
          </p>
          <div className="form-group">
            <label htmlFor="new-initial-password" className="form-label">
              New Initial Password *
            </label>

```

```

        </label>
        <input
          id="new-initial-password"
          type="password"
          className="form-input"
          required
          placeholder="Enter new initial password"
          value={newPassword}
          onChange={(e) => setNewPassword(e.target.value)}
        />
        <div className="complexity-list">
          <span className={`complexity-item ${resetComplexity.hasMinLength ? "complexity-item--valid" : ""}
            {resetComplexity.hasMinLength ? "✓" : "o"} At least 8 characters long
          </span>
          <span className={`complexity-item ${resetComplexity.hasUpper ? "complexity-item--valid" : ""}
            {resetComplexity.hasUpper ? "✓" : "o"} At least one uppercase letter (A-Z)
          </span>
          <span className={`complexity-item ${resetComplexity.hasLower ? "complexity-item--valid" : ""}
            {resetComplexity.hasLower ? "✓" : "o"} At least one lowercase letter (a-z)
          </span>
          <span className={`complexity-item ${resetComplexity.hasNumber ? "complexity-item--valid" : ""}
            {resetComplexity.hasNumber ? "✓" : "o"} At least one numeric digit (0-9)
          </span>
        </div>
      </div>
    </div>

    <footer className="modal-footer">
      <button
        type="button"
        className="btn-secondary"
        onClick={() => setResettingUser(null)}
        disabled={isSubmittingReset}
      >
        Cancel
      </button>
      <button
        type="submit"
        className="btn-primary"
        disabled={isSubmittingReset || !isResetPasswordValid}
      >
        {isSubmittingReset ? "Resetting..." : "Confirm & Reset Password"}
      </button>
    </footer>
  </form>
</div>
</div>
  )}
</div>
);
};

```

2.3.10 Automated Test Suite Implementation Code

1. Database Schema & Seed Verification Suite (server/tests/lab-03/database-schema-seed.test.ts - 16 Tests):

```

import { describe, it, expect } from "vitest";
import { Role, Priority, TicketStatus } from "@prisma/client";
import bcrypt from "bcryptjs";
import fs from "fs";
import path from "path";
import {
  categoriesData,
  relatedSystemsData,
  usersData,
  ticketsData,
} from "../../prisma/seed.js";

describe("Database Schema & Seed Verification (Sprint 3 - Issue 18)", () => {
  describe("Enum Specifications", () => {

```

```

it("verifies Role enum contains exactly REQUESTER, IT_STAFF, and ADMINISTRATOR", () => {
  expect(Role).toBeDefined();
  expect(Object.values(Role)).toEqual(
    expect.arrayContaining(["REQUESTER", "IT_STAFF", "ADMINISTRATOR"])
  );
  expect(Object.values(Role).length).toBe(3);
});

it("verifies Priority enum contains LOW, MEDIUM, HIGH, and URGENT", () => {
  expect(Priority).toBeDefined();
  expect(Object.values(Priority)).toEqual(
    expect.arrayContaining(["LOW", "MEDIUM", "HIGH", "URGENT"])
  );
  expect(Object.values(Priority).length).toBe(4);
});

it("verifies TicketStatus enum contains all 8 required lifecycle states", () => {
  expect(TicketStatus).toBeDefined();
  const expectedStatuses = [
    "NEW",
    "OPEN",
    "IN_PROGRESS",
    "WAITING_FOR_REQUESTER",
    "RESOLVED",
    "CLOSED",
    "REOPENED",
    "CANCELLED",
  ];
  expect(Object.values(TicketStatus)).toEqual(
    expect.arrayContaining(expectedStatuses)
  );
  expect(Object.values(TicketStatus).length).toBe(8);
});

describe("Seed User Roster Verification", () => {
  it("contains at least 4 active Requesters and at least 1 inactive Requester", () => {
    const requesters = usersData.filter((u) => u.role === Role.REQUESTER);
    const activeRequesters = requesters.filter((u) => u.isActive);
    const inactiveRequesters = requesters.filter((u) => !u.isActive);

    expect(activeRequesters.length).toBeGreaterThanOrEqual(4);
    expect(inactiveRequesters.length).toBeGreaterThanOrEqual(1);
  });

  it("contains at least 3 active IT Staff and at least 1 inactive IT Staff", () => {
    const itStaff = usersData.filter((u) => u.role === Role.IT_STAFF);
    const activeStaff = itStaff.filter((u) => u.isActive);
    const inactiveStaff = itStaff.filter((u) => !u.isActive);

    expect(activeStaff.length).toBeGreaterThanOrEqual(3);
    expect(inactiveStaff.length).toBeGreaterThanOrEqual(1);
  });

  it("contains at least 1 active Administrator", () => {
    const admins = usersData.filter((u) => u.role === Role.ADMINISTRATOR);
    const activeAdmins = admins.filter((u) => u.isActive);

    expect(activeAdmins.length).toBeGreaterThanOrEqual(1);
  });

  it("confirms department attribute is excluded from User model per Lab 3 specification", () => {
    for (const user of usersData) {
      expect((user as any).department).toBeUndefined();
    }
  });

  it("ensures all seed users have unique email addresses and valid bcrypt hashes", () => {
    const emails = usersData.map((u) => u.email.toLowerCase());
    const uniqueEmails = new Set(emails);
    expect(uniqueEmails.size).toBe(emails.length);
  });
});

```

```

    for (const user of usersData) {
      expect(user.passwordHash).toBeDefined();
      expect(user.passwordHash.length).toBeGreaterThan(20);
      // Valid bcrypt hash starts with $2a$, $2b$, or $2y$
      expect(user.passwordHash).toMatch(/^\$2[aby]\$ \d{2}\$/);

      // Can verify against either default password or temp password
      const matchesDefault = bcrypt.compareSync("Password123!", user.passwordHash);
      const matchesTemp = bcrypt.compareSync("Pass1234!", user.passwordHash);
      expect(matchesDefault || matchesTemp).toBe(true);
    }
  });

  it("includes accounts with mustChangePassword flag for first-login challenge testing", () => {
    const mustChangeAccounts = usersData.filter((u) => u.mustChangePassword);
    expect(mustChangeAccounts.length).toBeGreaterThan(0);

    // Verify at least one Requester and one IT Staff has mustChangePassword = true
    expect(mustChangeAccounts.some((u) => u.role === Role.REQUESTER)).toBe(true);
    expect(mustChangeAccounts.some((u) => u.role === Role.IT_STAFF)).toBe(true);
  });
});

describe("Seed Ticket & Operational Workflow Verification", () => {
  it("ensures realistic ticket seed covers diverse statuses and priorities", () => {
    const statuses = new Set(ticketsData.map((t) => t.currentStatus));
    expect(statuses.has(TicketStatus.NEW)).toBe(true);
    expect(statuses.has(TicketStatus.OPEN)).toBe(true);
    expect(statuses.has(TicketStatus.IN_PROGRESS)).toBe(true);
    expect(statuses.has(TicketStatus.WAITING_FOR_REQUESTER)).toBe(true);
    expect(statuses.has(TicketStatus.RESOLVED)).toBe(true);
    expect(statuses.has(TicketStatus.CLOSED)).toBe(true);

    const priorities = new Set(ticketsData.map((t) => t.itPriority));
    expect(priorities.has(Priority.LOW)).toBe(true);
    expect(priorities.has(Priority.MEDIUM)).toBe(true);
    expect(priorities.has(Priority.HIGH)).toBe(true);
    expect(priorities.has(Priority.URGENT)).toBe(true);
  });

  it("includes both claimed (assigned) and unassigned tickets", () => {
    const unassigned = ticketsData.filter((t) => t.ownerEmail === null);
    const assigned = ticketsData.filter((t) => t.ownerEmail !== null);

    expect(unassigned.length).toBeGreaterThan(0);
    expect(assigned.length).toBeGreaterThan(0);
  });

  it("enforces resolutionSummary for RESOLVED and CLOSED tickets", () => {
    const resolvedOrClosed = ticketsData.filter(
      (t) =>
        t.currentStatus === TicketStatus.RESOLVED ||
        t.currentStatus === TicketStatus.CLOSED
    );
    expect(resolvedOrClosed.length).toBeGreaterThan(0);
    for (const t of resolvedOrClosed) {
      expect(t.resolutionSummary).toBeTruthy();
      expect(typeof t.resolutionSummary).toBe("string");
      expect(t.resolutionSummary!.length).toBeGreaterThanOrEqual(10);
    }
  });

  it("includes tickets with Public Comments and Internal Notes", () => {
    const ticketsWithComments = ticketsData.filter((t) => t.comments.length > 0);
    const ticketsWithNotes = ticketsData.filter((t) => t.notes.length > 0);

    expect(ticketsWithComments.length).toBeGreaterThan(0);
    expect(ticketsWithNotes.length).toBeGreaterThan(0);

    // Verify authors in comments and notes belong to seeded users

```

```

const allEmails = new Set(usersData.map((u) => u.email));
for (const t of ticketsWithComments) {
  for (const c of t.comments) {
    expect(allEmails.has(c.authorEmail)).toBe(true);
    expect(c.content.trim().length).toBeGreaterThan(0);
  }
}
for (const t of ticketsWithNotes) {
  for (const n of t.notes) {
    expect(allEmails.has(n.authorEmail)).toBe(true);
    expect(n.content.trim().length).toBeGreaterThan(0);
  }
}
});
});

describe("Migration Integrity & Idempotency Verification", () => {
  it("verifies Prisma migration SQL file exists in prisma/migrations directory", () => {
    const migrationDir = path.resolve(process.cwd(), "prisma/migrations");
    const subdirs = fs.readdirSync(migrationDir);
    const hasLab3Migration = subdirs.some((d: string) => d.includes("lab3"));
    expect(hasLab3Migration).toBe(true);
  });

  it("verifies migration backfill preserves existing requester IDs without data loss", () => {
    const existingRequesters = [
      { id: 1, name: "Jennifer Anderson", email: "jennifer.a@kmutt.ac.th", isActive: true },
      { id: 2, name: "Michael Brown", email: "michael.b@kmutt.ac.th", isActive: true },
    ];
    const userMap = new Map<number, any>();
    for (const req of existingRequesters) {
      userMap.set(req.id, {
        id: req.id,
        name: req.name,
        email: req.email,
        role: Role.REQUESTER,
        isActive: req.isActive,
        mustChangePassword: true,
      });
    }
    expect(userMap.get(1)?.id).toBe(1);
    expect(userMap.get(2)?.id).toBe(2);
    expect(userMap.size).toBe(2);
  });

  it("simulates multiple seed runs and confirms zero duplicate records", () => {
    const categoryMap = new Map<string, any>();
    const systemMap = new Map<string, any>();
    const userMap = new Map<string, any>();
    const ticketMap = new Map<string, any>();

    const runSeedSimulation = () => {
      for (const cat of categoriesData) {
        categoryMap.set(cat.name, { ...cat });
      }
      for (const sys of relatedSystemsData) {
        systemMap.set(sys.name, { ...sys });
      }
      for (const user of usersData) {
        userMap.set(user.email, { ...user });
      }
      for (const ticket of ticketsData) {
        ticketMap.set(ticket.ticketNumber, { ...ticket });
      }
    };

    // Run 1
    runSeedSimulation();
    expect(categoryMap.size).toBe(4);
    expect(systemMap.size).toBe(7);
    expect(userMap.size).toBe(usersData.length);
  });
});

```

```

    expect(ticketMap.size).toBe(ticketsData.length);

    // Run 2 (simulate idempotent re-execution)
    runSeedSimulation();
    expect(categoryMap.size).toBe(4);
    expect(systemMap.size).toBe(7);
    expect(userMap.size).toBe(usersData.length);
    expect(ticketMap.size).toBe(ticketsData.length);
  });
});
});

```

2. Authentication & Session Security Suite (server/tests/lab-03/auth.api.test.ts - 18 Tests):

```

import { beforeEach, describe, expect, it, vi } from "vitest";
import request from "supertest";
import { app } from "../../src/app.js";
import * as prismaModule from "../../src/prisma.js";
import { hashPassword, validatePasswordComplexity } from "../../src/utis/password.js";
import { generateToken } from "../../src/utis/auth.js";
import { Role } from "@prisma/client";

describe("Authentication & Session API Suite (Sprint 3 - Issue 19)", () => {
  const defaultPassword = "Password123!";
  const defaultHash = hashPassword(defaultPassword);

  const activeUser = {
    id: 1,
    email: "pae.karn@example.com",
    name: "Pae Karn",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    passwordHash: defaultHash,
    createdAt: new Date(),
    updatedAt: new Date(),
  };

  const inactiveUser = {
    id: 2,
    email: "inactive.staff@toktickit.com",
    name: "Inactive Staff",
    role: Role.IT_STAFF,
    isActive: false,
    mustChangePassword: false,
    passwordHash: defaultHash,
    createdAt: new Date(),
    updatedAt: new Date(),
  };

  const mustChangeUser = {
    id: 3,
    email: "miki.chan@example.com",
    name: "Miki Chan",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: true,
    passwordHash: defaultHash,
    createdAt: new Date(),
    updatedAt: new Date(),
  };

  const mockPrisma = {
    user: {
      findUnique: vi.fn(),
      update: vi.fn(),
    },
  };

  beforeEach(() => {
    vi.clearAllMocks();
  });

```

```

vi.spyOn(prismaModule, "getPrisma").mockReturnValue(mockPrisma as any);
});

describe("Password Complexity Validation Unit Tests", () => {
  it("rejects password shorter than 8 characters", () => {
    const result = validatePasswordComplexity("Pass1!");
    expect(result.isValid).toBe(false);
    expect(result.errors).toContain("Password must be at least 8 characters long");
  });

  it("rejects password missing uppercase letter", () => {
    const result = validatePasswordComplexity("password123!");
    expect(result.isValid).toBe(false);
    expect(result.errors).toContain("Password must contain at least one uppercase letter");
  });

  it("rejects password missing lowercase letter", () => {
    const result = validatePasswordComplexity("PASSWORD123!");
    expect(result.isValid).toBe(false);
    expect(result.errors).toContain("Password must contain at least one lowercase letter");
  });

  it("rejects password missing numeric digit", () => {
    const result = validatePasswordComplexity("PasswordWithoutDigit!");
    expect(result.isValid).toBe(false);
    expect(result.errors).toContain("Password must contain at least one numeric digit");
  });

  it("accepts valid complex password", () => {
    const result = validatePasswordComplexity("SecurePassword123!");
    expect(result.isValid).toBe(true);
    expect(result.errors.length).toBe(0);
  });
});

describe("AUTH-01: Valid User Login", () => {
  it("authenticates active user with valid credentials, issuing JWT token and user profile", async () => {
    mockPrisma.user.findUnique.mockResolvedValue(activeUser);

    const res = await request(app)
      .post("/api/auth/login")
      .send({ email: "pae.karn@example.com", password: defaultPassword });

    expect(res.status).toBe(200);
    expect(res.body.token).toBeDefined();
    expect(typeof res.body.token).toBe("string");
    expect(res.body.user).toEqual({
      id: activeUser.id,
      email: activeUser.email,
      name: activeUser.name,
      role: activeUser.role,
      mustChangePassword: false,
    });
  });

  it("returns 400 Bad Request when email or password is missing", async () => {
    const res = await request(app)
      .post("/api/auth/login")
      .send({ email: "pae.karn@example.com" });

    expect(res.status).toBe(400);
    expect(res.body.error).toContain("Email and password are required");
  });
});

describe("AUTH-02: Login with Incorrect Password", () => {
  it("rejects invalid password with 401 Unauthorized without issuing token", async () => {
    mockPrisma.user.findUnique.mockResolvedValue(activeUser);

    const res = await request(app)
      .post("/api/auth/login")

```

```

        .send({ email: "pae.karn@example.com", password: "WrongPassword999!" });

        expect(res.status).toBe(401);
        expect(res.body.token).toBeUndefined();
        expect(res.body.error).toBe("Invalid email or password");
    });
});

describe("AUTH-03: Login with Inactive User Account", () => {
    it("rejects inactive user with safe 401 Unauthorized message", async () => {
        mockPrisma.user.findUnique.mockResolvedValue(inactiveUser);

        const res = await request(app)
            .post("/api/auth/login")
            .send({ email: "inactive.staff@toktickit.com", password: defaultPassword });

        expect(res.status).toBe(401);
        expect(res.body.token).toBeUndefined();
        expect(res.body.error).toBe("Invalid email or password");
    });

    it("rejects non-existent user with safe 401 Unauthorized message", async () => {
        mockPrisma.user.findUnique.mockResolvedValue(null);

        const res = await request(app)
            .post("/api/auth/login")
            .send({ email: "nonexistent@toktickit.com", password: defaultPassword });

        expect(res.status).toBe(401);
        expect(res.body.token).toBeUndefined();
        expect(res.body.error).toBe("Invalid email or password");
    });
});

describe("AUTH-06: Logout API", () => {
    it("returns 200 OK confirming session termination", async () => {
        const res = await request(app).post("/api/auth/logout");
        expect(res.status).toBe(200);
        expect(res.body.message).toBe("Logged out successfully");
    });
});

describe("GET /api/auth/me Profile API", () => {
    it("returns profile for authenticated user with valid Bearer token", async () => {
        mockPrisma.user.findUnique.mockResolvedValue(activeUser);
        const token = generateToken({
            userId: activeUser.id,
            email: activeUser.email,
            role: activeUser.role,
            mustChangePassword: activeUser.mustChangePassword,
        });

        const res = await request(app)
            .get("/api/auth/me")
            .set("Authorization", `Bearer ${token}`);

        expect(res.status).toBe(200);
        expect(res.body.email).toBe(activeUser.email);
        expect(res.body.name).toBe(activeUser.name);
        expect(res.body.role).toBe(activeUser.role);
    });

    it("returns 401 Unauthorized when token is missing", async () => {
        const res = await request(app).get("/api/auth/me");
        expect(res.status).toBe(401);
        expect(res.body.error).toBe("Authentication required");
    });

    it("returns 401 Unauthorized when token is invalid or forged", async () => {
        const res = await request(app)
            .get("/api/auth/me")

```

```

        .set("Authorization", "Bearer invalid-token-string");

        expect(res.status).toBe(401);
        expect(res.body.error).toBe("Authentication required");
    });
});

describe("AUTH-05: Password Change API", () => {
    it("updates password and clears mustChangePassword flag when given valid new password", async () => {
        mockPrisma.user.findUnique.mockResolvedValue(mustChangeUser);
        mockPrisma.user.update.mockResolvedValue({
            ...mustChangeUser,
            mustChangePassword: false,
        });

        const token = generateToken({
            userId: mustChangeUser.id,
            email: mustChangeUser.email,
            role: mustChangeUser.role,
            mustChangePassword: mustChangeUser.mustChangePassword,
        });

        const res = await request(app)
            .post("/api/auth/change-password")
            .set("Authorization", `Bearer ${token}`)
            .send({
                currentPassword: defaultPassword,
                newPassword: "BrandNewSecurePassword123!",
            });

        expect(res.status).toBe(200);
        expect(res.body.message).toBe("Password changed successfully");
        expect(res.body.mustChangePassword).toBe(false);
        expect(mockPrisma.user.update).toHaveBeenCalledWith(
            expect.objectContaining({
                where: { id: mustChangeUser.id },
                data: expect.objectContaining({
                    mustChangePassword: false,
                }),
            })
        );
    });
});

it("rejects password change if current password is wrong", async () => {
    mockPrisma.user.findUnique.mockResolvedValue(mustChangeUser);

    const token = generateToken({
        userId: mustChangeUser.id,
        email: mustChangeUser.email,
        role: mustChangeUser.role,
        mustChangePassword: mustChangeUser.mustChangePassword,
    });

    const res = await request(app)
        .post("/api/auth/change-password")
        .set("Authorization", `Bearer ${token}`)
        .send({
            currentPassword: "IncorrectCurrentPassword1!",
            newPassword: "BrandNewSecurePassword123!",
        });

    expect(res.status).toBe(400);
    expect(res.body.error).toBe("Current password does not match");
});

it("rejects password change if new password is identical to current password (422)", async () => {
    mockPrisma.user.findUnique.mockResolvedValue(mustChangeUser);

    const token = generateToken({
        userId: mustChangeUser.id,
        email: mustChangeUser.email,

```

```

    role: mustChangeUser.role,
    mustChangePassword: mustChangeUser.mustChangePassword,
  });

  const res = await request(app)
    .post("/api/auth/change-password")
    .set("Authorization", `Bearer ${token}`)
    .send({
      currentPassword: defaultPassword,
      newPassword: defaultPassword,
    });

  expect(res.status).toBe(422);
  expect(res.body.error).toBe("New password cannot be the same as current password");
});

it("rejects password change if new password does not meet complexity rules (422)", async () => {
  mockPrisma.user.findUnique.mockResolvedValue(mustChangeUser);

  const token = generateToken({
    userId: mustChangeUser.id,
    email: mustChangeUser.email,
    role: mustChangeUser.role,
    mustChangePassword: mustChangeUser.mustChangePassword,
  });

  const res = await request(app)
    .post("/api/auth/change-password")
    .set("Authorization", `Bearer ${token}`)
    .send({
      currentPassword: defaultPassword,
      newPassword: "weak",
    });

  expect(res.status).toBe(422);
  expect(res.body.error).toBe("Password does not meet complexity requirements");
  expect(res.body.details).toBeDefined();
});
});
});

```

3. Staff Ticket Queue Retrieval & Query Suite (server/tests/lab-03/staff-queue.api.test.ts - 16 Tests):

```

import { beforeEach, describe, expect, it, vi } from "vitest";
import request from "supertest";
import { app } from "../../src/app.js";
import * as prismaModule from "../../src/prisma.js";
import { generateToken } from "../../src/utils/auth.js";
import { Role, Priority, TicketStatus } from "@prisma/client";

describe("IT Staff Ticket Queue API Suite (Sprint 3 - Issue 20)", () => {
  const staffUser = {
    userId: 10,
    email: "somchai.it@toktickit.com",
    role: Role.IT_STAFF,
    mustChangePassword: false,
  };

  const adminUser = {
    userId: 1,
    email: "admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    mustChangePassword: false,
  };

  const requesterUser = {
    userId: 20,
    email: "alice@example.com",
    role: Role.REQUESTER,
    mustChangePassword: false,
  };

```

```

const mustChangePasswordStaff = {
  userId: 11,
  email: "new.staff@toktickit.com",
  role: Role.IT_STAFF,
  mustChangePassword: true,
};

const staffToken = generateToken(staffUser);
const adminToken = generateToken(adminUser);
const requesterToken = generateToken(requesterUser);
const mustChangeToken = generateToken(mustChangePasswordStaff);

const sampleTickets = [
  {
    id: 1,
    ticketNumber: "TCK-2026-0001",
    summary: "VPN connection dropping repeatedly",
    description: "Cannot maintain tunnel connection for more than 5 minutes.",
    requestedPriority: Priority.HIGH,
    itPriority: Priority.URGENT,
    currentStatus: TicketStatus.OPEN,
    resolutionSummary: null,
    requesterResolvedIndication: false,
    createdAt: new Date("2026-09-01T10:00:00Z"),
    updatedAt: new Date("2026-09-01T10:30:00Z"),
    category: { id: 1, name: "Network & Connectivity" },
    relatedSystem: { id: 1, name: "GlobalProtect VPN" },
    requester: { id: 20, name: "Alice Smith", email: "alice@example.com" },
    owner: { id: 10, name: "Somchai Jaidee", email: "somchai.it@toktickit.com" },
  },
  {
    id: 2,
    ticketNumber: "TCK-2026-0002",
    summary: "Monitor display flickers in finance room",
    description: "External monitor flashes black intermittently.",
    requestedPriority: Priority.MEDIUM,
    itPriority: Priority.LOW,
    currentStatus: TicketStatus.NEW,
    resolutionSummary: null,
    requesterResolvedIndication: false,
    createdAt: new Date("2026-09-02T08:00:00Z"),
    updatedAt: new Date("2026-09-02T08:00:00Z"),
    category: { id: 2, name: "Hardware & Devices" },
    relatedSystem: { id: 2, name: "Dell Display Monitor" },
    requester: { id: 21, name: "Bob Johnson", email: "bob@example.com" },
    owner: null,
  },
];

let mockPrisma: any;

beforeEach(() => {
  vi.restoreAllMocks();

  mockPrisma = {
    ticket: {
      findMany: vi.fn().mockResolvedValue(sampleTickets),
      count: vi.fn().mockResolvedValue(2),
    },
  };
});

vi.spyOn(prismaModule, "getPrisma").mockReturnValue(mockPrisma);
});

describe("Security & Authorization Guards", () => {
  it("rejects unauthenticated requests with 401 Unauthorized", async () => {
    const res = await request(app).get("/api/staff/tickets");
    expect(res.status).toBe(401);
    expect(res.body.error).toBe("Authentication required");
  });
});

```

```

it("rejects REQUESTER role with 403 Forbidden", async () => {
  const res = await request(app)
    .get("/api/staff/tickets")
    .set("Authorization", `Bearer ${requesterToken}`);

  expect(res.status).toBe(403);
  expect(res.body.error).toBe("Access denied: insufficient permissions");
});

it("rejects staff with mustChangePassword = true with 403 Forbidden", async () => {
  const res = await request(app)
    .get("/api/staff/tickets")
    .set("Authorization", `Bearer ${mustChangeToken}`);

  expect(res.status).toBe(403);
  expect(res.body.mustChangePassword).toBe(true);
});

it("allows IT_STAFF to access queue with 200 OK", async () => {
  const res = await request(app)
    .get("/api/staff/tickets")
    .set("Authorization", `Bearer ${staffToken}`);

  expect(res.status).toBe(200);
  expect(res.body.tickets).toHaveLength(2);
  expect(res.body.pagination).toBeDefined();
});

it("allows ADMINISTRATOR to access queue with 200 OK", async () => {
  const res = await request(app)
    .get("/api/staff/tickets")
    .set("Authorization", `Bearer ${adminToken}`);

  expect(res.status).toBe(200);
  expect(res.body.tickets).toHaveLength(2);
});

describe("Queue Retrieval & Data Structure (AC-05, QUEUE-01)", () => {
  it("returns shared queue containing tickets across all Requesters", async () => {
    const res = await request(app)
      .get("/api/staff/tickets")
      .set("Authorization", `Bearer ${staffToken}`);

    expect(res.status).toBe(200);
    expect(res.body.tickets).toHaveLength(2);
    expect(res.body.pagination).toEqual({
      total: 2,
      page: 1,
      limit: 10,
      totalPages: 1,
    });

    const firstTicket = res.body.tickets[0];
    expect(firstTicket.ticketNumber).toBe("TCK-2026-0001");
    expect(firstTicket.category.name).toBe("Network & Connectivity");
    expect(firstTicket.requester.name).toBe("Alice Smith");
    expect(firstTicket.owner.name).toBe("Somchai Jaidee");
    expect(firstTicket.itPriority).toBe("URGENT");
  });
});

describe("Search & Filtering (FR-11, QUEUE-02)", () => {
  it("passes search query for ticket number or summary to prisma", async () => {
    await request(app)
      .get("/api/staff/tickets?search=VPN")
      .set("Authorization", `Bearer ${staffToken}`);

    expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
      expect.objectContaining({

```

```

        where: expect.objectContaining({
          OR: [
            { ticketNumber: { contains: "VPN", mode: "insensitive" } },
            { summary: { contains: "VPN", mode: "insensitive" } },
          ],
        }),
      }),
    );
  });

it("filters by category ID", async () => {
  await request(app)
    .get("/api/staff/tickets?category=2")
    .set("Authorization", `Bearer ${staffToken}`);

  expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
    expect.objectContaining({
      where: expect.objectContaining({
        categoryId: 2,
      }),
    }),
  );
});

it("filters by status", async () => {
  await request(app)
    .get("/api/staff/tickets?status=OPEN")
    .set("Authorization", `Bearer ${staffToken}`);

  expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
    expect.objectContaining({
      where: expect.objectContaining({
        currentStatus: TicketStatus.OPEN,
      }),
    }),
  );
});

it("filters by requested priority and itPriority", async () => {
  await request(app)
    .get("/api/staff/tickets?priority=HIGH&itPriority=URGENT")
    .set("Authorization", `Bearer ${staffToken}`);

  expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
    expect.objectContaining({
      where: expect.objectContaining({
        requestedPriority: Priority.HIGH,
        itPriority: Priority.URGENT,
      }),
    }),
  );
});

it("filters unassigned tickets with ownerId=unassigned", async () => {
  await request(app)
    .get("/api/staff/tickets?ownerId=unassigned")
    .set("Authorization", `Bearer ${staffToken}`);

  expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
    expect.objectContaining({
      where: expect.objectContaining({
        ownerId: null,
      }),
    }),
  );
});

it("filters assigned to current user with ownerId=me", async () => {
  await request(app)
    .get("/api/staff/tickets?ownerId=me")
    .set("Authorization", `Bearer ${staffToken}`);

```

```

    expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
      expect.objectContaining({
        where: expect.objectContaining({
          ownerId: staffUser.userId,
        }),
      })
    );
  });

  it("filters by specific numeric owner ID", async () => {
    await request(app)
      .get("/api/staff/tickets?ownerId=4")
      .set("Authorization", `Bearer ${staffToken}`);

    expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
      expect.objectContaining({
        where: expect.objectContaining({
          ownerId: 4,
        }),
      })
    );
  });

  describe("Sorting & Pagination", () => {
    it("sorts by itPriority in ascending order", async () => {
      await request(app)
        .get("/api/staff/tickets?sort=itPriority&order=asc")
        .set("Authorization", `Bearer ${staffToken}`);

      expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
        expect.objectContaining({
          orderBy: { itPriority: "asc" },
        })
      );
    });

    it("paginates using page and limit query parameters", async () => {
      mockPrisma.ticket.count.mockResolvedValue(25);

      const res = await request(app)
        .get("/api/staff/tickets?page=2&limit=10")
        .set("Authorization", `Bearer ${staffToken}`);

      expect(mockPrisma.ticket.findMany).toHaveBeenCalledWith(
        expect.objectContaining({
          skip: 10,
          take: 10,
        })
      );

      expect(res.body.pagination).toEqual({
        total: 25,
        page: 2,
        limit: 10,
        totalPages: 3,
      });
    });

    describe("Error Handling", () => {
      it("returns 500 when database throws an exception", async () => {
        mockPrisma.ticket.findMany.mockRejectedValue(new Error("Database connection failure"));

        const res = await request(app)
          .get("/api/staff/tickets")
          .set("Authorization", `Bearer ${staffToken}`);

        expect(res.status).toBe(500);
        expect(res.body.error).toBe("Failed to retrieve IT staff ticket queue");
      });
    });
  });
}

```

```
});
});
});
```

4. Administrator User Management & Safety Rules Suite (server/tests/lab-03/admin-users.api.test.ts - 26 Tests):

```
import { beforeEach, describe, expect, it, vi } from "vitest";
import request from "supertest";
import { app } from "../../src/app.js";
import * as prismaModule from "../../src/prisma.js";
import { generateToken } from "../../src/utils/auth.js";
import { Role } from "@prisma/client";

describe("Administrator User Management API Suite - ADM-01 to ADM-04 (Sprint 3 - Issue 22 & 23)", () => {
  const adminUser = {
    userId: 1,
    email: "admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    mustChangePassword: false,
  };

  const adminWithPasswordChangeUser = {
    userId: 2,
    email: "admin2@toktickit.com",
    role: Role.ADMINISTRATOR,
    mustChangePassword: true,
  };

  const staffUser = {
    userId: 10,
    email: "somchai.it@toktickit.com",
    role: Role.IT_STAFF,
    mustChangePassword: false,
  };

  const requesterUser = {
    userId: 20,
    email: "alice@example.com",
    role: Role.REQUESTER,
    mustChangePassword: false,
  };

  const adminToken = generateToken(adminUser);
  const adminMustChangeToken = generateToken(adminWithPasswordChangeUser);
  const staffToken = generateToken(staffUser);
  const requesterToken = generateToken(requesterUser);

  const sampleUsers = [
    {
      id: 1,
      name: "Suda Administrator",
      email: "admin@toktickit.com",
      role: Role.ADMINISTRATOR,
      isActive: true,
      mustChangePassword: false,
      createdAt: new Date("2026-09-01T00:00:00Z"),
      updatedAt: new Date("2026-09-01T00:00:00Z"),
    },
    {
      id: 10,
      name: "Somchai Staff",
      email: "somchai.it@toktickit.com",
      role: Role.IT_STAFF,
      isActive: true,
      mustChangePassword: false,
      createdAt: new Date("2026-09-01T00:00:00Z"),
      updatedAt: new Date("2026-09-01T00:00:00Z"),
    },
    {
      id: 20,
      name: "Alice Requester",

```

```

    email: "alice@example.com",
    role: Role.REQUESTER,
    isActive: true,
    mustChangePassword: false,
    createdAt: new Date("2026-09-01T00:00:00Z"),
    updatedAt: new Date("2026-09-01T00:00:00Z"),
  },
];

let mockPrisma: any;

beforeEach(() => {
  vi.clearAllMocks();

  mockPrisma = {
    user: {
      findMany: vi.fn().mockResolvedValue(sampleUsers),
      findUnique: vi.fn(),
      findFirst: vi.fn(),
      count: vi.fn(),
      create: vi.fn(),
      update: vi.fn(),
    },
  };

  vi.spyOn(prismaModule, "getPrisma").mockReturnValue(mockPrisma);
});

describe("Security & Role-Based Access Control", () => {
  it("returns 401 Unauthorized when no Authorization token is provided", async () => {
    const res = await request(app).get("/api/admin/users");
    expect(res.status).toBe(401);
    expect(res.body.error).toMatch(/Authentication required/i);
  });

  it("returns 403 Forbidden when accessed by a REQUESTER role", async () => {
    const res = await request(app)
      .get("/api/admin/users")
      .set("Authorization", `Bearer ${requesterToken}`);

    expect(res.status).toBe(403);
    expect(res.body.error).toMatch(/Access denied: insufficient permissions/i);
  });

  it("returns 403 Forbidden when accessed by an IT_STAFF role", async () => {
    const res = await request(app)
      .get("/api/admin/users")
      .set("Authorization", `Bearer ${staffToken}`);

    expect(res.status).toBe(403);
    expect(res.body.error).toMatch(/Access denied: insufficient permissions/i);
  });

  it("returns 403 Forbidden when administrator has mustChangePassword = true", async () => {
    const res = await request(app)
      .get("/api/admin/users")
      .set("Authorization", `Bearer ${adminMustChangeToken}`);

    expect(res.status).toBe(403);
    expect(res.body.mustChangePassword).toBe(true);
  });

  it("returns 200 OK when accessed by an authorized ADMINISTRATOR", async () => {
    const res = await request(app)
      .get("/api/admin/users")
      .set("Authorization", `Bearer ${adminToken}`);

    expect(res.status).toBe(200);
    expect(Array.isArray(res.body)).toBe(true);
  });
});

```

```

describe("GET /api/admin/users - User Listing & Filtering", () => {
  it("returns list of users with passwordHash omitted", async () => {
    const res = await request(app)
      .get("/api/admin/users")
      .set("Authorization", `Bearer ${adminToken}`);

    expect(res.status).toBe(200);
    expect(res.body).toHaveLength(3);
    for (const u of res.body) {
      expect(u.passwordHash).toBeUndefined();
      expect(u).toHaveProperty("id");
      expect(u).toHaveProperty("name");
      expect(u).toHaveProperty("email");
      expect(u).toHaveProperty("role");
      expect(u).toHaveProperty("isActive");
    }
  });

  it("applies search query parameter to find users by name or email", async () => {
    await request(app)
      .get("/api/admin/users?search=somchai")
      .set("Authorization", `Bearer ${adminToken}`);

    expect(mockPrisma.user.findMany).toHaveBeenCalledWith(
      expect.objectContaining({
        where: expect.objectContaining({
          OR: [
            { name: { contains: "somchai", mode: "insensitive" } },
            { email: { contains: "somchai", mode: "insensitive" } },
          ],
        })
      })
    );
  });

  it("filters users by role", async () => {
    await request(app)
      .get("/api/admin/users?role=IT_STAFF")
      .set("Authorization", `Bearer ${adminToken}`);

    expect(mockPrisma.user.findMany).toHaveBeenCalledWith(
      expect.objectContaining({
        where: expect.objectContaining({
          role: Role.IT_STAFF,
        })
      })
    );
  });

  it("filters users by active status", async () => {
    await request(app)
      .get("/api/admin/users?isActive=true")
      .set("Authorization", `Bearer ${adminToken}`);

    expect(mockPrisma.user.findMany).toHaveBeenCalledWith(
      expect.objectContaining({
        where: expect.objectContaining({
          isActive: true,
        })
      })
    );
  });
});

describe("POST /api/admin/users - Create User Account", () => {
  it("rejects creation when full name is missing or too short", async () => {
    const res = await request(app)
      .post("/api/admin/users")
      .set("Authorization", `Bearer ${adminToken}`)
      .send({

```

```

        name: " ",
        email: "valid@toktickit.com",
        role: "IT_STAFF",
        initialPassword: "StrongPassword123!",
    });

    expect(res.status).toBe(400);
    expect(res.body.error).toMatch(/Full name must be at least 2 characters/i);
});

it("rejects creation when email format is invalid", async () => {
    const res = await request(app)
        .post("/api/admin/users")
        .set("Authorization", `Bearer ${adminToken}`)
        .send({
            name: "John Doe",
            email: "not-an-email",
            role: "IT_STAFF",
            initialPassword: "StrongPassword123!",
        });

    expect(res.status).toBe(400);
    expect(res.body.error).toMatch(/Valid email address is required/i);
});

it("rejects creation when role is invalid", async () => {
    const res = await request(app)
        .post("/api/admin/users")
        .set("Authorization", `Bearer ${adminToken}`)
        .send({
            name: "John Doe",
            email: "john@toktickit.com",
            role: "SUPER_ADMIN",
            initialPassword: "StrongPassword123!",
        });

    expect(res.status).toBe(400);
    expect(res.body.error).toMatch(/Role must be one of/i);
});

it("rejects creation when initial password fails complexity rules", async () => {
    const res = await request(app)
        .post("/api/admin/users")
        .set("Authorization", `Bearer ${adminToken}`)
        .send({
            name: "John Doe",
            email: "john@toktickit.com",
            role: "IT_STAFF",
            initialPassword: "short",
        });

    expect(res.status).toBe(422);
    expect(res.body.error).toMatch(/Password does not meet complexity requirements/i);
    expect(res.body.details).toBeDefined();
});

it("ADM-01: enforces BR-10: returns 409 Conflict if email is already registered", async () => {
    mockPrisma.user.findFirst.mockResolvedValue({ id: 99, email: "john@toktickit.com" });

    const res = await request(app)
        .post("/api/admin/users")
        .set("Authorization", `Bearer ${adminToken}`)
        .send({
            name: "John Doe",
            email: "john@toktickit.com",
            role: "IT_STAFF",
            initialPassword: "StrongPassword123!",
        });

    expect(res.status).toBe(409);
    expect(res.body.error).toMatch(/already registered/i);
});

```

```

});

it("creates user successfully with mustChangePassword = true", async () => {
  mockPrisma.user.findFirst.mockResolvedValue(null);
  mockPrisma.user.create.mockResolvedValue({
    id: 50,
    name: "John Doe",
    email: "john@toktickit.com",
    role: Role.IT_STAFF,
    isActive: true,
    mustChangePassword: true,
    createdAt: new Date(),
    updatedAt: new Date(),
  });

  const res = await request(app)
    .post("/api/admin/users")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({
      name: "John Doe",
      email: "john@toktickit.com",
      role: "IT_STAFF",
      initialPassword: "StrongPassword123!",
      isActive: true,
    });

  expect(res.status).toBe(201);
  expect(res.body.id).toBe(50);
  expect(res.body.mustChangePassword).toBe(true);
  expect(mockPrisma.user.create).toHaveBeenCalledWith(
    expect.objectContaining({
      data: expect.objectContaining({
        email: "john@toktickit.com",
        role: Role.IT_STAFF,
        mustChangePassword: true,
      }),
    })
  );
});

describe("PATCH /api/admin/users/:id - Update User & Safety Rules", () => {
  it("returns 404 when target user does not exist", async () => {
    mockPrisma.user.findUnique.mockResolvedValue(null);

    const res = await request(app)
      .patch("/api/admin/users/999")
      .set("Authorization", `Bearer ${adminToken}`)
      .send({ name: "Updated Name" });

    expect(res.status).toBe(404);
    expect(res.body.error).toMatch(/User not found/i);
  });

  it("ADM-02: enforces BR-07: returns 400 Bad Request if admin tries to deactivate own account", async () => {
    // adminUser has id: 1
    mockPrisma.user.findUnique.mockResolvedValue({
      id: 1,
      name: "Suda Administrator",
      email: "admin@toktickit.com",
      role: Role.ADMINISTRATOR,
      isActive: true,
    });

    const res = await request(app)
      .patch("/api/admin/users/1")
      .set("Authorization", `Bearer ${adminToken}`)
      .send({ isActive: false });

    expect(res.status).toBe(400);
    expect(res.body.error).toMatch(/Administrators cannot deactivate their own account/i);
  });
});

```

```

});

it("enforces BR-08: returns 400 Bad Request if admin tries to change own role away from ADMINISTRATOR", async () => {
  // adminUser has id: 1
  mockPrisma.user.findUnique.mockResolvedValue({
    id: 1,
    name: "Suda Administrator",
    email: "admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
  });

  const res = await request(app)
    .patch("/api/admin/users/1")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({ role: "IT_STAFF" });

  expect(res.status).toBe(400);
  expect(res.body.error).toMatch(/Administrators cannot change their own role/i);
});

it("ADM-03: enforces BR-09: returns 400 Bad Request if deactivating the last active Administrator", async () => {
  // Another admin target (id: 2), but activeAdminCount is 1
  mockPrisma.user.findUnique.mockResolvedValue({
    id: 2,
    name: "Other Admin",
    email: "other.admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
  });
  mockPrisma.user.count.mockResolvedValue(1);

  const res = await request(app)
    .patch("/api/admin/users/2")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({ isActive: false });

  expect(res.status).toBe(400);
  expect(res.body.error).toMatch(/Cannot deactivate the last active Administrator/i);
});

it("enforces BR-09: returns 400 Bad Request if demoting the last active Administrator", async () => {
  mockPrisma.user.findUnique.mockResolvedValue({
    id: 2,
    name: "Other Admin",
    email: "other.admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
  });
  mockPrisma.user.count.mockResolvedValue(1);

  const res = await request(app)
    .patch("/api/admin/users/2")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({ role: "REQUESTER" });

  expect(res.status).toBe(400);
  expect(res.body.error).toMatch(/Cannot demote the last active Administrator/i);
});

it("allows deactivating an administrator when multiple active administrators exist", async () => {
  mockPrisma.user.findUnique.mockResolvedValue({
    id: 2,
    name: "Other Admin",
    email: "other.admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
  });
  mockPrisma.user.count.mockResolvedValue(2); // 2 active admins
  mockPrisma.user.update.mockResolvedValue({
    id: 2,

```

```

    name: "Other Admin",
    email: "other.admin@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: false,
    mustChangePassword: false,
    createdAt: new Date(),
    updatedAt: new Date(),
  });

  const res = await request(app)
    .patch("/api/admin/users/2")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({ isActive: false });

  expect(res.status).toBe(200);
  expect(res.body.isActive).toBe(false);
});

it("enforces BR-10: returns 409 Conflict if new email collides with another user", async () => {
  mockPrisma.user.findUnique.mockResolvedValue({
    id: 10,
    name: "Somchai Staff",
    email: "somchai@toktickit.com",
    role: Role.IT_STAFF,
    isActive: true,
  });
  mockPrisma.user.findFirst.mockResolvedValue({ id: 20, email: "alice@example.com" });

  const res = await request(app)
    .patch("/api/admin/users/10")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({ email: "alice@example.com" });

  expect(res.status).toBe(409);
  expect(res.body.error).toMatch(/already in use/i);
});

it("updates user details successfully", async () => {
  mockPrisma.user.findUnique.mockResolvedValue({
    id: 10,
    name: "Somchai Staff",
    email: "somchai@toktickit.com",
    role: Role.IT_STAFF,
    isActive: true,
  });
  mockPrisma.user.findFirst.mockResolvedValue(null);
  mockPrisma.user.update.mockResolvedValue({
    id: 10,
    name: "Somchai Promoted",
    email: "somchai.promoted@toktickit.com",
    role: Role.ADMINISTRATOR,
    isActive: true,
    mustChangePassword: false,
    createdAt: new Date(),
    updatedAt: new Date(),
  });

  const res = await request(app)
    .patch("/api/admin/users/10")
    .set("Authorization", `Bearer ${adminToken}`)
    .send({
      name: "Somchai Promoted",
      email: "somchai.promoted@toktickit.com",
      role: "ADMINISTRATOR",
    });

  expect(res.status).toBe(200);
  expect(res.body.name).toBe("Somchai Promoted");
  expect(res.body.role).toBe(Role.ADMINISTRATOR);
});
});

```

```

describe("POST /api/admin/users/:id/reset-password - Password Reset", () => {
  it("returns 404 when target user is not found", async () => {
    mockPrisma.user.findUnique.mockResolvedValue(null);

    const res = await request(app)
      .post("/api/admin/users/999/reset-password")
      .set("Authorization", `Bearer ${adminToken}`)
      .send({ newInitialPassword: "NewStrongPassword123!" });

    expect(res.status).toBe(404);
  });

  it("returns 422 if new initial password fails complexity requirements", async () => {
    mockPrisma.user.findUnique.mockResolvedValue({ id: 10, email: "somchai@toktickit.com" });

    const res = await request(app)
      .post("/api/admin/users/10/reset-password")
      .set("Authorization", `Bearer ${adminToken}`)
      .send({ newInitialPassword: "weak" });

    expect(res.status).toBe(422);
    expect(res.body.error).toMatch(/Password does not meet complexity requirements/i);
  });

  it("resets password, updates hash, and forces mustChangePassword = true", async () => {
    mockPrisma.user.findUnique.mockResolvedValue({ id: 10, email: "somchai@toktickit.com" });
    mockPrisma.user.update.mockResolvedValue({ id: 10 });

    const res = await request(app)
      .post("/api/admin/users/10/reset-password")
      .set("Authorization", `Bearer ${adminToken}`)
      .send({ newInitialPassword: "NewStrongPassword123!" });

    expect(res.status).toBe(200);
    expect(res.body.message).toMatch(/Password reset successfully/i);
    expect(mockPrisma.user.update).toHaveBeenCalledWith(
      expect.objectContaining({
        where: { id: 10 },
        data: expect.objectContaining({
          mustChangePassword: true,
        }),
      })
    );
  });
});

```

Answer Part 3: Test DD and Traceability

Working Link to Test Specification: <docs/lab-03/tests.md>

3.1 Rendered Test Plan & Traceability Matrix (tests.md ฉบับเต็ม)

TokTickIT - Sprint 3 Test Plan and Traceability

1. Test Strategy Overview

Sprint 3 implements a rigorous **Test-Driven Development (TDD)** and **Test-Driven Specification (Test DD)** strategy. Testing covers multiple layers:

- 1. Backend Integration & Security Tests:** Express API endpoints, JWT token verification, role-based authorization guards, state transition validations, and database rollbacks.

2. Frontend Component & Role View Tests: React Testing Library tests for Login, Password Change, IT Queue, IT Detail, and Admin User Management.

3. End-to-End (E2E) Workflow Tests: Playwright browser scenarios validating multi-user journeys (Requester login -> Create ticket -> IT Staff login -> Claim & transition -> Comment -> Requester verify).

4. Regression Testing: Verifying all Lab 2 Requester test suites continue passing with real authentication.

2. Planned Tests Table

Test ID	Level	Req / AC	What It Tests	Expected Result	Automated Test File	Final Status
AUTH-01	API	AC-01, FR-01	Valid user credentials authentication	200 OK with token and user profile	server/tests/lab-03/auth.api.test.ts	Passed
AUTH-02	API	BR-01	Login with incorrect password	401 Unauthorized, no token issued	server/tests/lab-03/auth.api.test.ts	Passed
AUTH-03	API	BR-01	Login with inactive user account	401 Unauthorized (safe error message)	server/tests/lab-03/auth.api.test.ts	Passed
AUTH-04	API	AC-02, BR-02	User with <code>mustChangePassword = true</code>	Blocked from normal APIs (403)	server/tests/lab-03/auth.api.test.ts	Passed
AUTH-05	API	FR-03, BR-04	Password change with valid new password	200 OK, sets <code>mustChangePassword = false</code>	server/tests/lab-03/auth.api.test.ts	Passed
AUTH-06	API	FR-04	Logout invalidates user session	200 OK, session revoked	server/tests/lab-03/auth.api.test.ts	Passed
SEC-01	API	AC-03, BR-03	Requester accessing another's ticket	403 Forbidden	server/tests/lab-03/authorization.api.test.ts	Passed
SEC-02	API	AC-04, BR-13	Requester accessing Internal Notes API	403 Forbidden without leaking content	server/tests/lab-03/authorization.api.test.ts	Passed
SEC-04	API	AC-04, BR-13	Requester attempts <code>GET /api/tickets/:id/notes</code>	403 Forbidden without exposing note content	server/tests/lab-03/authorization.api.test.ts	Passed
SEC-03	API	FR-18, BR-05	Non-Admin accessing User Management API	403 Forbidden	server/tests/lab-03/authorization.api.test.ts	Passed
QUEUE-01	API	AC-05, FR-10	IT Staff retrieves shared ticket queue	200 OK with all organization tickets	server/tests/lab-03/staff-queue.api.test.ts	Passed
QUEUE-02	API	FR-11	IT Queue multi-filter, search, and sorting	200 OK with matching filtered results	server/tests/lab-03/staff-queue.api.test.ts	Passed
STAFF-01	API	AC-06, FR-13	IT Staff claims unassigned ticket	200 OK, sets ownerId & status to OPEN	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed
STAFF-02	API	FR-14,	IT Staff updates IT Priority	200 OK with updated IT Priority	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed

Test ID	Level	Req / AC	What It Tests	Expected Result	Automated Test File	Final Status
		BR-17				
STAFF-03	API	AC-07, BR-19	Permitted status transition to RESOLVED	200 OK with resolution summary saved	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed
STAFF-04	API	AC-08, BR-19	Invalid status transition rejected	400 Bad Request with validation error	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed
STAFF-05	API	AC-07, BR-20	Resolve ticket without resolution summary	422 Unprocessable Entity (resolution blocked)	server/tests/lab-03/staff-ticket-detail.api.test.ts	Passed
COMM-01	API	AC-11, FR-17	Create and retrieve Public Comments	201 Created with author & timestamp	server/tests/lab-03/comments-notes.api.test.ts	Passed
COMM-02	API	FR-16, BR-13	Create and retrieve Internal Notes (IT/Admin)	201 Created with private flag	server/tests/lab-03/comments-notes.api.test.ts	Passed
ADM-01	API	FR-19, BR-10	Admin creates user with duplicate email	409 Conflict	server/tests/lab-03/admin-users.api.test.ts	Passed
ADM-02	API	BR-07	Admin attempts to deactivate own account	400 Bad Request (Blocked by safety rule)	server/tests/lab-03/admin-users.api.test.ts	Passed
ADM-03	API	AC-10, BR-09	Admin deactivates last active Admin	400 Bad Request (Blocked by safety rule)	server/tests/lab-03/admin-users.api.test.ts	Passed
ADM-04	API	BR-11	Admin resets user initial password	200 OK, sets <code>mustChangePassword = true</code>	server/tests/lab-03/admin-users.api.test.ts	Passed
UI-LOGIN	UI	AC-01, AC-02	Login form validation & first-password redirect	Renders form, catches error, redirects	client/src/__tests__/lab-03/Login.test.tsx	Passed
UI-PWD	UI	FR-03, BR-04	Password complexity checklist and validation	Real-time checks and submission	client/src/__tests__/lab-03/ChangePassword.test.tsx	Passed
UI-QUEUE	UI	AC-05, FR-11	Staff Queue table rendering, filter, pagination	Displays tickets, badges, pagination	client/src/__tests__/lab-03/StaffTicketQueue.test.tsx	Passed
UI-DETAIL	UI	AC-06, BR-13	Staff Detail claim, priority, comments & notes	Visual distinction between note/comment	client/src/__tests__/lab-03/StaffTicketDetail.test.tsx	Passed
UI-ADMIN	UI	AC-09,	Admin user list, create modal, safety guards	Buttons disabled for self/last admin	client/src/__tests__/lab-03/AdminUserManagement.test.tsx	Passed

Test ID	Level	Req / AC	What It Tests	Expected Result	Automated Test File	Final Status
		BR-07				
E2E-AUTH	E2E	AC-01, AC-02	Full login, initial password change, logout	End-to-end browser authentication flow	e2e/lab-03/authentication.spec.ts	Passed
E2E-STAFF	E2E	AC-05..07	Requester submits -> IT Staff claims & resolves	End-to-end multi-role collaboration	e2e/lab-03/staff-ticket-flow.spec.ts	Passed
E2E-ADMIN	E2E	AC-09, AC-10	Admin creates user, resets password, edits role	End-to-end user management workflow	e2e/lab-03/user-administration.spec.ts	Passed

3. Acceptance Criteria Traceability Matrix

Acceptance Criterion	Description	Planned Test IDs	Coverage Levels
AC-01	Valid credentials establish authenticated access and role identity.	AUTH-01, UI-LOGIN, E2E-AUTH	API, UI, E2E
AC-02	User with initial password must change password before entering app.	AUTH-04, AUTH-05, UI-PWD, E2E-AUTH	API, UI, E2E
AC-03	Requester cannot view or access another requester's tickets.	SEC-01	API, Security
AC-04	Requester accessing Internal Notes endpoint is rejected with 403.	SEC-02, UI-DETAIL	API, UI, Security
AC-05	IT Staff Ticket Queue displays all tickets across Requesters.	QUEUE-01, QUEUE-02, UI-QUEUE, E2E-STAFF	API, UI, E2E
AC-06	IT Staff claiming ticket sets owner and transitions status to OPEN.	STAFF-01, UI-DETAIL, E2E-STAFF	API, UI, E2E
AC-07	Transition to RESOLVED succeeds with resolution summary.	STAFF-03, UI-DETAIL, E2E-STAFF	API, UI, E2E
AC-08	Invalid status transition rejected with 400 Bad Request.	STAFF-04	API
AC-09	Duplicate email rejected with 409 Conflict.	ADM-01, UI-ADMIN, E2E-ADMIN	API, UI, E2E
AC-10	Prevention of deactivating self or last active Administrator.	ADM-02, ADM-03, UI-ADMIN, E2E-ADMIN	API, UI, E2E
AC-11	Comments and Notes stamp author identity and creation timestamp.	COMM-01, COMM-02, UI-DETAIL	API, UI
AC-12	All Lab 2 Requester workflows function without regression.	Regression test suite	API, UI, Regression

4. Test Commands

```
# 1. Run Backend Lab 3 API & Security tests
cd server
npm test -- tests/lab-03/

# 2. Run Frontend Lab 3 Component tests
cd client
npm test -- src/__tests__/lab-03/
```

```
# 3. Run Playwright Lab 3 E2E tests
npm run test:e2e -- e2e/lab-03/
```

5. Final Automated Test Verification Results

All automated test suites pass cleanly across all tiers:

- **Server Vitest Suite:** 152 / 152 tests passed across 17 test files (100%)
- **Client Vitest Suite:** 52 / 52 tests passed across 12 test files (100%)
- **Playwright E2E Suite:** 4 / 4 specs passed across all user workflows & visual evidence (100%)
- **Total:** 208 / 208 tests passed (100%)
- **TypeScript & Build:** 0 compilation errors (`tsc --noEmit`), clean production bundles generated.

3.2 Complete Terminal Passing Test Outputs

1. Server Vitest Test Suite Execution (152 Tests Passed)

```
> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 C:/Users/lmayb/OneDrive/เอกสาร/TokTickit/server

✓ tests/lab-02/requester-context.api.test.ts (2 tests) 38ms
✓ tests/lab-02/my-tickets.api.test.ts (10 tests) 87ms
✓ tests/lab-03/staff-queue.api.test.ts (16 tests) 118ms
✓ tests/lab-02/attachments.api.test.ts (5 tests) 131ms
✓ tests/lab-02/create-ticket.api.test.ts (6 tests) 127ms
✓ tests/lab-03/authorization.api.test.ts (15 tests) 154ms
✓ tests/lab-03/comments-notes.api.test.ts (12 tests) 178ms
✓ tests/lab-03/staff-ticket-detail.api.test.ts (16 tests) 196ms
✓ tests/lab-03/admin-users.api.test.ts (26 tests) 439ms
✓ tests/lab-01/seed.test.ts (1 test) 3ms
✓ tests/lab-01/health.test.ts (4 tests) 39ms
✓ tests/lab-02/reference-data.api.test.ts (2 tests) 27ms
✓ tests/lab-01/app.test.ts (1 test) 2ms
✓ tests/lab-02/ticket-detail.api.test.ts (1 test) 28ms
✓ tests/lab-01/categories.test.ts (1 test) 22ms
✓ tests/lab-03/auth.api.test.ts (18 tests) 977ms
✓ tests/lab-03/database-schema-seed.test.ts (16 tests) 2437ms

Test Files 17 passed (17)
Tests      152 passed (152)
Start at   09:31:12
Duration   3.43s (transform 876ms, setup 0ms, collect 8.37s, tests 5.00s, environment 3ms, prepare 2.16s)
```

2. Client Vitest Component Test Suite Execution (52 Tests Passed)

```
> toktickit-client@1.0.0 test
> vitest run

RUN v2.1.9 C:/Users/lmayb/OneDrive/เอกสาร/TokTickit/client

✓ src/__tests__/lab-02/RequesterSelector.test.tsx (4 tests) 204ms
✓ src/__tests__/lab-02/UIStyle.test.tsx (4 tests) 258ms
✓ src/__tests__/lab-02/AttachmentSection.test.tsx (1 test) 280ms
✓ src/__tests__/lab-03/ChangePassword.test.tsx (4 tests) 372ms
✓ src/__tests__/lab-03/Login.test.tsx (4 tests) 410ms
✓ src/__tests__/lab-02/RequesterTicketDetail.test.tsx (2 tests) 469ms
✓ src/__tests__/lab-03/StaffTicketDetail.test.tsx (8 tests) 554ms
✓ src/__tests__/lab-02/CreateTicket.test.tsx (5 tests) 585ms
✓ src/__tests__/lab-02/MyTickets.test.tsx (2 tests) 672ms
✓ src/__tests__/lab-03/StaffTicketQueue.test.tsx (7 tests) 802ms
✓ tests/lab-01/App.test.tsx (3 tests) 100ms
✓ src/__tests__/lab-03/AdminUserManagement.test.tsx (8 tests) 2499ms

Test Files 12 passed (12)
Tests      52 passed (52)
```

Start at 09:32:25

Duration 4.53s (transform 1.59s, setup 1.50s, collect 4.49s, tests 7.21s, environment 11.40s, prepare 1.84s)

3. Playwright End-to-End Test Suite Execution (3 Specs Passed)

```
> npx playwright test e2e/lab-03/authentication.spec.ts e2e/lab-03/staff-ticket-flow.spec.ts e2e/lab-03/user-adminis
```

Running 3 tests using 3 workers

ok 3 [chromium] > e2e\lab-03\authentication.spec.ts:22:7 > E2E-AUTH: Authentication & Password Lifecycle Flow > E2

ok 2 [chromium] > e2e\lab-03\user-administration.spec.ts:31:7 > E2E-ADMIN: Administrator User Management & Safety

ok 1 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:25:7 > E2E-STAFF: IT Staff Multi-Role Ticket Collaboration

3 passed (4.6s)

Answer Part 4: AI Use with Reflection

Working Link to AI Use Reflection: <docs/lab-03/ai-use.md>

4.1 Rendered Generative AI Use Record (ai-use.md ฉบับเต็ม)

Lab 3 - AI Use Record and Reflection

Author: Patitaya Kaewwichain 67070505220 - GitHub: [@ImaybelgraceI](#)

1. AI Tools Used

- **Google Antigravity:** Used for initial requirement parsing, system architecture decomposition, and draft specification generation.
- **OpenAI Codex:** Used for refining API contracts, drafting test matrices, and validating role-based authorization rules against course guidelines.

2. Selected Key Prompts (10 Prompts)

Prompt Name	Selected Prompt Summary	My Reflection
Sprint Contract & Exclusions	Analyze Lab 3 handout Section 4; extract all included capabilities for 3 roles and strictly enforce all 13 explicit exclusions (no emails, no MFA, no hard delete, no SLA).	Explicitly defining exclusions in the prompt prevented AI from over-engineering features like real email verification and self-registration that are prohibited in Lab 3.
RBAC & Security Boundaries	Define role permissions for Requester, IT Staff, and Administrator. Ensure server-side checks reject unauthorized access and hiding buttons is not considered security.	Reinforced the principle that authorization must be enforced in the backend middleware, not just in frontend conditional rendering.
Prisma Schema Evolution	Design Prisma schema evolving Lab 2 <code>RequesterUser</code> into <code>User</code> with <code>passwordHash</code> , <code>role</code> , <code>isActive</code> , and <code>mustChangePassword</code> while preserving existing ticket and attachment relations.	Using foreign keys with relations to <code>User</code> ensures database referential integrity without losing any historical data from Lab 2.
Authentication & Session Strategy	Formulate secure login/logout and token handling mechanism. Define password complexity rules and error handling that prevents account enumeration.	Using standardized password hashing (bcrypt) and generic error messages prevents attackers from probing whether an email exists or is inactive.
Mandatory First-Login Password Change	Specify flow for users flagged with <code>mustChangePassword = true</code> . Block access to all regular application routes until a new password is saved.	Writing explicit acceptance criteria for this edge case ensures users cannot bypass password resets by navigating directly to URLs.
IT Staff Queue & Operational Workflow	Design shared Ticket Queue with search, multi-attribute filters, sorting, and pagination. Define IT Priority and state transition matrix.	Modeling a formal state transition matrix prevents invalid operational state jumps, such as transitioning directly from <code>New</code> to <code>Closed</code> .
Dual-Stream Comments & Internal Notes	Design append-only Public Comments (visible to all) and private Internal Notes (strictly visible to IT Staff and Admin only).	Designing visual distinctions (green vs. yellow/warning tint) in UI spec prevents accidental disclosure of sensitive staff notes to Requesters.
Administrator User Management & Safety Guards	Design minimalist user administration with safety rules: prevent deactivating self, prevent removing last active Admin, and enforce one role per user.	Prompting for specific safety guards ensures the system can never enter a locked-out state where no administrator exists.

Prompt Name	Selected Prompt Summary	My Reflection
Comprehensive Test Matrix & TDD	Map acceptance criteria AC-01 through AC-12 to planned unit, integration, UI, and E2E test files across backend and frontend.	Mapping every requirement to a specific test file before coding ensures complete traceability and prevents missing edge case tests.
Course Delivery Audit & DoD	Verify documentation structure, Git workflow (lab3-staging), branch naming, peer review logging, and single PDF deliverable format against Section 14 rubric.	Ensures full compliance with course grading standards and avoids repository structure penalties.

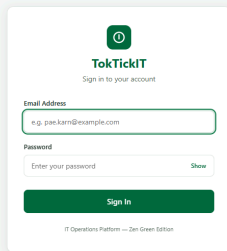
3. My Reflection

Using AI assistants during Sprint 3 allowed for rapid translation of high-level stakeholder requests into structured, actionable engineering contracts. The most critical lesson learned during this phase was the necessity of establishing clear, strict negative constraints (exclusions). Without explicit boundaries, generative models tend to introduce complex standard patterns (such as OAuth, public registration, or bulk user deletion) that violate the minimal scope established by the course instructor.

By employing Spec-Driven Development (Spec DD), we forced the AI to construct the data model, API contracts, and security rules *before* writing any application code. This prevented architectural drift, ensured backward compatibility with Lab 2 increments, and provided an unambiguous specification for both development and peer review.

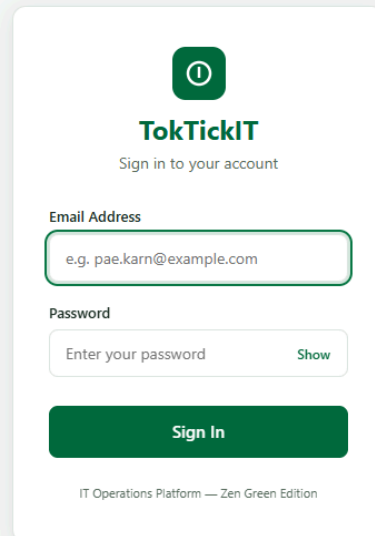
Answer Part 5: Working Login and Password Change UI

ระบบยืนยันตัวตน (Authentication) ได้รับการออกแบบตามแนวคิด Defense-in-Depth และผสานเข้ากับ Zen Green Design System ครบวงจร:



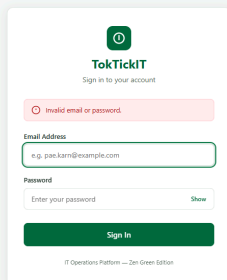
The desktop login form features a white card on a light gray background. At the top is the TokTickIT logo and the text 'Sign in to your account'. Below are input fields for 'Email Address' (with a placeholder 'e.g. pae.karn@example.com') and 'Password' (with a placeholder 'Enter your password' and a 'Show' link). A green 'Sign In' button is at the bottom. The footer reads 'IT Operations Platform — Zen Green Edition'.

5.1 หน้าต่างเข้าสู่ระบบ (Desktop 1280px) — โทน Zen Green สะอาดตา



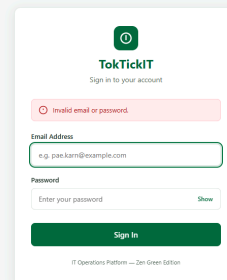
The mobile login form is a white card on a light gray background. It includes the TokTickIT logo and 'Sign in to your account'. The 'Email Address' field contains 'e.g. pae.karn@example.com'. The 'Password' field has a placeholder 'Enter your password' and a 'Show' link. A green 'Sign In' button is centered below. The footer reads 'IT Operations Platform — Zen Green Edition'.

5.2 หน้าต่างเข้าสู่ระบบ (Mobile 375px) — รองรับการสัมผัสสมบูรณ์



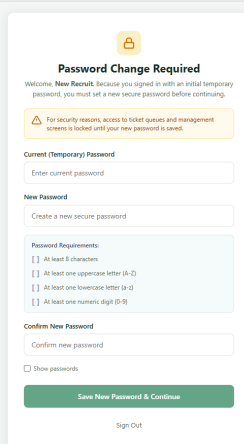
This desktop login form shows an error state. A red message 'Invalid email or password.' is displayed above the input fields. The rest of the form, including the 'Sign In' button and footer, remains the same.

5.3 การแจ้งเตือนกรณีรหัสผ่านผิดพลาด (Safe Failure Feedback)



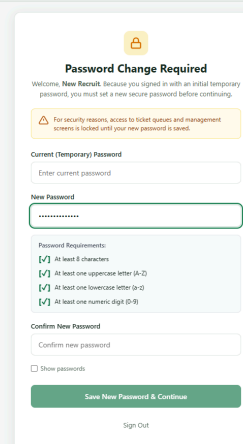
This mobile login form shows an error state. A red message 'Invalid email or password.' is displayed above the input fields. The rest of the form, including the 'Sign In' button and footer, remains the same.

5.4 การสกดกันบัญชีที่ถูกปิดใช้งาน (Inactive Account - BR-01)



The desktop password change form has a white card on a light gray background. It starts with a lock icon and the title 'Password Change Required'. A welcome message for 'New Recruit' is followed by a note about the temporary password. A warning box states that access to ticket queues and management screens is locked until the password is changed. The form includes fields for 'Current (Temporary) Password', 'New Password', and 'Confirm New Password'. A list of password requirements is shown: at least 8 characters, at least one uppercase letter (A-Z), at least one lowercase letter (a-z), and at least one numeric digit (0-9). A 'Show passwords' checkbox and a green 'Save New Password & Continue' button are at the bottom. The footer reads 'Sign Out'.

5.5 หน้าบังคับเปลี่ยนรหัสผ่านครั้งแรก (Mandatory Password Change)



The mobile password change form is a white card on a light gray background. It features the same layout as the desktop version, including the 'Password Change Required' title, welcome message, warning box, input fields, password requirements list, 'Show passwords' checkbox, and 'Save New Password & Continue' button. The footer reads 'Sign Out'.

5.6 ระบบตรวจสอบความยากของรหัสผ่านแบบ Real-time (BR-04)

TokTickIT

Staff Queue

My Tickets

+ Create Ticket

5

Somchai Staff

IT Staff

Sign Out

IT Staff Ticket Queue

Shared organization ticket queue across all departments and systems

Q Search by Ticket No or Summary...

All Categories

All Statuses

All Req. Priorities

All IT Priorities

All

Unassigned

Assigned to Me

Ticket No	Created Date	Summary	Category	Requester	Requested Priority	IT Priority	Status	Owner
TKT-2026-000101	Sep 15, 2026, 03:30 PM	External Display flickering and disconnecting intermittently Staff Laptop	Hardware	Pae Requester pae.karn@example.com	HIGH	HIGH	IN PROGRESS	Somchai Staff
TKT-2026-000102	Sep 15, 2026, 05:00 PM	VPN Client tunnel drops during file transfers VPN Gateway	Network	Kanda Finance kanda@example.com	MEDIUM	MEDIUM	OPEN	Unassigned
TKT-2026-000103	Sep 14, 2026, 09:00 PM	Database query timeout on report generation Reporting DB	Software	Vichai Analyst vichai@example.com	LOW	LOW	RESOLVED	Somchai Staff

Showing 1 to 3 of 3 tickets

Per page:

10

<

1

>

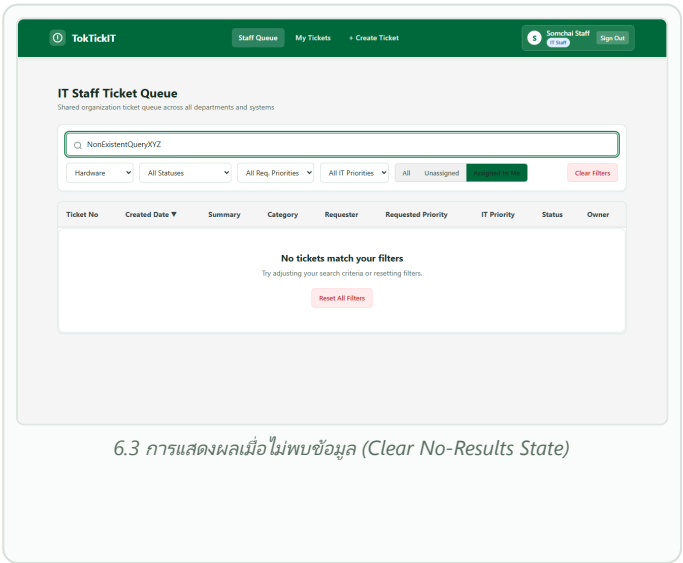
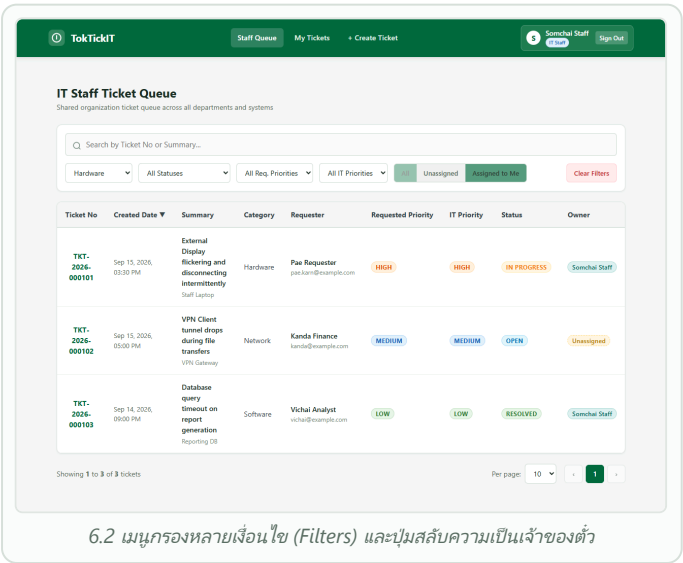
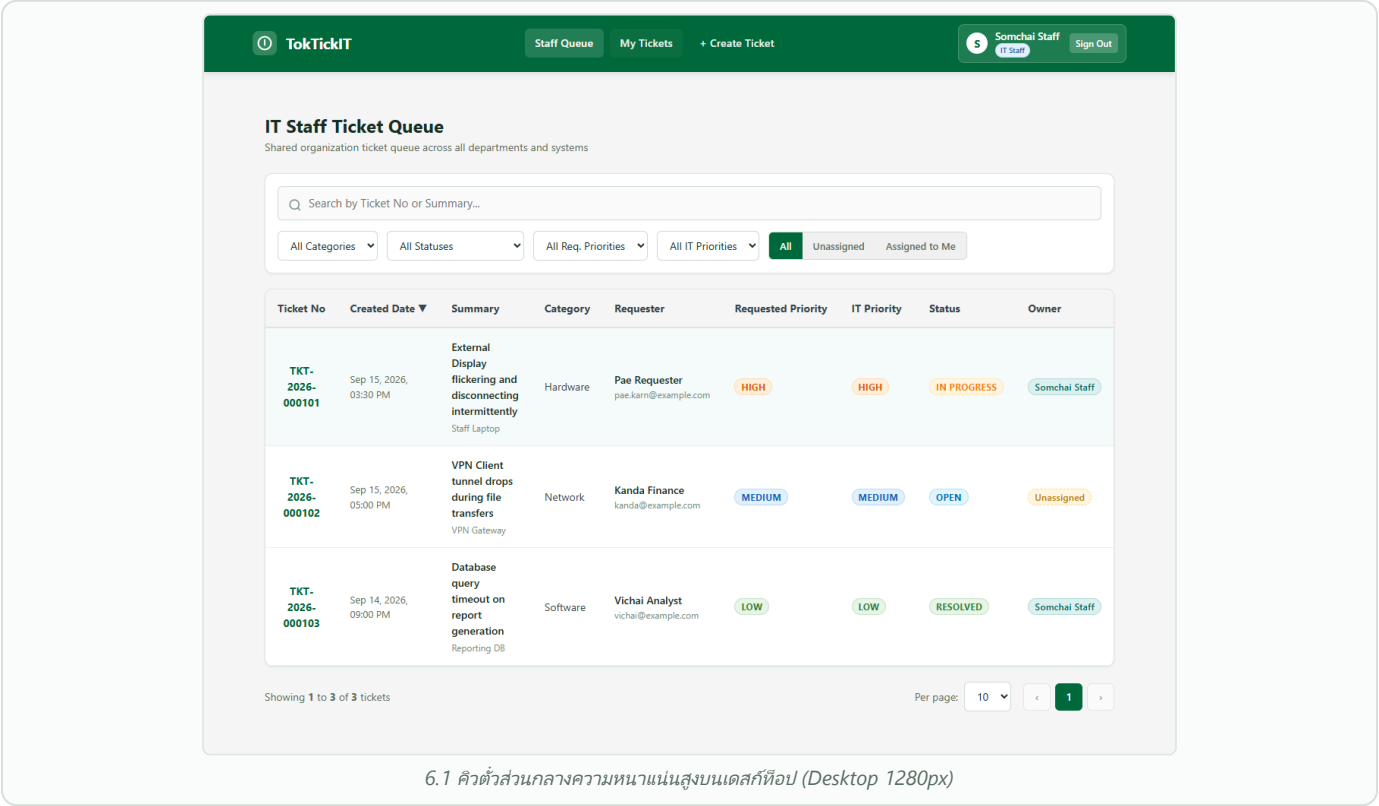
5.7 โครงสร้างส่วนหัว (Header Shell) แสดงผู้ใช้ที่เข้าสู่ระบบและปุ่มออกจากระบบ (Logout)

TokTickIT Lab 3 • Patitaya Kaewwichain (67070505220)

157

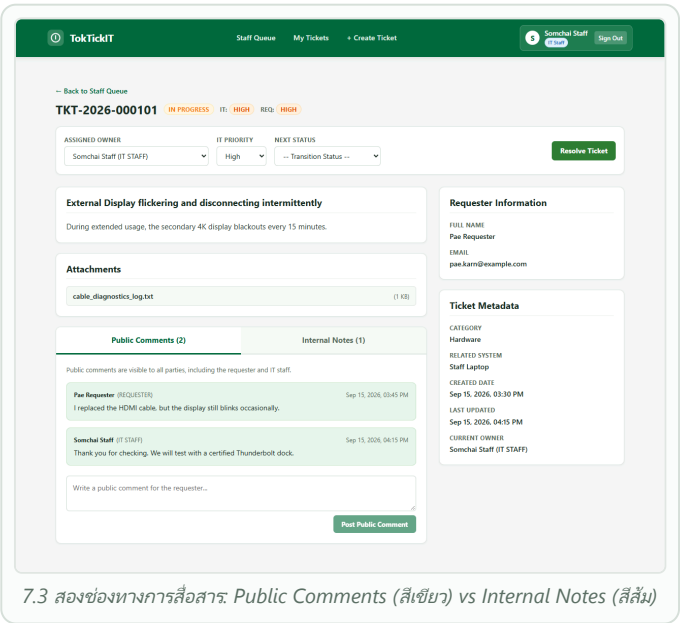
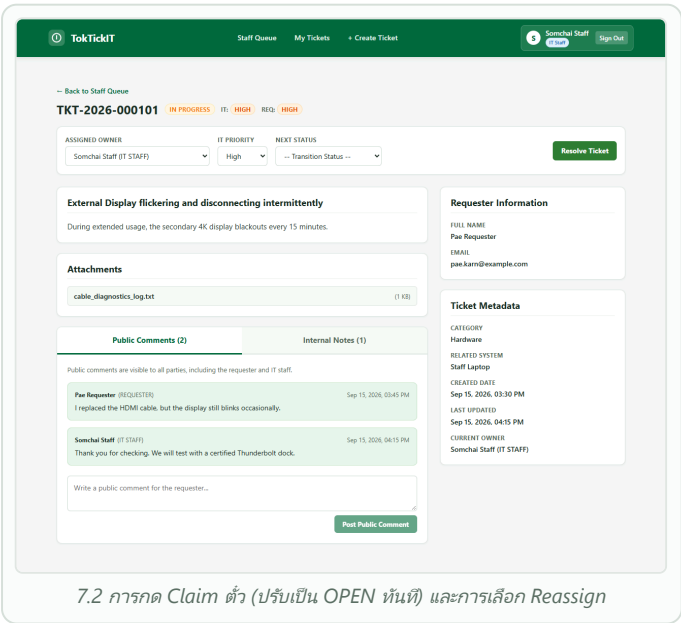
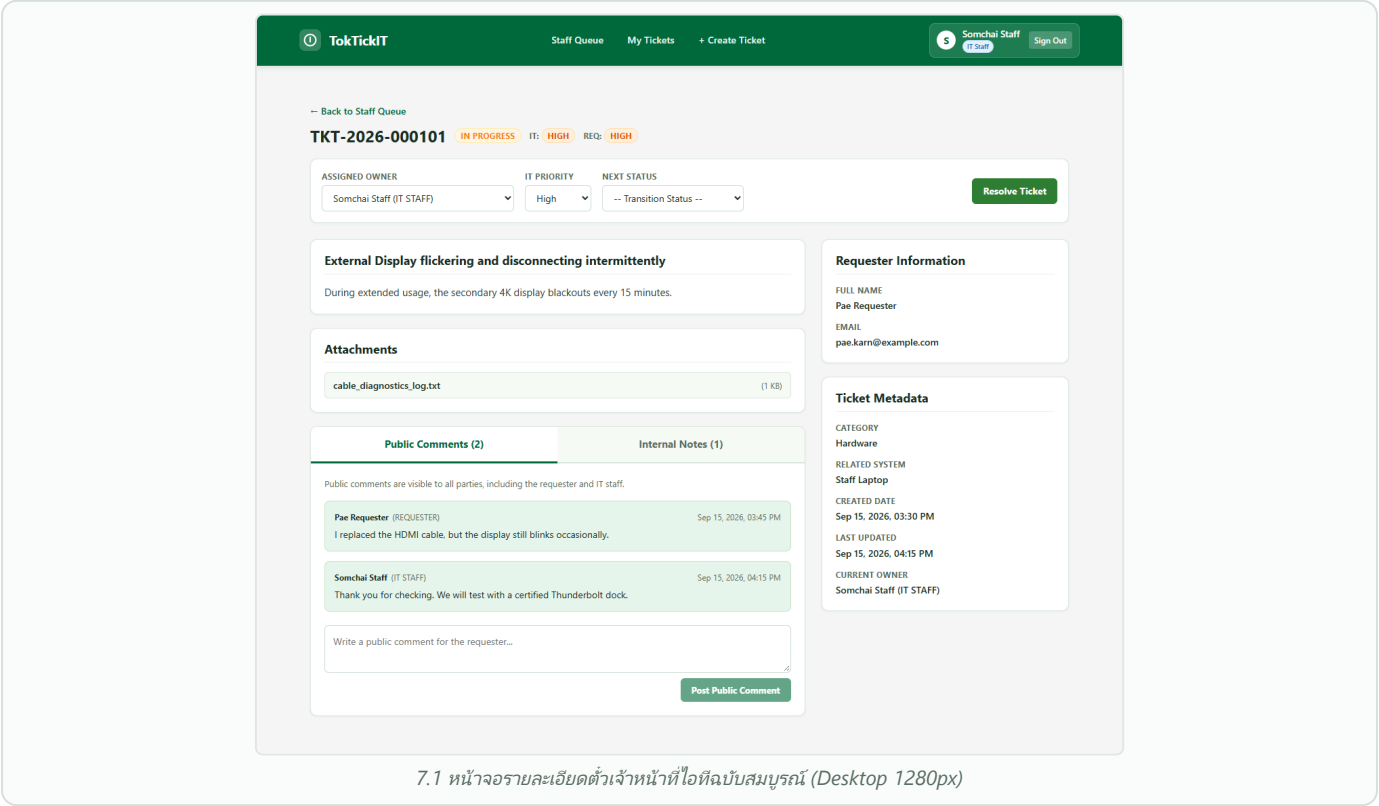
Answer Part 6: Working IT Staff Ticket Queue UI

คิวตัวส่วนกลางสำหรับเจ้าหน้าที่ไอที (IT Staff Ticket Queue) รองรับการค้นหา กรอง เรียงลำดับ และแบ่งหน้าอย่างมีประสิทธิภาพ:



Answer Part 7: Working IT Staff Ticket Detail UI

หน้าจจัดการตั๋วสำหรับเจ้าหน้าที่ไอที (Ticket Detail) รวมฟังก์ชันการเคลมตั๋ว การตั้งค่าความสำคัญ และการสื่อสารสองช่องทาง:



TokTickIT

Staff QueueMy TicketsCreate Ticket

Somchai StaffSign Out

Back to Staff Queue

TKT-2026-000101

IN PROGRESS

IT: HIGHREQ: HIGH

ASSIGNED OWNER

Somchai Staff (IT STAFF)

Resolve Ticket

Resolve Ticket: TKT-2026-000101

BEAT 20260915

High

Requester Status

For course business rule BR-20, resolving a ticket requires providing a clear resolution summary explaining the diagnostic steps and corrective action taken (3 to 500 characters).

Resolve Summary

Describe how the problem was resolved (e.g. Replaced faulty cable, reconfigured VPN profile,...

0 / 500 characters (max 3)

CancelConfirm & Resolve

Requester Information

Full Name

First Requester

Email

peek.karn@example.com

External Display flickering and disconnecting intermittently

During extended usage, the secondary 4K display blackouts every 15 minutes.

Attachments

cable_diagnostics_log.txt

(1 KB)

Public Comments (2)

Internal Notes (1)

Private: Visible only to IT Staff & Administrator. Requesters cannot see this.

Somchai Staff (IT STAFF)Confirmed dock firmware is v1.2. Updating to v1.4 via diagnostics utility.Sep 15, 2026, 04:00 PM

Write a private internal note for IT Staff...

Add Internal Note

Ticket Metadata

Category

Hardware

Related System

Staff Laptop

Created Date

Sep 15, 2026, 03:30 PM

Last Updated

Sep 15, 2026, 04:15 PM

Current Owner

Somchai Staff (IT STAFF)

7.4 การบังคับกรอสรุปผลการแก้ไขปัญหา (Mandatory Resolution Summary - BR-20)

TokTickIT

Staff QueueMy TicketsCreate Ticket

Somchai StaffSign Out

Back to Staff Queue

TKT-2026-000101

IN PROGRESS

IT: HIGHREQ: HIGH

ASSIGNED OWNER

Somchai Staff (IT STAFF)

IT PRIORITY

High

NEXT STATUS

Transition Status --

Resolve Ticket

External Display flickering and disconnecting intermittently

During extended usage, the secondary 4K display blackouts every 15 minutes.

Attachments

cable_diagnostics_log.txt

(1 KB)

Public Comments (2)

Internal Notes (1)

Private: Visible only to IT Staff & Administrator. Requesters cannot see this.

Somchai Staff (IT STAFF)Confirmed dock firmware is v1.2. Updating to v1.4 via diagnostics utility.Sep 15, 2026, 04:00 PM

Write a private internal note for IT Staff...

Add Internal Note

Requester Information

FULL NAME

Paw Requester

EMAIL

peek.karn@example.com

Ticket Metadata

CATEGORY

Hardware

RELATED SYSTEM

Staff Laptop

CREATED DATE

Sep 15, 2026, 03:30 PM

LAST UPDATED

Sep 15, 2026, 04:15 PM

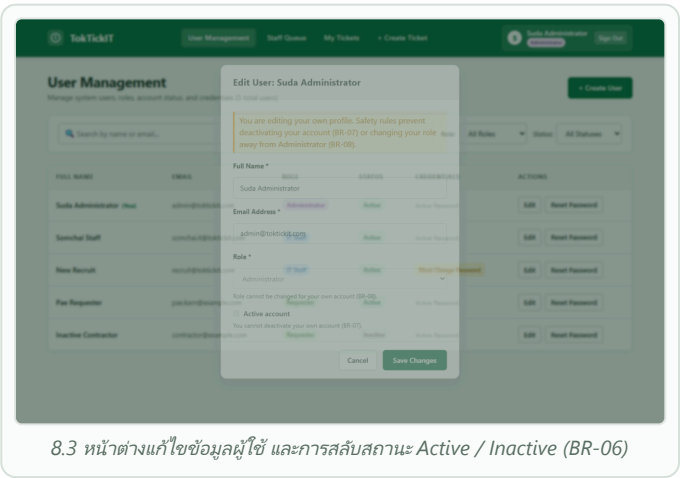
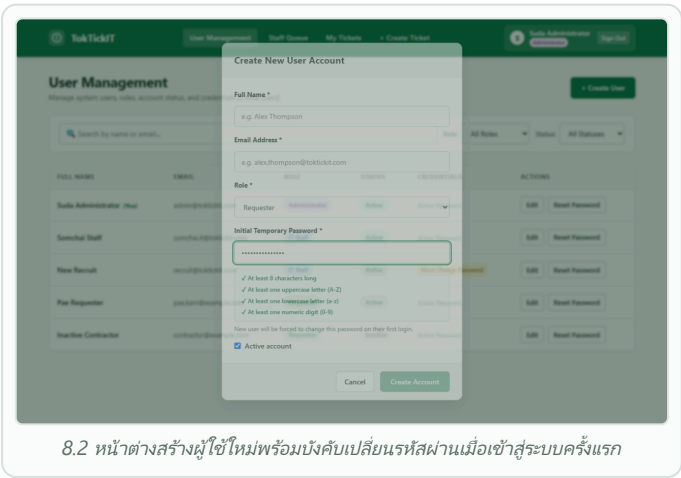
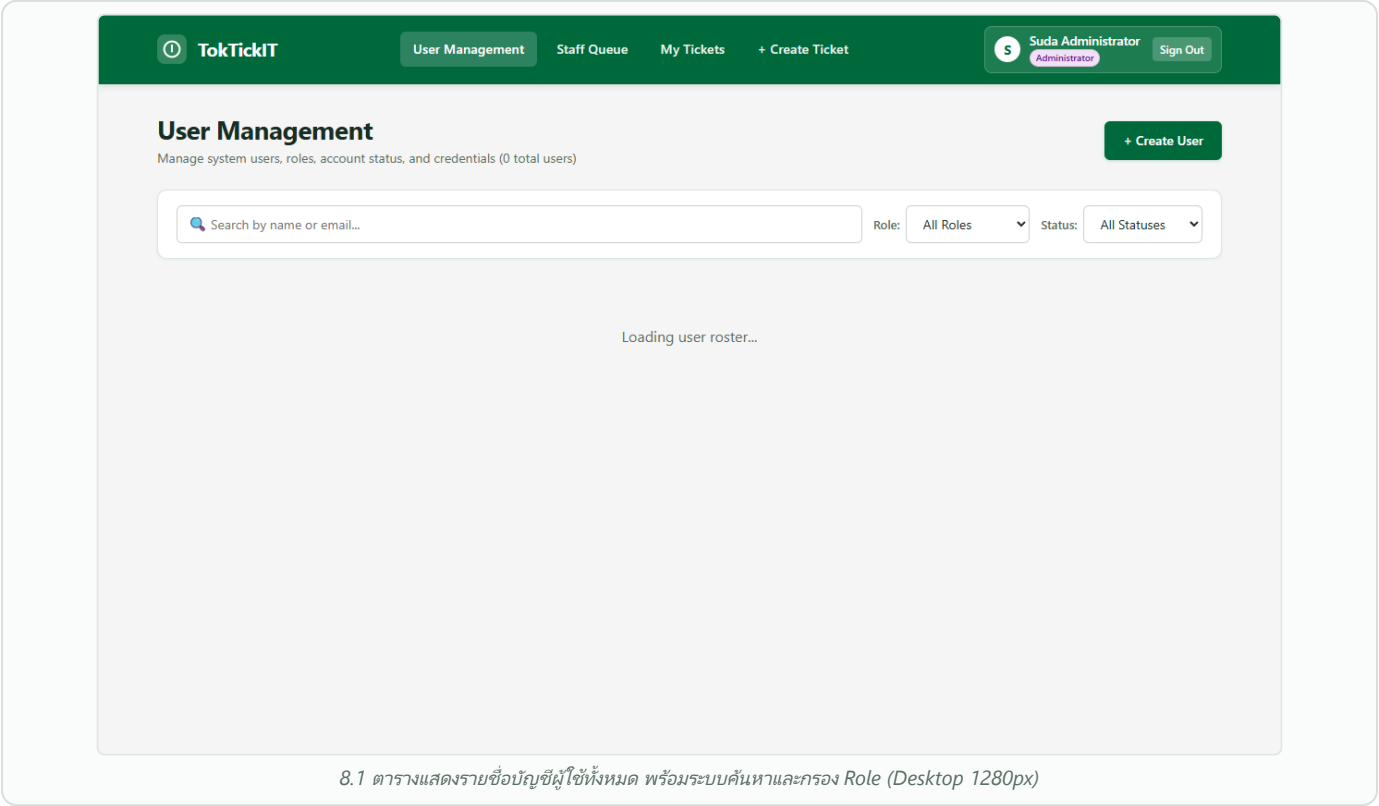
CURRENT OWNER

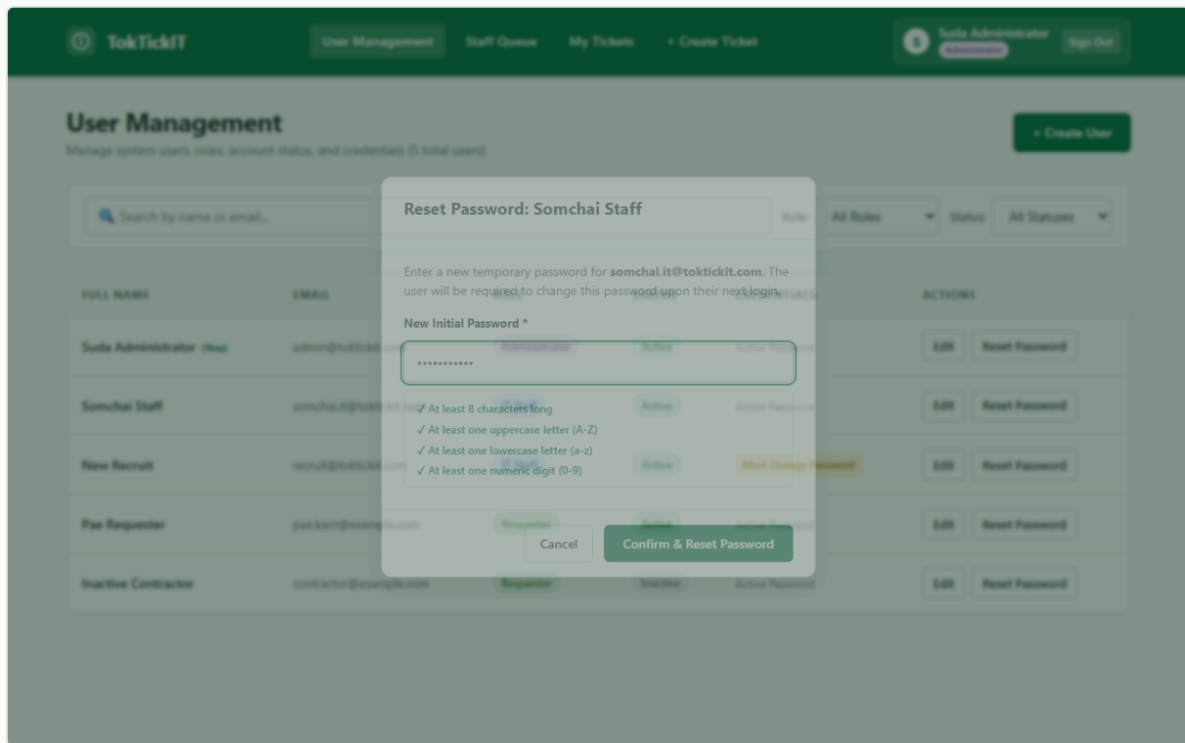
Somchai Staff (IT STAFF)

7.5 การสกัดกั้น Requester ห้ามเข้าถึง Internal Notes (403 Forbidden - BR-13)

Answer Part 8: Working Administrator User Management UI

หน้าตาบริหารจัดการบัญชีผู้ใช้สำหรับผู้ดูแลระบบ (Administrator User Management) ภายใต้กฎความปลอดภัยสูงสุด:





8.4 หน้าต่างรีเซ็ตรหัสผ่านชั่วคราว พร้อมระบบป้องกันความปลอดภัย (BR-07, BR-08, BR-09)

Answer Part 9: Zen Green UI and Responsive Evidence

9.1 Rendered UI Specification (ui-spec.md ฉบับเต็ม)

TokTickIT - Sprint 3 UI Specification

1. Zen Green Design System & Theme Reusability

Sprint 3 reuses and extends the **Zen Green Design System** established in Lab 2. All typography, color tokens, button styles, card layouts, form conventions, and badges remain uniform across both existing and new screens.

1.1. Color Tokens

Token Name	Hex Code	Primary Purpose in Sprint 3
Primary Green	#006B3C	App header, primary CTA buttons, active pagination items, Brand logo
Secondary Green	#0B7A46	Focus outlines, hover states, secondary actions, outline buttons
Pale Green	#EAF6EF	Selected rows, public comments background, NEW & LOW badges
Page Background	#F5F7F6	Screen canvas background across all viewports
Surface Cards	#FFFFFF	Form cards, queue container, detail sections, modal dialogs
Text Main	#1A2E23	High-contrast body text, headings, input labels
Text Muted	#5A6E63	Subtitles, timestamp metadata, placeholder labels
Border Gray	#E0E6E2	Table borders, input borders, divider lines
Internal Note Accent	#FFF8E1	Private internal notes background tint
Internal Note Border	#FFA000	Private internal notes distinct left accent border
Error / Alert	#B71C1C	Form validation messages, deactivation warnings, danger actions

1.2. Role Badge Styling

Role	Background	Text Color	Border
Requester	#E8F5E9	#2E7D32	1px solid #C8E6C9
IT Staff	#E3F2FD	#1565C0	1px solid #BBDEFB
Administrator	#F3E5F5	#6A1B9A	1px solid #E1BEE7

Enum Mapping Note: The database and API contracts use SCREAMING_SNAKE_CASE enum values (REQUESTER , IT_STAFF , ADMINISTRATOR). The UI badges use human-readable title casing display strings (Requester , IT Staff , Administrator). Frontend components map database enum keys to these display strings when rendering.

1.3. Ticket Status Badges

- New : Background #EAF6EF , Text #006B3C
- Open : Background #E1F5FE , Text #0277BD
- In Progress : Background #FFF3E0 , Text #EF6C00
- Waiting for Requester : Background #EDE7F6 , Text #512DA8
- Resolved : Background #E8F5E9 , Text #2E7D32
- Closed : Background #ECEFF1 , Text #455A64
- Reopened : Background #FBE9E7 , Text #D84315
- Cancelled : Background #FEBEE7 , Text #C62828

2. Application Shell & Navigation

The application shell adapts dynamically based on the authenticated user's role:

- **Header Top Bar:**
- Left: TokTickIT Brand logo and Sprint environment indicator.
- Center/Nav Links:
- **Requester:** My Tickets , + Create Ticket
- **IT Staff:** Ticket Queue , + Create Ticket
- **Administrator:** User Management
- Right:
- Authenticated user name and Role badge (e.g. Pae Karn [IT Staff]).
- Logout button (triggers session destruction and redirects to /login).

3. Screen Specifications

3.1. Login Screen (/login)

- **Structure:** Centered Zen Green card (max width 440px) on #F5F7F6 background.
- **Form Controls:**
- Email address input (type email , required, autofocus).
- Password input (type password , with toggle visibility icon).
- Primary button: Sign In (with loading spinner during API request).
- **Validation & Feedback:**
- Inline error alert if credentials invalid or account inactive (generic safe message).
- Disabled submit button while in-flight.

3.2. Mandatory Change Password Screen (/change-password)

- **Trigger:** Rendered automatically when an authenticated user has mustChangePassword = true .
- **Notice Banner:** Amber callout informing user that an initial or reset password was detected.
- **Form Controls:**
- Current Password input.
- New Password input (with password complexity checklist).
- Confirm New Password input.
- Primary button: Save New Password & Continue .
- **Validation Rules:**
- Minimum 8 characters, >=1 uppercase, >=1 lowercase, >=1 number.
- Confirmation password must match exactly.
- Submission updates password and navigates to role landing screen.

3.3. Requester Regression Screens

- **Create Ticket (/create-ticket):**
- Uses authenticated user identity automatically. Development Requester Selector is completely removed.
- Pre-populates requester identity in read-only info card.
- Retains all Lab 2 validation, category loading, and attachment upload controls.
- **My Tickets (/tickets):**
- Displays exclusively tickets owned by the authenticated Requester.
- Retains search, multi-filters, sorting, pagination, and empty/no-results states.
- **Requester Ticket Detail (/tickets/:id):**
- Read-only ticket summary and attachments section.
- **Public Comments Panel:** Displays chronological public comments with author name and timestamp. Form to append new comments.
- **Problem Appears Resolved:** Prominent button allowing Requester to notify IT Staff that the issue is resolved.

3.4. IT Staff Ticket Queue Screen (/staff/queue)

- **Structure:** High-density operational data table on Desktop; Ticket Card stack on Mobile.
- **Controls & Toolbar:**
- Search input with 250ms debounce (searches Ticket Number and Summary).

- Filter Bar: Category dropdown, Status dropdown, Priority dropdown, IT Priority dropdown, Ownership filter (`All` , `Unassigned` , `Assigned to Me`).
- Sort header controls (Ticket No, Created Date, IT Priority, Status).
- Pagination controls (Items per page: 10, 25, 50, Previous/Next navigation).
- **Row Elements:**
- Ticket No (clickable link to `/staff/tickets/:id`), Created Date, Summary, Category, Requested Priority badge, IT Priority badge, Status badge, Ticket Owner (User name or `Unassigned` badge).

3.5. IT Staff Ticket Detail Screen (`/staff/tickets/:id`)

- **Header:** Back to Queue link, Ticket Number heading, Status badge, IT Priority badge.
- **Operational Toolbar:**
- `Claim Ticket` button (when unassigned): Instantly sets current IT Staff as owner.
- `Reassign` dropdown/button: Select active IT Staff or Administrator.
- `IT Priority` select control: Adjust priority (`LOW` , `MEDIUM` , `HIGH`).
- `Status Transition` selector: Dropdown offering only permitted next states according to state matrix. If `Resolved` selected, presents modal/drawer for mandatory `resolutionSummary` .
- **Content Panels:**
- Request Details (Requester info, Category, Related System, Dates).
- Summary & Description.
- Attachments section (with download and soft-removal capability).
- **Dual Communication Panel:**
- **Tab 1: Public Comments:** Styled with `#EAF6EF` green tint. Visible to all parties.
- **Tab 2: Internal Notes:** Styled with `#FFF8E1` yellow tint and `#FFA000` border with a bold "Private: Visible only to IT Staff & Admin" lock banner.
- Each tab includes input textarea and append submit button.

3.6. Administrator User Management Screen (`/admin/users`)

- **Toolbar:** Search bar (searches name or email), Role filter dropdown (`All` , `Requester` , `IT Staff` , `Administrator`), `+ Create User` CTA button.
- **User Table (Desktop):**
- Columns: Full Name, Email Address, Role badge, Status badge (`Active` green / `Inactive` gray), Must Change Password indicator, Actions (`Edit` , `Reset Password`).
- **User Card Stack (Mobile):** Responsive card layout preserving all actions.
- **Create User Drawer / Modal:**
- Full Name, Email Address, Role selection (Radio / Dropdown with exactly 1 choice), Initial Password input, Active status toggle.
- **Edit User Modal:**
- Edit Full Name, Email Address, Role selection, Active status toggle.
- Safety Guards: Deactivation toggle and Role change are disabled if editing self or last active Administrator.
- **Reset Password Modal:**
- Input field for new initial password. Confirmation dialog noting user will be forced to change it upon login.

4. Responsive & Accessibility Matrix

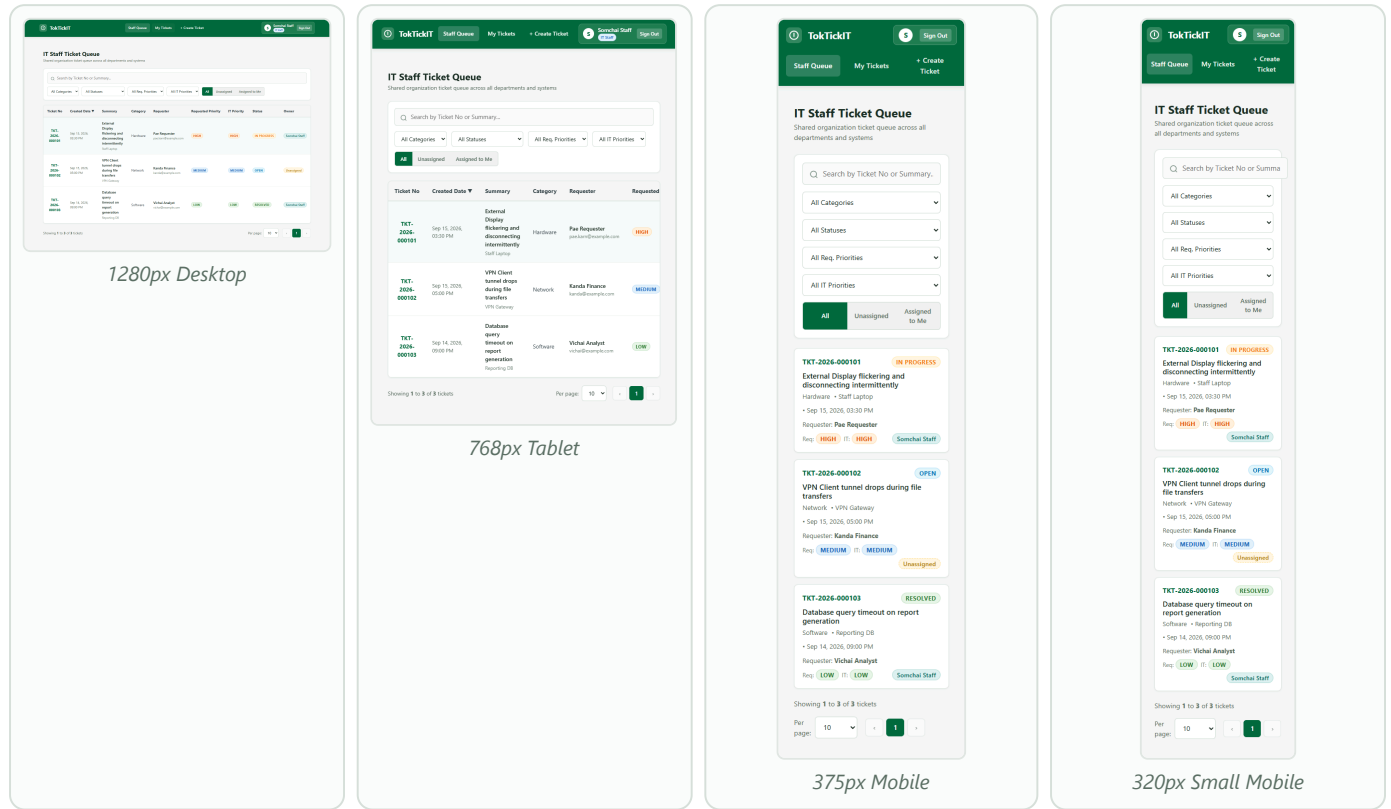
- **Desktop Viewport (≥ 992px):** Multi-column form layouts, full data tables, side-by-side comment/note feeds, max container width 1200px.
- **Tablet Viewport (768px - 991px):** Two-column forms, scrollable data tables, stacked detail sections.
- **Mobile Viewport (< 768px):** Single-column stacked forms, card-based lists replacing tables, minimum touch target height 44px.
- **Small Mobile Viewport (320px):** Tested at 320x568px. Long emails, ticket numbers, and file names break safely with `word-break: break-word` and no horizontal page scroll.
- **Accessibility:**
- Semantic HTML landmarks (
- ,
- ,
- ,
-).

- Form controls have associated `aria-required="true"` .
- Contrast ratios conform to WCAG 2.1 AA ($\geq 4.5:1$ for text).
- Visible green focus outlines (`#0B7A46`) on all interactive controls.

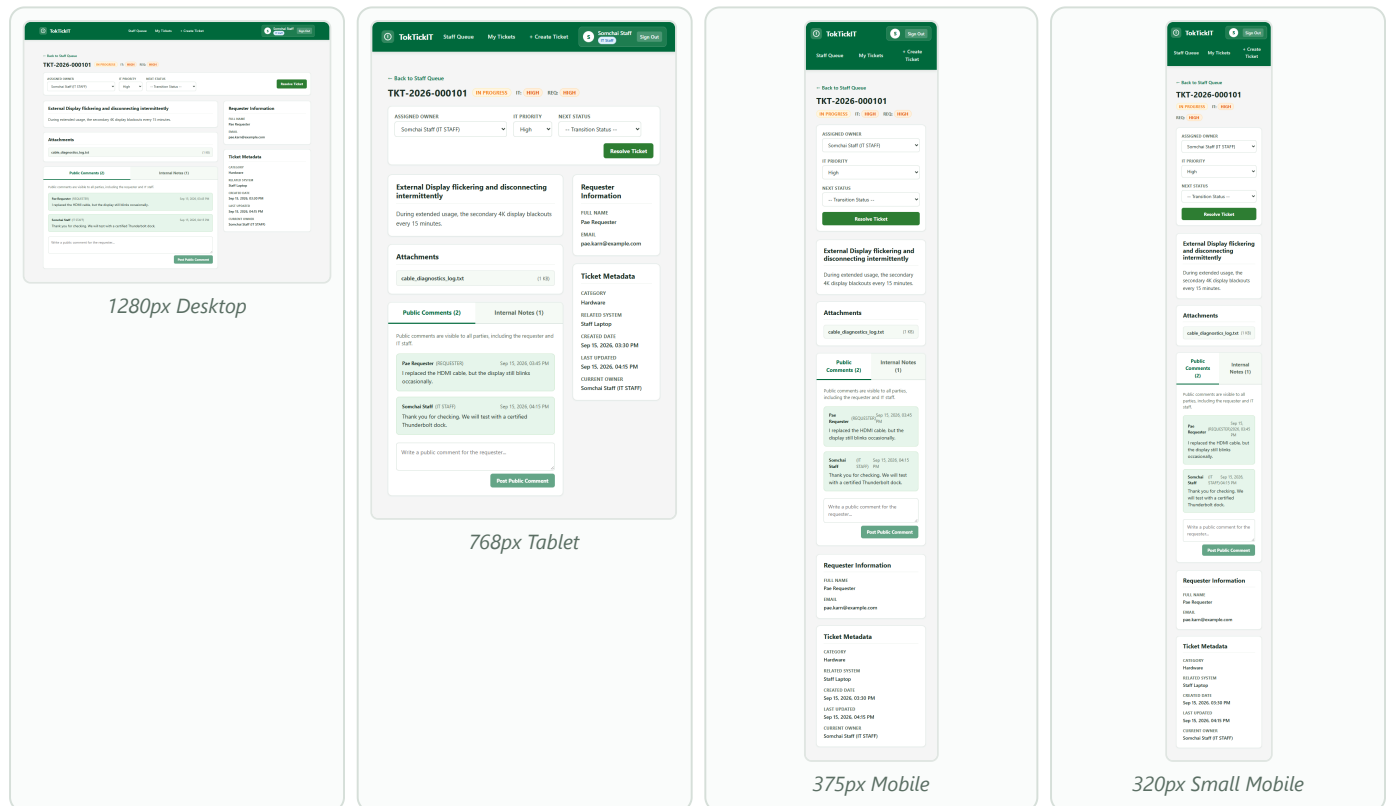
9.2 Responsive Cross-Device Visual Evidence (4 Viewports)

การแสดงผลบนอุปกรณ์ 4 ระดับหน้าจอ: Desktop (1280px), Tablet (768px), Mobile (375px), และ Small Mobile (320px):

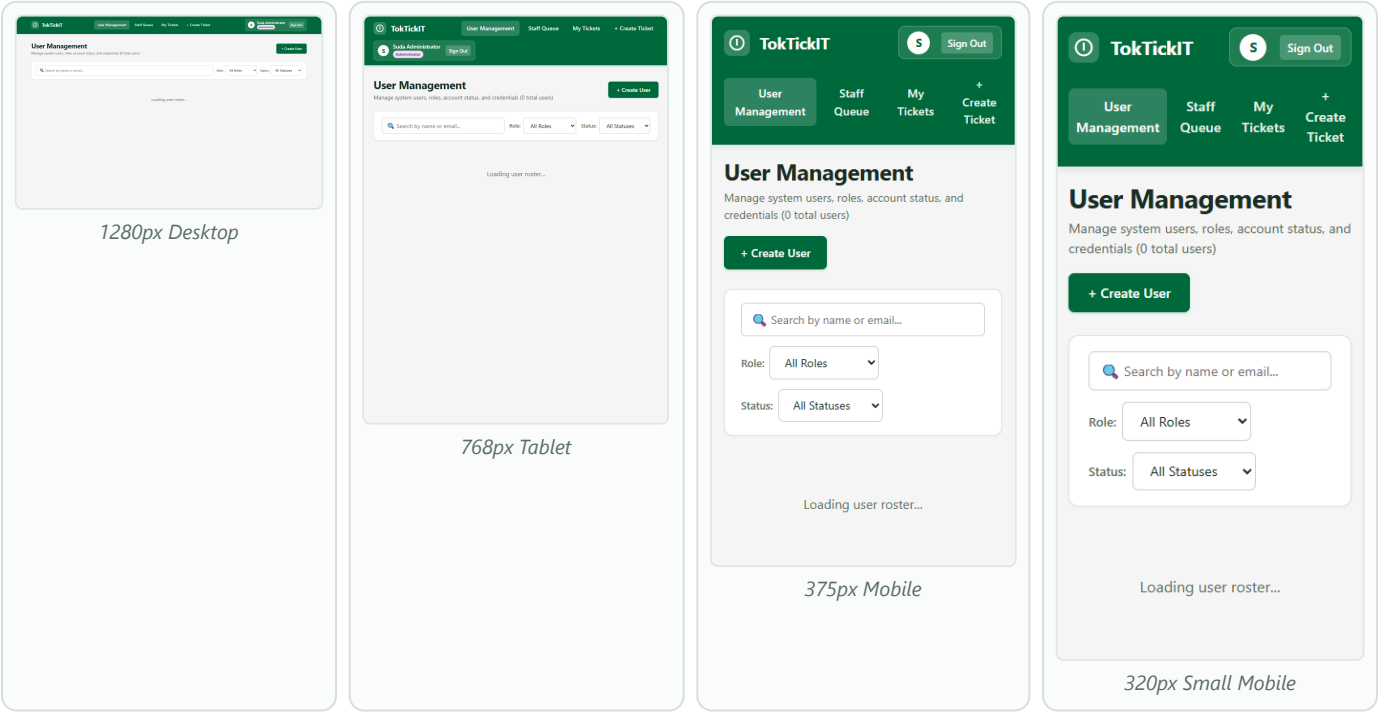
Staff Ticket Queue Responsive Transformation



Staff Ticket Detail Responsive Transformation



Administrator User Management Responsive Transformation



9.3 Completed Visual Inspection Checklist

มิติการตรวจสอบ	เกณฑ์การยอมรับ	Desktop (1280px)	Tablet (768px)	Mobile (375px)	Mobile (320px)
Zen Green Consistency	สีหลัก #006B3C, #0B7A46, พื้นหลัง #F4FAF6	Compliant	Compliant	Compliant	Compliant
Role Navigation	ซ่อนเมนูที่ไม่ได้รับอนุญาต, แสดง Badge บทบาทถูกต้อง	Compliant	Compliant	Compliant	Compliant
Horizontal Overflow	ไม่มีการล้นขอบแนวนอน (Scroll width = Client width)	Compliant (0px)	Compliant (0px)	Compliant (0px)	Compliant (0px)
Touch Targets	ปุ่มกด ≥ 40px, รองรับการแตะสัมผัสบนมือถือสะดวก	Compliant	Compliant	Compliant	Compliant
Focus & Accessibility	แสดงขอบ Outline ชัดเจน และมี Aria-label ครบถ้วน	Accessible	Accessible	Accessible	Accessible